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Complex systems with time-varying interactions

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COMPLEX SYSTEMS WITH TIME-VARYING INTERACTIONS

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DOCTOR BY THE UNIVERSITAT DE LES ILLES BALEARS

WE DECLARE:

That the thesis titled " Complex systems with time-varying interactions ", presented by Annalisa Caligiuri to obtain a doctoral degree, has been completed under our supervision.

For all intents and purposes, we hereby sign this document.

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Palma de Mallorca,

To my family, friends, and
all the people, acts, and serendipities
that helped me through this journey.

Abstract

The nonlinear interactions among components of complex systems give rise to intricate dynamics. Understanding such behaviors requires not only modeling the internal dynamics of the components, but also capturing how their interaction structure evolves over time. This evolution can be random, chaotic, and driven by external environmental factors. Developing tools to characterize such evolving patterns is essential for uncovering the mechanisms that shape complex system behavior.

Several approaches have been proposed to study the temporal evolution of complex systems, including methods from time series analysis, dynamical systems theory, and temporal network models. These tools have advanced our understanding of systems with time-varying interactions. However, many aspects of the dynamics of networks remain poorly understood, and the development of general tools for their analysis is still lacking. Moreover, it is possible that other types of dynamical behaviors are present but remain hidden, as current methods are not yet capable of detecting or characterizing them.

This thesis addresses these challenges by first examining how environmental changes affect the dynamics of complex systems. The analysis identifies the conditions under which temporal variability in the environment influences system behavior, and to what extent these effects manifest in the dynamics. In the second part, a new approach to network dynamics is introduced, based on the assumption that a temporal network can be viewed as a trajectory in graph space. Time series analysis metrics—such as empirical autocorrelation and Lyapunov exponent estimation—are applied to characterize this evolution for both labelled and unlabelled temporal networks. Finally, the proposed framework is applied to empirical brain network data, providing insights into how the predictability of neural dynamics varies across different pathological conditions. These contributions provide a unified framework for analyzing the dynamics of systems with time-varying interactions, offering methodological tools that can be applied across domains where evolving structures play a central role.

Riassunto della tesi

Le interazioni non lineari tra i componenti dei sistemi complessi danno origine a dinamiche intricate. Comprendere tali comportamenti richiede non solo la modellizzazione delle dinamiche interne dei componenti, ma anche la capacità di catturare come la loro struttura di interazione evolve nel tempo. Questa evoluzione può essere casuale, caotica e influenzata da fattori ambientali esterni. Sviluppare strumenti per caratterizzare tali schemi in evoluzione è essenziale per svelare i meccanismi che modellano il comportamento dei sistemi complessi.

Sono stati proposti diversi approcci per studiare l'evoluzione temporale dei sistemi complessi, inclusi metodi dell'analisi delle serie temporali, la teoria dei sistemi dinamici e i modelli di reti temporali. Questi strumenti hanno migliorato la nostra comprensione dei sistemi con interazioni variabili nel tempo. Tuttavia, molti aspetti della dinamica delle reti rimangono poco compresi, e lo sviluppo di strumenti generali per la loro analisi è ancora carente. Inoltre, è possibile che esistano altri tipi di comportamenti dinamici che rimangono nascosti, poiché i metodi attuali non sono ancora in grado di rilevarli o caratterizzarli.

Questa tesi affronta queste sfide esaminando innanzitutto come i cambiamenti ambientali influenzano la dinamica dei sistemi complessi. L'analisi identifica le condizioni in cui la variabilità temporale dell'ambiente influisce sul comportamento del sistema e in quale misura questi effetti si manifestano nella dinamica. Nella seconda parte, viene introdotto un nuovo approccio alla dinamica delle reti, basato sull'assunzione che una rete temporale possa essere vista come una traiettoria nello spazio dei grafi. Metriche dell'analisi delle serie temporali—come l'autocorrelazione empirica e la stima dell'esponente di Lyapunov—vengono applicate per caratterizzare questa evoluzione sia per reti temporali etichettate che non etichettate. Infine, il framework proposto viene applicato a dati empirici di reti cerebrali, fornendo spunti su come la prevedibilità della dinamica neurale vari in diverse condizioni patologiche.

Questi contributi offrono un framework unificato per analizzare la dinamica dei sistemi con interazioni variabili nel tempo, fornendo strumenti metodologici applicabili in diversi ambiti in cui le strutture evolutive svolgono un ruolo centrale.

Resum de la tesi

Les interaccions no lineals entre els components dels sistemes complexos donen lloc a dinàmiques intricades. Entendre aquests comportaments requereix no només modelar la dinàmica interna dels components, sinó també captar com evoluciona la seva estructura d'interacció al llarg del temps. Aquesta evolució pot ser aleatòria, caòtica i estar impulsada per factors ambientals externs. Desenvolupar eines per caracteritzar aquests patrons evolutius és essencial per descobrir els mecanismes que configuren el comportament dels sistemes complexos.

S'han proposat diversos enfocaments per estudiar l'evolució temporal dels sistemes complexos, incloent-hi mètodes d'anàlisi de sèries temporals, teoria de sistemes dinàmics i models de xarxes temporals. Aquestes eines han avançat la nostra comprensió dels sistemes amb interaccions que varien en el temps. No obstant això, molts aspectes de la dinàmica de les xarxes encara són poc coneguts, i el desenvolupament d'eines generals per a la seva anàlisi encara és insuficient. A més, és possible que hi hagi altres tipus de comportaments dinàmics que romanguin ocults, ja que els mètodes actuals encara no poden detectar-los o caracteritzar-los.

Aquesta tesi aborda aquests desafiaments examinant primer com els canvis ambientals afecten la dinàmica dels sistemes complexos. L'anàlisi identifica les condicions en què la variabilitat temporal de l'entorn influeix en el comportament del sistema, i fins a quin punt aquests efectes es manifesten en la dinàmica. A la segona part, s'introdueix un nou enfocament de la dinàmica de xarxes, basat en la hipòtesi que una xarxa temporal es pot veure com una trajectòria a l'espai dels grafs. Es fan servir mètriques d'anàlisi de sèries temporals—com ara l'autocorrelació empírica i l'estimació de l'exponent de Lyapunov—per caracteritzar aquesta evolució tant en xarxes temporals etiquetades com no etiquetades. Finalment, s'aplica el marc proposat a dades empíriques de xarxes cerebrals, proporcionant informació sobre com varia la predictibilitat de la dinàmica neuronal en diferents condicions patològiques.

Aquests resultats ofereixen un marc unificat per analitzar la dinàmica dels sistemes amb interaccions que canvien en el temps, aportant eines metodològiques aplicables a àmbits on les estructures evolutives tenen un paper central.

Resumen de la tesis

Las interacciones no lineales entre los componentes de los sistemas complejos dan lugar a dinámicas intrincadas. Comprender estos comportamientos requiere no solo modelar la dinámica interna de los componentes, sino también captar cómo su estructura de interacción evoluciona con el tiempo. Esta evolución puede ser aleatoria, caótica y estar impulsada por factores ambientales externos. Desarrollar herramientas para caracterizar estos patrones evolutivos es esencial para descubrir los mecanismos que moldean el comportamiento de los sistemas complejos.

Se han propuesto varios enfoques para estudiar la evolución temporal de los sistemas complejos, incluidos métodos del análisis de series temporales, teoría de sistemas dinámicos y modelos de redes temporales. Estas herramientas han mejorado nuestra comprensión de los sistemas con interacciones variables en el tiempo. Sin embargo, muchos aspectos de la dinámica de las redes siguen siendo poco comprendidos, y aún falta el desarrollo de herramientas generales para su análisis. Además, es posible que existan otros tipos de comportamientos dinámicos que permanecen ocultos, ya que los métodos actuales aún no pueden detectarlos o caracterizarlos.

Esta tesis aborda estos desafíos examinando primero cómo los cambios ambientales afectan la dinámica de los sistemas complejos. El análisis identifica las condiciones bajo las cuales la variabilidad temporal del entorno influye en el comportamiento del sistema, y hasta qué punto estos efectos se manifiestan en la dinámica. En la segunda parte, se introduce un nuevo enfoque para la dinámica de redes, basado en la suposición de que una red temporal puede verse como una trayectoria en el espacio de grafos. Se aplican métricas del análisis de series temporales—como la autocorrelación empírica y la estimación del exponente de Lyapunov—para caracterizar esta evolución tanto en redes temporales etiquetadas como no etiquetadas. Finalmente, el marco propuesto se aplica a datos empíricos de redes cerebrales, proporcionando información sobre cómo varía la predictibilidad de la dinámica neuronal en diferentes condiciones patológicas.

Estas contribuciones proporcionan un marco unificado para analizar la dinámica de sistemas con interacciones variables en el tiempo, ofreciendo herramientas metodológicas aplicables en dominios donde las estructuras evolutivas juegan un papel central.

Preface

This thesis is the outcome of a journey of academic growth, nurtured by interactions with researchers of all ages, by conversations about science, and by enriching collaborations. It marks the end of one path and the beginning of another, which I hope will be even more rewarding.

The quotations opening each chapter are drawn from Christopher Nolan's films. Like this thesis, his works explore science-related themes and approach time, space, and interactions in unconventional ways.

Chapter 2 opens with a quote from *Inception*, where dreams serve as a channel to implant ideas into another's mind—an analogy to opinion dynamics, where adoption arises through interaction.

Chapter 3 turns to periodic and chaotic dynamics in networks and the challenge of distinguishing them from randomness. In *Following*, the main character shadows strangers to reconstruct their lives, just as we track system trajectories to uncover whether they reflect randomness or underlying order.

Chapters 4 and 5 address the loss of identity in networks and unpredictability, themes that resonate with *Memento*. As in the film's fragmented memory, systems without identifiable components become difficult to interpret and forecast.

The final quote in the Conclusion is taken from *Dunkirk*, where parallel timelines intersect to create a complex narrative. This mirrors the essence of complex systems, where dynamics emerge from the interplay of entities, their interactions, and their environment.

List of publications

The list of articles on which this thesis is based is presented below.

(i) Caligiuri, Annalisa, and Tobias Galla. "Noisy voter models in switching environments." *Physical Review E* 108, no. 4 (2023): 044301.

(ii) Caligiuri, Annalisa, Victor M. Eguíluz, Leonardo Di Gaetano, Tobias Galla, and Lucas Lacasa. "Lyapunov exponents for temporal networks." *Physical Review E* 107, no. 4 (2023): 044305.

(iii) Caligiuri, Annalisa, Tobias Galla, and Lucas Lacasa. "Characterizing the dynamics of unlabeled temporal networks." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 35.5 (2025).

(iv) Caligiuri, Annalisa, David Papo, Görsev Yener, Bahar Güntekin, Tobias Galla, Lucas Lacasa, and Massimiliano Zanin. "Predictability of temporal network dynamics in normal ageing and brain pathology." *arXiv preprint* arXiv:2502.16557 (2025) (accepted for publication in the Journal of Complex Networks).

Other articles written during the PhD period that are not part of this thesis include:

- Caligiuri, Annalisa, et al. "Time-varying ecological interactions characterise equilibrium and stability." *arXiv preprint* arXiv:2506.22123 (2025).

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Chapter 1

Introduction

"I'm not afraid of death. I'm an old physicist — I'm afraid of time."
Dr. Brand, *Interstellar* (movie)

Time, as perceived at the macroscopic level, flows linearly, resembling a river composed of volatile events. How can such a seemingly simple concept give rise to the diverse dynamics observed in complex systems?

A complex system is commonly described as a collection of numerous interacting components whose nonlinear relationships give rise to emergent behaviors—system-level dynamics that cannot be reduced to or predicted from the properties of the individual parts alone [1–3]. While this is a broad and qualitative characterization, many efforts across disciplines have attempted to formalize the notion of complexity or classify complex systems more precisely. The works cited, among others, propose different frameworks and taxonomies that reflect the diverse phenomena captured under the umbrella of complexity science.

Despite these differences, complex systems often share key features such as *nonlinearity* [4, 5], *emergence* [6], *adaptability* [7], and *self-organization* [8]—properties that distinguish them from merely complicated or large systems. Nonlinearity implies that small perturbations may produce disproportionate effects, such as the cascading failure of a power grid or the rapid spread of disease. Emergence refers to macroscopic behavior that cannot be deduced from the properties of individual parts—ranging from flocking in birds to collective decision-making in societies. Adaptability characterizes systems that change their internal state in response to environmental fluctuations, often through feedback mechanisms that couple internal and external dynamics.

Examples of such systems abound in nature, society, and technology. The human brain is a prototypical complex system, where large-scale patterns of thought and perception arise from dynamic and nonlinear interactions among billions of neurons [9]. Ecosystems—comprising species, resources, and environments—adapt and evolve over time in response to internal dynamics and external shocks, often displaying tipping points and regime shifts [10, 11]. In social systems, collective behaviors such as opinion formation, cooperation, and migration emerge from individual decisions shaped by social networks and environmental cues [12–14]. Urban systems and transportation networks exhibit growth, feedback, and self-organization, producing macroscopic patterns like congestion, zoning, or scaling laws [15, 16]. Financial markets, likewise, are governed by adaptive agents whose decisions interact through evolving networks of transactions and regulations, often resulting in volatility and instability [17, 18]. Finally, engineered systems such as the power grid or the Internet combine intentional design with emergent behaviors, showing sensitivity to local disruptions and large-scale cascading failures [19–21].

When the components of a complex system evolve over time, dynamics become central to understanding the system's behavior [22]. Over the past decades, several approaches have been developed to characterize and model the temporal evolution of complex systems. Tools from dynamical systems theory, stochastic processes, and statistical physics have been adapted to describe how macroscopic behaviors emerge from microscopic rules, especially in systems exhibiting chaos, noise, or memory [23, 24]. In parallel, network science has extended from static to temporal networks, providing a framework to capture systems whose interaction topology evolves over time [25, 26]. This has enabled the study of spreading processes and synchronization on time-varying graphs, as well as the analysis of temporal motifs and adaptive connectivity, across domains including neuroscience, epidemiology, social systems, and infrastructure networks [27].

This thesis contributes to this line of research by exploring a variety of approaches to understanding complex systems with time-varying interactions, ranging from systems with fixed interaction structures exposed to a stochastic environment, to systems in which the structure itself—the network of interactions—evolves over time. These investigations leverage tools from dynamical systems theory, stochastic processes, and network science, contributing both conceptual insights and methodological advances.

The thesis is structured around two main topics. The first part focuses on dynamics taking place on systems in which the underlying interaction network remains fixed, but the external environment changes over time, modulating the frequency and intensity of interactions. In this context, we investigate how temporal variability in the environment—modeled as stochastic switching between discrete states—affects the dynamics of systems such as the noisy voter model. By characterizing the interplay between demographic noise and environmental fluctuations, we uncover rich dynamical behaviors, including shifts in phase transitions and the emergence of multimodal distributions.

The second part of the thesis focuses on the dynamics of the network itself, considering systems in which the pattern of interactions evolves over time, either due to endogenous dynamics or external drivers. In this setting, temporal networks are reinterpreted as trajectories through graph space, and their evolution is analyzed using dynamical observables. Some of these tools, such as temporal autocorrelation functions, have recently been proposed to capture memory and recurrence in temporal networks [28]. Building on these ideas, this thesis introduces a generalization of Lyapunov exponents to the realm of dynamics of temporal networks. We also extend both autocorrelation analysis and sensitivity-to-initial-conditions indicators—originally formulated for labelled networks—to the unlabelled setting, where node identities are hidden or absent. As an application, we characterize differences in the predictability of EEG-derived brain network dynamics across different conditions.

The structure of the thesis is as follows. Chapter 2, lays the theoretical groundwork, covering deterministic and stochastic dynamics, with special emphasis on chaos, Lyapunov exponents, and models of population dynamics in random environments. This chapter also briefly introduces key aspects of network science and temporal networks, providing the formalism needed to study time-varying interaction structures.

In Chapter 3, we examine how external environments that switch between discrete states influence the behavior of systems governed by stochastic dynamics. Focusing on noisy voter models, we investigate how demographic noise and temporal variability combine to produce rich phenomena such as phase transitions, polarization, and multimodal opinion states.

In Chapters 4 and 5, we turn the attention inward, from the environment to the system itself—specifically to the evolution of its interaction structure. Here, temporal networks are reinterpreted as trajectories in graph space. In Chapter 4, we introduce dynamical observables such as autocorrelation and Lyapunov exponents for labelled temporal networks, enabling the quantification of memory and sensitivity to initial conditions in empirical and synthetic data. In Chapter 5, we extend these ideas to the unlabelled setting, where node identities are not preserved.

In Chapter 6, we apply the theoretical framework to a real-world problem: the predictability of brain network dynamics in aging and neurodegenerative disease. By analyzing EEG-derived temporal networks, we quantify the degree to which brain activity becomes less predictable in pathological conditions.

Finally, Chapter 7 contains our main conclusions of this thesis and outlines possible extensions of the presented results, highlighting both methodological developments and new potential applications.

Chapter 2

Methods

"Theory will take you only so far."
Ernest Lawrence, *Oppenheimer* (movie)

2.1 Dynamical systems theory

Dynamical systems theory studies the evolution of sets of variables over time [4, 29, 30]. The types of evolution studied in this field are often deterministic, meaning that, in principle, given an exact initial condition of the system, we can determine its state at any future time step. Formally, given a system whose state is represented by a vector $\mathbf{x}(t) \in \mathbb{R}^d$, its continuous-time evolution in *configuration space* (the set of all possible states the system can occupy) is described by a system of ordinary differential equations (ODEs):

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(\mathbf{x}(t), t). \tag{2.1}$$

where $\mathbf{f}$ is a function governing the evolution of the system's state over time. If the system evolves in discrete time, $t \in \mathbb{N}$, its dynamics are instead given by a map of the form:

$$\mathbf{x}(t+1) = \mathbf{f}(\mathbf{x}(t), t). \tag{2.2}$$

In this thesis, we will mainly analyze maps. These are particularly interesting objects because rich behaviors can arise even in just one dimension [31]. Moreover, we will focus on autonomous maps, i.e., cases where $\mathbf{f}(\mathbf{x}(t))$ does not explicitly depend on time.

A map is defined as linear if $\mathbf{f}(\mathbf{x})$ respects the superposition principle, which states that for any two vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ and any scalars α and β:

$$\mathbf{f}(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha\mathbf{f}(\mathbf{x}_1) + \beta\mathbf{f}(\mathbf{x}_2). \tag{2.3}$$

An essential question in dynamical systems theory is understanding how a map behaves in configuration space. This task is often challenging due to the map's dimensionality and nonlinearity. If the map is linear—*i.e.* the function $\mathbf{f}$ is given by

$$\mathbf{f}(\mathbf{x}) = A\,\mathbf{x}, \tag{2.4}$$

where A is a constant $n \times n$ matrix—then its long-term behavior can be fully characterized by the eigenvalues and eigenvectors of A, which determine the stability and nature of trajectories in configuration space. A general approach is to study the behavior near a map's *fixed points* $\mathbf{x}^*$, obtained as the solutions of:

$$\mathbf{f}(\mathbf{x}^*) = \mathbf{x}^*. \tag{2.5}$$

A fixed point $\mathbf{x}^*$ is said to be *asymptotically stable* if trajectories starting near $\mathbf{x}^*$ not only remain close to it for all future iterations but also converge to the fixed point [32, 33].

Mathematically, a fixed point $\mathbf{x}^*$ is asymptotically stable if:

(i) For every small positive number ε, there exists a $\delta > 0$ such that if $\|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$, then $\|\mathbf{x}(t) - \mathbf{x}^*\| < \varepsilon$ for all future times t.

(ii) Additionally, $\lim_{t\to\infty} \mathbf{x}(t) = \mathbf{x}^*$.

The linear stability analysis provides information about the local behavior near a fixed point by examining the eigenvalues of the Jacobian matrix $\mathcal{J}_{ij} = \frac{\partial f_i}{\partial x_j}\Big|_{\mathbf{x}^*}$. If all the eigenvalues have absolute values strictly less than 1 ($|\lambda_i| < 1$) for discrete maps, the fixed point is asymptotically stable [34]. However, nonlinear terms disregarded in this approach may have a significant effect on the behavior of the map. The *Lyapunov stability criterion* [32, 35] is a general theorem that provides a sufficient condition for a map to be asymptotically stable around a fixed point:

Lyapunov Stability Criterion

Let $\mathbf{x}^*$ be a fixed point of the autonomous discrete-time system

$$\mathbf{x}(t+1) = \mathbf{f}(\mathbf{x}(t)), \tag{2.6}$$

where $\mathbf{f} : D \to \mathbb{R}^n$ is locally Lipschitz, meaning that, locally, nearby points are mapped to nearby outputs, in a domain $D \subset \mathbb{R}^n$, with $\mathbf{x}^* \in D$. Suppose there exists a continuous function $V : D \to \mathbb{R}$ satisfying the following conditions:

- $V(\mathbf{x}^*) = 0$ and $V(\mathbf{x}) > 0$ for all $\mathbf{x} \in D \setminus \{\mathbf{x}^*\}$; that is, V is **positive definite** with respect to $\mathbf{x}^*$.
- $V(\mathbf{f}(\mathbf{x})) - V(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in D$; that is, V is **non-increasing** along trajectories of the system.

Then, the fixed point $\mathbf{x}^*$ is *stable*. Furthermore, if

$$V(\mathbf{f}(\mathbf{x})) - V(\mathbf{x}) < 0 \quad \text{for all } \mathbf{x} \in D \setminus \{\mathbf{x}^*\}, \tag{2.7}$$

then $\mathbf{x}^*$ is *asymptotically stable*. This criterion provides a method to determine stability without explicitly solving the system. The problem now shifts to finding the function $V(\mathbf{x})$, but in general, it provides a systematic approach to assessing stability.

2.1.1 Chaotic dynamics

Asymptotic stability is just one of the possible behaviors a system can exhibit. Even low-dimensional maps can display a wide variety of dynamics, including chaos. In the last century, chaotic dynamics have received significant attention, and several definitions of chaos have been proposed [36–39]. Despite this variety, chaotic systems are generally characterized by aperiodic dynamics and sensitive dependence on initial conditions. We will now provide a brief overview of the first property in the context of chaotic maps and discuss the latter in more detail in the next section.

Aperiodic orbits – A defining feature of chaotic systems is the presence of aperiodic dynamics—trajectories that never settle into a repeating cycle, yet remain bounded and deterministic. Unlike periodic or quasi-periodic behavior, aperiodic orbits do not repeat after any finite number of steps, reflecting the system's intrinsic unpredictability. This irregularity is not due to noise or randomness but arises from the system's own nonlinear structure.

2.1.1.1 Sensitive dependence on initial conditions and Lyapunov exponents

Sensitive dependence on initial conditions is a key property of chaotic systems, often used to determine their chaotic nature. A system is said to exhibit sensitive dependence on initial conditions if two trajectories starting from nearby points diverge exponentially over time. To study the stability of a trajectory starting from $\mathbf{x}(0)$, we can observe the evolution of another trajectory originating nearby at $\mathbf{x}'(0)$.

In chaotic systems, two nearby trajectories typically diverge exponentially over time. This sensitive dependence on initial conditions can be characterized by the expression:

$$\delta(t) = \delta(0)e^{\lambda t} \tag{2.8}$$

where $\delta(t)$ is the distance between trajectories at time t, $\delta(0)$ is their initial separation, and λ is the *Lyapunov exponent*, which quantifies the average rate of divergence.

Since divergence can vary depending on the region of phase space one often distinguishes between *local* and *global* Lyapunov exponents. Locally, the rate of separation may depend on the initial condition $\mathbf{x}_0$. However, under the assumption of *ergodicity*—that is, when time averages over a single long trajectory are equivalent to ensemble averages over the configuration space—the long-time average of these local rates becomes representative of the system as a whole. This leads to the definition of the (global) Lyapunov exponent:

$$\lambda = \lim_{t\to\infty} \lim_{\delta(0)\to 0} \frac{1}{t} \ln\left(\frac{\delta(t)}{\delta(0)}\right). \tag{2.9}$$

For multidimensional systems, there are as many Lyapunov exponents as the dimension of the system, each corresponding to the average exponential rate of divergence or convergence along a particular direction in configuration space.

Among them, the *maximum Lyapunov exponent* plays a central role, as it quantifies the fastest rate of separation between nearby trajectories and is commonly used as a practical indicator of chaos. It is important to note that the Lyapunov spectrum is not invariant under topological conjugacy, as it depends on the metric. However, it is preserved under smooth conjugacies [40, 41]. We clarify this concept below where we define the topological conjugacy. The full spectrum of Lyapunov exponents provides a detailed fingerprint of the system: all-negative exponents imply asymptotic convergence to a fixed point or limit cycle, a single positive exponent indicates chaos, and multiple positive exponents correspond to hyperchaos [42, 43].

For one-dimensional discrete-time ergodic systems, where the evolution is given by $x_{t+1} = f(x_t)$, starting from x_0, the Lyapunov exponent is defined as:

$$\lambda = \lim_{t\to\infty} \frac{1}{t} \sum_{i=0}^{t-1} \ln |f'(x_i)| \tag{2.10}$$

where $f'(x_i)$ is the derivative of the map evaluated at x_i, the points in the orbit. This captures the average rate of separation of nearby trajectories in discrete time, quantifying the system's sensitivity to initial conditions when the dynamics, i.e., the function f, is known.

Topological conjugacy– Let $f : \mathbf{X} \to \mathbf{X}$ and $g : \mathbf{Y} \to \mathbf{Y}$ be two discrete-time dynamical systems defined on topological spaces $\mathbf{X}$ and $\mathbf{Y}$, respectively. The maps

$$\mathbf{x}(t+1) = f(\mathbf{x}(t)) \quad \text{and} \quad \mathbf{y}(t+1) = g(\mathbf{y}(t)) \tag{2.11}$$

are said to be *topologically conjugate* if there exists a homeomorphism $h : \mathbf{X} \to \mathbf{Y}$ (i.e., a continuous, invertible map with a continuous inverse) such that

$$g = h \circ f \circ h^{-1}. \tag{2.12}$$

Topological conjugacy is the natural equivalence relation in topological dynamics, and it preserves all qualitative (topological) aspects of the dynamics, such as the existence and type of fixed points and periodic orbits, topological transitivity, mixing, and the presence of chaos. Intuitively, the homeomorphism h is a smooth change of coordinates that does not alter the temporal structure of the map; it simply re-expresses trajectories in a different coordinate system. That is why the qualitative type of the dynamics and its key properties—such as stability, recurrence, and chaos—are preserved under topological conjugacy. If, in addition, h is a diffeomorphism (i.e., differentiable with differentiable inverse), the systems are said to be *smoothly conjugate* [44]. In this case, differentiable invariants such as Lyapunov exponents are preserved.

2.1.2 The Wolf and Kantz algorithms

When the equations of motion are known, several algorithms exist to compute the spectrum of Lyapunov exponents [40, 45, 46]. At a chosen point in configuration space, select n infinitesimal perturbations that are mutually orthogonal in the tangent space. As the system evolves, these vectors deform into an n-dimensional parallelepiped whose volume changes exponentially in time. The precise rate of that exponential growth or decay is given by the sum of the n largest Lyapunov exponents. For example, the main idea behind the Benettin algorithm [45] is to study the evolution of a parallelepiped of volume V^D, where D is the dimension of the configuration space, in the tangent space, the space spanned by D infinitesimal perturbations to a trajectory. However, in real data, where there is no access to the underlying equations of dynamics, these algorithms cannot be applied.

Wolf algorithm

One of the first examples of algorithms applicable to real data was proposed by Wolf et al. [47]. In this paragraph, we will demonstrate how to compute the largest Lyapunov exponent using this method. To analyze the dynamics of the system from a scalar time series $x(t)$, we reconstruct its state space using the *embedding procedure* [48], a technique grounded in Takens' theorem [49]. This involves transforming the scalar signal into a set of multidimensional vectors that represent the underlying system's trajectory. Specifically, we construct an m-dimensional attractor by forming vectors of the form

$$\mathbf{x}(t) = \{x(t), x(t+l), x(t+2l), \ldots, x(t+(m-1)l)\}, \tag{2.13}$$

where l is the *delay time* and m is the *embedding dimension*. These vectors approximate the system's state at time t, allowing us to reconstruct its dynamics in a higher-dimensional space. We consider the initial point $\mathbf{x}(t_0)$ at time $t = t_0$ and search along the trajectory for the point that minimizes the Euclidean distance to it in the reconstructed m-dimensional space. This point is referred to as the *nearest neighbor*. Their distance at time t_0 will be d_0. At time $t = t_0 + \tau$, the points have evolved such that their distance is d_τ. So the local expansion rate will be equal to

$$\ell(\mathbf{x}(t_0)) = \frac{1}{\tau} \log\left(\frac{d_\tau}{d_0}\right). \tag{2.14}$$

At this point, the new initial condition we consider is $\mathbf{x}(t_0 + \tau)$. We repeat the procedure by selecting a point that minimizes both the distance to $\mathbf{x}(t_0 + \tau)$ the angular separation, that is, the angle between two vectors, relative to the time-τ image of the nearest neighbor of $\mathbf{x}(t_0)$. If the angular separation is not small, the system will need to realign the vector, and the distance may not grow as quickly as expected. This slower growth can lead to an underestimation of the Lyapunov exponent. Iterating the process, the first Lyapunov exponent λ_1 is obtained by averaging the local expansion rate in Eq. 2.14 over the initial points:

$$\lambda_1 = \frac{1}{n} \sum_{w=0}^{n} \ell(\mathbf{x}(t_0 + w\tau)) \tag{2.15}$$

In Fig. 2.1, we reproduce a schematic view of the Wolf algorithm for the derivation of the first and largest Lyapunov exponent. The algorithm can be extended to find the second Lyapunov exponent, λ_2.

A few observations need to be made. Firstly, the time τ should not be too large, as this may lead to an underestimation of the Lyapunov exponents. This happens because, beyond a certain point, the attractor's folding limits the increase in distance, area, and volume. The Wolf algorithm can only detect positive Lyapunov exponents, since in the case of negative exponents, initially close points converge after a few time steps. Moreover, the accuracy of the measure is limited by several factors, primarily the amount of data available and the potential presence of measurement error. If the time series results from the combination of the system's true behavior and measurement inaccuracies, the data may be noisy. This can affect the identification of nearest neighbors, as noise increases the minimum distance between nearby vectors and can obscure the underlying dynamics. For a more detailed discussion on the topic, see [47].

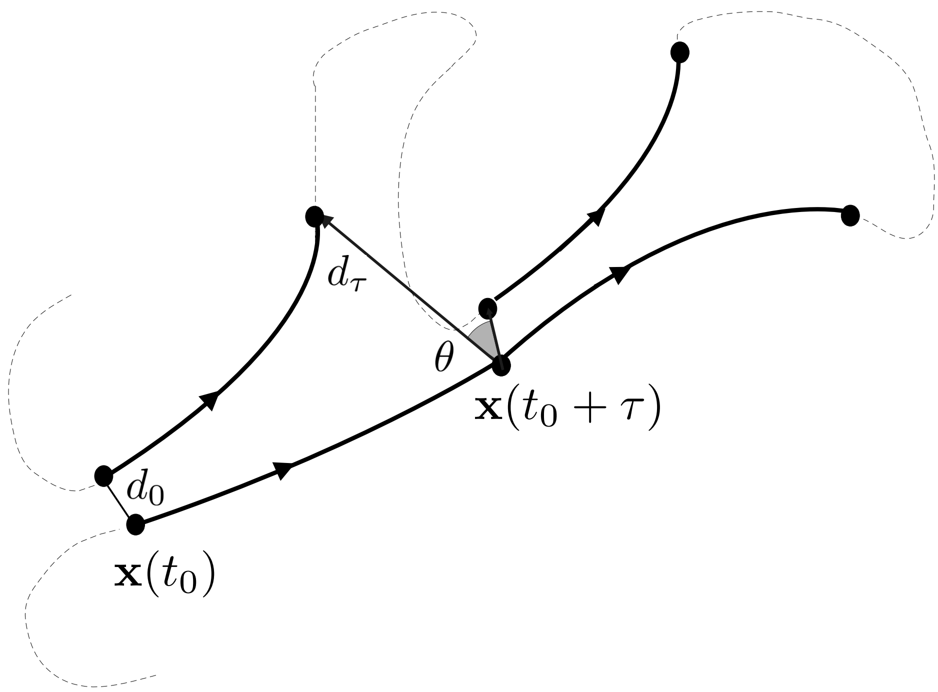

FIGURE 2.1: **Illustration of the Wolf Algorithm for Estimating the Largest Lyapunov Exponent**. The Wolf algorithm estimates the largest Lyapunov exponent by tracking the divergence of nearby trajectories in configuration space. Initially, a point $\mathbf{x}(t_0)$ on the reference trajectory is selected. A nearby trajectory is then chosen to minimize the distance to the reference point. On the next iteration, the reference point evolves to $\mathbf{x}(t_0 + \tau)$, and a new point is selected by examining all points in the time series to minimize both the distance and the angular separation θ relative to the evolved reference point. This process is repeated iteratively, with the divergence of the trajectories over time used to estimate the Lyapunov exponent, which quantifies the rate at which nearby trajectories diverge in chaotic systems.

The Wolf algorithm is the precursor to a series of algorithms introduced to obtain Lyapunov exponents from experimental data. However, some limitations are evident, as we previously mentioned. The limiting factors in computing the Lyapunov exponent using the Wolf algorithm include, first, the need to find a suitable neighbour to the point on the fiducial trajectory. If this is not possible —for example, when the minimization of distance and angular separation, as mentioned earlier, is not optimal—, one has to either accept an inadequete neighbour or discard it, which results in an accumulation of orientation errors. Another drawback is that the resulting Lyapunov exponent is highly dependent on the embedding dimension m.

Other algorithms are based on recreate the dynamics through a fit on the data, though this may be computationally demanding [50]. The flowchart of the Wolf algorith is illustrated in Figure A.1.

Kantz algorithm

Kantz et al. [51] proposed another algorithm to detect the maximum Lyapunov exponent from a time series of scalar data. In the standard procedure for estimating Lyapunov exponents, one typically starts from a one-dimensional observable—such as a scalar time series—which represents a projection of the original multidimensional trajectory. To reconstruct the underlying dynamics, an embedding technique is employed to reconstruct the attractor in a higher-dimensional space using a chosen embedding dimension m. A key improvement introduced by the Kantz algorithm over the Wolf algorithm is that it bypasses the need for embedding the time series into a higher-dimensional space. Unlike traditional approaches that require selecting an embedding dimension to reconstruct the attractor from a scalar time series, the Kantz method operates directly on the scalar observations.

Given a scalar time series x_t, which corresponds to the projection of the full trajectory—composed of points $\mathbf{x}_t$—onto a one-dimensional space, we can define the distance between two initially nearby states after a time τ as:

$$d(\mathbf{x}_t, \mathbf{x}_i, \tau) = |x_{t+\tau} - x_{i+\tau}| \,. \tag{2.16}$$

This distance is calculated between the values of the scalar time series at times $t+\tau$ and $i+\tau$, respectively. Since the algorithm works directly with the observable x_t, which is typically a single component of the full state vector $\mathbf{x}_t$, the quantity $d(\mathbf{x}_t, \mathbf{x}_i, \tau)$ reflects the divergence between two nearby trajectories as observed through their projection onto a one-dimensional subspace, without requiring reconstruction of the full configuration space.

We define U_t to denote an ϵ-neighborhood around the projection of the reference point x_t.

To obtain an estimation of the maximum Lyapunov exponent λ_1, we proceed as follows: we start from a point on the reference trajectory, say x_t, and select all the neighbours falling into U_t. At this point, we compute the average distance, previously defined, between the neighbors and the reference trajectory as they evolve. We then compute the logarithm of the average distance at time τ. We now average over time t, that is over different initial conditions on the reference trajectory, thus we obtain:

$$S(\tau) = \frac{1}{T}\sum_{t=1}^{T} \ln\left(\frac{1}{|U_t|}\sum_{i\in U_t} d(\mathbf{x}_t, \mathbf{x}_i, \tau)\right) . \tag{2.17}$$

For intermediate values of τ, $S(\tau)$ increases with a slope corresponding to the maximal Lyapunov exponent.

In Fig. 2.2, we observe that the distances between the reference trajectory and several initially close trajectories increase exponentially, but only up to a finite time τ. One drawback of this algorithm arises when no points are found within the ϵ-neighborhood of an initial condition. This creates a bias toward denser regions of configuration space, leading to an inaccurate estimation of the Lyapunov exponent.

The main difference compared to the Wolf algorithm here is the averaging over different initial conditions. This results in smaller statistical fluctuations and a faster convergence to the true Lyapunov exponent, which makes it more suitable for small datasets. Figure A.2 shows the flowchart of the Kantz algorithm.

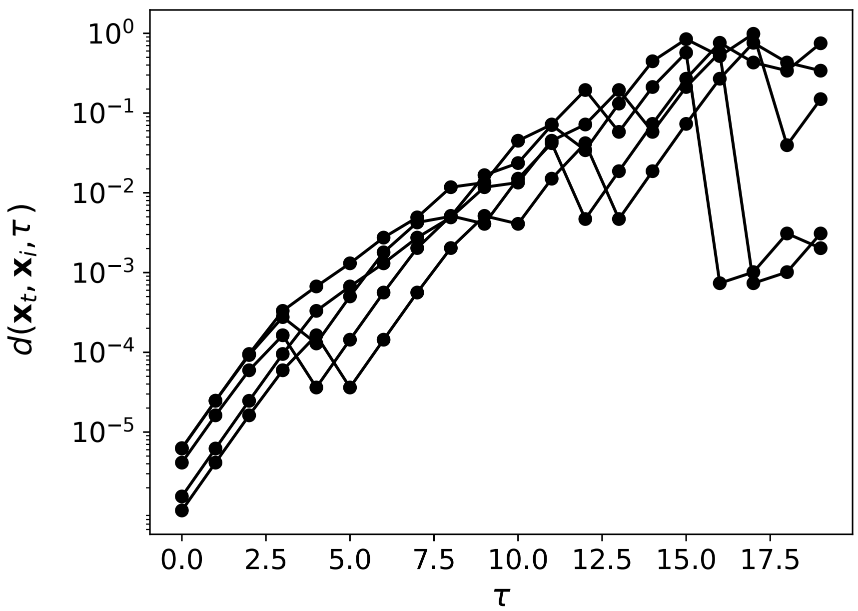

FIGURE 2.2: **Schematic representation of the Kantz algorithm for estimating the largest Lyapunov exponent.** The diagram illustrates the initial distance between a reference trajectory $\mathbf{x}_t$ and its neighbors $\mathbf{x}_i$ within the ϵ-neighborhood at $t = 0$. As time τ increases, the distance $d(\mathbf{x}_t, \mathbf{x}_i, \tau)$ increases exponentially over time, and the logarithm of the average is computed to estimate the maximal Lyapunov exponent.

2.1.3 Examples of chaotic maps

Two classic and well-studied examples of maps exhibiting chaotic behavior are the logistic map and the tent map. In the following paragraph, we explore their dynamics, similarities, and some of the possible routes to chaos.

Logistic map

The *Logistic map* was first introduced in the mid 20th century [52] and later popularized by Robert May in his famous article [53] as a model for population growth. The map is given by the equation:

$$x_{t+1} = rx_t(1 - x_t) \tag{2.18}$$

where:

- x_t represents the population size at time step n,
- r is a constant parameter, called the growth rate or reproduction rate,
- x_{t+1} represents the population size at the next time step,
- The term $(1 - x_t)$ reflects the limiting factor, which reduces growth as the population approaches the environment's carrying capacity (i.e., $x_n = 1$ for normalized populations).

The logistic map is a simple, non-linear recurrence relation. Initially, the population grows exponentially when it is far from the carrying capacity (when x_n is small). However, as the population increases, the term $(1 - x_n)$ becomes significant, slowing growth until it stabilizes when x_n reaches its carrying capacity. The parameter r controls the growth rate: when r is small, the population stabilizes quickly; when r exceeds a certain threshold, the population exhibits chaotic behavior. In more detail, when $r < 1$ the logistic map has just one fixed point $x^* = 0$, that corresponds to the intersection point between the bisecting line and the graph of $f_r(x) = rx(1 - x)$. As $r > 1$, the bisecting line intersects the logistic map at two points, $x_1^* = 0$ and $x_2^* = 1 - \frac{1}{r}$, with the second one being the only asymptotically stable point for $r < 3$. For $r > 3$, as the parameter increases, the logistic map undergoes a period-doubling bifurcation, as we see from *panel b* in Fig. 2.3. In this regime, the system transitions from a stable fixed point to a 2-cycle, then to a 4-cycle, an 8-cycle, and so on. This sequence of period-doubling bifurcations continues as r increases, with each new bifurcation doubling the period of the orbit, until the parameter reaches its critical value $r_\infty \approx 3.56994$, known as the accumulation point. Beyond this point, the logistic map exhibits an alternation of chaotic behavior and periodic windows, continuing up to the maximum possible value $r = 4$, known as the Ulam point, which corresponds to the fully chaotic regime of the logistic map. At this point, the Lyapunov exponent of the map reaches the maximum value possible, equal to $\lambda = \ln 2$, indicating sensitivity to initial conditions.

Tent map

Another widely studied map is the *tent map*:

$$x_{n+1} = \begin{cases} \mu x_n, & 0 \le x_n < 0.5 \\ \mu(1 - x_n), & 0.5 \le x_n \le 1 \end{cases}. \tag{2.19}$$

This is a piecewise-linear map that, as μ increases from 1 to 2, follows a route to chaos (see Fig. 2.4). Although the logistic map at the Ulam point and the tent

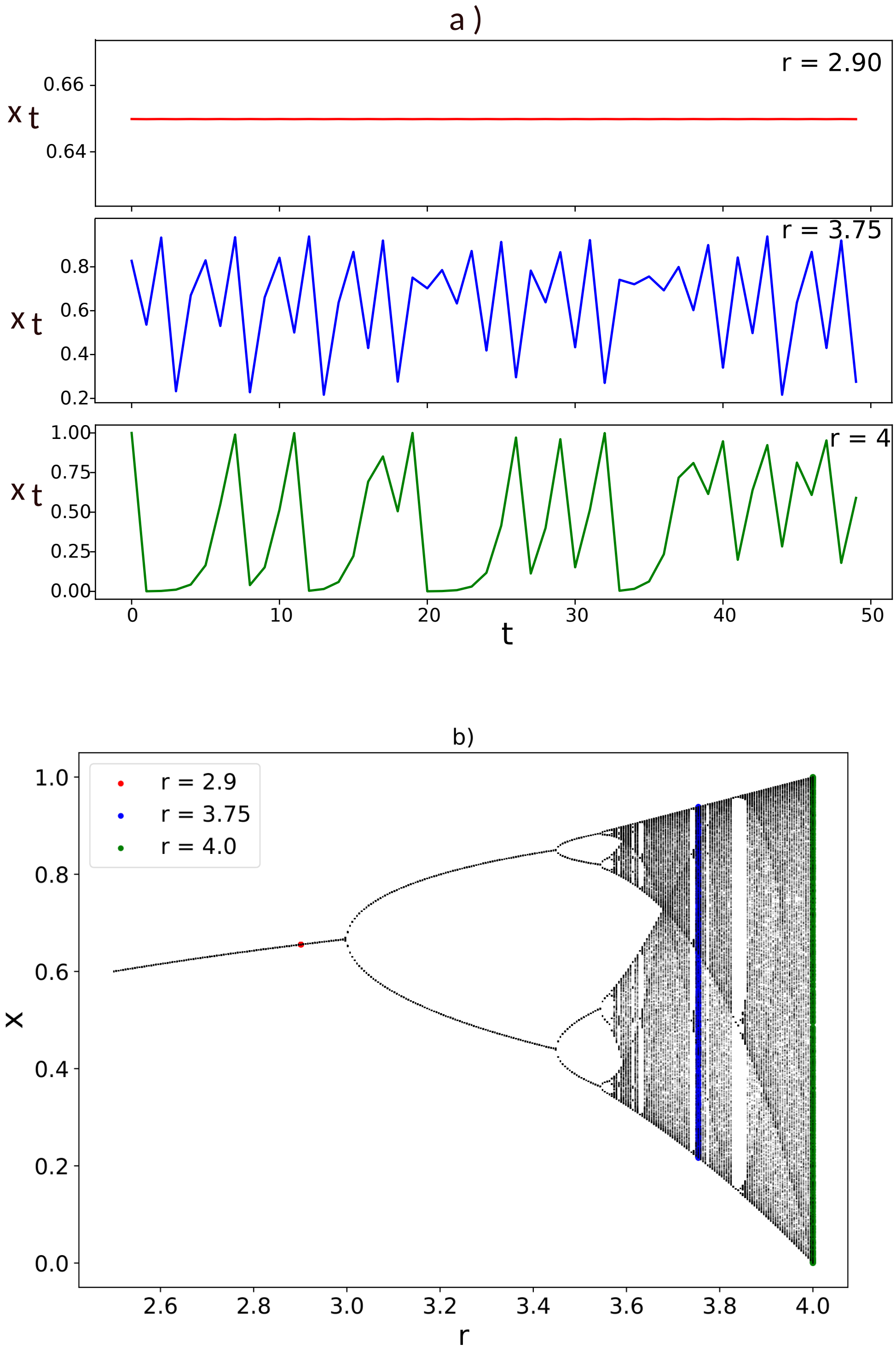

FIGURE 2.3: **Behaviour of the Logistic map**. In panel (a), we show three different trajectories for different values of the parameter r, sampled after a transient. At the top, where $r = 2.9$, the dynamics reach a fixed point equal to $1 - \frac{1}{r} \approx 0.65$; for $r = 3.75$, the dynamic is chaotic, but the configuration space does not yet show the maximum expansion. At $r = 4$ (bottom figure in panel (a)), the Ulam point, the dynamic maps the unit interval onto itself. In *panel (b)*, we show the bifurcation diagram of the logistic map. The bifurcation diagram is built by iterating the logistic map for a range of values of the parameter r. For each r, we discard a transient period to allow the system to settle into its long-term behavior, then plot the successive values of x_n over many iterations. This reveals the fixed points for $r < 3$, the period doubling, and the alternation of chaotic regions and periodic windows for $r > 3.56$. The highlighted points correspond to the trajectories plotted in *panel (a)*.

map for $\mu = 2$ are topologically conjugate, sharing the same Lyapunov exponent, their behavior in reaching that point is slightly different.

Indeed, while the logistic map follows a period-doubling route to chaos, the tent map displays a different route. For $\mu < 1$, the only stable fixed point is at $x = 0$. When $\mu > 1$, the dynamics become more complex: the map begins to stretch and fold the interval, and a subset of the unit interval becomes invariant under the dynamics. More precisely, the system maps a set of disjoint intervals, approximately located between $\mu - \mu^2/2$ and $\mu/2$, into themselves. This means that trajectories, regardless of their initial condition (except for a set of zero measure), will eventually fall into these intervals and remain confined there under iteration. As μ increases, these invariant intervals grow and begin to overlap. When $\mu > \sqrt{2}$, the disjoint intervals start merging. At $\mu = 2$, the map becomes fully chaotic, and the attractor is the entire unit interval, meaning that almost every initial condition will generate an orbit that densely fills $[0, 1]$.

The logistic map at $r = 4$ and the tent map at $\mu = 2$ are topologically conjugate via the homeomorphism

$$h(x) = \frac{2}{\pi} \arcsin\left(\sqrt{x}\right). \tag{2.20}$$

In this special case both maps have Lyapunov exponent $\ln 2$.

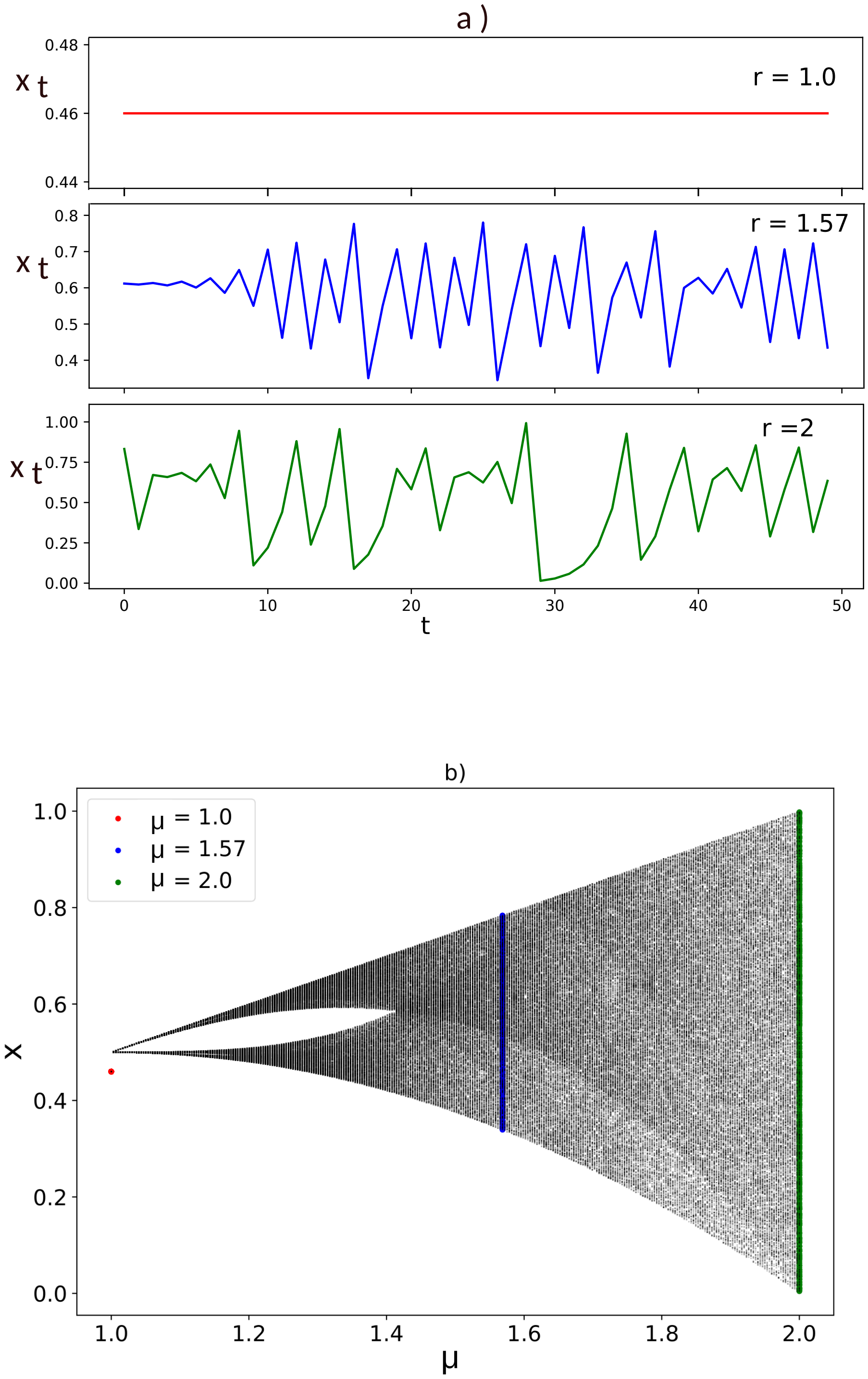

FIGURE 2.4: **Behaviour of the Tent map**. In panel (a), we show three different trajectories for different values of the parameter μ, sampled after a transient. At the top, where $\mu = 1$, every point lower than $x = 0.5$ is a fixed point; for $\mu = 1.57$, the dynamic is chaotic, but the trajectory does not span the full unit interval yet. At $\mu = 2$ (bottom figure in panel (a)), the dynamic maps the unit interval onto itself. In *panel (b)*, we show the bifurcation diagram of the tent map. The bifurcation diagram is built by iterating the tent map for a range of values of the parameter μ. For each μ, we discard a transient period to allow the system to settle into its long-term behavior, then plot the successive values of x_n over many iterations. This reveals the fixed points for $\mu = 1$, the mapping of two sets onto themselves for $\mu < \sqrt{2}$, and chaotic regions. The highlighted points correspond to the trajectories plotted in *panel (a)*.

2.2 Stochastic Dynamics

In Section 2.1, we considered systems that can, in principle, be perfectly predicted. However, in reality, most systems exhibit limited predictability. Chaotic systems, although fully deterministic, are notoriously difficult to forecast due to their sensitive dependence on initial conditions and our inability to measure these conditions precisely, as we saw in Sec. 2.1.1.1. Other systems are only partially predictable, as random fluctuations disturb an otherwise deterministic motion. These systems are known as *stochastic processes* [54–56].

A *stochastic process* is a family of random variables $\{X_t\}_{t\in T}$, where t typically represents time, and T is the index set (for example, $T = \mathbb{N}$ for discrete time or $T = [0, \infty)$ for continuous time).

Each random variable X_t represents the state of the process at time t. The values taken by X_t belong to a state space, which can be real numbers, vectors, or other sets depending on the context. We now focus on a particular and fundamental type of random process known as white noise.

2.2.1 White noise

White noise is a basic example of a stochastic process. In its discrete-time version, it consists of a sequence of random variables $\{X_t\}_{t\in\mathbb{N}_0}$, where each X_t is independent.

More precisely, X_t and X_s are statistically independent for all $t \neq s$, meaning that their joint distribution factorizes: $P(X_t, X_s) = P(X_t)P(X_s)$.

This definition implies that the joint distribution of any finite subset $\{X_{t_1}, X_{t_2}, \ldots, X_{t_n}\}$ factorizes as the product of the individual distributions:

$$P(X_{t_1}, \ldots, X_{t_n}) = \prod_{i=1}^{n} P(X_{t_i}). \tag{2.21}$$

White noise appears in a variety of real-world systems: it is used in music for sound synthesis and audio testing, it represents primordial fluctuations in the cosmic microwave background, and it models background noise in EEG recordings [57].

Further properties and implications of this process will be discussed in later sections.

2.2.2 Master equation

White noise is an example of a memoryless process, where successive values are completely uncorrelated. More generally, stochastic processes may also display temporal dependencies, meaning that the present state is statistically influenced by past states. The master equation provides a general framework to describe the time evolution of such processes—encompassing both memoryless dynamics and those with temporal correlations [54, 58]. It follows the probabilities of a system occupying each discrete state and employs a transition rate matrix to capture how past occupancies influence future behavior.

Mathematically, the master equation for an N-state system reads as:

$$\frac{dP_i(t)}{dt} = \sum_{j=1}^{N} \left[W_{ij}P_j(t) - W_{ji}P_i(t) \right], \quad \text{for } i = 1, 2, \ldots, N, \tag{2.22}$$

where $P_i(t)$ is the probability of being in state i at time t, and W_{ij} is the transition rate from state j to state i. We will now examine a specific example of a process that can be modeled using the master equation: *birth-death process*.

2.2.2.1 Birth–death process

In a birth–death process [59, 60], the system transitions between discrete states $n = 0, 1, 2, \ldots$, each representing the number of individuals in a population. In this paragraph, we consider the case in which the population can reach a maximum capacity of individuals equal to N.

The only allowed transitions are:

- From state n to $n + 1$ (a *birth*), with rate λ_n,
- From state n to $n - 1$ (a *death*), with rate μ_n.

In principle, the transition rates $\lambda_n(t)$ and $\mu_n(t)$ may depend on the current number of individuals n and on time t, allowing the model to incorporate density-dependent or time-varying effects. Moreover, since the system has a maximum capacity of N individuals, we must impose the conditions:

$$\lambda_N(t) = 0, \quad \mu_0(t) = 0 \quad \text{for every } t. \tag{2.23}$$

which ensure that the population cannot grow beyond N or decrease below 0. The master equation for this process, with time-dependent transition rates, reads:

$$\begin{aligned}\frac{dP_n(t)}{dt} &= \lambda_{n-1}(t)P_{n-1}(t) + \mu_{n+1}(t)P_{n+1}(t) - (\lambda_n(t) + \mu_n(t))\, P_n(t), \\ &\text{for } n = 0, 1, \ldots, N.\end{aligned} \tag{2.24}$$

Each term in this equation represents a contribution to the probability flux into or out of state n:

- $\lambda_{n-1}(t)P_{n-1}(t)$: birth from $n-1$ to n,
- $\mu_{n+1}(t)P_{n+1}(t)$: death from $n+1$ to n,
- $-\left(\lambda_n(t) + \mu_n(t)\right) P_n(t)$: departure from n via birth or death.

Boundary conditions:

- $\lambda_{-1}(t)P_{-1}(t) = 0$: no transitions from $n = -1$,
- $\mu_{N+1}(t)P_{N+1}(t) = 0$: no transitions to $n = N + 1$.

When the rates are constant over time, the system forms a finite Markov chain with $N+1$ states, where the probability of each event depends only on the previous state. The master equation thus becomes a finite set of coupled autonomous ordinary differential equations for the probabilities $P_n(t)$.

Steady-state solution for the process with time-independent rates

To obtain the steady-state distribution P_n^{ss}, we use the detailed balance condition:

$$\lambda_{n-1}\, P_{n-1}^{\mathrm{ss}} = \mu_n\, P_n^{\mathrm{ss}}, \quad \text{for } n = 1, 2, \ldots, N. \tag{2.25}$$

This relation follows from the requirement that, in the long-time limit, the net probability current between any two adjacent states must vanish. In one-step birth–death processes with a single discrete degree of freedom and nearest-neighbor transitions, stationarity implies that the current is constant across all links; with absorbing boundaries ($\mu_0 = 0$, $\lambda_N = 0$), this constant must be zero, enforcing detailed balance. This property is special to such one-dimensional processes: in systems with more than one degree of freedom (e.g., three species, corresponding to a two-dimensional state space), stationary states can sustain nonzero probability currents circulating along closed loops, so detailed balance need not hold. Concretely, defining

$$J_n \;\equiv\; \lambda_{n-1}\, P_{n-1}^{\mathrm{ss}} - \mu_n\, P_n^{\mathrm{ss}}, \tag{2.26}$$

the steady–state master equation imposes $J_n - J_{n+1} = 0$ for all n, and the reflecting-boundary conditions $J_0 = J_{N+1} = 0$ then force $J_n = 0$ at every link. Hence each pairwise flux balances as $\lambda_{n-1}P_{n-1}^{\mathrm{ss}} = \mu_n P_n^{\mathrm{ss}}$.

This yields a recursive solution:

$$P_n^{\mathrm{ss}} = \left(\prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k} \right) P_0^{\mathrm{ss}}. \tag{2.27}$$

The normalization condition $\sum_{n=0}^{N} P_n^{\mathrm{ss}} = 1$ gives:

$$P_0^{\mathrm{ss}} = \left[1 + \sum_{n=1}^{N} \prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k} \right]^{-1}. \tag{2.28}$$

Thus, the full steady-state distribution is:

$$P_n^{\mathrm{ss}} = \left(\prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k} \right) \left[1 + \sum_{m=1}^{N} \prod_{k=1}^{m} \frac{\lambda_{k-1}}{\mu_k} \right]^{-1}, \quad \text{for } n = 0, 1, \ldots, N. \tag{2.29}$$

2.2.3 Fokker-Planck equation

Starting from the master equation in Eq. (2.22),

$$\frac{dP_i(t)}{dt} = \sum_{j=1}^{N} \left[W_{ij}P_j(t) - W_{ji}P_i(t) \right], \quad i = 1, 2, \ldots, N, \tag{2.30}$$

the dynamics of a system are described as a jump process between discrete states i, with transition rates W_{ji}. When the state space is large, however, it is convenient to approximate this discrete description by a continuous one. To this end, we associate each state i with a position $x_i = i\Delta x$ on a continuous axis and expand the master equation using the Kramers–Moyal expansion [61, 62].

This expansion rewrites the time evolution of the probability density as an infinite series of spatial derivatives,

$$\frac{\partial P(x,t)}{\partial t} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \frac{\partial^n}{\partial x^n} \left[D^{(n)}(x) P(x,t) \right], \tag{2.31}$$

where the coefficients $D^{(n)}(x)$ are given by the conditional moments of the transition probabilities.

For a process with jumps of length $r\Delta x$, the n-th Kramers–Moyal coefficient reads

$$D^{(n)}(x) = \frac{1}{n!} \sum_r (r\Delta x)^n \, w(x + r\Delta x \mid x), \tag{2.32}$$

or, in discrete notation,

$$D^{(n)}(x_i) = \frac{1}{n!} \sum_j (x_j - x_i)^n \, W_{ji}. \tag{2.33}$$

Truncating the expansion after the second-order term (the *diffusion approximation*) yields the Fokker–Planck equation (FPE) [63]:

$$\frac{\partial P(x,t)}{\partial t} = -\frac{\partial}{\partial x} \left[A(x) P(x,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[B(x) P(x,t) \right], \tag{2.34}$$

with drift and diffusion coefficients

$$A(x) = D^{(1)}(x) = \sum_j (x_j - x) \, W_{j,x}, \qquad B(x) = 2D^{(2)}(x) = \sum_j (x_j - x)^2 \, W_{j,x}. \tag{2.35}$$

Here $A(x)$ captures the mean rate of change (deterministic tendency) and $B(x)$ quantifies the variance of fluctuations. Thus, the master equation → Kramers–Moyal expansion → Fokker–Planck equation provides a systematic path from an exact discrete process to a continuous approximation.

Connection to Stochastic Differential Equations

The Fokker–Planck equation has a trajectory-based counterpart: the Itô stochastic differential equation (SDE),

$$dx = A(x)\,dt + \sqrt{B(x)}\,dW_t, \tag{2.36}$$

where W_t is a *Wiener process* (Brownian motion). This stochastic process is defined by independent Gaussian increments with mean zero and variance equal to Δt,

$$W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t). \tag{2.37}$$

and continuous but nowhere differentiable sample paths. The term $\sqrt{B(x)}\,dW_t$ therefore represents Gaussian white noise with variance proportional to dt.

In this formulation, the deterministic drift $A(x)$ governs the average dynamics, while the Wiener-driven noise encodes random fluctuations. The two descriptions are equivalent: the SDE captures the evolution of individual sample paths, whereas the Fokker–Planck equation describes the time evolution of the probability density over an ensemble of such paths. Here we adopt the Itô interpretation, consistent with the Kramers–Moyal expansion.

Linear-Noise Approximation

Consider a population evolving according to a birth–death process. The state of the system is described by the integer-valued random variable $n(t)$, denoting the number of individuals at time t. The system size Ω sets the natural scale of the population (e.g. the carrying capacity), and it is convenient to introduce the rescaled concentration variable

$$x(t) = \frac{n(t)}{\Omega}.$$

When Ω is large but finite, the dynamics of $x(t)$ can be decomposed into a deterministic trajectory plus fluctuations,

$$x(t) = \phi(t) + \Omega^{-1/2}\,\xi(t), \tag{2.38}$$

where $\phi(t)$ is the deterministic macroscopic concentration (order $\mathcal{O}(\Omega^0)$) and $\xi(t)$ describes the stochastic fluctuations (order $\mathcal{O}(\Omega^0)$, scaled by $\Omega^{-1/2}$).

Expanding the master equation of the birth–death process in powers of $\Omega^{-1/2}$ yields a hierarchy of approximations:

- At order Ω^0: one obtains the deterministic mean-field equation

$$\dot{\phi}(t) = A(\phi(t)), \tag{2.39}$$

  where $A(\phi)$ is the macroscopic drift determined by the birth and death rates.

- At order $\Omega^{-1/2}$: one derives a Fokker–Planck equation for the fluctuations $\xi(t)$,

$$d\xi = J(\phi)\,\xi\,dt + \sqrt{B(\phi)}\,dW_t, \tag{2.40}$$

with $J(\phi) = \partial_x A(\phi)$ the Jacobian of the drift and $B(\phi)$ the noise intensity matrix arising from demographic stochasticity.

In this way, the linear-noise approximation describes the probability distribution of population fluctuations as Gaussian, centered on the deterministic trajectory $\phi(t)$. It thus provides a systematic bridge between the exact master equation of the birth–death process and its deterministic mean-field description, capturing leading-order stochastic corrections in a controlled way. In Chapter 3 we will see how this procedure is used in a concrete example.

Stationary Solutions of the FPE

In the long-time limit, the Fokker–Planck equation may admit stationary solutions $P_{\rm st}(x)$, provided the dynamics are ergodic and probability is confined. Setting $\partial_t P = 0$ reduces the FPE to a conservation law for the probability current,

$$\frac{dJ(x)}{dx} = 0, \qquad J(x) = A(x)P_{\rm st}(x) - \frac{1}{2}\frac{d}{dx}\left[B(x)P_{\rm st}(x)\right], \tag{2.41}$$

implying that in the scalar case the stationary current $J(x) \equiv J^*$ is constant. The general solution then follows as

$$P_{\rm st}(x) = \frac{\exp\left(\int^x \frac{2A(z)}{B(z)}\,dz\right)}{B(x)} \left[C_0 - 2J^* \int^x \exp\left(-\int^y \frac{2A(z)}{B(z)}\,dz\right) dy\right], \tag{2.42}$$

with C_0 fixed by boundary conditions.

Reflecting boundaries

For reflecting boundaries, probability current vanishes ($J^* = 0$), and Eq. (2.42) simplifies to the equilibrium distribution

$$P_{\rm st}(x) = \frac{C}{B(x)} \exp\left(\int^x \frac{2A(y)}{B(y)}\,dy\right), \tag{2.43}$$

where C is a normalization constant.

Absorbing boundaries

At absorbing boundaries, the density vanishes ($P_{\rm st}(x_{\rm b}) = 0$), and unless probability is reinjected, the only stationary state is the trivial $P_{\rm st} \equiv 0$. Non-trivial alternatives include:

- *Quasi-stationary distributions (QSD):* distributions conditioned on survival, governed by an eigenvalue problem.

- *Non-equilibrium steady states with source and sink:* if a source replenishes probability, a steady state with constant current $J^* \neq 0$ is possible, described by Eq. (2.42).

2.3 Populations dynamics in Stochastic Environments

In this section, we examine a class of stochastic models that describe how populations evolve under the influence of a changing environment. Such populations—whether made up of animals, humans, bacteria, cells, genes, ideas, or agents in social or technological systems [64, 65]—are subject not only to intrinsic randomness from processes like birth, death, migration, and competition, but also to extrinsic fluctuations driven by environmental variability [66–68]. These environmental changes can alter interaction rates, resource availability, and survival probabilities, leading to dynamics that reflect both internal stochasticity and the unpredictable nature of the surroundings.

The environment acts at two levels: it alters individual traits such as reproduction or survival, and it modulates interactions by changing contact patterns or competition intensity. These effects are interlinked, as shifts in one level can feed back on the other.

To illustrate these ideas, we consider a population of size N, whose individuals occupy discrete states ℓ, and which interacts with an environment that switches randomly between discrete states σ. The population dynamics depend on the current environmental state σ, while the environment transitions between states at rates that may depend on the population state ℓ. The joint probability $p(\ell, \sigma, t)$ of finding the population in state ℓ and the environment in state σ at time t evolves according to the master equation

$$\frac{d}{dt}p(\ell,\sigma,t) = \sum_{\ell'} R^{(\sigma)}_{\ell' \to \ell} p(\ell',\sigma,t) + \lambda \sum_{\sigma'} A_{\sigma' \to \sigma}(\ell) p(\ell,\sigma',t). \tag{2.44}$$

Here, $R^{(\sigma)}$ governs population transitions under environment σ, $A(\ell)$ governs environment switching which may depend on ℓ, and λ sets the relative timescale of environmental changes. This formulation captures the coupled dynamics of population and environment, allowing analysis of how stochastic environmental fluctuations affect population behavior.

2.3.1 Population dynamics in fast- and slow-switching environments

In systems where both the environment and the system have discrete states and evolve in continuous time, the *adiabatic limit* corresponds to the regime of infinitely fast environmental switching (i.e., $\lambda \to \infty$) [69, 70]. In this limit, the environment is assumed to instantly equilibrate to a stationary distribution $\rho^*(\sigma|\ell)$, conditioned on the current state ℓ of the system. We can factorize the joint probability as $p(\ell, \sigma, t) = \rho^*(\sigma \mid \ell)\, \Pi(\ell, t).$, and the marginal dynamics of the system, $\Pi(\ell, t)$, evolve according to an effective master equation averaged over the environmental fluctuations. This simplifies the analysis but ignores the impact of the environment's finite response time. When the environment switches rapidly but not infinitely fast (large but finite λ), corrections to the adiabatic approximation arise. These can be systematically derived via an expansion in λ^{-1}, leading to *reduced dynamics* that include both leading and sub-leading terms. These corrections capture non-instantaneous feedback effects from the environment and enrich the behavior of the system compared to the ideal adiabatic case, potentially giving rise to memory-like effects or modified transition rates that would not be captured by a naive averaging over environmental states.

In the opposite regime of a *slow environment* (i.e., $\lambda \to 0$), the environment evolves on a much slower timescale than the system. In this limit, the environmental state σ remains effectively constant over the timescale of the system's dynamics. As a result, the system reaches a stationary distribution conditioned on the current environmental state. On longer timescales, as the environment switches between states, the overall distribution $\Pi(\ell, t)$ exhibits a mixture of these "almost stationary" distributions, weighted by the probability of the environment being in each state.

2.3.2 Limit of infinite population: Piecewise-deterministic Markov Process (PDMP)

For intermediate values of the environmental switching rate λ and in the limit of large population size, the dynamics of populations in randomly switching environments can be effectively described by a class of hybrid systems known as *Piecewise-deterministic Markov Processes* (PDMPs) [71, 72]. These processes combine continuous evolution within each environmental state with random jumps between states, offering a natural framework to capture the interplay between internal dynamics and external noise.

Formally, a Piecewise-deterministic Markov Process is defined on a state space composed of a continuous variable x and a discrete variable σ. The evolution of the process proceeds through a combination of deterministic and stochastic dynamics. While the variable $\sigma(t)$ remains constant, $x(t)$ evolves according to a deterministic ordinary differential equation:

$$\dot{x}(t) = \mathbf{F}_\sigma(x(t)). \tag{2.45}$$

At random times, the process undergoes a jump in the *discrete state*, with transition probabilities governed by a rate function $r(\sigma, \sigma' \mid x)$, which may depend on the current value of the *continuous variable*. The actual rate of a jump from σ to σ' is given by $\lambda r(\sigma, \sigma' \mid x)$, where $\lambda > 0$ is a global scaling parameter controlling the frequency of the stochastic transitions.

In this way, the PDMP alternates between deterministic evolution of $x(t)$ under a fixed discrete state, and stochastic jumps in $\sigma(t)$ that modify the governing dynamics. The continuous and discrete variables can be coupled: the discrete state determines the vector field governing the motion of $x(t)$, while the jump rates of $\sigma(t)$ depend on the current value of $x(t)$. Throughout this work, we restrict our attention to time-homogeneous PDMPs, in which both the vector fields $F_\sigma(x)$ and the jump rates $r(\sigma, \sigma' \mid x)$ are independent of time.

The joint evolution of the system's probability density $p_\sigma(x,t)$, which represents the probability of being in discrete state σ and at continuous position x at time t, is governed by the Liouville–Master equation:

$$\begin{aligned}\partial_t p_\sigma(x,t) = & -\partial_t \left[F_\sigma(x)\, p_\sigma(x,t)\right] \\ & + \lambda \sum_{\sigma' \neq \sigma} \left[r(\sigma',\sigma \mid x)\, p_{\sigma'}(x,t) - r(\sigma,\sigma' \mid x)\, p_\sigma(x,t)\right]\end{aligned} \tag{2.46}$$

In the stationary regime, the probability density $p^*_\sigma(x)$ satisfies

$$0 = -\frac{\partial}{\partial x}\left[F_\sigma(x)\, p^*_\sigma(x)\right] + \lambda \sum_{\sigma' \neq \sigma} \left[r(\sigma',\sigma \mid x)\, p^*_{\sigma'}(x) - r(\sigma,\sigma' \mid x)\, p^*_\sigma(x)\right]. \tag{2.47}$$

In the case of two environmental states, $\sigma \in \{0,1\}$, the stationary distribution can be obtained explicitly. Assuming $F_0(x) < 0$ and $F_1(x) > 0$ for $x \in (x^*_0, x^*_1)$, one finds

$$p^*_0(x) = \frac{\mathcal{N}}{-F_0(x)}\, g(x), \qquad p^*_1(x) = \frac{\mathcal{N}}{F_1(x)}\, g(x), \tag{2.48}$$

with

$$g(x) = \exp\left[-\lambda \int^x du \left(\frac{r(0,1 \mid u)}{F_0(u)} + \frac{r(1,0 \mid u)}{F_1(u)}\right)\right], \tag{2.49}$$

and normalization constant $\mathcal{N}$ determined by

$$\int_{x^*_0}^{x^*_1} dx\, \left[p^*_0(x) + p^*_1(x)\right] = 1. \tag{2.50}$$

However, for systems with three or more states, no general analytical method is available [73]. As a result, one must typically rely on numerical simulations or approximations to study the stationary behavior, as we will see in the next chapter.

2.4 Correlations in time series

When we perform a measurement on a system evolving in time, the equations governing its dynamics are often unknown. Usually, during a measurement, we sample values of a quantity —an observable— of the system, from which we obtain a *time series*. From this time series, we aim to extract information about the system—whether its dynamics are deterministic with noise contamination, intrinsically stochastic, or exhibit features like chaotic behavior [74, 75]. In Section 2.1.2, we presented a method for analyzing real-time series with unknown governing equations and identifying chaotic behavior. This section discusses the *autocorrelation function* as a tool for detecting temporal correlations in time series.

2.4.1 Autocorrelation

The autocorrelation function $R(\tau)$ quantifies the correlation between entries of a time series $\mathbf{X}$ separated by a fixed time lag τ. Given a scalar time series X, the empirical autocorrelation at lag τ is defined as

$$\hat{R}(\tau) = \frac{1}{N-\tau}\sum_{t=1}^{N-\tau}(X_t - u_t)(X_{t+\tau} - u_t), \tag{2.51}$$

where u_t is the sample mean. In many contexts, the autocorrelation is normalized by the variance. Here, we refer to the unnormalized version of the autocorrelation. For stationary signals, $u_t = u$ is constant.

The theoretical autocorrelation is

$$R(\tau) = \mathbb{E}[(X_t - \mu)(X_{t+\tau} - \mu)], \tag{2.52}$$

where $\mu = \mathbb{E}[X_t]$ is the expected value. In what follows, we always refer to the empirical autocorrelation.

This definition can be extended to the case when each entry in the time series has multiple components. For example, suppose the time series consists of N time steps, and each entry $\hat{\mathbf{X}}_t \in \mathbb{R}^d$ is a d-dimensional vector. Then, the autocorrelation matrix function is defined as:

$$\hat{\mathbf{R}}(\tau) = \frac{1}{N-\tau}\sum_{t=1}^{N-\tau}(\hat{\mathbf{X}}_t - \hat{\mathbf{u}})(\hat{\mathbf{X}}_{t+\tau} - \hat{\mathbf{u}})^T. \tag{2.53}$$

Here, $\hat{\mathbf{u}}$ represents the mean of the signal over time, i.e., $\hat{\mathbf{u}} = \frac{1}{N}\sum_{t=1}^{N}\hat{\mathbf{X}}_t$, and the process is assumed to be stationary. The result $\hat{\mathbf{R}}(\tau) \in \mathbb{R}^{d\times d}$ is a $d \times d$ matrix capturing the cross-correlations between the d components of the signal at a time lag τ.

The autocorrelation function is a symmetric function, and its maximum value occurs at $\tau = 0$.

2.4.1.1 Properties of the autocorrelation function for specific stochastic processes

As mentioned earlier, the autocorrelation function extracts information about the temporal correlations within a time series. Its behavior at different lag times τ reveals various aspects of the system's nature and dynamics. Next, we illustrate this with examples of autocorrelation applied to specific dynamics.

Autocorrelation for white noise– For a white noise signal (see Section 2.2.1) with standard deviation σ, the autocorrelation function is zero for all nonzero lag times $\tau \neq 0$, and equal to the variance σ^2 at $\tau = 0$. This is because the values X_t are uncorrelated with each other by definition, meaning that $\mathbb{E}[X_t X_{t+\tau}] = 0$ for $\tau \neq 0$. Consequently, in the limit of large N, the empirical autocorrelation also vanishes:

$$R(\tau) = \frac{1}{N-\tau} \sum_{t=1}^{N-\tau} X_t X_{t+\tau} \longrightarrow 0 \quad \text{for } \tau \neq 0. \tag{2.54}$$

At zero lag, however, the autocorrelation reduces to the average of the squared values, which equals the variance:

$$R(0) = \frac{1}{N} \sum_{t=1}^{N} X_t^2 = \sigma^2. \tag{2.55}$$

Therefore, the autocorrelation function of a white noise process is

$$R(\tau) = \begin{cases} \sigma^2, & \tau = 0, \\ 0, & \tau \neq 0. \end{cases} \tag{2.56}$$

A typical behavior of the autocorrelation function for a discrete white noise signal is shown in panel a) of Fig. 2.5.

Autocorrelation for processes with memory– We now consider a class of stochastic processes that exhibit memory over a finite number of past steps. A discrete autoregressive process of memory ρ [76], denoted as DAR(ρ), is described by the equation:

$$X_t = Q_t X_{t-Z_t} + (1 - Q_t) Y_t \tag{2.57}$$

where

- Q_t is a sequence of Bernoulli random variables with parameter p;
- Y_t is a sequence of i.i.d. random variables taking values in a, by definition, discrete subset of the real line with probability distribution $P(Y)$;
- Z_t is a sequence of independent random variables taking values in $\{1, 2, \ldots, \rho\}$ with $P(Z_t = i) = \alpha_i$, where $\alpha_i \geq 0$ and $\sum_{i=1}^{\rho} \alpha_i = 1$.

The autocorrelation function for this process remains constant for $\tau \leq \rho$ and decays exponentially for $\tau > \rho$, as derived in the following paragraph and illustrated in panel d) of Fig. 2.5.

Autocorrelation of DAR(ρ).

Let $\mu = \mathbb{E}[Y_t]$ and define the centered process $X'_t = X_t - \mu$. From

$$X'_t = Q_t\, X'_{t-Z_t} + (1 - Q_t)\,(Y_t - \mu), \tag{2.58}$$

with $Q_t \sim \text{Bernoulli}(p)$ independent of (Y_t, Z_t) and $Z_t \in \{1, \ldots, \rho\}$ with $\mathbb{P}(Z_t = i) = \alpha_i$, we obtain for $\tau \geq 1$

$$\gamma_\tau = R(\tau) = \mathbb{E}\left[Q_t\, X'_{t-Z_t}\, X'_{t-\tau}\right] = p \sum_{i=1}^{\rho} \alpha_i\, \gamma_{\tau-i}, \tag{2.59}$$

where $\gamma_{-\tau} = \gamma_\tau$ and $\gamma_0 = \mathrm{Var}(X'_t)$. Moreover,

$$\gamma_0 = \mathbb{E}\Big[Q_t(X'_{t-Z_t})^2\Big] + \mathbb{E}\Big[(1-Q_t)(Y_t-\mu)^2\Big] = p\,\gamma_0 + (1-p)\,\mathrm{Var}(Y_t), \quad (2.60)$$

so

$$\gamma_0 = \mathrm{Var}(Y_t). \quad (2.61)$$

Hence, the normalized autocorrelation $\varphi_\tau = \gamma_\tau/\gamma_0$ satisfies the Yule–Walker–type recursion

$$\varphi_0 = 1, \qquad \varphi_\tau = p\sum_{i=1}^{\rho} \alpha_i\,\varphi_{\tau-i}, \quad \tau \geq 1. \quad (2.62)$$

Uniform-memory case. If Z_t is uniform on $\{1,\ldots,\rho\}$, i.e. $\alpha_i = 1/\rho$, then (2.62) becomes

$$\varphi_\tau = \frac{p}{\rho}\sum_{i=1}^{\rho} \varphi_{\tau-i}. \quad (2.63)$$

This yields a flat "plateau" for the first ρ lags:

$$\varphi_1 = \varphi_2 = \cdots = \varphi_\rho = \frac{p/\rho}{1-p+p/\rho}. \quad (2.64)$$

For $\tau > \rho$, φ_τ follows the same recursion and is therefore a linear combination of exponentials determined by the characteristic equation

$$\lambda^\rho - \frac{p}{\rho}\big(\lambda^{\rho-1} + \cdots + \lambda + 1\big) = 0, \quad (2.65)$$

so asymptotically

$$\varphi_\tau \sim C\,r_*^\tau, \quad |r_*| < 1, \quad (2.66)$$

i.e. exponential decay.

Special case $\rho = 1$. The model reduces to DAR(1) and (2.62) gives

$$\varphi_\tau = p^\tau. \quad (2.67)$$

These properties match the classic description: a constant autocorrelation up to lag ρ (for uniform α_i), followed by exponential decay [76].

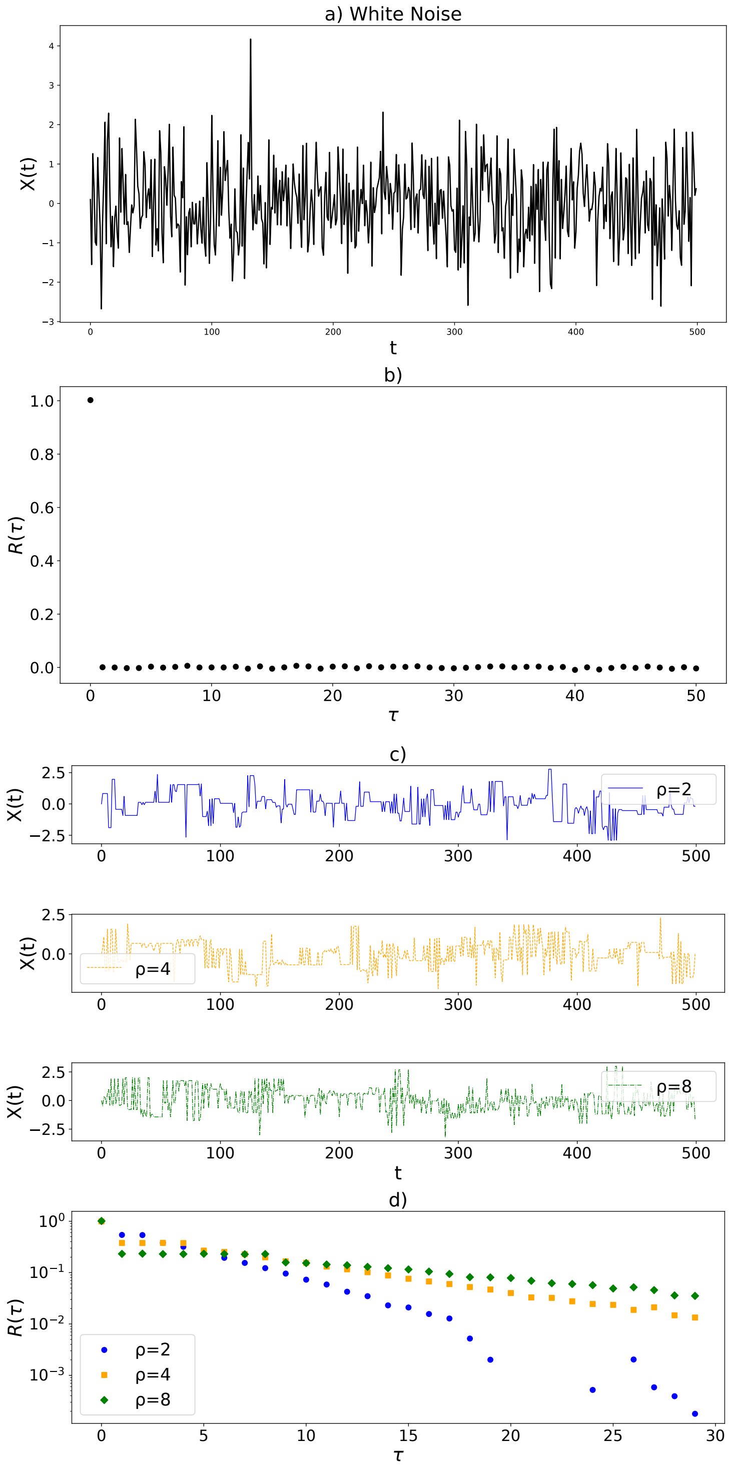

FIGURE 2.5: **Autocorrelation Function of white noise signal and DAR process.** Panel a) shows a time series of white noise generated with a mean of 0 and a standard deviation of 1. Panel b) displays the autocorrelation function of the white noise, demonstrating that it is near zero for all lags τ except at $\tau = 0$, where it equals the variance, indicating the lack of temporal correlation between the values. In panel c) we present time series of DAR processes with different memory lengths, denoted by $\rho = 2, 4, 8$, illustrating the process's dynamic behavior where previous values are probabilistically copied based on the memory length. Panel d) shows the autocorrelation for these processes. While for $1 < \tau < \rho$, the autocorrelation function is constant; the exponential decay becomes visible for $\tau > \rho$, with the memory length ρ influencing the speed of the decay. The autocorrelation is plotted on a logarithmic scale, and different markers and colors represent different values of ρ.

2.5 Network Science: Modeling Interactions in Complex Systems

Network science provides a framework for modeling complex systems composed of numerous interacting elements. By representing these systems as graphs, where nodes represent individual components and edges describe their interactions, we can gain insight into the underlying structures that drive system behavior. This section covers essential concepts in network theory, explores various graph models used to represent real-world systems

2.5.1 Basic concepts

A *network*, or *graph*, is a mathematical structure consisting of objects connected by various types of interactions or relationships [77–79].

Mathematically, a *graph* is represented as $G = (V, E)$, where:

- V is the set of vertices or nodes.
- E is the set of edges (or links), which are pairs of vertices.

Based on the properties of the edges, a graph can belong to different types. In this thesis, we will work with *simple* graphs, which may be weighted or unweighted.

A *simple graph* is an undirected graph containing no graph loops (a link connecting a node to itself) or multiple edges.

A *weighted graph* is a graph in which each edge is assigned a numerical value w, or weight, representing some characteristic of the relationship between the vertices (such as distance, cost, or time).

Other types of graphs include directed graphs, where edges have a direction [80]; and multilayer networks, which consist of multiple interconnected layers, each representing a different type of interaction or context, with node sets that may or may not be the same across layers [81, 82]; and hypergraphs, where a single edge (called a hyperedge) can connect more than two nodes [83, 84].

An important distinction when working with graphs is whether they are considered *labelled* or *unlabelled*. In a labelled graph, each node is assigned a unique identifier, and two graphs are regarded as different if their labellings differ, even if their connectivity patterns are the same. By contrast, in an unlabelled graph, only the structure of connections matters: two graphs that can be transformed into one another by simply relabelling the nodes are treated as equivalent. This distinction is crucial in network analysis, since many structural properties are invariant under relabelling, while others depend explicitly on node identities.

In the rest of this section, we will introduce the basic concepts for *labelled* graphs, which form the standard framework in most of network science. We will return to the distinction between labelled and unlabelled representations, and its implications, in Chapter 5.

An *adjacency matrix* is a 2D array used to represent a labelled simple graph. It is a square matrix $\hat{\mathbf{A}}$ where the rows and columns correspond to the vertices of the graph. The element a_{ij} represents the presence (or absence) of an edge between vertices i and j. In an unweighted graph, a_{ij} is 1 if there is an edge between vertices i and j, and 0 if there is no edge. In a weighted graph, a_{ij} stores the weight of the edge between vertices i and j.

An important characteristic of a node is its *degree*, which represents the number of edges connected to it. Mathematically, for a node v_i, its *degree* d_i is equal to the sum of the elements in the corresponding row or column of the adjacency matrix.

$$d_i = \sum_j a_{ij}. \tag{2.68}$$

In Fig. 2.6, we illustrate these basic concepts.

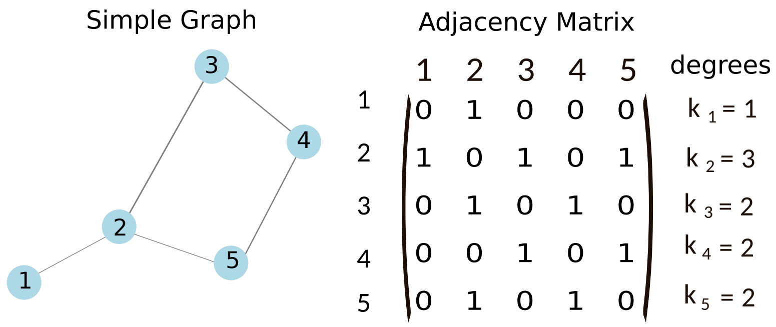

FIGURE 2.6: **Example of a simple graph**. A simple graph with its adjacency matrix and the degree, d_i, for each node. The graph has 5 nodes, and the adjacency matrix displays the connections between the nodes.

2.5.2 Models of graphs

After introducing the basic concepts of graph theory, it is interesting to present models for constructing specific types of networks and analyzing their properties. The straightforward types of graphs are the complete and the empty graphs. A *complete graph* is one in which every pair of nodes is connected by an edge, while an *empty graph* has no edges at all, containing only isolated nodes. These are two extremes of a graph model called a *k-regular* graph, where each node has exactly degree *k*.

These ordered types of graphs are not the most common in reality. Indeed, it is more likely that network models of real systems exhibit some form of probabilistic features, characterizing what are known as *random graphs*.

Erdős–Rényi random graph- The first models of *random graphs* were introduced by Paul Erdős and Alfred Rényi [85]. These models come in two main variants: in the $G(n, M)$ model, a graph is chosen uniformly at random from all graphs with n nodes and M edges, while in the $G(n, p)$ model, each possible edge between n nodes is included independently with probability p. As the number of edges increases and tends to infinity, these two models become increasingly similar. However, the differences at finite n have an impact on their statistical properties. First of all, the number of edges in the $G(n, M)$ model is exactly M, whereas in the $G(n, p)$ model, the number of edges is not fixed. Instead, it follows a binomial distribution with an expected number of edges given by

$$\mathbb{E}[M] = \frac{n(n-1)}{2} p. \tag{2.69}$$

To make the two models more comparable, one could set M in the $G(n, M)$ model to match this expected value. However, a key difference remains: in the $G(n, M)$ model, the edges are not placed independently, leading to correlations between them, whereas in the $G(n, p)$ model, each edge is chosen independently.

2.6 Temporal networks

As mentioned in the introduction, a key feature of complex systems is their evolution over time. Accordingly, the mathematical model representing such systems should be able to capture this temporal aspect. *Temporal networks* are a class of networks whose structure evolves over time—their links undergo a dynamic process that causes them to rewire, appear, or disappear [25, 26, 86]. Depending on the type of process we want to model, we would choose one of the different possible representations.

Contact sequence – A *contact sequence* represents a temporal network as a list of interactions, each described by a triplet (i, j, t), indicating that nodes i and j were connected at time t [87]. This format captures the exact timing of individual events and is well suited for modeling processes involving brief or instantaneous interactions, such as those in email networks.
It is also possible to represent the system as a sequence of static networks, each associated with a time step from the list of times. To each of these networks—often referred to as *snapshots*—one can assign an adjacency matrix $A(t)$ representing the network structure at that specific time.

Interval graphs – In this representation [88], each edge is active over a time interval. Each link is described by a quadruplet $(i, j, t_{\text{start}}, t_{\text{end}})$, meaning that nodes i and j are connected throughout the time interval from t_{start} to t_{end}. Interval graphs are useful when interactions have a duration, such as phone calls, meetings, or physical proximity.

In Fig. 2.7, we illustrate the two types of representations. We will mainly focus on using the first type of representation—the snapshots—while potentially incorporating link persistence by considering their presence across consecutive snapshots.

Temporal networks arise in a variety of real-world systems, each exhibiting characteristic types of dynamics. In email communication networks, interactions are typically bursty [89, 90], with links appearing and disappearing as people exchange messages in short, intense intervals. Air traffic networks display highly regular temporal patterns, with connections forming and dissolving according to daily flight schedules [91]. In brain functional connectivity networks, the links between regions fluctuate rapidly depending on cognitive tasks or resting states, reflecting transient synchronization [92]. Financial systems, such as interbank lending networks, show dynamic link formation that intensifies during periods of market stress, revealing patterns of systemic risk [93]. Similarly, in proximity networks recorded in schools, links reflect face-to-face interactions that follow strong daily and weekly rhythms driven by class schedules[94].

The study of these systems helped shape the field of temporal network analysis. The main techniques developed in this field focus on analyzing the system's dynamics across different scales. The microscopic scale examines links and their persistence or periodicity over time [95]. The mesoscopic scale focuses on the detection of communities in temporal networks [96], while the macroscopic scale analyzes the time evolution of global features of the network, such as mean degree, clustering coefficient, and others [97].

2.6.1 Temporal networks as Network trajectories

Temporal network analysis primarily focuses on extending traditional network science tools to temporal networks—for example, by redefining connectivity in time-evolving systems [98] or studying disease spread across these dynamic structures [99, 100]. A less explored but equally intriguing perspective was introduced by [28] and later expanded in other studies [101–103], where a temporal network is viewed as the trajectory traced by a network in graph space. The mechanism driving this trajectory may be deterministic—described by a dynamical system of graphs, where the network evolves according to a function of its past states—or stochastic, involving random processes such as the rewiring of links. An important aspect of this perspective is that it builds on this interpretation to apply established methods from dynamical systems theory and time series analysis to temporal graphs. This unified approach enables to capture and analyze network properties at various scales (for example, individual connections, community structures, and overall network trends) all within one methodological framework. For example, as we will see, extending the autocorrelation function to temporal networks makes it possible to detect periodicity both at the level of individual links and at a global scale as well [28].

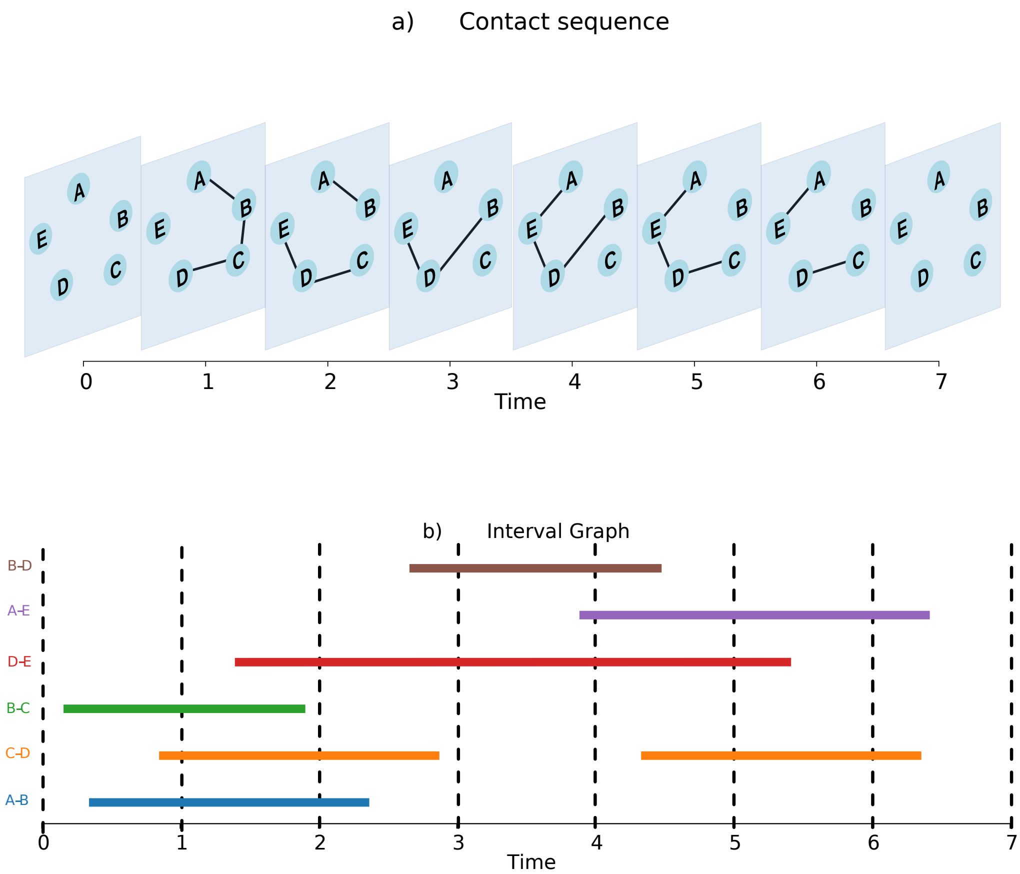

FIGURE 2.7: **Illustration of two common representations of temporal networks.** (a) Contact sequence: each interaction is instantaneous and recorded as a triplet (i, j, t). (b) Interval graph: each interaction spans a time interval and is recorded as a quadruplet $(i, j, t_{\text{start}}, t_{\text{end}})$.

Chapter 3

Complex Systems in Switching Environments

Saito: If you can steal an idea, why can't you plant one there instead?
Arthur: Okay, this is me, planting an idea in your mind. I say: don't think about elephants. What are you thinking about?
Inception (movie)

3.1 Introduction

The dynamics of populations evolving in fluctuating environments are central to a wide range of fields, from ecology and evolutionary biology to social systems and technological adoption. In Section 2.3, we presented a general modeling framework for such systems, in which individuals interact and evolve under transition rules that depend on a randomly switching external environment. This class of models accounts for both internal stochasticity—such as birth, death, imitation, or decision-making—and external variability, such as changes in climate, resources, or social context.

This chapter focuses on a specific application of that framework to voter models, which are widely used to describe opinion dynamics in interacting populations.

We begin with an overview of classical *voter models* and their extensions (Section 3.2), including the *noisy voter model* (nVM) [104–107], which introduces spontaneous opinion changes and exhibits noise-driven phase transitions [108, 109]. We also review the effects of *zealots* and other sources of asymmetry [110, 111].

The core of the chapter focuses on two new model classes presented in Section 3.3:

- **nVM with switching noise-to-herding ratio**: the relative strength of imitation and noise alternates between environmental states, modeling changing levels of conformity or randomness.

- **nVM with switching influencers**: a subset of agents influences the population but changes opinion state over time, introducing time-dependent external bias.

We analyze both models in the *fast- and slow-switching limits* (Sections 3.3.2.1–3.3.2.2), where analytical results can be derived. In the *infinite-population limit*, the dynamics reduce to a *piecewise-deterministic Markov process* (Section 2.3.2). In the case where the environment switches between two states, we derive the exact stationary distribution analytically. For environments with three or more switching states, we solve the stationary distribution numerically using the algorithm presented in Appendix B.

Overall, this chapter illustrates how internal noise, finite-size effects, and structured environmental variability interact in shaping collective outcomes in opinion dynamics models.

3.2 Opinion Dynamics Models

Humans are complex agents whose decision-making processes involve cognition, emotions, memory, and social context. Capturing these processes in full detail—from brain function to collective behavior—remains far beyond current modeling capabilities. However, simplified models have been proposed to study how individual interactions can lead to collective outcomes, such as consensus formation, polarization, or fragmentation in a population [13, 112, 113].

These models fall under the umbrella of *opinion dynamics*, a field that seeks to understand how social influence, randomness, and network structure shape the evolution of opinions over time. Despite their simplicity, opinion dynamics models can reproduce a wide range of emergent behaviors observed in real societies. They typically represent individuals as nodes in a network, each holding a discrete or continuous opinion, and update these opinions through rules inspired by social imitation, bounded confidence, or reinforcement learning [114, 115].

Classic examples include the *voter model*, where agents adopt the opinion of a neighbor; the *majority rule model*, where individuals conform to local majority opinions [116]; and the *bounded confidence models*, where interaction occurs only between agents with sufficiently similar views [117, 118]. Variants introduce noise, external influence, heterogeneous agents, or time-varying network structures to better capture real-world complexities [119]. In the following sections, we review the voter model and its variants, laying the groundwork for the study of noisy voter models in switching environments presented later in this chapter.

3.2.1 Voter Models

The *voter model* [104, 105] is a simple yet powerful framework to describe how individuals influence one another's opinions through social interactions. Each individual in a finite population of size N holds one of two possible opinions, labeled A or B. These individuals are placed on a network, which may be a complete graph or have a more structured topology with local interactions. The dynamics proceed as follows: at each event, an individual is selected at random and adopts the opinion of a randomly chosen neighbor. This rule captures the mechanism of *social imitation*, which is central to many processes of opinion formation.

Originally introduced by probabilists [105], the voter model has found broad applications in various domains, including the modeling of opinion dynamics, language competition, and population genetics [120–126]. It has also been the subject of extensive study in statistical physics, where it serves as a minimal model for systems with coarsening dynamics [127, 128], and where it has inspired field-theoretic treatments and investigations into phase transitions and universality classes [129, 130].

3.2.1.1 Voter Model in well-mixed populations

In well-mixed populations, the state of the system can be described by the number i of individuals holding opinion A; the number of those holding opinion B is then $N - i$. The system evolves through state changes in which individuals may switch from A to B, or vice versa, through different mechanisms. In the basic voter model, the only mechanism is imitation: an individual adopts the opinion of a randomly chosen neighbor.

This imitation process can be formulated as a set of reaction rules. Let A and B denote individuals holding each opinion. Then the elementary update events can be represented as

$$A + B \xrightarrow{\frac{1}{2}} A + A, \quad A + B \xrightarrow{\frac{1}{2}} B + B, \tag{3.1}$$

where in each interaction between an A- and a B-individual, one adopts the other's opinion with equal probability. These reactions preserve the total number of individuals N, but change the count i of individuals in state A by ± 1, depending on the direction of the update.

The rate of each reaction depends on the number of interacting pairs. Assuming a well-mixed population (i.e., a complete graph as the interaction network), the probability of selecting an A-individual and a B-neighbor in a single random draw is $i(N-i)/N^2$. In a continuous-time formulation, we multiply this probability by the total number of update attempts per unit time. Using the standard convention that each of the N individuals has one update attempt on average per unit time, the total attempt rate is N, which removes one factor of N from the denominator. This yields the overall transition rates:

$$T_i^+ = \frac{i(N-i)}{N} \quad \text{(a B-agent copies an A-agent)} \tag{3.2}$$

$$T_i^- = \frac{i(N-i)}{N} \quad \text{(an A-agent copies a B-agent).} \tag{3.3}$$

These expressions follow from the above probabilities by assuming, on average, one interaction attempt per individual per unit time. As a result, the process is symmetric and unbiased: both opinions have an equal chance of eventually taking over the entire population. When the interaction network is connected, the voter model exhibits absorbing states in which all individuals share the same opinion—either full consensus on A or on B. In finite systems, these consensus states are always eventually reached due to stochastic fluctuations. However, in infinite systems, the dynamics and time to reach consensus are strongly dependent on the underlying network topology. On regular low-dimensional lattices (e.g., in $d = 1$ or $d = 2$), the system undergoes a coarsening process, where domains of like-minded agents grow over time and interfaces between them shrink due to interfacial noise [129]. In contrast, for $d > 2$ or in infinite systems, the process may fail to fully order [13]. The behavior of the voter model depends strongly on the interaction network topology. In heterogeneous networks, different update rules lead to different consensus times due to degree asymmetries [131–133]. In small-world networks, long-range shortcuts hinder domain coarsening, leading to metastable states with coexisting opinions; consensus is eventually reached in finite systems, but the disordered state persists in the infinite-size limit [134].

3.2.1.2 Noisy Voter Model

In the standard voter model, agents change their opinion only by copying that of a neighbor with a different one. In the noisy voter model, a variant of the original, agents can also switch opinions randomly, independent of their neighbors [106, 107]. In the noisy voter model, the imitation process is governed by a herding rate h, modifying the reactions to

$$A + B \xrightarrow{h/2} A + A, \quad A + B \xrightarrow{h/2} B + B. \tag{3.4}$$

To this imitation mechanism, the model adds a second process: random opinion changes occurring at a noise rate a, represented by the spontaneous flip reactions

$$A \xrightarrow{a} B, \quad B \xrightarrow{a} A. \tag{3.5}$$

The resulting dynamics combine both imitation and noise, leading to a competition between consensus driven by social influence and disorder introduced by random fluctuations. The interplay between the herding parameter h and the noise parameter a determines whether the system converges to consensus or remains in a fluctuating coexistence of opinions.

Unlike the traditional model, the nVM does not feature absorbing states and instead exhibits a finite-size phase transition [106, 108, 109]. When noise levels are high relative to the strength of imitation, the system stabilizes in a unimodal distribution, reflecting coexistence of opinions. However, if the noise falls below a critical value ($a < h/N$) the opinion distribution becomes bimodal [109]. In this regime, the system typically resides near one of the consensus states, occasionally jumping between them due to stochastic fluctuations.

3.2.1.3 Noisy Voter Model with Zealots

The models discussed so far assume homogeneous populations, where all agents follow the same update rules. This assumption is relaxed in works such as [110, 112], where a subset of agents—known as zealots—are introduced. These agents either maintain a fixed opinion or have a persistent preference for one opinion over the other, regardless of their interactions. The impact of zealots on the voter model has been extensively studied, for instance in [110, 111, 135–137], revealing that their presence can break the symmetry of the steady-state distribution. As a result, the system tends to become biased toward the opinion held by the majority of zealots. In the presence of zealots with a fixed opinion, Eqs. 3.4 and 3.5

are modified to account for the possibility that a voter may also change opinion by copying a zealot of a different opinion. This introduces additional imitation reactions such as:

$$x_A + B \xrightarrow{h} x_A + A, \quad x_B + A \xrightarrow{h} x_B + B, \tag{3.6}$$

where x_A and x_B denote zealot agents holding opinions A and B, respectively. Zealots influence others through imitation but do not change their own opinion, and are not subject to noise.

Let i be the number of non-zealot A-agents in a population of N non-zealot agents, and let Z_A and Z_B be the number of zealots with opinion A and B, respectively. The total population size is therefore $N + Z_A + Z_B$, but only the N non-zealots are subject to opinion change.

In an all-to-all interaction setting, the transition rates for non-zealot agents include both imitation (herding) and spontaneous flip (noise) contributions:

$$T_i^+ = \frac{h(i + Z_A)(N - i)}{N + Z_A + Z_B} + a(N - i) \tag{3.7}$$

$$T_i^- = \frac{h(N - i + Z_B)i}{N + Z_A + Z_B} + ai \tag{3.8}$$

Here:

- The first term in each expression represents imitation: a B-agent may copy a non-zealot or a zealot holding opinion A, and vice versa for an A-agent copying opinion B.
- The second term reflects noise: spontaneous opinion flips, which affect only the non-zealot agents.

The presence of zealots breaks the symmetry of the original model, leading to a bias in the dynamics toward opinion A. As a result, the steady-state distribution shifts away from the symmetric center at $i = N/2$, and the population tends to align more strongly with the dominant zealot opinion.

To capture these effects, the dynamics can be reformulated as a noisy voter model without zealots but with effective parameters:

$$a^{\pm} = a + \frac{h}{2N}(z_A + z_B) \pm \frac{h}{2N}(z_A - z_B), \tag{3.9}$$

$$h_{\text{eff}} = \frac{N}{N + z_A + z_B} h; \tag{3.10}$$

here, the influence of zealots manifests as increased noise parameters and a directional bias favoring the majority opinion among zealots.

The impact on the stationary distribution depends on whether the number of zealots is balanced or unbalanced. In the *balanced case* ($z_A = z_B$), the system remains symmetric, but the stationary distribution becomes narrower and more peaked around coexistence due to the increased effective noise. In the *unbalanced case* ($z_A \neq z_B$), the symmetry is broken: the distribution shifts toward the opinion favored by the majority of zealots. The peak near that opinion becomes more pronounced, and the probability of the population spending time near the opposing state is significantly reduced. Thus, the stationary behavior reflects both the strength and asymmetry of the external bias imposed by the zealots.

3.3 Noisy Voter Models in Switching Environments

In the remainder of this chapter, we investigate how time-dependent features affect the dynamics of the noisy voter model, as presented in [138]. In the framework of statistical physics, this setup belongs to a class of population dynamics influenced by environmental variability, as investigated in works such as [68, 69, 139–146]. Additionally, recent developments on voter models in fluctuating environments can be found in [147], where a three-state constrained voter model under dynamic influence is analyzed. We also point to [148], where a voter model alternates between noisy and noiseless phases. Reference [149] addresses the role of correlated external perturbations in a voter-like system. Specifically, we focus on two distinct types of temporal variability: (i) changes in the relative strength of imitation and spontaneous opinion change, and (ii) time-dependent external influences acting on the population.

More specifically, concerning point (i), we explore variations of the noisy voter model (nVM) in which the ratio between the noise and herding rates changes randomly between two distinct values. This leads to alternating phases: some where the herding mechanism dominates and promotes consensus, and others where the disordering influence of spontaneous opinion changes prevails.

Regarding point (ii), we consider the presence of agent groups that do not respond to the herding dynamics (similar to zealots), but which are able to change their opinion state randomly over time. We term these agents *influencers*. This label is intended broadly: it does not solely refer to individual actors, but encompasses a variety of external influences on standard voter model agents. These may include media, advertising, social platforms, or the emergence of new information and events (such as the revelation of a political scandal) that can sway public opinion. A key aspect of our model is that the influence exerted by these agents is not constant but varies over time.

The aim of this section is to investigate how fluctuations in the relative noise rate and in external influences shape the stationary distribution of opinion shares within a population. Central to this analysis is the interplay between demographic noise (arising from the finite size of the system), decision noise (random individual opinion changes), and time-varying external factors. Understanding how these different sources of randomness interact provides insight into the mechanisms driving collective opinion dynamics under fluctuating conditions.

To address these questions, we use a number of different approaches from statistical physics. In the limit of infinite populations, and thus discarding demographic randomness, the system reduces to a so-called piecewise-deterministic Markov process (PDMP) as introduced in Section 2.3.2. The stationary distribution of such a process can be obtained analytically for the case of two environmental states, see for example [73]. As a by-product of our work we develop a numerical scheme to obtain the stationary state of models with three or more environmental states. Advancing the method of [142] we also compute corrections to the infinite-population limit. This can be used to approximate the stationary distribution of the system with large but finite populations.

Separately, analytical progress is also possible in the the adiabatic limit of fast switching environment [68]. The opposite extreme, very slow environmental changes, can also be addressed analytically.

In the following, we first introduce the model, with particular focus on how environmental fluctuations are implemented (Section 3.3.1). We then outline the analytical methods used to study the system in regimes of very fast or very slow environmental switching, as well as in the large-population limit (Section 3.3.2). Section 3.3.3 presents results for the voter model with a fluctuating noise parameter, including both analytical findings for limiting cases and simulation results for intermediate switching rates. The variant with time-dependent influencers is examined in Section 3.3.4. A summary of the main results and a brief outlook are provided in Section 3.4.

3.3.1 Model definitions

As introduced earlier in the chapter, we consider a finite population of N individuals. At any given time, each individual can be in one of two states, which we label as A and B.

The composition of the population evolves in continuous time via reactions that convert an individual of type A into type B, or vice versa. An individual can change state through three different mechanisms: the already mentioned (i) imitation mechanism and (ii) spontaneous change; and, in addition, in this model we have (iii) interaction with an influencer, which can also lead to an opinion change. The model operates on a fully connected graph, that is any one of the N individuals can copy the state of any other individual in item (i). Similarly, the interaction with the influencers is also all-to-all, in the sense that in item (iii) any influencer can, in principle, affect any of the N individuals in the population.

In order to model processes (i) and (ii) we follow the conventions of existing literature on the nVM [106, 109, 136, 150]. The external influence [process (iii)] is represented by 'forces' driving the individuals towards one of the opinion states. We model these forces as a group of size αN (with $\alpha \geq 0$ a model parameter). We re-iterate that influencers are not necessarily to be thought of individuals, there is therefore no strict need to limit αN to integer values. Instead, α characterises the total strength of all influencers, relative to that of the N agents in the population. Not all influencers need to act towards the same state (A or B). Instead, at any one time a fraction z of the αN influencers acts towards A, and a fraction $1 - z$ acts in the direction of opinion B. Naturally, z is restricted to the interval $[0, 1]$.

We assume that the fraction z fluctuates in time. More precisely, and to allow for a compact notation, we think of the population dynamics as subject to an external environment, which can take states $\sigma = 0, 1, \ldots, S - 1$. This environment determines the fraction z of influencers acting in the direction of opinion A (that is z is a function of σ), and it can also affect the noise rate in the dynamics. We will now describe this in detail.

The per capita rates, in environment σ, for an agent in state A to change to state B and for the reverse process respectively are given by

$$\begin{aligned}\pi_{A\to B,\sigma}(i) &= a_\sigma + h\left[\frac{N-i}{(1+\alpha)N} + \frac{\alpha N(1-z_\sigma)}{(1+\alpha)N}\right],\\ \pi_{B\to A,\sigma}(i) &= a_\sigma + h\left[\frac{i}{(1+\alpha)N} + \frac{\alpha N z_\sigma}{(1+\alpha)N}\right].\end{aligned} \tag{3.11}$$

The quantity a is the rate of spontaneous opinion changes. We assume that this parameter can take different values in the different environmental states, as indicated by the subscript σ. From the above expressions it is clear that only ratio of the noise and the herding parameters is relevant for the stationary state. We can therefore set $h = 1$ throughout. This amounts to fixing the time scales of the processes in Eq. (3.11). For the time being we will keep the value of h general though, as this allows us to track the origin of different terms in the dynamics.

The square brackets in the rates represent processes (i) and (iii) described above. A focal individual chooses an interaction partner either from the population of N agents, or from the set of αN external influencers, and then adopts the opinion of this interaction partner. A change of the composition of the population occurs only if the interaction partners is in the opinion state opposite to that of the focal individual. The expression $(1+\alpha)N$ in the denominator in Eq. (3.11) is the total number of possible interaction partners, hence $(N-i)/[(1+\alpha)N]$ is the probability that the interaction partner is an individual from the population and in opinion state B. Similarly, $\alpha N(1-z_\sigma)/[(1+\alpha)N]$ is the probability that the interaction partner is an external influencer promoting opinion B.

The expressions in Eq. (3.11) are per capita rates. The total rate of converting individuals of type B to type A (or vice versa respectively) in the population are then

$$\begin{aligned}T^+_{i,\sigma} &= (N-i)\pi_{B\to A,\sigma}(i),\\ T^-_{i,\sigma} &= i\pi_{A\to B,\sigma}(i).\end{aligned} \tag{3.12}$$

These are the rates with which transitions $i \to i+1$ and $i \to i-1$ occur in the population if the environment is in state σ.

It remains to specify the dynamics of the environmental state. We assume that the environment undergoes a Markovian process governed by rates $\lambda\mu_{\sigma\to\sigma'}(i)$. The $\mu_{\sigma\to\sigma'}(i)$ are the elements of a stochastic matrix (with $\sum_{\sigma'}\mu_{\sigma\to\sigma'}(i) = 1$ for all σ). We set $\mu_{\sigma\to\sigma}(i) = 0$. In the present work the rates $\mu_{\sigma\to\sigma'}$ do not depend on the state of the population, i. However, to develop the general formalism we will allow for such a dependence in principle whenever possible.

As introduced in Sec. 2.3.1, the parameter λ controls the time scale of the environmental dynamics. The limit $\lambda \to 0$ corresponds to 'slow switching', while $\lambda \to \infty$ corresponds to 'fast switching'.

We write $P(i,\sigma,t)$ for the probability to find the system in state (i,σ) at time t, that is the probability to have i individuals of opinion A in the population, and the environment in state σ. The time dependence of P is omitted below to make the notation more compact. We then have the following master equation

$$\frac{d}{dt}P(i,\sigma) = (E-1)[T^-_{i,\sigma}-1]P(i,\sigma) + (E^{-1}-1)[T^+_{i,\sigma}-1]P(i,\sigma) \\ +\lambda\sum_{\sigma'}[\mu_{\sigma'\to\sigma}(i)P(i,\sigma') - \mu_{\sigma\to\sigma'}(i)P(i,\sigma)], \tag{3.13}$$

where we have defined the raising operator E, acting on functions of i as $Ef(i) = f(i+1)$. Its inverse is E^{-1}, i.e., we have $E^{-1}f(i) = f(i-1)$.

3.3.2 Theoretical Analysis

3.3.2.1 Fast-switching limit

In the regime of extremely rapid environmental switching ($\lambda \to \infty$), the environmental process relaxes much faster than the population dynamics, and can therefore be approximated as being at its stationary distribution. In this limit, the system effectively experiences an averaged environment corresponding to the mean over all possible environmental states σ. We denote this stationary distribution by $\rho^*_\sigma(i)$. It satisfies the condition

$$\sum_{\sigma'}[\mu_{\sigma'\to\sigma}\rho^*_{\sigma'}(i) - \mu_{\sigma\to\sigma'}\rho^*_\sigma(i)] = 0 \tag{3.14}$$

for every σ. Following [69] and what is mentioned in 2.3.1, the probability density of the system can be factorized as

$$P(i,\sigma) = P(i)\rho^*_\sigma(i), \tag{3.15}$$

and, by substituting this into Eq. 3.13, we obtain that the dynamics of the population in the fast-switching limit are governed by the effective rates

$$\overline{T}^\pm_i \equiv \sum_\sigma \rho^*_\sigma(i)\, T^\pm_{i,\sigma}, \tag{3.16}$$

which are simply the average of the original rates over the environmental states at stationarity.

For our system these effective rates are

$$\begin{aligned}\overline{T}^+_i &= \left[\overline{a} + h\frac{i}{(1+\alpha)N} + \frac{\alpha N h\overline{z}}{(1+\alpha)N}\right](N-i) \\ \overline{T}^-_i &= \left[\overline{a} + h\frac{N-i}{(1+\alpha)N} + \frac{\alpha N h(1-\overline{z})}{(1+\alpha)N}\right]i,\end{aligned} \tag{3.17}$$

where we have written

$$\overline{f} = \sum_\sigma \rho^*_\sigma(i) f_\sigma(i). \tag{3.18}$$

We have suppressed the potential i-dependence of objects of this type.

If model parameters are such that $\overline{a} \neq 0$ then there are no absorbing states for this effective birth-death process. The stationary distribution is given by (see e.g. [111]),

$$\overline{P}_i^* = \frac{\prod_{k=1}^{N} \overline{\gamma}_{i-k}}{1 + \sum_{\ell=1}^{N} \sum_{k=1}^{\ell} \overline{\gamma}_k}, \tag{3.19}$$

where $\overline{\gamma}_i = \overline{T}_i^+ / \overline{T}_{i-1}^-$.

3.3.2.2 Slow-switching limit

In the case of slow environmental switching, and assuming that the switching rates $\mu_{\sigma \to \sigma'}$ are independent of i -so that the environment evolves independently of the population and can reach its own stationary distribution-, the stationary distribution can be written as a weighted sum of the stationary distributions $P^*(i|\sigma)$ corresponding to the system in each fixed environment $\sigma \in \{0, 1\}$. These conditional distributions are derived from equations similar to Eq. (3.19), but evaluated in a fixed environment and using the rates $T_{i,\sigma}^{\pm}$ in place of the averaged rates $\overline{T}_i^{\pm}$. We then have

$$P^*(i) = \sum_{\sigma} \rho_{\sigma}^* P^*(i|\sigma). \tag{3.20}$$

3.3.2.3 Rate equations and piecewise-deterministic Markov process

Piecewise-deterministic Markov process in the limit of infinite populations

In the limit of an infinitely large population, demographic stochasticity becomes negligible, and the system's evolution between environmental switches is governed by deterministic dynamics. This results in a piecewise-deterministic Markov process (PDMP), as described previously in Section 2.3.2.

Defining $\phi = i/N$ and $\mathcal{T}_{\sigma}^{\pm}(\phi) = T_{i,\sigma}^{\pm}/N$, and taking the limit $N \to \infty$, the evolution of the system between environmental transitions is described by

$$\dot{\phi} = \mathcal{T}_{\sigma}^{+}(\phi) - \mathcal{T}_{\sigma}^{-}(\phi). \tag{3.21}$$

In our model, this can be rewritten as

$$\dot{\phi} = v_{\sigma}(\phi), \tag{3.22}$$

where

$$v_{\sigma}(\phi) \equiv a_{\sigma}(1 - 2\phi) + \frac{h\alpha}{1+\alpha}(z_{\sigma} - \phi). \tag{3.23}$$

As before, the environmental state σ evolves according to a Markov process with transition rates $\lambda \mu_{\sigma \to \sigma'}$.

Each term in Eq. (3.23) has a clear interpretation when the environment is held fixed. The first term, $a_{\sigma}(1 - 2\phi)$, tends to drive the system toward a balanced state $\phi = 1/2$, corresponding to an equal split between opinions A and B. This term captures random spontaneous opinion shifts, with equal rates in both directions.

In the absence of any external influence, and for $a_\sigma > 0$, this would lead the population to converge to $\phi = 1/2$ in the long run.

The second term in Eq. (3.23) represents the effect of external influencers. A fraction z_σ of influencers promotes opinion A, while the remaining $1 - z_\sigma$ support opinion B. This contribution pushes the system toward $\phi = z_\sigma$, and its strength depends on the herding coefficient h and the factor $\alpha/(1+\alpha)$, which accounts for the relative influence of the αN external agents among the $N + \alpha N$ total interaction partners. When $h\alpha \gg (1+\alpha)a_\sigma$, the influence of external sources dominates, and the spontaneous opinion-change term becomes negligible.

It is also worth noting that peer interactions within the population do not affect the deterministic dynamics described by Eq. (3.23) [13, 109]. This is a characteristic feature of the voter model (VM) and arises from the symmetry in opinion adoption: when individuals with different opinions interact, each is equally likely to adopt the other's view.

The deterministic dynamics in Eq. (3.23) has a unique stable fixed point, given by

$$\phi^*_\sigma = \frac{a_\sigma + h\frac{\alpha}{1+\alpha} z_\sigma}{2a_\sigma + h\frac{\alpha}{1+\alpha}}. \tag{3.24}$$

This fixed point always lies in the interval $[0, 1]$. It is located at the boundaries 0 or 1 only when $a_\sigma = 0$, $\alpha > 0$, and $z_\sigma \in \{0, 1\}$, i.e., when there are no spontaneous changes, a non-zero group of influencers is present, and all of them favor the same opinion.

Throughout most of this work, we exclude the special cases in which distinct environmental states yield the same fixed point. That is, we generally assume $\phi^*_\sigma \neq \phi^*_{\sigma'}$ for $\sigma \neq \sigma'$ and for $\alpha, h \neq 0$. Without loss of generality, we can then label the environmental states such that $\phi^*_0 < \phi^*_1 < \cdots < \phi^*_{S-1}$. Consequently, the dynamics of the PDMP is confined to the interval (ϕ^*_0, ϕ^*_{S-1}), bounded by the smallest and largest fixed points on the ϕ axis.

Stationary distribution

The PDMP defined in Sec. 3.3.2.3, governed by Eqs. (3.22) and the dynamics of the environmental process can be described by the following Liouville-master equation for the probability $\Pi(\phi, \sigma)$ to find the system in state (ϕ, σ),

$$\frac{d}{dt}\Pi(\phi,\sigma) = -\frac{\partial}{\partial\phi}\left[v_\sigma(\phi)\Pi(\phi,\sigma)\right] + \lambda\sum_{\sigma'}[\mu_{\sigma'\to\sigma}(\phi)\Pi(\phi,\sigma') - \mu_{\sigma\to\sigma'}(\phi)\Pi(\phi,\sigma)]. \tag{3.25}$$

In slight abuse of notation we have written $\mu_{\sigma\to\sigma'}(\phi)$ for the transition rates of the environmental process if the population is in state ϕ.

The stationary state of the PDMP is defined by $\frac{d}{dt}\Pi(\phi,\sigma) = 0$ for all ϕ,σ. In this state we have

$$\frac{\partial}{\partial\phi}\left[v_\sigma(\phi)\Pi(\phi,\sigma)\right] = \lambda\sum_{\sigma'}\left[\mu_{\sigma'\to\sigma}(\phi)\Pi(\phi,\sigma') - \mu_{\sigma\to\sigma'}(\phi)\Pi(\phi,\sigma)\right]. \quad (3.26)$$

Special case of two environmental states

If there are only two environmental states, $\sigma \in \{0,1\}$ then the stationary state can be found explicitly [73, 142], and takes the following form,

$$\begin{aligned}\Pi(\phi,0) &= \frac{\mathcal{N}}{-v_0(\phi)}g(\phi),\\ \Pi(\phi,1) &= \frac{\mathcal{N}}{v_1(\phi)}g(\phi),\end{aligned} \quad (3.27)$$

where $\phi \in (\phi_0^*, \phi_1^*)$, and

$$g(\phi) = \exp\left[-\lambda\int^\phi du\left(\frac{\mu_{0\to1}(u)}{v_0(u)} + \frac{\mu_{1\to0}(u)}{v_1(u)}\right)\right]. \quad (3.28)$$

We note that $v_0(\phi) < 0$ and $v_1(\phi) > 0$ for $\phi \in (\phi_0^*, \phi_1^*)$. The constant $\mathcal{N}$ in Eq. (3.27) is determined by normalisation, $\int_{\phi_0^*}^{\phi_1^*} du\,[\Pi(u,0)+\Pi(u,1)] = 1$.

Systems with more than two environmental states

For systems with three or more environmental states we do not know of any method to find the stationary distribution of the resulting PDMP analytically. However, is possible to numerically integrate the system in Eq.(3.26).

To deal with singularities in Eq. (3.26) at the fixed points ϕ_σ^* one can divide the interval $0 < \phi < 1$ into $S-1$ subintervals $\phi_\sigma^* < \phi < \phi_{\sigma+1}^*$ ($\sigma = 0,\dots,S-2$), and perform a numerical integration in each of these intervals. One then needs to ensure continuity of all functions $\Gamma_\sigma(\phi) = v_\sigma(\phi)\Pi(\phi,\sigma)$ at the boundaries. Further details can be found in Appendix B.

3.3.2.4 Leading-order effects of noise

The PDMP framework incorporates environmental noise but neglects intrinsic stochasticity within the population at fixed environmental states. This description becomes exact in the limit of an infinite population size, $N\to\infty$. To study the leading-order effects of intrinsic noise, one can perform a systematic expansion of the master equation (3.13) in powers of $1/N$, following the approach outlined in Sec. **??**.

At leading order, this expansion recovers the PDMP. The next-order correction yields a description in terms of a set of stochastic differential equations with piecewise structure [68, 69, 142]. These take the form $\dot{x} = v_\sigma(x) + \sqrt{w_\sigma(x)/N}\,\eta(t)$, where $\eta(t)$ denotes Gaussian white noise with zero mean and unit variance (i.e., $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$). The noise intensity is governed by the function

$$w_\sigma(x) = \mathcal{T}_\sigma^+(x) + \mathcal{T}_\sigma^-(x), \tag{3.29}$$

as given in [142]. As before, the environmental state σ evolves as a Markov process with transition rates $\lambda\mu_{\sigma\to\sigma'}$.

To go further, one can apply the linear-noise approximation, as detailed in Sec. ??. This involves expressing the system's state as $i/N = \phi + N^{-1/2}\xi$, where ϕ follows the deterministic PDMP trajectory for a given realization of the environmental process (i.e., $\dot{\phi} = v_\sigma(\phi)$). Expanding to linear order in ξ leads to the equation

$$\dot{\xi}(t) = v'_\sigma(\phi)\,\xi + \sqrt{w_\sigma(\phi)}\,\zeta(t), \tag{3.30}$$

where $\zeta(t)$ is again white Gaussian noise, and $v'_\sigma(\phi) = dv_\sigma(\phi)/d\phi$.

An approximation for the stationary distribution of the original system described by Eq. (3.13) is then given by

$$\Pi(x) = \sum_\sigma \int d\phi\, d\xi\, \left[\Pi(\phi,\sigma)\,\Pi(\xi|\phi)\,\delta\big(x-\phi-N^{-1/2}\xi\big)\right], \tag{3.31}$$

where $\Pi(\phi,\sigma)$ is the stationary distribution of the PDMP, and $\Pi(\xi|\phi)$ is a Gaussian distribution with zero mean and variance $s^2(\phi)$, given by

$$s^2(\phi) = -\frac{1}{2}\frac{\sum_\sigma \Pi(\sigma|\phi)\,w_\sigma(\phi)}{\sum_\sigma \Pi(\sigma|\phi)\,v'_\sigma(\phi)}. \tag{3.32}$$

This expression was derived in [142] for systems with two environmental states, but it applies more generally, as discussed in Appendix B.4.

3.3.3 Noisy voter model with switching noise parameter

3.3.3.1 Setup

In this section, we will examine the simple case of $\alpha = 0$, i.e., the system is not affected by any influencers. The rates in environmental state σ are then

$$\begin{aligned} T^+_{i,\sigma} &= \left(a_\sigma + h\frac{i}{N}\right)(N-i), \\ T^-_{i,\sigma} &= \left(a_\sigma + h\frac{N-i}{N}\right)i. \end{aligned} \tag{3.33}$$

Despite the absence of influencers, the system operates within a switching environment, as the noise parameter a_σ fluctuates in time. We study the case of two environmental states $\sigma = 0, 1$. We label the states such that $a_0 < a_1$. The rates for the environmental switches in our analysis are assumed not to depend on the population state i. Therefore, the stationary distribution for the environmental state σ will be simply

$$\rho_0^* = \frac{\mu_{1\to 0}}{\mu_{1\to 0} + \mu_{0\to 1}}, \quad \rho_1^* = \frac{\mu_{0\to 1}}{\mu_{1\to 0} + \mu_{0\to 1}}. \tag{3.34}$$

Throughout our analysis, we assume $\mu_{0\to 1} = \mu_{1\to 0}$, and consequently we have $\rho_0^* = \rho_1^* = 1/2$.

We will first investigate the slow-switching and fast-switching limits. The total rate for events in the population in environment σ is $T_{i,\sigma}^+ + T_{i,\sigma}^- = a_\sigma N + 2hi(N-i)/N$, and therefore takes values between $a_\sigma N$ and $(a_\sigma + h/2)N$. The environment can therefore be considered slow when $\lambda \ll a_0 N$. Similarly, the environment is fast relative to the population when $\lambda \gg N(a_1 + h/2)$.

3.3.3.2 Slow-switching limit

Figure 3.1 shows the stationary distribution in the slow-switching regime, comparing theoretical predictions with simulation results across different population sizes N.

We identify three distinct distribution shapes. For small populations (e.g., $N = 15$), the distribution is bimodal, with peaks at the consensus states, indicating a tendency toward full polarization. In contrast, for large populations (e.g., $N = 55$), the distribution becomes unimodal and centered, suggesting that the system predominantly resides in states where both opinions coexist in similar proportions.

Up to this point, the behavior resembles that of the standard two-state voter model, where one typically observes a transition from bimodal distributions at small N to unimodal ones as N increases [109]. However, our model reveals an additional intermediate phase not present in the conventional voter model. At moderate population sizes (e.g., $N = 40$), the stationary distribution becomes trimodal, featuring two peaks at the extremes and a third one at the center. This pattern corresponds to a dynamical regime in which the system alternates between periods of coexistence and periods of strong polarization.

This intermittent behavior is further illustrated by the time series in Fig. 3.2. In broad terms, it reflects a societal context in which public opinion typically includes both viewpoints, but where certain events can transiently amplify the influence of herding relative to random fluctuations, leading to episodes of polarization.

3.3.3.3 Fast-switching limit

In the limit of fast environmental switching we have effective transition rates

$$\begin{aligned} \overline{T}_i^+ &= \left(\overline{a} + h\frac{i}{N}\right)(N-i), \\ \overline{T}_i^- &= \left(\overline{a} + h\frac{N-i}{N}\right) i \end{aligned} \tag{3.35}$$

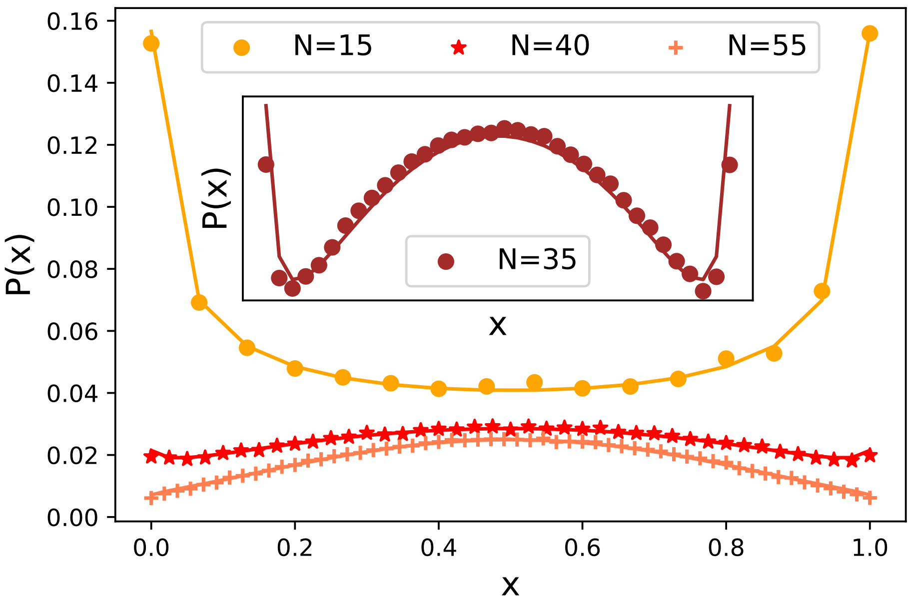

FIGURE 3.1: **Voter model with slow-switching noise rate.** Stationary distribution from simulations (symbols) and from theory [lines, from Eq. (3.20)]. Model parameters are $a_0 = 0.02$, $a_1 = 0.05$, $h = 1$ and $\lambda = 0.02$. In the inset, we highlight the new trimodal shape ($N = 35$). Each distribution is from 10^6 entries sampled every 50 units of time, after an initial transient of 1000 units of time

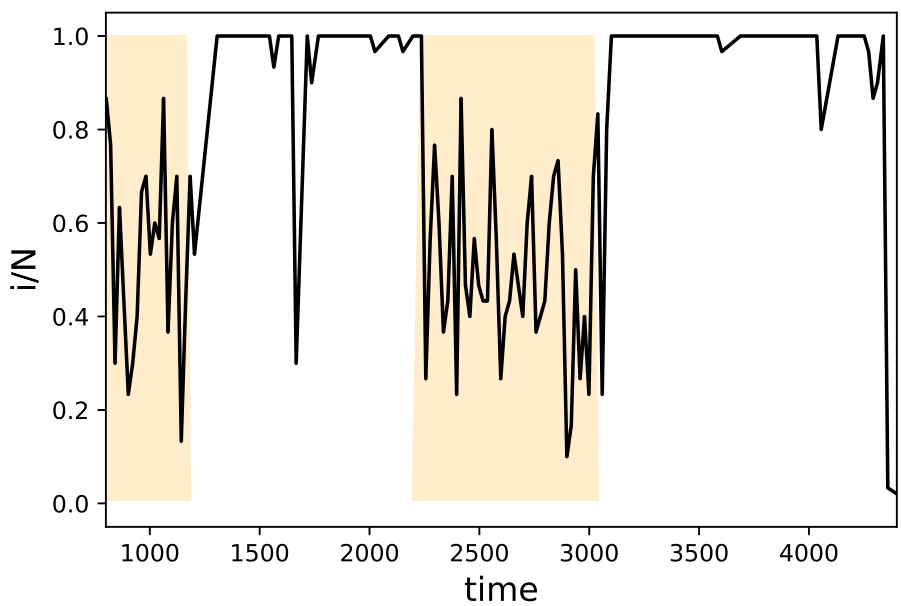

FIGURE 3.2: **Time series of the fraction of agents in state A from a simulation of the voter model with switching noise parameter**. Shaded segments indicate high noise rate, white background low noise rate. Model parameters are $a_0 = 0.001$, $a_1 = 0.1$, $h = 1$, $\lambda = 0.001$ and $N = 30$.

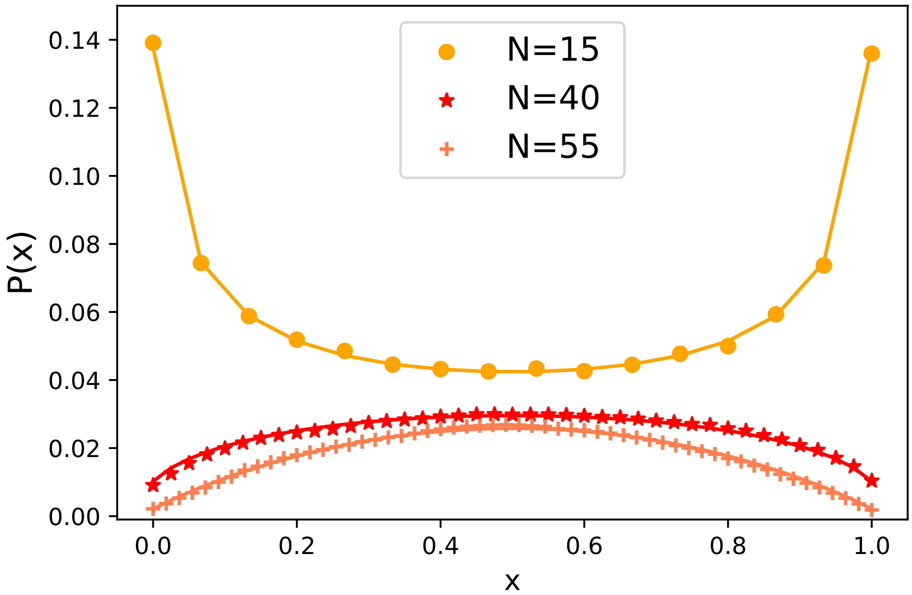

FIGURE 3.3: **Voter model with fast-switching noise rate.** Stationary distribution from simulations (symbols), and from theory [lines, Eq. (3.19), with noise parameter $\bar{a} = (a_0 + a_1)/2$]. Model parameters: $a_0 = 0.02$, $a_1 = 0.05$, $h = 1$ and $\lambda = 100$. Each distribution is from 5×10^6 samples; time between subsequent samples is $\Delta t = 0.01$, after a transient of 500 units of time.

This describes a conventional noisy VM [109], with noise parameter $\bar{a}$ and herding parameter h. The stationary distribution is bimodal if $N < h/\bar{a}$, and unimodal otherwise, as shown in Fig. 3.3. The transition between these two regimes occurs without an intermediate trimodal shape.

3.3.3.4 Simulations for intermediate switching rates

When the time scales for population and environmental switches are comparable, an analytical characterisation is not easily available. Nonetheless, we can conduct simulations, varying the value of λ to interpolate between the slow-switching regime in Sec. 3.3.3.2 to fast switching in Sec. 3.3.3.3.

Figure 3.4 shows the resulting phase diagram in the (λ, N)-plane, at fixed values of the remaining model parameters. For slow switching (low values of λ), the stationary distribution exhibits three different shapes (bimodal, trimodal, and unimodal) as N increases. For faster environmental dynamics (higher values of λ), the trimodal shape disappears, resulting in the well-known finite-size transition between unimodal and bimodal states in a nVM with an effective noise constant.

3.3.4 Noisy voter model with switching influencers

In this section we focus on the impact of fluctuating groups of influencers on the nVM. The state of the influencers plays the role of the external environment.

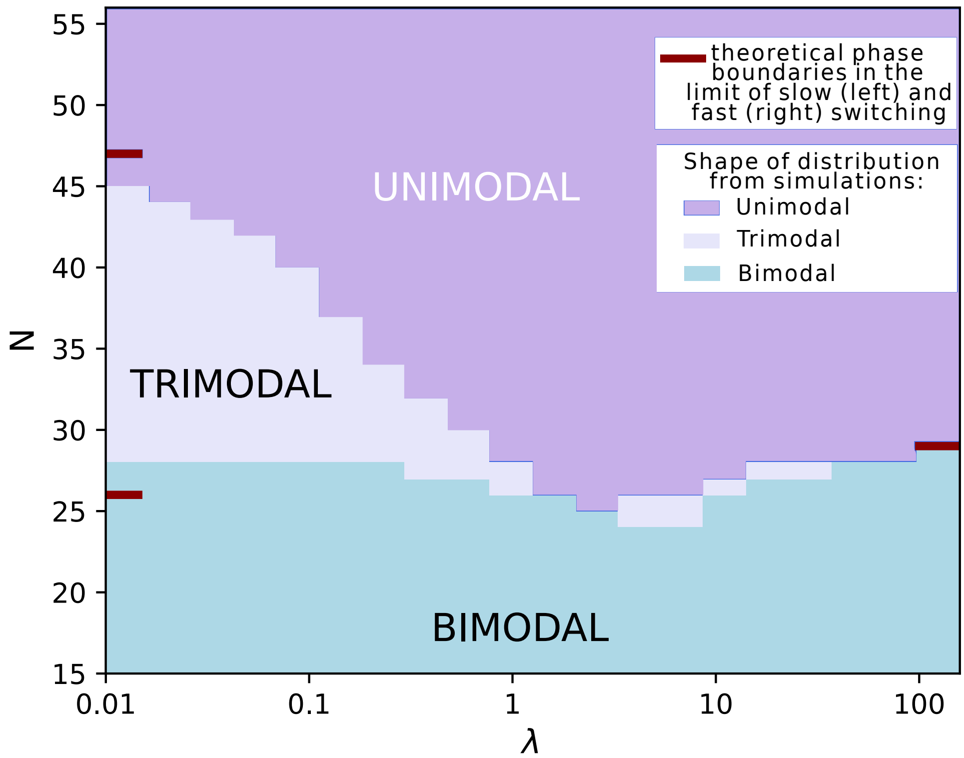

FIGURE 3.4: **Phase diagram for the voter model with switching noise parameter**. The coloured shading indicates the shape of the stationary distribution as found in simulations. The red lines on the left and right show the phase boundaries in the limits of slow switching [left, found from evaluating the expression in Eq. (3.20)], and fast switching [right, from Eq. (3.19)]. For each pair of values for N and λ, we obtain the stationary distribution $P^*(i)$ ($i = 0, \ldots, N$). Using the expected symmetry $P^*(i) = P^*(N-i)$, we classify a distribution as unimodal when $P^*(0) < P^*(1)$, as trimodal when $P^*(0) > P^*(1)$ and there is a local maximum in the interval $[N/2-1, N/2+1]$, and as bimodal otherwise. Model parameters are $a_0 = 0.02$, $a_1 = 0.05$, $h = 1$. Time between subsequent samples is $\Delta t = 1/\lambda$, for each distribution we take $10^5 - 10^7$ samples after a transient of 500 units of time.

We assume that $a_\sigma \equiv a$ and $h_\sigma \equiv 1$ across environmental states. We begin by examining the two-state scenario, which allows us to obtain an explicit solution for stationary distribution of the model. If there are more than two environmental states we resort to numerical integration to solve Eq. (3.26).

3.3.4.1 Two states of the group of influencers

We consider a model with two environmental states and in which all influencers form one group of total strength αN. At any one time, they act coherently either in favour of opinion A or of opinion B. As before we write $\sigma \in \{0,1\}$ for the two environmental states, and $\lambda\mu_{0\to1}$ and $\lambda\mu_{1\to0}$ for the switching rates. We have $z_0 = 0$ and $z_1 = 1$. The stationary state of the environmental dynamics is given by Eq. (3.34).

3.3.4.2 Fast-switching limit

We begin by analyzing the fast-switching limit, $\lambda \to \infty$. In this regime, the effective rates given in Eq. (3.17) reduce to

$$\begin{aligned}\overline{T}_i^+ &= \left[a^+ + \frac{i}{(1+\alpha)N}\right](N-i),\\ \overline{T}_i^- &= \left[a^- + \frac{N-i}{(1+\alpha)N}\right]i,\end{aligned} \tag{3.36}$$

where we define

$$a^+ \equiv a + \frac{\alpha\overline{z}}{1+\alpha}, \tag{3.37}$$

$$a^- \equiv a + \frac{\alpha(1-\overline{z})}{1+\alpha}, \tag{3.38}$$

with $\overline{z} = \rho_0^* z_0 + \rho_1^* z_1$, as also introduced in Eq. (3.18).

These expressions in Eq. (3.36) remain valid for any number of environmental states, provided $a_\sigma \equiv a$ and $h = 1$, with the generalization $\overline{z} = \sum_\sigma \rho_\sigma^*(i) z_\sigma$. The resulting dynamics corresponds to a noisy voter model, potentially asymmetric due to differing values of a^+ and a^-.

In the special case $\overline{z} = 1/2$, the model becomes symmetric, with $a^+ = a^- = a + \alpha/[2(1+\alpha)]$. The effective herding rate is then $1/(1+\alpha)$. For the symmetric noisy voter model, a finite-size transition between bimodal and unimodal distributions occurs when the ratio of the noise parameter to the herding rate equals $1/N$ [13, 109, 151]. This condition yields the critical population size

$$N_c = \frac{1}{a(1+\alpha) + \alpha/2}. \tag{3.39}$$

Simulation results supporting this prediction are shown in Fig. 3.5.

In this model, the total influence of external agents is equivalent to that of αN standard individuals. For a fixed α, this means that the total weight of influencers is less than the weight of a single normal agent when the population size satisfies

$N < N_1 \equiv 1/\alpha$. In this regime, it is no longer meaningful to treat influencers as discrete agents.

We now briefly consider the asymmetric case, $\bar{z} \neq 1/2$. In this case, the stationary distribution is no longer symmetric [i.e., the distribution will not fulfill $P^*(i) = P^*(N-i)$ for all i]. We therefore study the shape of the distribution near its left and right ends of the domain separately. As parameters are varied, the 'slope' of the distribution near the left end changes when $P(i=0) = P(i=1)$. This is the case if and only if $\overline{T}_0^+ = \overline{T}_1^-$. This in turn leads to

$$a(1+\alpha)(N-1) + \alpha\left[N - \bar{z}(N+1)\right] - \frac{N-1}{N} = 0. \tag{3.40}$$

For given a, α and $\bar{z}$ we denote the physically relevant solution of this equation by N_c^{left}. An analogous equation is obtained from setting $\overline{T}_N^- = \overline{T}_{N-1}^+$,

$$a(1+\alpha)(N-1) + \alpha\left[\bar{z}(N+1) - 1\right] - \frac{N-1}{N} = 0. \tag{3.41}$$

We denote the solution of this equation by N_c^{right}. The resulting behaviour of the asymmetric model is illustrated in Fig. 3.6 (a). In the example shown we have $\bar{z} = 0.85$ so that influencers tend to favour opinion A. In this setup one finds $N_c^{\text{left}} < N_c^{\text{right}}$. For relatively small populations ($N < N_c^{\text{left}}$) the stationary distribution is bimodal, but with a higher peak at $x = 1$ than at $x = 0$. As N is increased, the left edge of the distribution (near $x = 0$) first changes slope, and a distribution which is strictly increasing in x results for $N_c^{\text{left}} < N < N_c^{\text{right}}$. Finally, when $N > N_c^{\text{right}}$ the distribution is unimodal, but with its maximum closer to $x = 1$ than to $x = 0$. Fig. 3.6(b) shows the resulting phase diagram in the (α, N) plane, indicating the transitions between a bimodal phase in small populations, a phase with a strictly increasing functional shape for the stationary distribution at intermediate population sizes, and finally a unimodal phase for large populations.

We end this section of the fast-switching limit with a brief discussion of Reference [149] where a related model is studied. In this work the numbers of up/down zealots are drawn randomly (with no temporal correlations) before each microscopic update. In our setup the total number of influencers is constant. This is closest to the case in which the numbers of up and down zealots in [149] have the same variance as one another, and in which these two random numbers are fully anticorrelated. Increasing the number of influencers in our model, or analogously the number of effective zealots in [149], while keeping all other parameters fixed, leads to a more unimodal stationary distribution, as illustrated in Fig. 3.5 and 3.6 of the present work, and in Fig. 1 of [149].

3.3.4.3 Limit of large populations

In the limit $N \to \infty$ the internal noise in the population becomes irrelevant, and a PDMP results. The velocities in the two environments are given in Eq. (3.23). Using the expressions in Sec. 3.3.2.3 the stationary distribution for the model with two environmental states can be obtained for any choice of the rates $\lambda\mu_{0\to 1}$ and

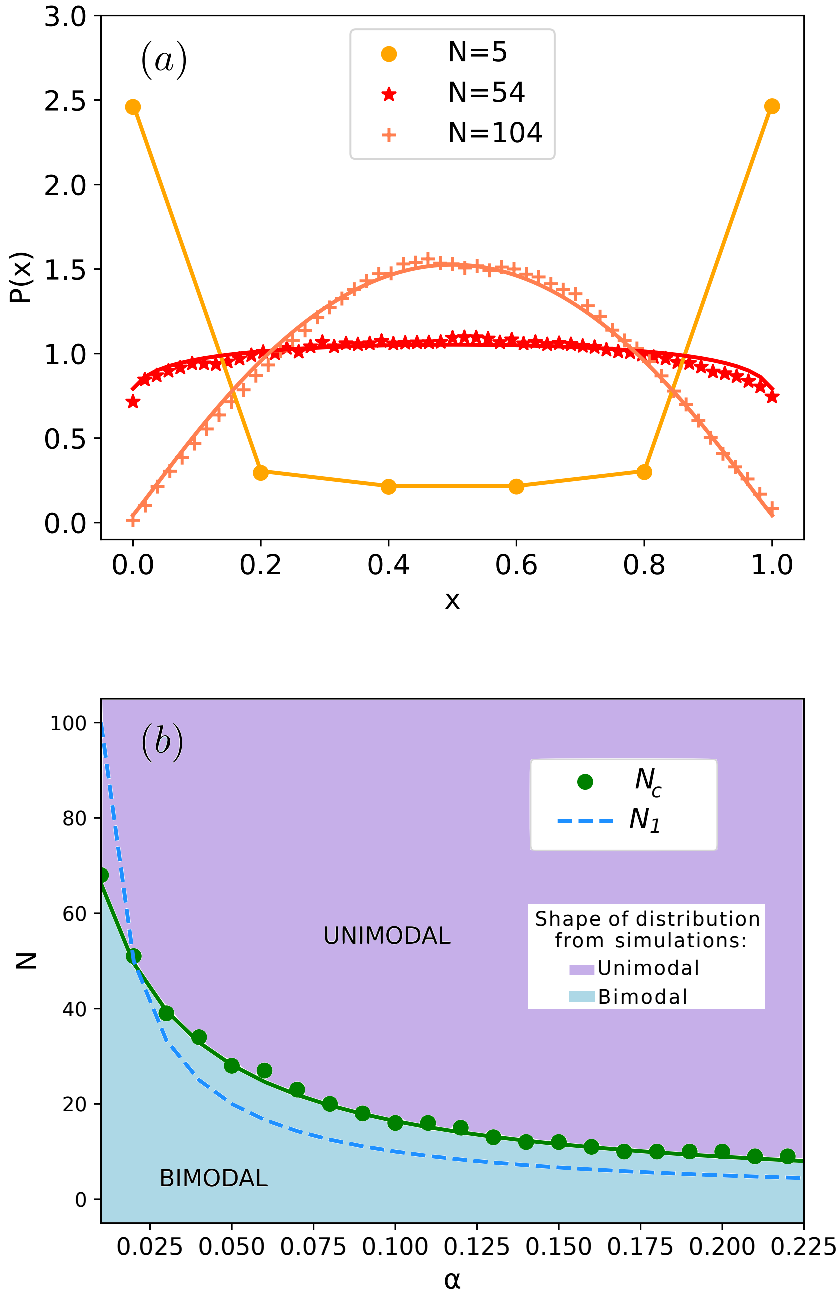

FIGURE 3.5: **Transition between bimodal and unimodal stationary distributions in the model with two external states and fast switching influencers.** Panel (a): Stationary distributions for different values of N and $\alpha = 0.02$. For $N = 5$ the stationary distribution is bimodal; for $N = 54$ and $N = 104$ it is unimodal. Panel (b): Location of the phase transition, $N_c(\alpha)$, as a function of the weight α of the influencers. The prediction from Eq. (3.39) is shown as a solid line, markers are from simulations. The coloured shading indicates the shape of the stationary distribution as found in simulations. Below $N_c(\alpha)$ the stationary distribution has a bimodal shape, above it is unimodal. The dashed line shows $N_1 = 1/\alpha$. Below this line the total weight of influencers is less than that of one normal agent. Model parameters are $\lambda\mu_{0\to1} = \lambda_{\mu_1\to0} = 50$, $z_1 = 1 - z_0 = 0$, $a = 0.01$. Time between subsequent samples is $\Delta t = 0.1$, we take 10^6 samples after a transient of one unit of time.

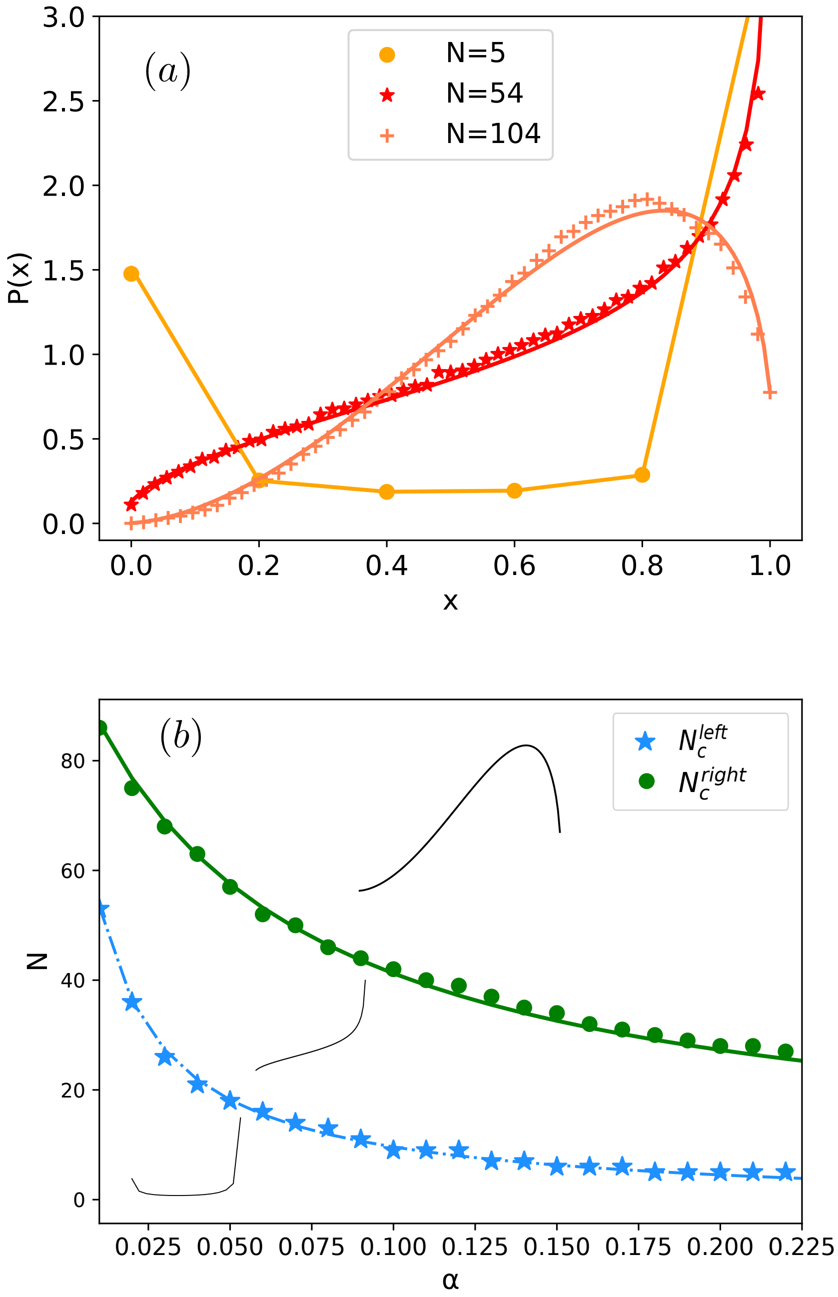

FIGURE 3.6: **Model with asymmetric influencers in the fast-switching limit.** Panel (a) shows the shapes of the stationary distribution for x in a model with two environmental states, $\bar{z} = 0.85$ and fast switching, for different sizes of the population. Remaining model parameters are $a = 0.01$, $\alpha = 0.02$. Markers are from simulations, lines from the analytical theory in the fast-switching limit. Time between subsequent samples in simulations is $\Delta t = 0.1$, we take 10^6 samples after a transient of one unit of time. Panel (b) shows N_c^{left} and N_c^{right} from Eqs. (3.40) and (3.41) respectively (lines). Markers are from simulations. For $N < N_c^{\text{left}}$ the distribution is bimodal and asymmetric, in the area between the lines it is strictly increasing in x, and for $N > N_c^{\text{right}}$ the distribution has a unimodal asymmetric shape.

$\lambda\mu_{1\to 0}$. We here restrict the discussion to the case $\mu_{0\to 1} = \mu_{1\to 0} = 1$, but keep the time scale separation λ general. We then find

$$\Pi^*(\phi) = \mathcal{C}\left[(\phi - \phi_0^*)(\phi_1^* - \phi)\right]^{\lambda/\lambda_c - 1}, \tag{3.42}$$

where $\mathcal{C}$ is a normalisation constant, and where

$$\lambda_c = 2a + \frac{\alpha}{1+\alpha}. \tag{3.43}$$

The fixed points ϕ_0^* and ϕ_1^* are obtained from Eq. (3.24). The stationary distribution becomes singular at $\phi = \phi_0^*$ and $\phi = \phi_1^*$ respectively for $\lambda < \lambda_c$.

An example is shown in Fig. 3.7. For $\lambda < \lambda_c$ the distribution is bimodal as shown in panel (a). For $\lambda = \lambda_c$ the distribution is mostly flat [panel (b)], and for $\lambda > \lambda_c$ a unimodal state results [panel (c)].

These results can be understood from the form of the flow fields $v_\sigma(\phi) = a(1-2\phi) + \alpha(z_\sigma - \phi)/(1+\alpha)$ obtained from Eq. (3.23). In each environment σ, the variable ϕ thus moves towards the fixed point ϕ_σ^* on a characteristics time scale given by $[2a + \alpha/(1+\alpha)]^{-1}$. The inverse of this time scale sets the value λ_c for the switching rate, separating the unimodal and bimodal regimes. Thus, for $\lambda < \lambda_c$ the environmental switching is slower than the relaxation of the population in any fixed environment. This relaxation can therefore proceed before the next switch occurs, and hence probability accumulates near the fixed points. The distribution of ϕ is bimodal, and if inspected at a given time, the population is likely to be found near the consensus state favoured by the influencers in the environmental state at that time.

If on the other hand $\lambda > \lambda_c$ then the environment switches quickly, before the population can approach either fixed point. The system frequently reverses its direction of motion, and the most likely states of the variable ϕ are those in the interiour of the interval from 0 to 1. As a result, the stationary distribution is peaked in the middle (unimodal). Both opinion states are typically found in the population at any given time.

The resulting phase diagram is shown in Fig. 3.8. The system is in the unimodal state above the phase line, and in the bimodal state below the line.

3.3.4.4 Lowest-order correction to the PDMP

For the model with $\mu_\pm = 1$ and $\bar{z} = 0$ we find from Eq. (3.32)

$$s^2(\phi) = \phi(1-\phi)\left(\frac{h}{(1+\alpha)\lambda_c} + 1\right). \tag{3.44}$$

This can be used to approximate the stationary distribution following [142]. An illustration is shown in Fig. 3.7 where the red lines show the resulting predictions for a model with $N = 200$ agents. Intrinsic noise in the model with finite populations smoothens the distribution compared to that in the PDMP limit, but the main characteristics of being bimodal or unimodal are preserved. Nevertheless, the finite size of the population results in a notable alteration in the bimodal

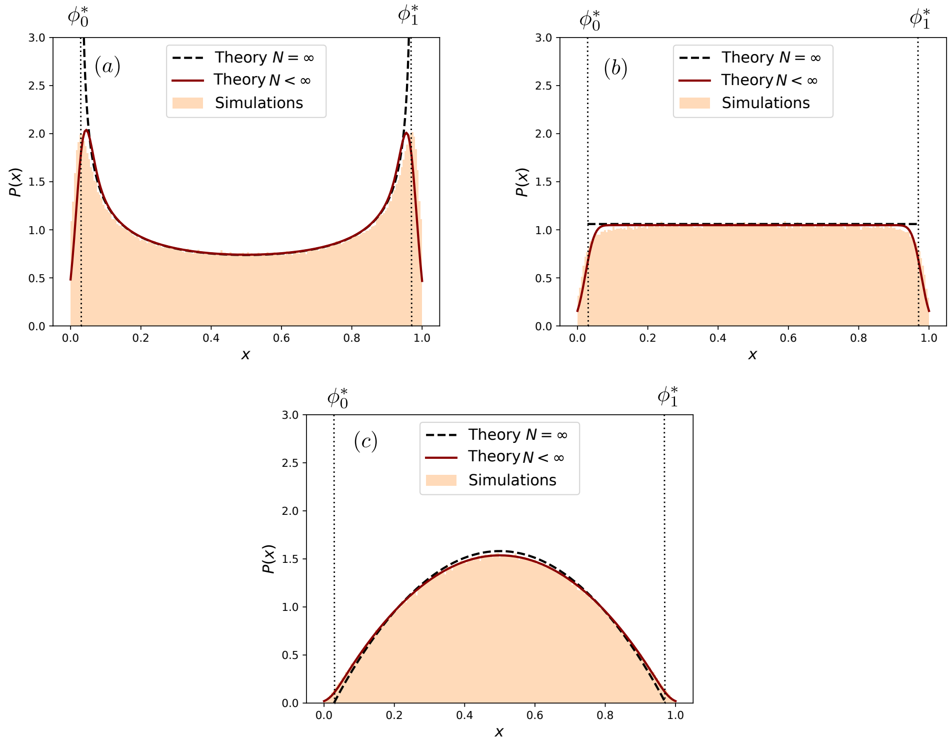

FIGURE 3.7: **Stationary distribution of the model with influencers switching between two states.** The black dashed lines in each panel are from Eq. (3.42) (PDMP limit), solid red lines are from the numerical integration of Eq. (3.31), capturing leading-order corrections to PDMP limit. The shaded histograms are from simulations of the full model. In all panels $a = 0.01$, $\alpha = 0.5$, $N = 200$ and $\mu_{0\to1} = \mu_{1\to0} = 1$. The switching rates are (a) $\lambda = 0.2$, (b) $\lambda = \lambda_c \approx 0.35$ and (c) $\lambda = 0.7$, where λ_c is obtained from Eq. (3.43). Time between subsequent samples is $\Delta t = 5$, for each distributions we take 10^6 samples after a transient of 50 units of time.

phase. With intrinsic noise the regions $x < \phi_0^*$ and $x > \phi_1^*$ become populated. These parts of phase space are not accessible by the PDMP. Thus, intrinsic stochasticity enhances the polarization of the population.

3.3.5 Three states of the group of influencers

In this section the group of influencers switches among three states $\sigma = 0, 1, 2$. As before we assume that there is a state in which all influencers support opinion B ($z_0 = 0$), and another in which all influencers favour opinion A ($z_2 = 1$). In the intermediate state, $\sigma = 1$, we assume that a fraction δ of influencers supports A, and a fraction $1 - \delta$ acts in favour of B. Thus, $z_1 = \delta$. Switches between these three states are taken to occur in a Markov process as follows

$$0 \underset{\lambda/2}{\overset{\lambda}{\rightleftharpoons}} 1 \underset{\lambda}{\overset{\lambda/2}{\rightleftharpoons}} 2. \tag{3.45}$$

Thus, the environment switches out of state 0 and to state 1 with rate λ, and similarly for switches $2 \to 1$. The total rate of leaving state $\sigma = 1$ is also λ, split

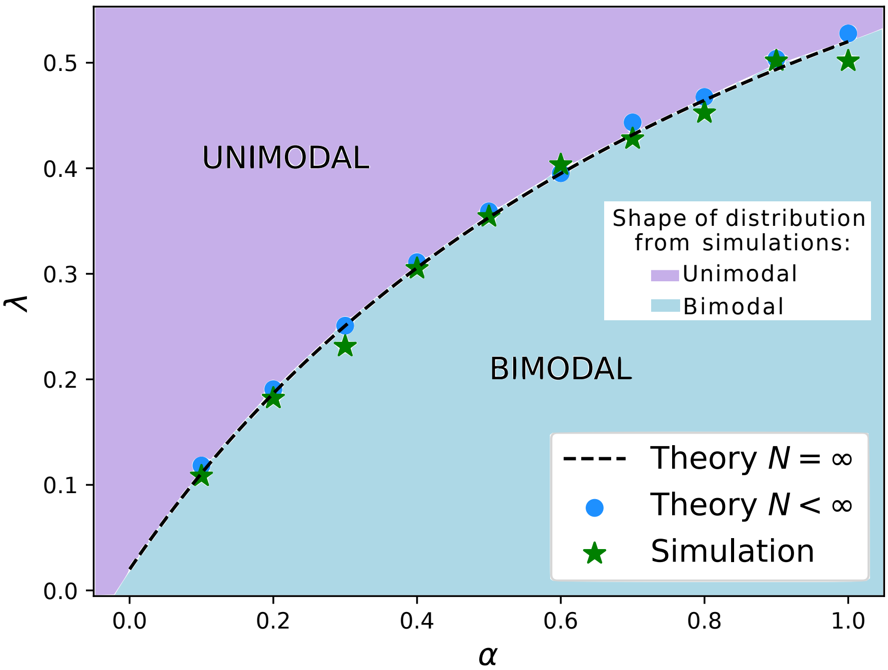

FIGURE 3.8: **Phase diagram for the model with two states for the group of influencers.** The dashed line is λ_c obtained in the PDMP limit [Eq. (3.43)]. The coloured shading indicates the shape of the stationary distribution as found in simulations. It separates a phase in which the stationary distribution is bimodal (below the line) from the other phase in which the distribution is unimodal (above the line). Green asterisks are from simulations of the individual-based model with $a = 0.01$ and $N = 500$. Blue dots indicate the phase boundary obtained from the theory which takes into account leading-order corrections to the PDMP [Eq. (3.31)]. Model parameters are $\bar{z} = 1$, $\mu_{0\to1} = \mu_{1\to0} = 1$, $a = 0.01$.

equally for transitions to states 0 and 2, respectively.

We first discuss the model in the PDMP limit, that is for infinite populations, $N \to \infty$. The stationary state is then to be obtained from Eq. (3.26). In the present setup this can be reduced into a system of two coupled ODE (see Appendix B) but we are unable to obtain an analytical solution. However, as also explained in Appendix B one can proceed numerically.

It is useful to note that the presence of three environmental states does not affect the relaxation time scale in any fixed environment. This is due to our assumption $a_\sigma \equiv a$ in all three states. Therefore, λ_c continues to be given by the expression in Eq. (3.43).

Results are shown in Fig. 3.9. We first focus on the black dashed lines showing the stationary distribution in the PDMP limit. When environmental switching is slower than the relaxation in the population [$\lambda < \lambda_c$ shown in panel (a)] the distribution has three sharp singularities, positioned at the fixed-point values ϕ_0^*, ϕ_1^* and ϕ_2^* obtained from Eq. (3.24) (we attribute minor numerical deviations to discretisation effects). For $\lambda > \lambda_c$ on the other hand [panel (c)], the distribution is asymmetrically unimodal with peak at ϕ_1^*. In panel (b), where $\lambda = \lambda_c$, the distribution also has a single maximum at $\phi = \phi_1^*$. In contrast with panel (c) though, the stationary density in the PDMP limit (black dashed line) remains non-zero at $\phi = \phi_0^*$ and ϕ_2^* respectively. In panel (c) the density tends to zero at the bondaries.

In Fig. 3.9 we also show results from the theory capturing the leading-order corrections in $1/N$ (solid lines). As seen intrinsic noise does not manifestly change the overall structure of the stationary distribution. Its main effect is to smoothen the singularities, and as expected there is now a non-zero probability of finding the system in the intervals $i/N \in [0, \phi_0^*]$ and $i/N \in [\phi_0^*, 1]$ respectively. These intervals are (by construction) unattainable by the PDMP.

3.3.6 Multiple states for groups of influencers

We now turn to systems in which the environment of the influencers can occupy more than three states. In this case, numerically solving Eq. (3.26) becomes increasingly challenging. We therefore rely on direct simulations of both the original individual-based model and its PDMP approximation.

3.3.6.1 Five environmental states

As a specific example, we consider an extension of the three-state system in Eq. (3.45) to a five-state environmental process:

$$0 \underset{\lambda/2}{\overset{\lambda}{\rightleftharpoons}} 1 \underset{\lambda/2}{\overset{\lambda/2}{\rightleftharpoons}} 2 \underset{\lambda/2}{\overset{\lambda/2}{\rightleftharpoons}} 3 \underset{\lambda}{\overset{\lambda/2}{\rightleftharpoons}} 4. \tag{3.46}$$

We set $z_\sigma = \sigma/4$ for $\sigma = 0, 1, \ldots, 4$, so that in state $\sigma = 0$ all influencers favor opinion B, while in state $\sigma = 4$ they fully support opinion A. Intermediate states interpolate linearly between these extremes: state $\sigma = 1$ exhibits a mild bias toward B, state $\sigma = 2$ is unbiased, and state $\sigma = 3$ is weakly biased toward A.

Figure 3.10 shows the stationary distributions obtained from simulations of the full individual-based model (shaded histograms) for a population size of

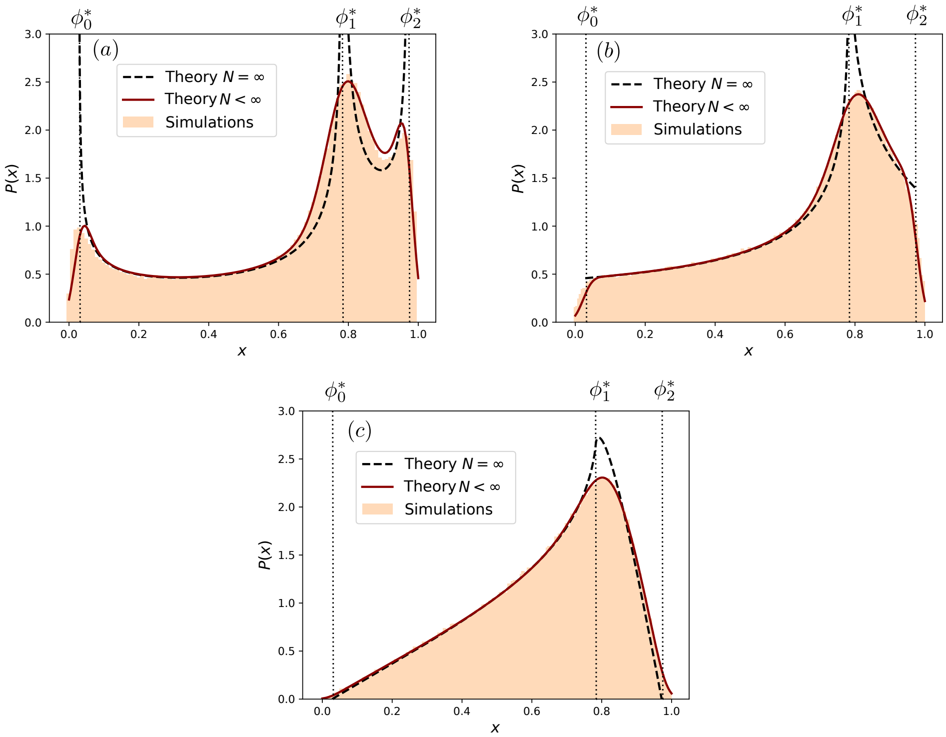

FIGURE 3.9: **Stationary distributions for the model with three states for the group of influencers**. In each panel the dashed line represents the PDMP limit, and is obtained from a numerical solution of Eq. (3.26). The solid lines are from the numerical integration of Eq. (3.31), capturing leading-order corrections in $1/N$. The shaded histograms are from simulations of the full model. The dotted lines are the values of ϕ_0^*, ϕ_1^* and ϕ_2^* found from Eq. (3.24). The environmental switching rate is $\lambda = 0.2$ in panel (a), $\lambda = \lambda_c \approx 0.35$ in (b), and $\lambda = 0.7$ in panel (c). In all panels $a = 0.01$, $\alpha = 0.5$, $N = 200$, $z_0 = 0$, $z_1 = 0.8$, $z_2 = 1$, and $\mu_{0\to1} = \mu_{2\to1} = 1$ and $\mu_{1\to0} = \mu_{1\to2} = 1/2$. Time between subsequent samples is $\Delta t = 5$; for each distributions we take 10^6 samples, after a transient of 50 units of time.

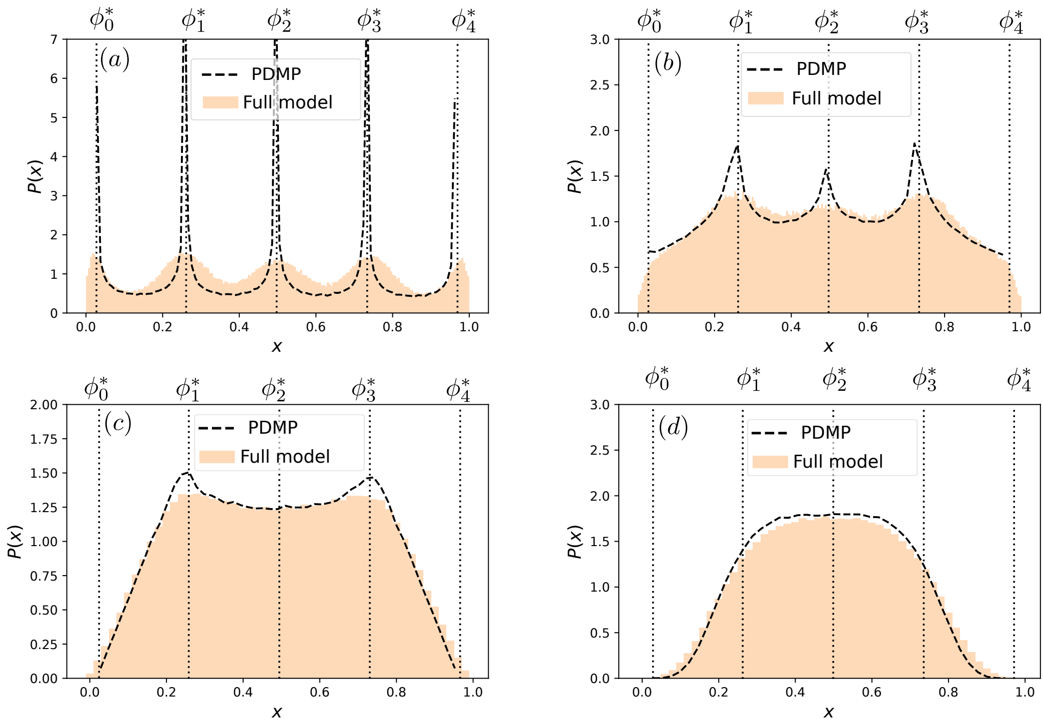

FIGURE 3.10: **Stationary distribution for the model with five environmental states.** The dashed lines in each panel are from numerical simulations of PDMP capturing the limit of infinite populations. Shaded histograms are from simulations of the full individual-based model with $N = 200$. The environmental dynamics is as in Eq. (3.46). The switching rates in panels (a)-(d) are $\lambda = 0.1$, $\lambda = \lambda_c \approx 0.35$, $\lambda = 0.7$ and $\lambda = 2$ respectively. Time between samples is $\Delta t = 1$ in (a)-(c), and $\Delta t = 0.2$ for panel (d). We take 10^7 samples after a transient of 10 units of time.

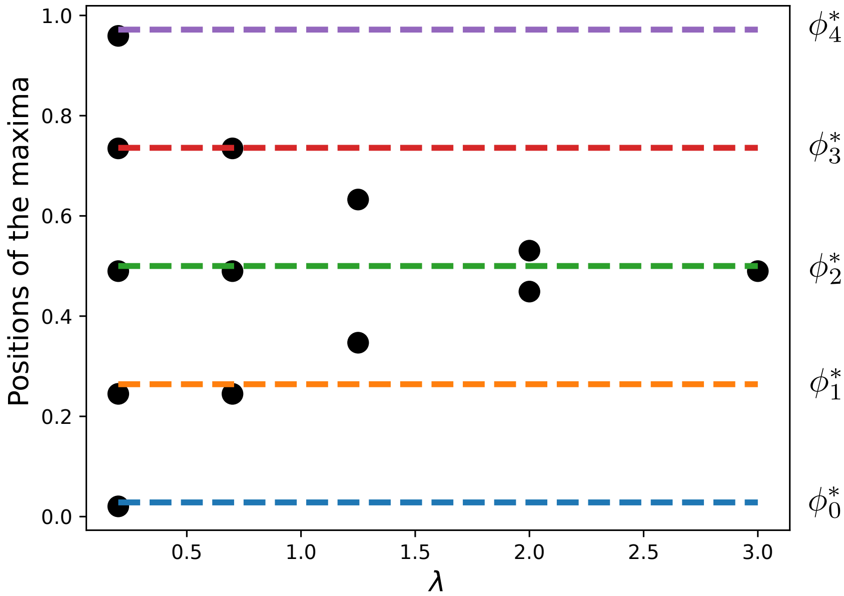

FIGURE 3.11: **Location of the maxima of the stationary distribution for the model with five environmental states (Fig. 3.10).**The figure shows the location of maxima in the stationary distribution of the PDMP for the model with five environmental states (Sec. 3.3.6.1). The markers are from simulations of the PDMP, the lines indicate the fixed points $\phi_0^*, \ldots, \phi_4^*$. Except for λ parameters are as in Fig. 3.10.

$N = 200$, alongside the corresponding PDMP results (dashed lines), for different values of the switching rate λ.

In panel (a), we consider the regime $\lambda < \lambda_c$, where the population dynamics is faster than environmental switching. The PDMP stationary distribution exhibits five distinct singularities located at the fixed points ϕ_σ^*. In the full model, intrinsic noise smooths these features but the underlying structure remains visible.

Panel (b) corresponds to the critical switching rate $\lambda = \lambda_c$. Here, the PDMP displays three peaks in the stationary distribution, which are also visible in the full model, though somewhat blurred due to internal fluctuations.

As λ increases further [panel (c)], the number of peaks in the stationary distribution reduces to two. Finally, in panel (d), the distribution becomes unimodal, indicating that the fast environmental switching effectively averages out the heterogeneity of the influencer states.

To illustrate the evolution of these peaks, Fig. 3.11 shows the positions of the maxima in the stationary distribution as a function of the switching rate λ. For small λ, five maxima are present, located at the fixed points ϕ_σ^*. At intermediate values of λ, only three remain—specifically at ϕ_1^*, ϕ_2^*, and ϕ_3^*. With further increase in λ, the central peak at ϕ_2^* vanishes. Eventually, the two remaining maxima approach each other and merge, resulting in a single peak at large λ, consistent with the unimodal regime.

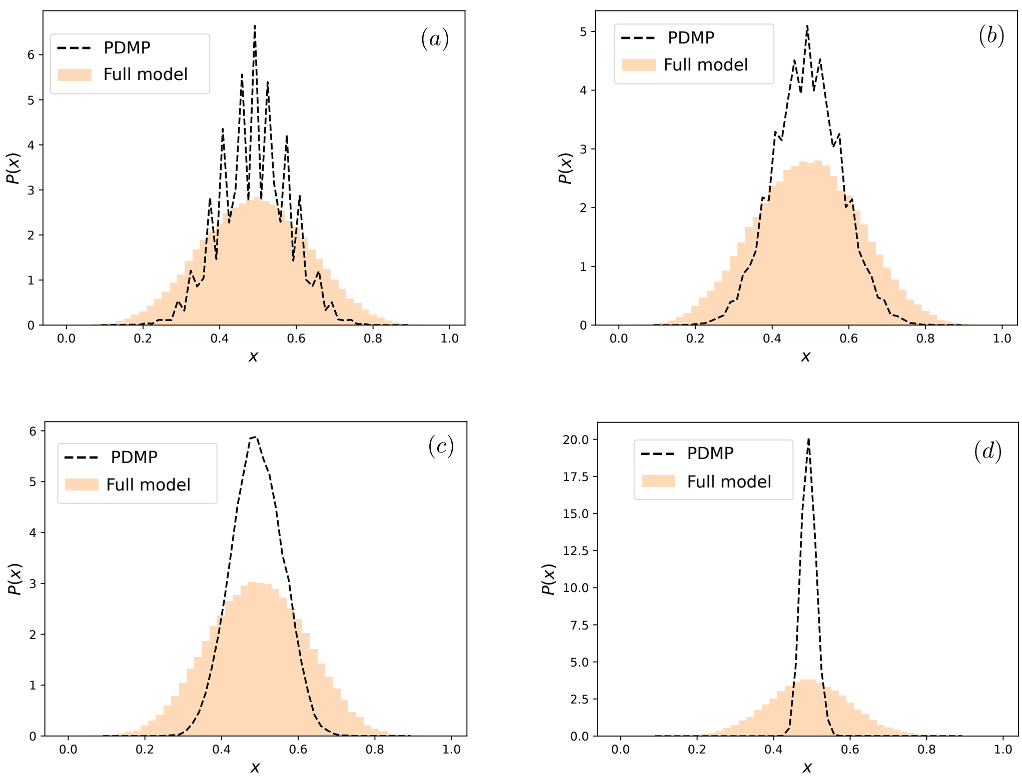

FIGURE 3.12: **Stationary distribution for the model with independent influencers.** The influencers follow the process in Eq. (3.47). We use $z_\sigma = \sigma/(\alpha N)$ with $\sigma = 0, ..., \alpha N$, with $\alpha N = 20$. Dashed lines in each panel are from numerical simulations of the PDMP, the shaded histogram is from simulations ($N = 200$). Model parameters are $a = 0.01$, $\alpha = 0.1$. Environmental switching rate is (a) $\lambda = 0.05$, (b) $\lambda_c \approx 0.11$, (c) $\lambda = 1$ and (d) $\lambda = 20$. Time between subsequent samples is 20 units of time in (a) and (b), 2 units in (c), and 0.2 units of time in (d). For each distributions we take 10^7 samples after a transient of duration $100/\lambda$.

3.3.6.2 Independent influencers

Next, we consider the case of independent influencers. The influencers are all taken to have the same strength, and each influencer can act in favour of opinion A, or of opinion B. In the example in Fig. 3.12 there are $\alpha N = 20$ influencers, and hence $S = \alpha N + 1$ environmental states $\sigma = 0, 1, ..., S-1$. In state σ there are σ influencers favouring A, and $S-1-\sigma$ influencers promoting B. Thus, $z_\sigma = \sigma/(S-1)$.

In state σ there are σ influencers who can switch to promoting B instead of A, and $S-1-\sigma$ influencers who can change from favouring B to favouring A. Thus, the rate of transitioning from state σ to state $\sigma-1$ is proportional to σ, and that of transitioning from σ to $\sigma+1$ is proportional to $S-1-\sigma$. We set $\mu_{\sigma\to\sigma-1} = \sigma/(S-1)$ and $\mu_{\sigma\to\sigma+1} = 1-\sigma/(S-1)$. Keeping in mind the overall multiplying factor λ, the environmental dynamics can then be summarised as

$$0 \underset{\lambda z_1}{\overset{\lambda(1-z_0)}{\rightleftharpoons}} \ldots \underset{\lambda z_\sigma}{\overset{\lambda(1-z_{\sigma-1})}{\rightleftharpoons}} \sigma \underset{\lambda z_{\sigma+1}}{\overset{\lambda(1-z_\sigma)}{\rightleftharpoons}} \ldots \underset{\lambda z_{S-1}}{\overset{\lambda(1-z_{S-2})}{\rightleftharpoons}} S-1. \tag{3.47}$$

Effectively, this means that each one of the αN individual influencers changes state with rate $\lambda/(S-1)$. We note that the division by $S-1$ is immaterial as any constant factors can be absorbed into the overall multiplier λ.

The resulting stationary distributions in Fig. 3.12 show some of the behaviour seen in the previous example in Fig. 3.10. For slow environmental switching the distribution has multiple maxima in the PDMP approximation. The number of extrema decreases with increasing switching rate of the environment, and ultimately only one single maximum remains [panels (c) and (d)]. Carrying out simulations of the full model for a population of the same size ($N = 200$) as in Fig. 3.10 we find in Fig. 3.12 that the stationary distribution is unimodal throughout. This is a consequence of the fact that the maxima of the PDMP for slow switching [Fig. 3.12(a)] are found relatively closely to each other. Intrinsic noise therefore 'washes out' this structure much more easily than in Fig. 3.10(a), where the maxima for the PDMP are more separated.

3.3.6.3 Details of the influencer dynamics matter

To characterise the relation between the distributions in Figs. 3.12 and 3.10 further, we study an intermediate scenario. As in Fig. 3.12 we allow for $\alpha N + 1 = 21$ states of the environment, and we use $z_\sigma = \sigma/(S-1)$. However, influencers no longer switch states independently from one another, but instead the environmental state is governed by a process more akin to that in Eq. (3.46). Specifically, we focus on

$$0 \underset{\lambda/2}{\overset{\lambda}{\rightleftharpoons}} 1 \underset{\lambda/2}{\overset{\lambda/2}{\rightleftharpoons}} \ldots \underset{\lambda/2}{\overset{\lambda/2}{\rightleftharpoons}} \alpha N-1 \underset{\lambda}{\overset{\lambda/2}{\rightleftharpoons}} \alpha N \tag{3.48}$$

The stationary distribution for the model with independent influencers [Fig. 3.12] was found to be unimodal throughout for the parameters we tested, and the corresponding PDMP has a unimodal envelope, modulated by local maxima for slow switching.

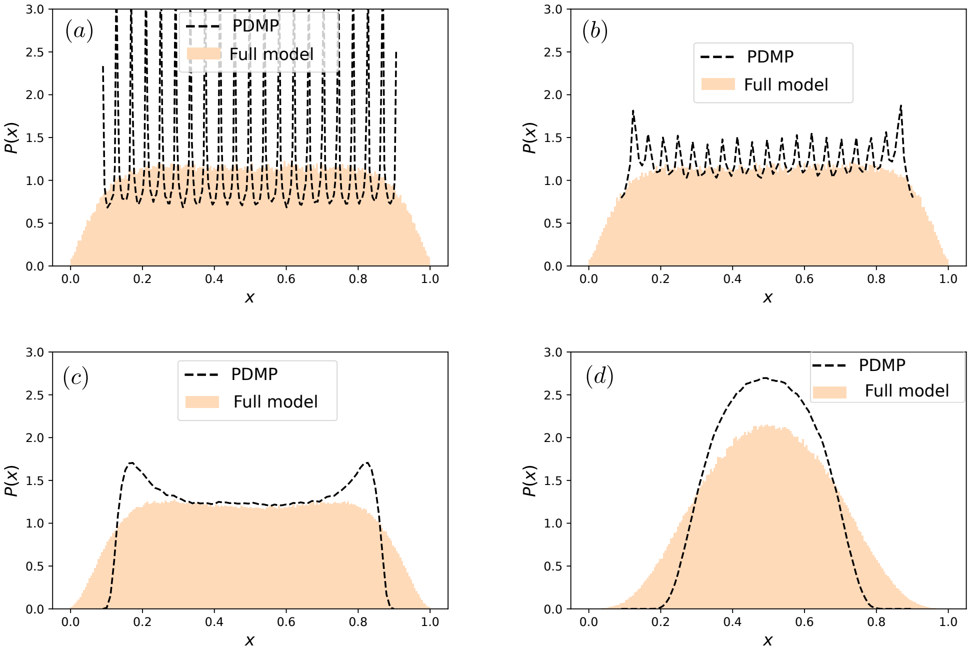

FIGURE 3.13: **Stationary distribution for the model with environmental dynamics as in Eq. (3.48).** Dashed lines are from numerical simulations of PDMP process, shaded histograms are from simulations of the full model ($N = 200$). In all the panels $a = 0.01$, $\alpha = 0.1$. The environmental switching rates are (a) $\lambda = 0.05$, (b) $\lambda = \lambda_c \approx 0.11$, (c) $\lambda = 1$ and (d) $\lambda = 20$. We use $z_\sigma = \sigma/(\alpha N)$ with $\sigma = 0, ..., \alpha N$; where $\alpha N = 20$. Time between subsequent samples 20 units of time in (a) and (b), two units of time in (c), and 0.2 units of time in (d). For each distribution we take 10^7 samples after a transient of length $100/\lambda$.

In contrast, if the environmental dynamics is as given in Eq. (3.48) the envelope of the stationary distribution of the PDMP is more flat at least for slow and intermediate environmental switching rates [Fig. 3.13 (a)-(c)]. We again find a modulation and the resulting maxima. The stationary distribution of the full model (i.e., including intrinsic noise) also has a broader shape than that in Fig. 3.12.

These differences in outcome can only be attributed to the differences in the environmental process [Eq. (3.47) vs Eq. (3.48)]. In the former case the environment has a proclivity to move from the more extreme states (those near $\sigma = 0$ and $\sigma = \alpha N$ respectively) towards the more balanced states (those with values of σ close to $\alpha N/2$). As a consequence the stationary distribution of the environmental process will be concentrated on the balanced states. In contrast, the dynamics in Eq. (3.48) results in a lower tendency for the influencers to populate the more balanced states. As result of that, in turn, the stationary distribution for the population of voters becomes more broad.

3.4 Conclusions

In this chapter, we have presented a comprehensive analysis of noisy voter models in the presence of temporally fluctuating environments. We began with a review of the standard noisy voter model (nVM) and its variant with zealots, as paradigmatic examples of binary opinion dynamics governed by a balance between spontaneous changes and social imitation.

The core of the chapter focused on extensions of these models in which external parameters change according to a discrete-time Markovian switching process. This approach was used to capture time-dependent features of the social interactions between agents or of the information available to them that influence the opinion dynamics in the population.

We examined two main classes of environmental fluctuations: changes in the ratio of herding to noise, and changes in the direction and strength of external influencers. For each of these cases, we considered different environmental switching regimes—ranging from fast to slow—analysed the behaviour of the model in these limiting cases, and examined the intermediate regime in the large-population limit through simulations and by numerically solving the equation for the stationary probability distribution.

Noisy voter model with switching noise parameter. In the first class of models, we allowed the herding-to-noise ratio to fluctuate over time. In the fast-switching limit ($\lambda \to \infty$), the dynamics can be effectively reduced to that of a standard noisy voter model with a constant average noise level. In this regime, one recovers the well-known finite-size transition between a unimodal stationary distribution for large populations and a bimodal distribution for small populations [13, 109].

In contrast, in the slow-switching limit, where the environment changes on a timescale slower than the population's relaxation dynamics, we identified an additional trimodal phase for intermediate population sizes. This phase is characterized by alternating periods of polarization and coexistence, as shown in

Fig. 3.1. The peaks in the stationary distribution correspond to fixed points associated with different environmental states. This dynamical feature is further illustrated in the time series in Fig. 3.2, where we observe fluctuations between consensus and mixed-opinion states.

Noisy voter model with switching influencers. In the second class of models, we introduced a population of influencers whose effect on the voter dynamics changes with the environment. When the environment switches between two symmetric influencer states, and herding/noise parameters remain constant, we again observe a transition between unimodal and bimodal distributions. In the fast-switching regime, this can be understood via an effective nVM with modified noise levels, resulting in the phase diagram shown in Fig. 3.5(b). The critical population size separating the two phases can be estimated analytically [Eq. (3.39)], and simulation results confirm the predicted behavior [Fig. 3.5(a)]. If the two influencer states are not symmetric, we observe an additional phase in which the stationary distribution becomes monotonic, with a single peak at one boundary [Fig. 3.6]. For very large populations, a piecewise-deterministic Markov process (PDMP) provides a good approximation of the system's dynamics. The transition between unimodal and bimodal regimes can be analyzed as a function of the environmental switching rate λ and the strength of influencer effects, as shown in Fig. 3.8.

Multiple environmental states. Increasing the number of environmental states further enriches the dynamics. In Section 3.3.6.1, we explored a five-state influencer model [Eq. (3.46)] with linearly spaced biases $z_\sigma = \sigma/4$. In the slow-switching regime, the PDMP displays multiple sharp peaks at the corresponding fixed points ϕ^*_σ, which are gradually reduced in number as the switching rate increases [Fig. 3.10]. For very fast switching, only a single peak remains, reflecting effective opinion coexistence. The progression in the number and location of maxima as a function of λ is summarized in Fig. 3.11.

We also considered variants in which groups of influencers evolve independently, or where environmental states are not symmetrically arranged. These models demonstrate that not only the number, but also the structure and symmetry of the environmental states matter. Even if the average influence remains constant, the details of the switching dynamics can significantly affect the shape and modality of the stationary distribution.

Throughout the chapter, we employed a combination of analytical techniques and numerical tools. In the infinite population limit, we derived deterministic dynamics using the PDMP formalism. To account for intrinsic stochasticity at finite N, we carried out a leading-order expansion of the master equation in powers of $1/N$, resulting in stochastic differential equations with state-dependent noise amplitudes [68, 69, 142]. Using a linear-noise approximation, we obtained an expression for the stationary distribution [Eq. (3.31)], including the variance $s^2(\phi)$ [Eq. (3.32)].

While these tools provide substantial insight, analytical results remain limited for systems with more than two environmental states. To address this, we proposed a numerical scheme to compute stationary distributions of PDMPs in such settings (see Appendix B). Although not a substitute for a full analytical treatment, this method avoids the need for time-consuming simulations and offers a practical alternative in many cases.

The results presented here contribute to the growing body of work extending voter-like models to fluctuating environments [147, 148]. These models not only enrich our understanding of opinion dynamics under time-dependent conditions, but also provide concrete examples of stochastic processes driven by random environments. Future directions include analytical characterizations of PDMPs with multiple environmental states, extensions to more than two opinions, incorporation of network structure for both voters and influencers, and the study of continuous rather than discrete environmental processes. Each of these directions opens up new challenges and possibilities, both for theory and for modelling real-world systems.

Chapter 4

Metrics to Characterise the Intrinsic Dynamics of Labelled Temporal Networks

"And when it stopped being random, that's when it started to go wrong."
Bill, *Following* (movie)

4.1 Introduction

In the previous chapter, we examined how the behavior of a dynamical system can be influenced by an external environment switching between two or more discrete states. The network structure remained fixed, while the dynamics were modulated by the current state of the environment, leading to different behaviors depending on the switching pattern.

In this chapter, we shift focus from how dynamics respond to a changing environment to how the interaction network itself evolves over time. Temporal networks provide a natural framework to model complex systems whose interaction architecture is not static, but dynamic—changing as the system evolves [25, 26]. This allows us to treat the network not just as a substrate for dynamics, but as a dynamical object in its own right.

Our central idea is that a temporal network can be understood as a single object following a trajectory in a high-dimensional *graph space*. Rather than viewing its evolution as a sequence of independent link changes, we consider it the output of a latent *graph dynamical system*. This perspective opens the door to applying tools from dynamical systems theory to analyze network evolution in a principled way [28, 101, 103].

The first part of this chapter (Sec. 4.2) introduces a generalization of the *autocorrelation function* (Sec. 2.4.1) to temporal networks, providing a way to quantify memory, oscillations, and structural persistence in network trajectories [28].

In the second part (Sec. 4.3), we present a novel contribution of this thesis: the definition and computation of *network Lyapunov exponents*, a tool to quantify sensitive dependence on initial conditions in the evolution of temporal networks [101]. Rather than tracking chaos at the level of individual components, we propose a method to measure how the network as a whole responds to small perturbations. The approach is inspired by classical algorithms used in time series analysis and adapted to work directly on observed sequences of graphs.

Together, these contributions provide new ways to characterize the intrinsic dynamics of temporal networks and extend the application of dynamical systems theory to the structural evolution of complex systems.

4.2 Autocorrelation Function for Labelled Temporal Networks

Understanding how structural patterns persist or fluctuate over time is central to the study of temporal networks. To this end, a version of the autocorrelation function specifically designed for labelled temporal networks was introduced in [28]. This framework captures temporal dependencies in the network's structure by treating its evolution as a time series of adjacency matrices. Two complementary formulations were proposed: a matrix-valued autocorrelation function and a scalar-valued version that aggregates the overall temporal correlations into a single quantity.

Consider a labelled, unweighted temporal network composed of n nodes and described by a sequence of adjacency matrices $\{\mathbf{A}(t)\}_{t=1}^{T}$, where each $\mathbf{A}(t) = \{A_{ij}(t)\}_{i,j=1}^{n}$ encodes the network's topology at time t. We define the *matrix autocorrelation function* $\mathbf{c}(\tau)$ as:

$$\mathbf{c}(\tau) := \frac{1}{T-\tau} \sum_{t=1}^{T-\tau} [\mathbf{A}(t) - \boldsymbol{\mu}] \cdot \left[\mathbf{A}(t+\tau)^{\top} - \boldsymbol{\mu}^{\top}\right], \tag{4.1}$$

where $\boldsymbol{\mu} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{A}(t)$ is the time-averaged adjacency matrix, $\mathbf{A}^{\top}$ denotes the matrix transpose.

The matrix $\mathbf{c}(\tau)$ quantifies temporal correlations across the network at lag τ, capturing not only the autocorrelation of individual links but also cross-correlations between different links. Each entry $c_{ij}(\tau)$ represents how the presence or absence of link (i, j) at time t relates to the presence or absence of other links at a delayed time $t + \tau$.

To obtain a scalar summary, we take the trace of the matrix, defining:

$$\tilde{c}(\tau) := \mathtt{tr}\left(\mathbf{c}(\tau)\right). \tag{4.2}$$

This scalar autocorrelation function measures the average similarity between network configurations at time t and at a delayed time $t + \tau$, over the whole observation window. In analogy with classical time series analysis, $\tilde{c}(\tau)$ plays the role of a lag-τ autocorrelation, acting as a scalar product between the centered adjacency matrix sequence and its time-shifted counterpart.

More explicitly, $\tilde{c}(\tau)$ can be decomposed into the sum of individual link autocorrelations:

$$\tilde{c}(\tau) = \sum_{i,j} \tilde{c}_{ij}(\tau), \tag{4.3}$$

where each term is given by:

$$\tilde{c}_{ij}(\tau) = \frac{1}{T-\tau} \sum_{t=1}^{T-\tau} \left(A_{ij}(t) - \mu_{ij}\right) \left(A_{ij}(t+\tau) - \mu_{ij}\right), \tag{4.4}$$

capturing the temporal autocorrelation of link (i, j) independently.

In this section, we adopt $\tilde{c}(\tau)$ as a reference measure of temporal correlation in labelled networks. Later (in Chapter 5), it will serve as a benchmark for comparison with autocorrelation functions designed for the unlabelled case, where node identities are no longer preserved. The matrix version $\mathbf{c}(\tau)$ provides a richer view by encoding temporal dependencies between links, while the scalar $\tilde{c}(\tau)$ summarizes the overall temporal coherence of the network's evolution.

In the following subsection, we apply these measures to two classes of artificial temporal networks: one exhibiting periodicity, and the other characterized by memory effects.

4.2.1 Noisy Periodic Temporal Networks

We evaluate the scalar autocorrelation function $\tilde{c}(\tau)$ for the noisy periodic temporal network model introduced in [28]. In this model, one first constructs a strictly periodic sequence of graphs. The period is denoted by T_{period}. Specifically, T_{period} independent Erdős-R'enyi graphs are generated, each with n nodes and connection probability p. These graphs are labelled as $G_1, \ldots, G_{T_{\text{period}}}$ and define the base sequence. If we consider only one sequence, without repetition, the signal becomes equivalent to white noise in network space. In this case, the autocorrelation function is maximal at $\tau = 0$ and close to zero for all $\tau > 0$, reflecting the absence of temporal correlations. If we instead repeat the sequence periodically in time, we obtain a periodic network. Formally, we set $G_{mT_{\text{period}}+s} = G_s$ for all $m = 1, 2, \ldots$ and $s = 1, \ldots, T_{\text{period}}$.

To introduce temporal variability, we perturb this perfectly periodic structure with noise. For each snapshot G_t with $t > T_{\text{period}}$, each unordered node pair (i, j) is independently selected with probability q for potential perturbation. If selected, the edge between i and j is redrawn: it is present with probability p and absent with probability $1 - p$, regardless of its prior state. This process introduces controlled stochasticity into the otherwise deterministic periodic sequence.

Figure 4.1(a) shows the scalar autocorrelation function $\tilde{c}(\tau)$ applied to a noisy periodic temporal network with $T = 400$ snapshots, $n = 20$ nodes, $T_{\text{period}} = 20$, $q = 0.2$, and $p = 0.2$. As expected, a clear secondary peak appears at $\tau = T_{\text{period}}$, indicating that the autocorrelation correctly detects the underlying periodicity despite the added noise.

To complement this, we show as insets in the same panel the full matrix-valued autocorrelation $\mathbf{c}(\tau)$ at three representative delay times: $\tau = 0$, $\tau = 10$, and $\tau = 20$. These matrices represent the element-wise autocorrelation between adjacency matrix entries at a given lag τ. At $\tau = 0$, the matrix reflects the variance structure of the network; at this lag, each link is highly correlated with itself and uncorrelated with others. At an intermediate delay ($\tau = 10$), correlations are reduced. At $\tau = 20$—equal to the period of the underlying trajectory—clear structural similarities with the $\tau = 0$ case re-emerge, visually confirming the periodicity detected by the scalar measure.

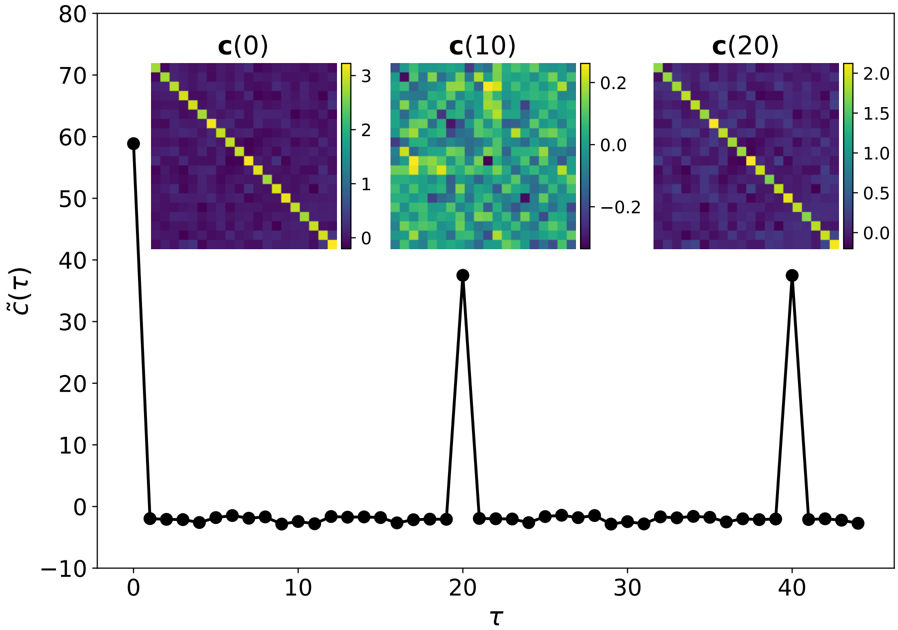

FIGURE 4.1: **Autocorrelation function for a noisy periodic labelled network.** Scalar autocorrelation $\tilde{c}(\tau)$ (Eq. 4.2) computed for a labelled noisy periodic network trajectory of $T = 400$ snapshots, with parameters $N = 20$, $T_{\text{period}} = 20$, $q = 0.2$, and $p = 0.2$. The secondary peak at $\tau = 20$ reflects the retrieval of the true period. Insets show the matrix-valued autocorrelation $\mathbf{c}(\tau)$ at three representative delays: $\tau = 0$, $\tau = 10$, and $\tau = 20$. The structure at $\tau = 20$ closely resembles that at $\tau = 0$, visually confirming the periodicity observed in the scalar function.

To systematically assess the robustness of the autocorrelation function in detecting periodicity under increasing levels of noise, we define an effective z-score:

$$z = \frac{\tilde{c}(T_{\text{period}}) - \mu}{\sigma}, \tag{4.5}$$

where μ and σ are the mean and standard deviation of the values $\tilde{c}(\tau)$, with $\tau = 1, .., T_{\text{period}} - 1$. This score quantifies how significantly the autocorrelation at lag T_{period} stands out from typical background values. A high z-score indicates a strong periodic signal despite noise.

In Figure 4.2(a), we show how the z-score decays as a function of noise level q. Each point is averaged over 10 independent realizations of the noisy periodic network. The scalar autocorrelation function $\tilde{c}(\tau)$ remains robust up to noise levels around $q \approx 0.9$, highlighting its effectiveness in detecting periodic structure in labelled temporal networks.

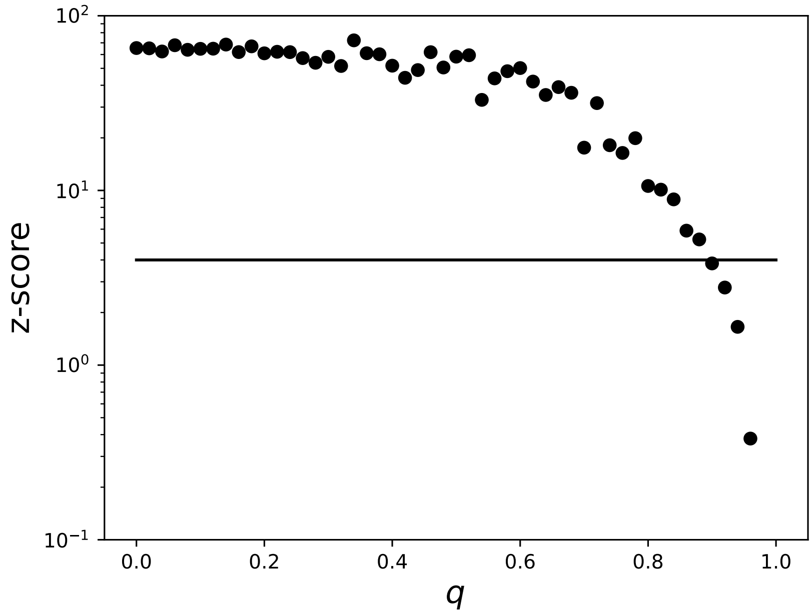

FIGURE 4.2: **Robustness of the labelled autocorrelation function against noise.** z-score [Eq. 4.5] as a function of the noise parameter q, for a labelled noisy periodic network of $T = 400$ snapshots, $n = 20$ nodes, $T_{\text{period}} = 10$, and $p = 0.2$. The horizontal line at $z = 4$ indicates a conservative threshold for reliable period detection.

4.2.2 Temporal Networks with Memory

We now test the behavior of the scalar autocorrelation function $\tilde{c}(\tau)$ on a model that generates network trajectories with built-in memory. Specifically, we use the Discrete Autoregressive Network model with memory order ρ, or DARN(ρ), introduced in [152], which can be seen as the network-level extension of the Discrete Autoregressive (DAR) process defined in Section 2.4.1. This model generates a sequence of undirected, unweighted graphs with n nodes, where the presence or absence of each edge $\ell = 1, \dots, n(n-1)/2$ at time t is governed by a binary random variable $x_{\ell,t} \in 0, 1$. Each edge evolves independently according to a stochastic process with memory.

At each time step, the value of $x_{\ell,t}$ is determined either by drawing a fresh sample from a Bernoulli distribution with parameter p (with probability $1-q$), or by copying one of its past ρ states (with probability q). This can be formally written as:

$$x_{\ell,t} = Q_{\ell,t} x_{\ell,t-Z_{\ell,t}} + (1 - Q_{\ell,t}) Y_{\ell,t}, \tag{4.6}$$

where $Q_{\ell,t}$ is a Bernoulli random variable equal to 1 with probability q, and $Y_{\ell,t}$ is another Bernoulli variable with parameter p. The lag variable $Z_{\ell,t}$ is drawn uniformly from $1, \dots, \rho$, and determines which past state is copied when $Q_{\ell,t} = 1$.

In Fig. 4.3, we report the scalar autocorrelation function $\tilde{c}(\tau)$ for labelled temporal networks generated by the DARN(ρ) model, for different values of the memory parameter ρ. As shown in panel (a), $\tilde{c}(\tau)$ displays a characteristic shape:

it remains approximately constant for $\tau \leq \rho$, reflecting the temporal correlations introduced by the memory mechanism, and then decays exponentially for $\tau > \rho$, analogous to the behavior observed for the scalar DAR process in *panel b* of Fig. 2.5. This confirms that the scalar autocorrelation function for labelled networks effectively captures memory in the trajectory.

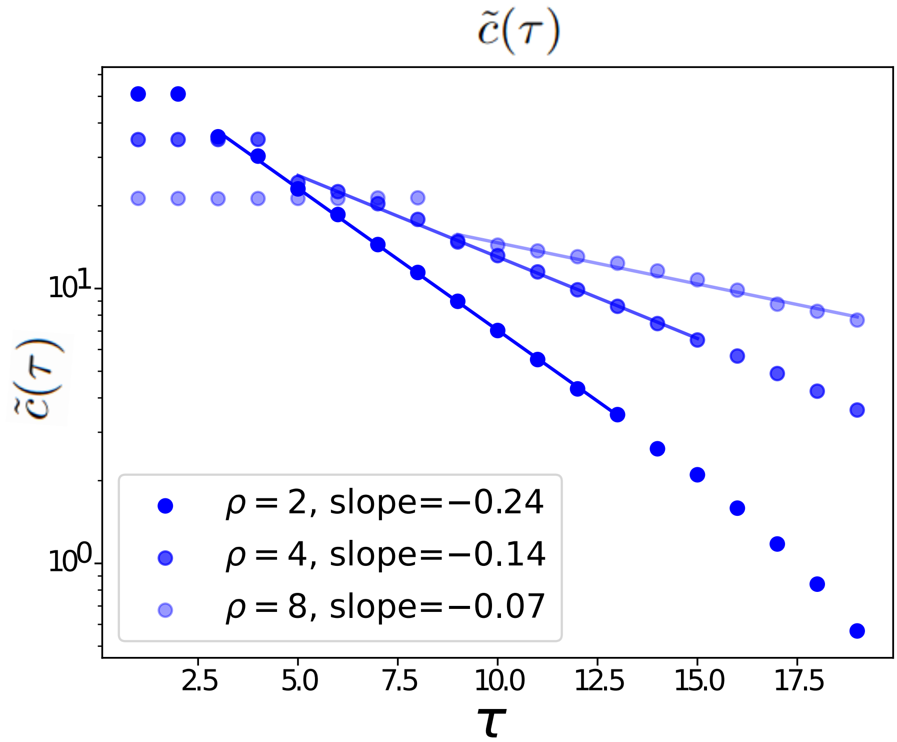

FIGURE 4.3: **Autocorrelation function detects memory in labelled DARN networks.** Semi-log plot of the scalar autocorrelation function $\tilde{c}(\tau)$ computed from a labelled network trajectory of $T = 10^3$ snapshots generated by a DARN(ρ) process with parameters $p = 0.5$, $q = 0.7$, and memory orders $\rho = 2,\ 4,\ 8$ (data is averaged over 10 realizations). The flat region for $\tau < \rho$ and the exponential decay for $\tau > \rho$ confirm the ability of $\tilde{c}(\tau)$ to detect temporal memory.

4.2.3 Empirical Labelled Temporal Network Trajectories

We conclude this analysis by applying the scalar autocorrelation function $\tilde{c}(\tau)$ to three real-world empirical temporal networks for which node labels are available: US domestic flights [153], face-to-face contact in a village in Malawi [154], and interactions at the 2009 Annual French Conference on Nosocomial Infections (SF2H, also referred to as SFHH) [155]. In the SFHH and Malawi datasets, nodes represent individuals and edges indicate face-to-face proximity events recorded via RFID sensors, with a temporal resolution of 20 seconds and 1 minute, respectively. In the US flights dataset, nodes represent airports and edges correspond to flights, active for the duration of each trip, with a temporal resolution of 1 hour.

Figure 4.4 panels (a), (b), and (c) summarize the behavior of $\tilde{c}(\tau)$ for these three labelled temporal networks. Panel (a) shows the scalar autocorrelation for the US domestic flight network, where a clear 24-hour periodicity is visible, reflecting the daily structure of flight activity. Panels (b) and (c) display $\tilde{c}(\tau)$ for the Malawi village and SFHH conference data, respectively. In both cases, the autocorrelation decays slowly, indicating the presence of long-range temporal correlations in human contact patterns, consistent with previous findings in [28].

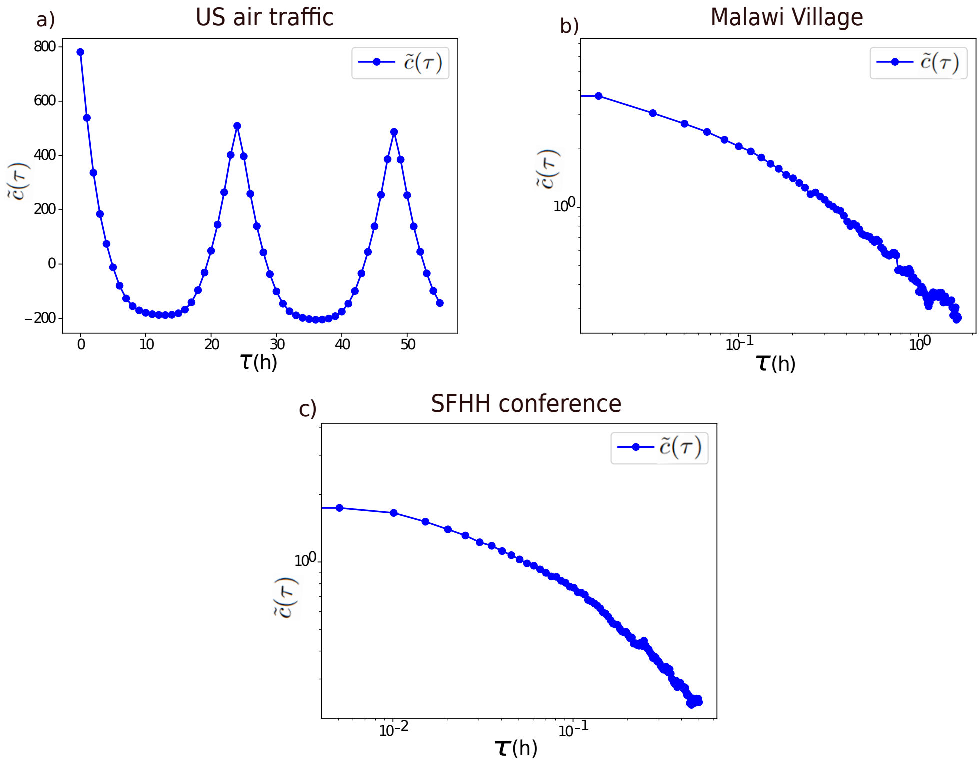

FIGURE 4.4: **Autocorrelation function for real-world labelled temporal networks.** Log-log plots of the scalar autocorrelation function $\tilde{c}(\tau)$ for three empirical labelled temporal networks. Panel (a) shows results for the US domestic flight network, computed up to $\tau = 55$ hours with a temporal resolution of 1 hour. A clear 24-hour periodicity is visible. Panel (b) shows the autocorrelation for face-to-face interactions in a Malawi village, up to $\tau = 1$ hour with 1-minute resolution. Panel (c) reports the same analysis for the SFHH conference dataset, up to $\tau = 30$ minutes with 20-second resolution. In the latter two cases, the slow decay of $\tilde{c}(\tau)$ indicates long-range temporal correlations in human contact patterns.

4.2.4 Other applications of the autocorrelation function for temporal networks

We conclude the section on the autocorrelation function for labelled temporal networks by briefly discussing other applications of this metric, as well as possible extensions to different classes of networked systems. The scalar and matrix-valued autocorrelation functions can serve as general-purpose tools for characterizing temporal regularities in time-evolving structures and have found use beyond the specific models analyzed so far.

In [28], the authors analyzed $\tilde{c}(\tau)$ in temporal networks generated by chaotic logistic maps, showing that while the fully chaotic regime ($r = 4$) leads to flat autocorrelation similar to white noise, the edge-of-chaos regime ($r = r_\infty$) reveals a complex, self-similar pattern with a nested hierarchy of regularly spaced peaks. These peaks are reminiscent of the infinite cascade of period-doubling modes ($T = 2^k$) characteristic of dynamics near the onset of chaos.

In a more recent study [156], an autocorrelation function was introduced specifically for temporal hypergraphs, enabling the detection of higher-order temporal correlations beyond pairwise interactions. Applying this framework to human interaction data revealed nontrivial mesoscopic organization and complex group-level memory, offering a richer understanding of the dynamics in systems where interactions occur in groups.

These examples illustrate the flexibility of the autocorrelation framework, which can be used to analyze both artificial and empirical temporal networks across a variety of domains. It is worth mentioning that other metrics have also been introduced to characterize temporal correlations in temporal networks. For instance, [157] proposes a similarity metric between network snapshots, called *temporality*, designed to capture the persistence of links over time. In [158], the authors introduce spectral methods based on static representations to identify dominant periodic time scales. Meanwhile, [159] quantifies long-timescale correlations using an autocorrelative measure of structural memory in synthetic networks with stochastic node dynamics.

In chapter 5, we will see how this notion can be extended to *unlabelled* temporal networks—where node identities are not accessible. Finally, we note that another promising direction is the extension of autocorrelation tools to other generalized network settings, such as signed networks [160].

4.3 Lyapunov Exponents for Labelled Temporal Networks

In this section, we further develop the perspective of temporal networks as trajectories in graph space by addressing the problem of dynamical instability and the quantification of chaos. Specifically, we aim to generalize the concept of Lyapunov exponents (Sec. 2.1.1.1) to the domain of TNs. Our goal is not to assess instability at the level of individual links or nodes, but rather to quantify how the collective evolution of the network responds to small perturbations in its initial configuration.

To do this, we propose a method to estimate Lyapunov exponents directly from an observed TN trajectory. Inspired by classical algorithms developed by Wolf, Kantz, and Rosenstein for time series analysis [47, 51, 74, 161], we adapt these ideas, already introduced in Sec. 2.1.2 to the context of graph-valued data. Importantly, our approach requires only access to a single empirical TN trajectory and does not rely on knowledge of the underlying GDS, although it is equally applicable when such a model is available. A key challenge in this setting is validation, as the concept of chaotic temporal networks is largely unexplored in the literature, with only a few recent efforts beginning to link temporal connectivity patterns to chaotic regimes ([162],[163]). To address this, we introduce synthetic generative models of chaotic temporal networks that serve as ground-truth testbeds. These models allow us to verify that our method reliably estimates the network Lyapunov exponent (nMLE) and responds appropriately to different dynamical regimes, including the presence of noise and the transition to stability.

The remainder of the section is structured as follows. In Sec. 4.3.1 we define the network maximum Lyapunov exponent (nMLE), along with its estimation algorithm. In Section 4.3.4, we apply the method to a baseline of random network dynamics. Section 4.3.5 presents a low-dimensional chaotic TN model used to validate our estimator against known ground-truth values. In Section 4.3.6, we explore a high-dimensional chaotic TN model, showing how the estimated nMLE reflects the system's dynamical complexity.

4.3.1 Theory and method

4.3.2 Lyapunov exponents for graph dynamical systems

In nonlinear time series analysis, the maximum Lyapunov exponent λ_{max} of a dynamical system quantifies how two trajectories that are initially close separate over time (Sec. 2.1). To briefly recall, the maximum Lyapunov exponent (λ_{max}) is defined as in Eq. 2.9

$$\lambda_{\text{max}} = \lim_{t\to\infty} \lim_{\delta_0\to 0} \frac{1}{t} \ln \frac{\delta_t}{\delta_0}, \tag{4.7}$$

where δ_0 is the initial distance between two trajectories and δ_t their separation at time t. In practical implementations, δ_0 is small but finite, and one tracks δ_t until it reaches a plateau due to the finite size of the attractor. This transition marks the end of the exponential growth phase. We refer to the corresponding time as the saturation time τ, i.e., the time beyond which a chaotic system no longer exhibits exponential separation due to bounded dynamics. In our setting, τ provides a natural cutoff for estimating λ_{max} from empirical data.

In the network setting, we consider the evolution of a graph over time in discrete steps, governed by some (possibly unknown) rule that specifies how the network changes from one time step to the next. For simplicity, we assume a fixed, labelled set of vertices that are distinguishable from one another, so that only the edges between them evolve in time. The resulting evolution can be represented as a trajectory consisting of a sequence of network snapshots, which together define the temporal network generated by the system. Each trajectory is given by the sequence of adjacency matrices $\mathcal{S} = (A_t)_{t\geq 0}$, $A_t = \{a_{ij;t}\}$; $i, j = 1, 2, \ldots, n$ where $a_{ij;t} = 1$ if the vertices i and j are connected at time t, and zero otherwise (this symmetric choice is for simple, undirected networks, but the method works essentially along the same lines for non-simple and directed networks, perhaps except for different normalization factors in the definition of the distance, see below). As such, we are considering labelled, unweighted networks of n nodes.

It is not clear a priori if the specific choice of the distance used to quantify the deviation between two originally close network trajectories is critical. We conjecture that, as long as this distance is based on the full adjacency matrix—rather than on a projection of it—results should hold independently of the specific metric. This is based on the fact that in dynamical systems, the $\lambda_{\max}$ is invariant under different choices of the underlying norm $||\cdot||$ (see Sec. 2.1 and [40]). There exist many graph distances [164]. For simplicity, we take an intuitive definition of such a distance which is based on the amount of edge overlap between two networks: given two networks with the same number of nodes n and adjacency matrices A and B,

$$d(A,B) = \frac{1}{2}\sum_{i,j} |a_{ij} - b_{ij}|, \tag{4.8}$$

where i and j take values from 1 to n[1]. In our setting, networks are simple and unweighted. In the particular case where A and B have the same number of edges, the distance defined in Eq. (4.8) is indeed a rewiring distance, i.e., it is given by the number of unique rewirings needed to transform A into B, and therefore d is a positive integer-valued function. One can then further normalize d as appropriate such that it is defined in $[0,1]$, as we will show later. Further details can be found in Appendix C, where we also introduce alternative distances.

4.3.3 Inference of network Lyapunov exponents

4.3.3.1 Local expansion rates

We aim to quantify the sensitivity to initial conditions in temporal networks by estimating the network counterpart of the maximum Lyapunov exponent, denoted as λ_{nMLE}. We consider the case where the underlying graph dynamical system (GDS) is unknown, and only a single observed trajectory of adjacency matrices is available, $\mathcal{S} = (A_0, A_1, A_2, \dots)$. This setup parallels classical problems in time series analysis, where one seeks to estimate the $\lambda_{\max}$ from a single scalar or vector sequence. Standard methods, such as those proposed by Wolf, Kantz, and Rosenstein [47, 51, 74, 161], rely on recurrence detection in phase space and tracking how initially close states diverge over time. We extend this philosophy to the context of graph-valued trajectories.

To begin, we select an arbitrary adjacency matrix A_t from the trajectory $\mathcal{S}$, treating it as the initial condition. We then search for a recurrence in the trajectory—another matrix $A_{t'}$ at a different time t'—such that $d(A_t, A_{t'}) < \epsilon$, where ϵ is a small threshold defined a priori.[2] This pair of similar networks allows us to monitor how initially close configurations diverge. The initial separation is recorded as $d_0 := d(A_t, A_{t'})$.

We then track the future evolution of both A_{t+k} and $A_{t'+k}$ in $\mathcal{S}$ for $k = 1, 2, \dots$, and compute their distance at each step: $d_k = \|A_{t+k} - A_{t'+k}\|$. This process yields a sequence of local expansion rates $\ell_1, \ell_2, \dots$, which describe the relative growth

[1]The pre-factor is used to have a normalized distance when networks are simple, undirected and have the same number of links, otherwise, a different normalization is needed.

[2]In practice, $A_{t'}$ is taken as the closest recurrence to A_t.

of separation between the two trajectories:

$$d_k = d_{k-1} \exp(\ell_k). \tag{4.9}$$

Each ℓ_k may be positive (local divergence), negative (local contraction), or zero. The values of ℓ_k generally depend on k, reflecting local variations in the dynamical expansion or contraction as the system explores different regions of its attractor [74].

We then define the *trajectory-averaged expansion rate* ℓ over a finite time τ (the saturation time introduced earlier) as

$$\ell = \frac{1}{\tau} \sum_{k=1}^{\tau} \ell_k = \frac{1}{\tau} \sum_{k=1}^{\tau} \ln \frac{d_k}{d_{k-1}}, \tag{4.10}$$

which simplifies to

$$\ell = \frac{1}{\tau} \ln \frac{d_\tau}{d_0}. \tag{4.11}$$

Under the assumption that the GDS is ergodic—i.e., that a sufficiently long trajectory explores the phase space thoroughly—ℓ is expected to converge to the network Maximum Lyapunov Exponent λ_{nMLE} as $\tau \to \infty$, independent of the specific choice of A_t. However, in practice, τ is finite and ergodicity may not hold. Consequently, ℓ is interpreted as a *local Lyapunov exponent* [165], and one must consider averaging over multiple initial conditions A_t to obtain a more global estimate. In our network setting, ergodicity is not necessarily guaranteed, and thus it will be useful to keep track of the local quantification, as we will also show below.

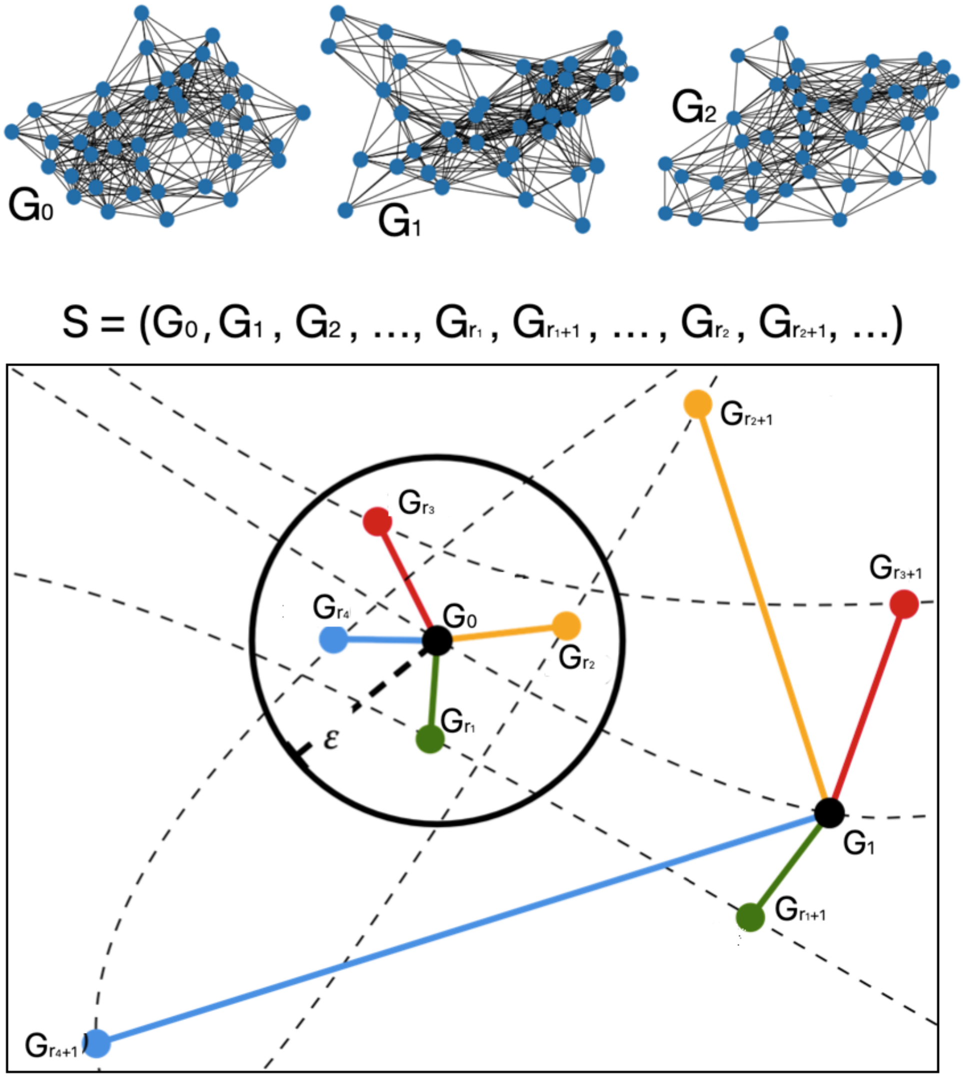

FIGURE 4.5: **Temporal Networks as Trajectories and Illustration of the Kantz Algorithm.** Illustration of a temporal network $\mathcal{S} = (G_0, G_1, G_2, ...)$ as a trajectory of a latent graph dynamical system (GDS) and the methodology to compute its network Maximum Lyapunov exponent (Kantz version) λ^K_{nMLE}. Each element of the trajectory is a network, i.e. a snapshot of the temporal network. λ^K_{nMLE} is estimated directly from $\mathcal{S}$ (i.e., without accessing the GDS directly) by looking at recurrences in $\mathcal{S}$ and quantifying the average expansion around different network snapshots. In this illustration, a ball of radius ϵ around an arbitrary G_0 is fixed, and four recurrences are found where $d(G_0, G_r) < \epsilon$. The initial distance $d(G_0, G_r)$ is averaged over the four recurrences and the average distance after one time step $d(G_1, G_{r+1})$ is computed. λ^K_{nMLE} is computed by averaging over time, volume and different initial conditions G_0 (see the text for details).

Generalized Wolf Method for Estimating the nMLE. We now introduce a generalization of Wolf's approach (see Sec. 2.1.2) aimed at estimating the network

Maximum Lyapunov Exponent (nMLE), denoted as λ^W_{nMLE}. The goal is to compute

$$\langle \ell \rangle = \left\langle \frac{1}{\tau} \ln \frac{d_\tau}{d_0} \right\rangle,$$

where the average is taken over pairs of states A_t and $A_{t'}$. In practice, we consider a set of w initial points A_t, indexed by $i = 1, \ldots, w$, and for each, we find one nearby point $A_{t'}$ such that $d(A_t, A_{t'}) < \epsilon$. This yields the estimate

$$\lambda^W_{\text{nMLE}} = \frac{1}{w} \sum_{i=1}^{w} \ell(i), \tag{4.12}$$

where $\ell(i)$ is the expansion rate associated with the i-th initial condition, computed as in Eq. (4.10). Note that λ^W_{nMLE} is the first moment of the distribution $P(\ell)$, which is obtained by sampling across different choices of A_t. Because ergodicity is not guaranteed, this value may depend on the selected A_t. For this reason, we adopt a cautious interpretation and only refer to λ^W_{nMLE} as a well-defined quantity when the distribution $P(\ell)$ is unimodal.

Algorithmically, this method offers the advantage of not requiring a fixed choice of τ: each initial condition is associated with a τ adapted to the available d_0 within the dataset $\mathcal{S}$. However, it remains pointwise in nature, as only one nearby $A_{t'}$ is used for each A_t. Consequently, this method does not necessarily capture the average expansion behavior around each initial condition.

Generalised Kantz approach to measuring the NMLE. In order to calculate such an average expansion rate (for a given choice of A_t) one would, for a fixed A_t, have to average over the expansion rates for choices of $A_{t'}$ in an ϵ-ball about A_t. This *volume-averaging* is the basis of Kantz's generalization [51, 161] of Wolf's algorithm [47], see Fig.4.5 for an illustration. For a given initial condition A_t, Kantz's method provides a trajectory and volume averaged expansion rate $\langle \frac{1}{\tau} \ln \frac{d_\tau}{d_0} \rangle_{\text{volume}}$. We will write Λ for the volume-averaged expansion rate. For fixed A_t, this could algorithmically be obtained as follows. One chooses N different $A_{t'}$ from the trajectory $\mathcal{S}$, all within distance ϵ from A_t. We label these $j = 1, \ldots, N$. For each $A_{t'}$ one then computes $\ell(j)$ via Eq. (4.10). Then one sets

$$\Lambda(A_t) = \frac{1}{N} \sum_{j=1}^{N} \ell(j), \tag{4.13}$$

where N is the number of initial conditions inside a ball of radius ϵ and centered at A_t that we have found in the sequence $\mathcal{S}$.

In practice, Kantz algorithm proceeds slightly differently. Instead of first computing the $\ell(j)$, and then averaging the expansion rates, the average over the $A_{t'}$ is instead computed at the level of distances. That is to say, one makes N choices of $A_{t'}$ as described above, and then obtains $d_k(j)$ for each $j = 1, \ldots, N$

and $k = 0, 1, 2, \ldots$ (k runs up to the relevant cut-off time). One then sets

$$\Lambda(A_t) = \frac{1}{\tau} \ln \frac{N^{-1} \sum_{j=1}^{N} d_\tau(j)}{N^{-1} \sum_{j=1}^{N} d_0(j)}, \tag{4.14}$$

where τ is a priori fixed for all j. The numerator in the logarithm represents the volume average (over choices of $A_{t'}$ in a ball about A_t) of d_τ, and the denominator is the volume average of d_0.

We stress that Eqs. (4.13) and (4.14) are mathematically different and do not necessarily lead to the same results. While Eq. (4.13) allows adapting the precise value τ (which depends on d_0) for each trajectory, Eq. (4.14) instead requires us to use a uniform τ across all trajectories. The latter expression looks at the expansion in time of an initially small volume centered on A_t, and is closer in spirit to capturing the underlying nonlinear dynamics than Eqs. (4.13). Accordingly, Eq. (4.14) is the basis of the Kantz' method.

The quantity Λ in Eq. (4.14) is still a local quantity, in the sense that it was computed for a phase space volume around a fixed choice of A_t. In principle, the local volume-averaged expansion rate could vary across different regions in phase space (if ergodicity is not assumed). To capture the global long-term behaviour we therefore additionally average over choices of A_t, and then finally obtain the global volume-averaged network maximum Lyapunov exponent

$$\lambda^K_{\text{nMLE}} = \left\langle \frac{1}{\tau} \ln \frac{\langle d_\tau \rangle_{\text{volume}}}{\langle d_0 \rangle_{\text{volume}}} \right\rangle_{A_t}, \tag{4.15}$$

where we have written $\langle \cdots \rangle_{A_t}$, for the average over initial conditions A_t. In practice, this is carried out by averaging over a set of w choices of A_t, i.e.,

$$\lambda^K_{\text{nMLE}} = \frac{1}{w} \sum_{j=1}^{w} \Lambda(A_t^{(j)}) \tag{4.16}$$

where $\Lambda(A_t^{(j)})$ stands for the expression in Eq. (4.14) for the fixed choice $A_t = A_t^{(j)}$. As we did for the Wolf version, we also keep track of the ensemble distribution $P(\Lambda)$ and only speak of a well-defined λ^K_{nMLE} when $P(\Lambda)$ is sufficiently peaked around its mean λ^K_{nMLE}.

We note that the value of the saturation time τ or the radius of the ball ϵ will have to be selected after some numerical exploration. Indeed, a better estimate of the Lyapunov exponent is obtained when the cut-off time τ is large. This, in turn, is the case when the initial distance between A_t and $A_{t'}$ is small, hence favouring choosing a relatively small value of ϵ. However, a small value of ϵ complicates the task of finding points $A_{t'}$ that are at most a distance ϵ away from A_t on the given trajectory. In practice, a trade-off is to be struck.

To summarise, from a given TN trajectory (i.e. a sequence of network snapshots) we first measure the local expansion rates $\{\ell_k\}_{k=1}^{\tau}$ via Eq. (4.9) for a fixed choice of A_t and $A_{t'}$. The set of ℓ_k obtained in this way provide information on the fluctuations of the local expansion rate (for fixed A_t and $A_{t'}$), and its trajectory-average ℓ. We can then proceed along two alternative routes. In the first approach (i) we average ℓ over different choices for the initial condition A_t [Eq. (4.12)], and obtain Wolf's approximation to the nMLE λ^W_{nMLE}. Alternatively, (ii) we can initially perform, for each initial condition A_t, an average over $A_{t'}$. This is done by computing the local expansion of a volume $\langle d_k \rangle_{\text{volume}}$ and then averaging this over time [Eq. (4.14)]. This is then repeated for different choices of A_t, and one hence obtains a distribution $P(\Lambda)$ describing the fluctuations across different points in the network phase space. Its mean provides Kantz's approximation to the nMLE λ^K_{nMLE}.

While these two approaches offer complementary estimates of the network maximal Lyapunov exponent (nMLE), their application to temporal networks comes with important limitations. First, the discrete and combinatorial nature of networks restricts the resolution of distance measurements between snapshots. Unlike real-valued dynamical systems, where infinitesimally close states can diverge smoothly, network distances often take values from a coarse set determined by structural features (e.g., number of differing links or eigenvalue shifts). As a result, detecting signatures of chaos—such as sustained exponential divergence—requires the trajectory to explore a sufficiently large and diverse region of graph space, which in practice demands large enough networks. Second, the Wolf algorithm described in Section 2.1.2 requires an initial choice of embedding dimension. In the following section, we set this dimension to one, based on the observation that the graph space explored by the trajectory can be mapped to the unit interval. In contrast, for higher-dimensional trajectories, we will restrict our analysis to Kantz's algorithm, which avoids the need for an explicit embedding and is thus better suited to such cases.
In the following sections, we present a validation of this method for random, low-dimensional, and high-dimensional chaotic temporal networks.

4.3.4 The white-noise equivalent of a temporal network: independent and identically distributed random graphs

Before addressing the case of chaotic dynamics, we briefly discuss the case of random network trajectories, with no correlations in time. One then expects no systematic expansion or compression in time, and the resulting Lyapunov exponent should hence vanish[3]. We here seek to verify that this is indeed the case for the procedures we have introduced to estimate the Lyapunov exponents of TNs. Studying this is of interest, among other reasons, because an empirically obtained time series may appear random. It is then important to be able to decide if the trajectory is consistent with an uncorrelated random trajectory in network space, or with a deterministic chaotic model.
Here we study the simple case where $\mathcal{S}$ is an independently drawn sequence

[3]By construction, $d_1 > d_0$ as we force $d_0 < \epsilon$, so in order to avoid spurious expansions at $k = 1$, in this section, we don't take into account d_0 in the estimation of finite Lyapunov exponents, i.e. our starting time is $k = 1$.

of Erdös-Rényi graphs with n nodes and in which the probability that any two nodes are connected is p. This is an analog to white noise for TNs, i.e., a situation in which the TN displays delta-distributed autocorrelation function as mentioned in Sec. 4.2.1.

At odds with a deterministic GDS, the distances between different points on a network trajectory are now random variables. More precisely, since all elements of $\mathcal{S}$ are the adjacency matrices of Erdös-Rényi graphs, the elements of these matrices are Bernoulli variables, taking values zero (with probability $1-p$) or one (with probability p). For independent adjacency matrices A and B, the possible values of $|a_{ij}-b_{ij}|$ are then zero with probability $p^2+(1-p)^2$, and one with probability $2p(1-p)$. Thus we have

$$d(A,B) \sim \text{Binomial}[n(n-1)/2, 2p(1-p)]. \tag{4.17}$$

Eq. (4.17) is numerically verified in panel (a) of Fig. 4.6.

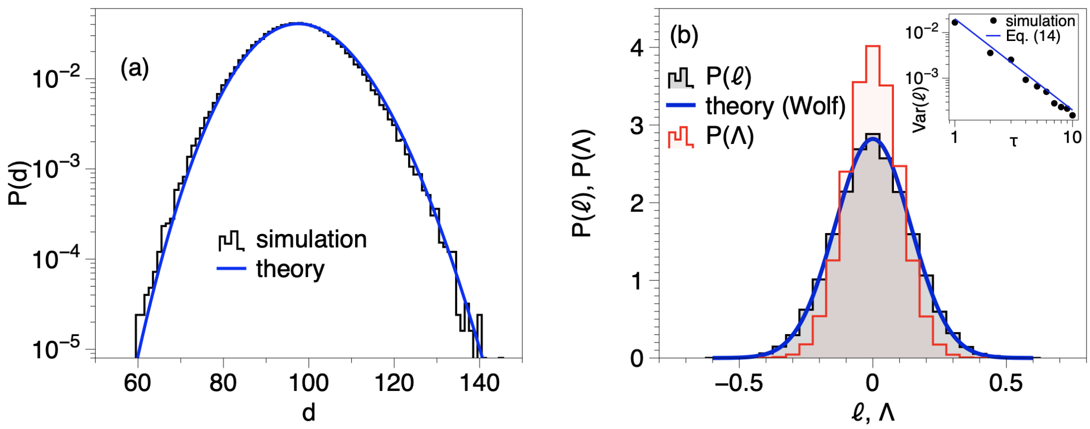

FIGURE 4.6: **Analysis of i.i.d. sequence of Erdos-Renyi graphs**. Panel (a): Probability distribution of the distance between consecutive networks in an i.i.d. sequence of $|\mathcal{S}| = 10^5$ Erdos-Renyi graphs with $n = 100$ nodes and $p = 0.01$ (black line). The plot is in semi-log, where a Gaussian shape appears as an inverted parabola, so we can better appreciate the tails. Blue solid line is Eq. (4.17). Panel (b): $P(\ell)$ (black) and $P(\Lambda)$ (red) associated with a sequence of $|\mathcal{S}| = 5 \cdot 10^4$ i.i.d. ER networks ($n = 100, p = 0.01$), for $\tau = 1$ and $w = \mathcal{O}(10^4)$ initial conditions. The mean of both distributions (nMLE) is essentially zero for both methods, but the dispersion around the mean is larger in Wolf's approach. The solid blue line is the theoretical prediction, i.e. a Gaussian distribution with mean 0 and variance as in Eq. (4.20) with $\tau = 1$. The inset of panel (b) shows how the variance of ℓ shrinks as τ increases (resulting in a much lower uncertainty around a zero nMLE): dots are different numerical simulations with $|\mathcal{S}| = 10^4$ and $w = 10^3$, for different τ; the blue line is Eq. (4.20).

The quantity ℓ in Eq. (4.11) is given by $\ell = \frac{1}{\tau}(\ln d_\tau - \ln d_0)$. As we have just established, d_0 and d_τ are independent binomial random variables following the distribution in Eq. (4.17). For large networks ($n \gg 1$) this distribution can be approximated as a Gaussian, with mean $\mu = qn(n-1)/2$ and variance $\sigma^2 = q(1-q)n(n-1)/2$, where we have written $q \equiv 2p(1-p)$. Writing $d_0 = \mu + \sigma z_0$, with z_0 a standard Gaussian random variable, we have

$$\ln d_0 \approx \ln \left[\mu(1 + \frac{\sigma}{\mu} z_0)\right] = \ln \mu + \frac{\sigma}{\mu} z_0 - \frac{1}{2}\frac{\sigma^2}{\mu^2} z_0^2 + \dots, \tag{4.18}$$

after an expansion in powers of σ/μ, where the latter quantity is of order $\mathcal{O}(1/n)$. The same expansion can be carried out for d_τ, and we therefore find

$$\ell = \frac{1}{\tau}\left[\frac{\sigma}{\mu}(z_\tau - z_0) + \frac{1}{2}\frac{\sigma^2}{\mu^2}(z_0^2 - z_\tau^2)\right] + \dots . \tag{4.19}$$

We note that the second term in the bracket is of sub-leading order in $1/n$. Hence, ℓ is to lowest order in $1/n$ approximately Gaussian, with mean zero and variance

$$\mathrm{Var}(\ell) = \frac{2}{\tau^2}\frac{\sigma^2}{\mu^2} = \frac{1}{\tau^2}\frac{4(1-q)}{qn(n-1)} \tag{4.20}$$

This theory has been numerically verified, and in panel (b) of Fig.4.6 we plot $P(\ell)$ both for $\tau = 1$ (outer panel) and Eq.4.20 for increasing values of τ (inset panel).

The case of Λ (Kantz-version) should intuitively converge even faster than ℓ (Wolf-version) since in this case, we are carrying out two averages instead of just one, i.e. $P(\Lambda)$ should have a smaller variance than $P(\ell)$, for a given τ.

This is confirmed in panel (b) of Fig. 4.6, where we also observe that both methods yield the same (correct) estimation of the nMLE, which in this case is approximately zero (both estimates are of the order of 10^{-6}). Note that the main panels of Fig. 4.6) are for the case $\tau = 1$, so it is a worst-case scenario: as τ increases $\mathrm{Var}(\ell)$ shrinks [Eq. (4.20)] and the uncertainty around the null shrinks accordingly [see inset of Fig. 4.6(b)].

We were not able to find a closed-form solution for $P(\Lambda)$ as averages inside the ϵ-ball are random variables whose distribution explicitly depends on the specific initial condition A_t: this calculation is left as an open problem. In any case, we conclude that an i.i.d. temporal network has a null $\lambda_{\max}$ Lyapunov exponent and the methodology (in both variants) correctly estimates it.

4.3.5 Low-dimensional chaotic networks

4.3.5.1 Network generation: the dictionary trick

To validate the method in the context of chaotic dynamics, we would ideally have access to chaotic network trajectories with a known ground-truth nMLE. This is, however, challenging: since the set of possible unweighted graphs is finite, it is not possible to construct a truly chaotic temporal network in the usual sense of the term. Nevertheless, there are ways to generate temporal trajectories that exhibit forms of pseudo-chaos. To address this limitation, in this section we present a procedure for constructing low-dimensional chaotic network trajectories by symbolising, in graph space, time series generated by low-dimensional chaotic maps. This approach, referred to as the 'dictionary trick' in [28], proceeds as follows:

- We begin by constructing a network dictionary $\mathcal{D}$, which is a set of networks enabling a mapping from real-valued scalars $x \in [0,1]$[4] to networks, in such a way that distances between scalars are preserved in the graph space. The dictionary $\mathcal{D}$ is ordered and equipped with a metric, ensuring that the distance between two scalars $|x - x'|$ corresponds to the distance between their associated graph symbols. More concretely, we define $\mathcal{D} = (G_1, G_2, \ldots, G_L)$ such that $d(G_p, G_q) \propto |p - q|$. The distance d can be normalized by the dictionary length so that $d \in [0,1]$.

- Once the dictionary is established, any one-dimensional time series can be transformed into a sequence of networks. In particular, chaotic time series with known MLEs can be symbolised in this way, producing network trajectories from which an independent estimate of the nMLE can be obtained.

Algorithmically, the dictionary is generated sequentially, starting with $G_1 \sim \mathrm{ER}(p)$ (an Erdős–Rényi graph with parameter p). Each subsequent graph G_{k+1} is built from G_k by rewiring a single edge, subject to the conditions that (i) the edge has not been rewired in any previous iteration, and (ii) the new edge did not exist in any previous iteration. This construction guarantees that $\mathcal{D}$ defines a partition of the unit interval: $[0,1] = \cup_{k=0}^{L-1}[k/L, (k+1)/L]$, where L is the total number of networks in the dictionary. As a result, the dictionary is metrical, in the sense that the graph distance between any two entries closely approximates the distance between the corresponding real numbers, for sufficiently large L.

With this dictionary, synthetic temporal network trajectories can be generated by symbolising the dynamics of any one-dimensional system over the unit interval. Specifically, a scalar value falling in the subinterval $[k/L, (k+1)/L]$ is mapped to the graph symbol G_{k+1}. The resulting temporal network $\mathcal{S}$ inherits, by construction, the properties of the original scalar time series — including chaoticity — and thus provides a way to construct chaotic temporal networks.

[4]We fix the interval to $[0,1]$ without loss of generality.

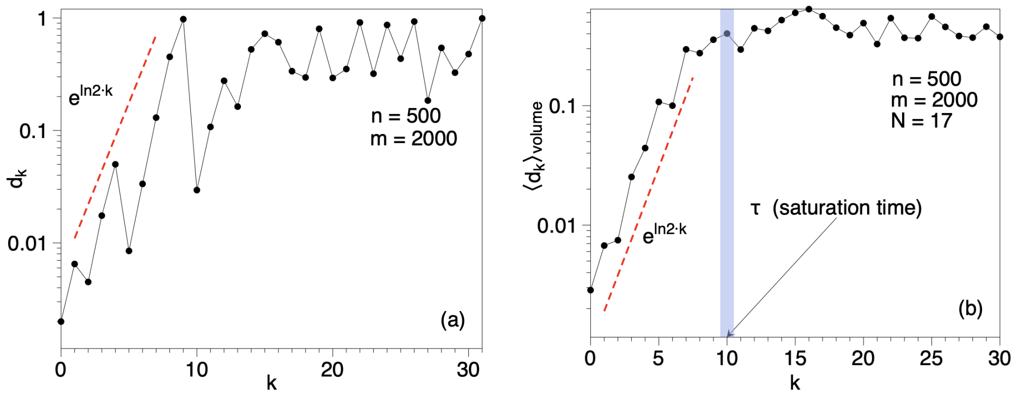

FIGURE 4.7: **Comparison of Wolf and Kantz Algorithms in Low-Dimensional Chaotic Temporal Networks.** Panel (a): Semi-log plot of the distance d_k as a function of iteration index k, for two initially close network trajectories sampled from $\mathcal{S}$. We can appreciate an initial exponentially expanding phase, followed by a saturation phase, although the local expansion rate strongly fluctuates. Panel (b) Volume-averaged distance $\langle d_k \rangle_{\text{volume}}$ as a function of time k, for $N = 17$ initial graph conditions inside a volume centered at an initial graph of $n = 500$ nodes and $m = 2000$ edges. Network dynamics evolve according to a logistic map as described in the text, whose true Lyapunov exponent is $\ln 2 \approx 0.693$. We can see how the volume enclosing the graphs on average expands exponentially fast –with an exponent close to $\ln 2$, as expected– until it reaches the attractor size, which happens at the saturation time $\tau \approx 10$.

4.3.5.2 Results for the logistic map

As a first validation, we consider the fully chaotic logistic map

$$x_{t+1} = 4x_t(1 - x_t), \quad x_t \in [0, 1], \tag{4.21}$$

which generates chaotic trajectories with a maximal Lyapunov exponent $\lambda_{\max} = \ln 2 \approx 0.693$, as discussed in Section 2.1.3. Using the dictionary trick, from a signal extracted from Eq. (4.21) we generate a temporal network trajectory $\mathcal{S}$ of $|\mathcal{S}| = 3000$ network snapshots. In this validation, networks have $n = 500$ nodes and $m = 2000$ edges.

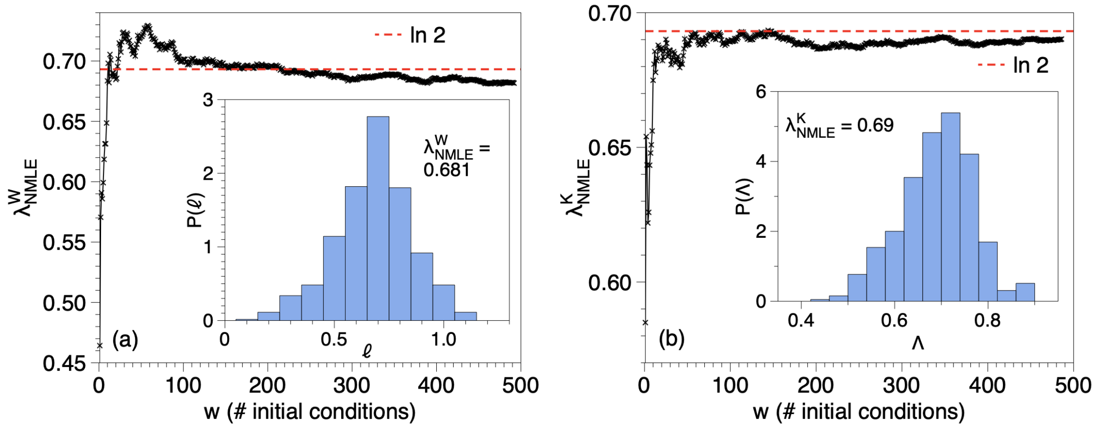

FIGURE 4.8: **Wolf and Kantz Algorithms for Low-Dimensional Chaotic Temporal Networks.** Panel (a): Approximation to λ^W_{nMLE} following Wolf's approach (see text), computed by averaging ℓ over w randomly sampled initial conditions [Eq.4.12], as a function of w. We can see that the exponent converges to the ground true exponent $\ln 2$ as w increases. Inset in (a): Probability distribution $P(\ell)$, sampled by estimating ℓ for $w = 500$ different initial graph conditions sampled randomly from $\mathcal{S}$. The mean of this empirical distribution is $\lambda^W_{\text{nMLE}} = \langle \ell \rangle_{A_t} \approx 0.681$, very close to the true exponent $\ln 2 \approx 0.693$. Panel (b): Same as panel (a), but using Kantz's approach (see the text), where we compute the volume and trajectory averaged expansion rate Λ for w initial conditions. We find $\lambda^K_{\text{nMLE}} = \langle \Lambda \rangle_{A_t} \approx 0.69$. Convergence properties are similar in both cases, with slightly better results for the Kantz version, as expected.

For illustration, in panel (a) of Fig. 4.7 we plot in semi-log scales the (properly normalized) distance d_k as a function of the iteration index k, for two initially close network trajectories sampled from $\mathcal{S}$. We can see an initial exponentially expanding phase (whose exponent is an estimation of ℓ) followed by a saturation, although the distance function shows strong fluctuations. To cope with these, in panel (b) of the same figure we plot the volume-averaged expansion $\langle d_k \rangle_{\text{volume}}$ vs k for a ball of radius $\epsilon = 0.005$ centered at a specific initial graph from $\mathcal{S}$. We can now clearly see the initial exponential phase followed by a cross-over to a saturation phase. The cross-over marks the saturation time τ where the distance reaches the attractor size. Note that the slope of the exponential expansion (i.e. the estimate of Λ) is close to $\ln 2$, the true $\lambda_{\max}$.

Figure 4.8 shows the estimated of the nMLE obtained both using Wolf's approach [panel (a)] and Kantz's approach [panel (b)]. These are from averaging ℓ (Wolf) and Λ (Kantz) over $w = 500$ initial graph conditions sampled from $\mathcal{S}$. In both cases, the average quickly stabilises for $w \approx 100$, and for $w = 500$ we obtain estimates $\lambda^W_{\text{nMLE}} \approx 0.681$ and $\lambda^K_{\text{nMLE}} \approx 0.69$, very close to the ground truth $\lambda_{\max} = \ln 2 \approx 0.693$.

Such behavior is different from the one observed for iid sequences: we make such comparison explicit in Fig.4.9, finding that purely chaotic and purely uncorrelated noise are easy to distinguish.

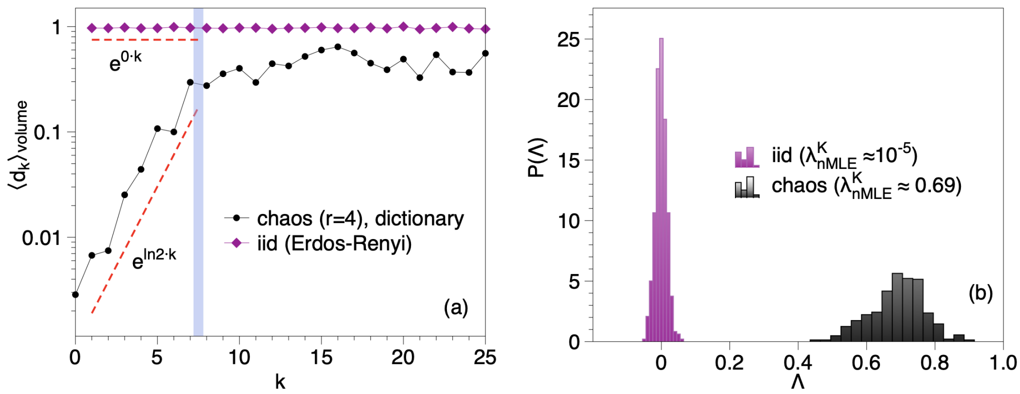

FIGURE 4.9: **Distinguishing Chaotic and Random Temporal Network Dynamics.**Panel (a): Comparison of the Volume-averaged distance $\langle d_k \rangle_{\text{volume}}$ as a function of time k, for the chaotic network trajectory of Fig.4.7 (black dots) and for an iid Erdos-Renyi network trajectory of the same size (purple diamonds). To make curves comparable, for the iid case we generated a sequence of ER networks with the same number of nodes and $p = 0.016$, and the distance in the iid case is normalized over the largest distance found in the iid sequence. The chaotic case shows an exponential expansion while the iid case shows a flat expansion. Panel (b): $P(\Lambda)$ for both cases (we use $\tau = 1$ for the iid case), showing that for iid the distribution $P(\Lambda)$ is clearly peaked at zero, and for the chaotic case the distribution is peaked close to $\ln 2$.

4.3.5.3 Noisy chaotic networks

To investigate how noise contamination affects the estimation of the nMLE, we generate a temporal network $\mathcal{S}$ from Eq. (4.21) using the dictionary trick. In this process, the original chaotic signal is first perturbed by a certain level of white Gaussian noise $\mathcal{N}(0, \sigma^2)$[5] before the network mapping. As in Section 4.3.4, we mitigate potential algorithmic biases by excluding $\langle d_0 \rangle$ from the computation of Λ. The results, shown in Fig. 4.10, reveal that noise generally shortens the exponential phase (i.e., the saturation time τ decreases). For low noise levels, this phase remains discernible, and the estimated nMLE remains consistent with that of the noise-free case. However, beyond a certain noise threshold, the chaotic signal becomes effectively obscured, the exponential phase disappears, and the nMLE appears to vanish. These findings align with both intuition and established phenomenology in the study of noisy chaotic time series [47, 51].

[5]Note that we discard realizations of the noise that cause the scalar variable to fall outside the unit interval.

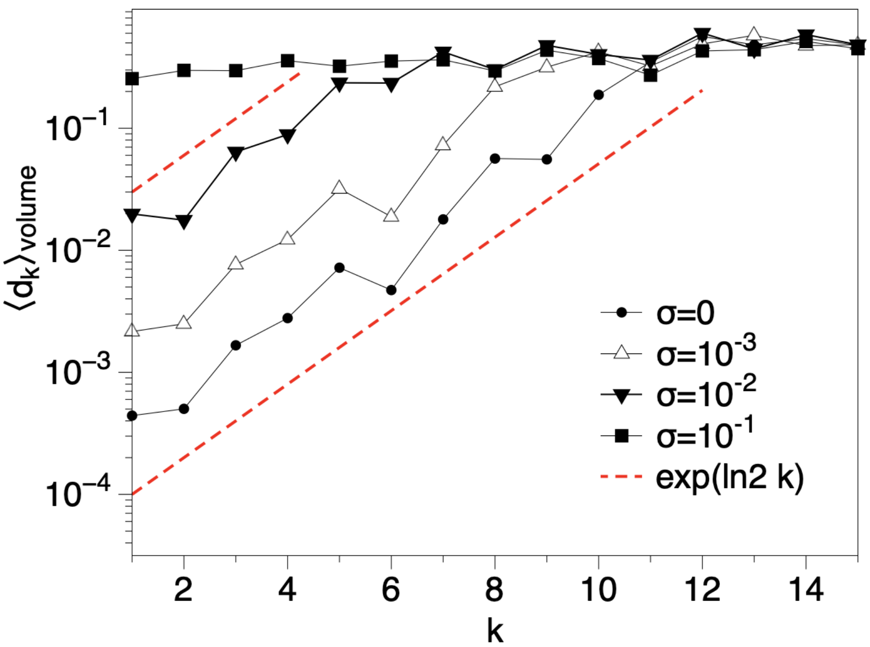

FIGURE 4.10: **Effect of noise on chaotic temporal networks**. Volume-averaged distance $\langle d_k \rangle_{\text{volume}}$ as a function of time k, for a network dynamics evolving according to a chaotic logistic map $x_{t+1} = 4x_t(1-x_t)$, polluted with extrinsic Gaussian noise $\mathcal{N}(0,\sigma^2)$ as described in the text, for four different noise intensities $\sigma = 0, 10^{-3}, 10^{-2}, 10^{-1}$. The exponential expansion phase –which systematically suggests the same exponent $\ln 2$, as expected– is gradually erased as the noise intensity increases.

4.3.5.4 Results for the parametric logistic map

Here we consider the logistic map $x_{t+1} = rx_t(1-x_t)$, $x \in [0,1]$. For each value of the parameter $r_\infty < r < 4$ –where $r_\infty \approx 3.569945672\ldots$ is the so-called accumulation point of the map–, using the dictionary trick we generate a long sequence of networks $\mathcal{S}_r$ with the desired chaoticity properties, and proceed to estimate the network Lyapunov exponent using the method detailed in Section 4.3.1. In panel (a) of Fig.4.11 we plot λ^W_{nMLE} vs $\lambda_{\max}$ of the map, for a range of values of the parameter r. The agreement is excellent in the region of parameters where the temporal network is chaotic.

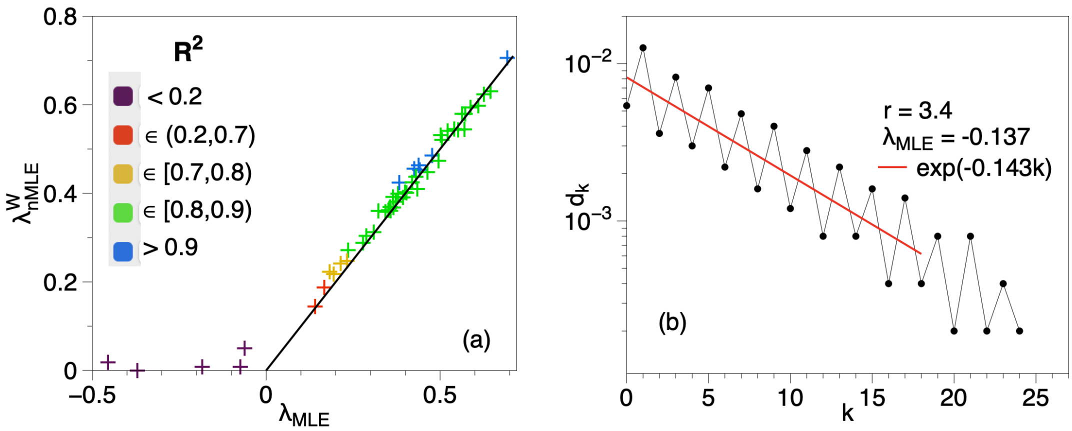

FIGURE 4.11: **Detecting negative lyapunov exponents in temporal network dynamics.** Panel (a): Scatterplot of $\lambda^W_{\mathrm{nMLE}}$, estimated from a temporal network $\mathcal{S}_r$ generated via the dictionary trick (see the text) from a logistic map $x_{t+1} = rx_t(1 - x_t)$ for a range of values of r, vs the ground true $\lambda_{\max}$. The solid line is the diagonal of perfect agreement $y = x$, highlighting the good agreement found in the chaotic region. The legend states the goodness of fit metric R^2 of the fit of d_k to an exponential function. The method is unable to capture negative Lyapunov exponents (observe that in those cases the R^2 of the exponential fit is very bad), but these cases can easily be identified as periodic orbits using the autocorrelation function [28], see text for details. Panel (b): Estimate of the negative nMLE using two initially close temporal networks generated via the dictionary trick from the logistic map at $r = 3.4$ (period-2 orbit), where one initial condition belongs to the period-2 attractor and the other is outside the attractor (see the text for details).

4.3.5.5 A note on negative Lyapunov exponents

The classical approach to estimate $\lambda_{\max}$ from a single trajectory, as implemented in the Wolf and Kantz algorithms—based on the recurrence properties of the trajectory—is, by construction, unable to capture negative MLEs. The reason is straightforward: once the system settles on a periodic attractor, the trajectory sequentially visits each element of the periodic orbit. Consequently, no recurrences are found that are both close in phase space yet sufficiently far from the initial condition of interest. For this same reason, our method to estimate the nMLE also fails in such cases, as confirmed in Fig. 4.11(a).

This limitation can be addressed through two alternative strategies.

First, it is well known that a periodic time series exhibits an autocorrelation function with peaks at integer multiples of the period. Therefore, from a practical standpoint, before attempting to estimate the nMLE of a given temporal network, it is advisable to apply the procedure described in Section 4.2, originally introduced in [28], in order to check for periodicity. Identifying a periodic signal typically[6] suggests the presence of a negative nMLE.

[6]Some pathological cases exist where a system may appear periodic but does not exhibit a negative $\lambda_{\max}$, such as when the attractor is disconnected and composed of several chaotic bands that are visited periodically.

Once this test is passed, the nMLE estimation method presented in this paper can be reliably applied.

Second, negative nMLEs can indeed be estimated when one has access to the latent graph dynamical system (GDS). In this case, a Wolf/Kantz-type approach is unnecessary, as one can directly generate temporal networks from nearby initial graph conditions. To illustrate this, Fig. 4.11(b) shows the evolution of the graph distance between two initially close networks evolving under the logistic map, for a parameter value where the orbit is periodic (the TNs are again generated via the dictionary trick). One initial condition is chosen on the periodic orbit, while the other is a nearby graph outside the attractor. As shown, the initial distance shrinks exponentially, and the slope provides an estimate of the nMLE, which is negative and in good agreement with the theoretical expectation.

Beyond checking for periodicity, there are several alternative ways to extract negative Lyapunov exponents from scalar time series when recurrence-based methods do not work. One class of methods improves on the limitation of local linearisation by explicitly including curvature in the reconstructed flow. For example, in [166] the authors show that fitting a second-order (or higher) Taylor expansion to the neighbourhood-to-neighbourhood map in delay space significantly improves the estimation of contracting directions. The additional nonlinear terms capture the local geometry of the stable manifold and remain robust under moderate observational noise. This makes it possible to obtain the full Lyapunov spectrum—including negative exponents—from a single scalar observable, as long as enough neighbours are available for stable polynomial regression. A second approach uses neural networks to learn a global surrogate of the one-step map, from which the Jacobian can be computed along the reconstructed trajectory. Yakovleva et al. [167] show that these network-based Jacobian estimators can accurately capture negative exponents, work with shorter data segments, and reduce the bias caused by manifold thickness and noise that affects purely geometric methods. Together, these approaches provide practical alternatives for scalar time series, complementing classical methods while overcoming their main limitations in strongly contracting directions.

4.3.6 High-dimensional chaotic networks

We now turn to the case of high-dimensional chaotic dynamics in temporal networks. To this end, we introduce a generative model based on *coupled map lattices* (CMLs), which are high-dimensional dynamical systems with discrete time and continuous state variables. These systems are widely employed to model complex spatio-temporal dynamics across diverse fields such as turbulence [168, 169], financial markets [170], biological populations [171], and even quantum field theory [172].

A particularly useful variant is the *globally coupled map* (GCM) [173], a mean-field version of a CML in which the local diffusive coupling is replaced by an all-to-all interaction. The dynamics of a GCM with m units is given by

$$x_i(t+1) = (1-\alpha)F[x_i(t)] + \frac{\alpha}{m}\sum_{j=1}^{m} F[x_j(t)], \quad i = 1, 2, \ldots, m, \tag{4.22}$$

where $F(x) = 4x(1-x)$ is the logistic map on the interval $x \in [0,1]$, and $\alpha \in [0,1]$ controls the coupling strength. In the uncoupled case ($\alpha = 0$), each unit evolves independently, and the full system consists of m uncoupled chaotic maps. In this regime, all Lyapunov exponents equal $\ln 2$, giving a maximum Lyapunov exponent of $\lambda_{\max} = \ln 2$.

At the opposite extreme, when $\alpha = 1$, the system becomes fully synchronized: $x_i(t) = x_j(t)$ for all i, j at every time t, and the dynamics again reduce to the one-dimensional logistic map with $\lambda_{\max} = \ln 2$. Notably, complete synchronization is known to occur for $\alpha > 1/2$ [173].

For intermediate values of α, the system exhibits a variety of macroscopic phases [173]. In particular, for weak coupling $\alpha < 0.2$, the GCM enters the so-called *turbulent state*, characterized by high-dimensional chaos. In the case of CMLs with diffusive coupling, a scaling law has been derived for the maximum Lyapunov exponent [174]:

$$\lambda_{\max} = \log 2 - \beta\alpha^{1/p}, \tag{4.23}$$

where p denotes the type of nonlinearity in $F(x)$ (e.g., $p = 2$ for the logistic map, $p = 1$ for the tent map). For GCMs, the applicability of this scaling is more nuanced. In particular, when the mean-field coupling becomes effectively *thermalized* (i.e., independent of x) [175, 176], Eq. (4.23) holds with $\beta = 1$, but this condition is only met in specific cases, such as for tent maps. In what follows, we focus on the turbulent regime $\alpha \in [0, 0.2]$.

Temporal Networks from GCM Trajectories

We interpret the collection $\{x_i\}_{i=1}^{m}$ as representing the (weighted) edge set of a fully connected undirected network of n nodes and $m = n(n-1)/2$ edges. For each edge i, we generate a time series $\{x_i(t)\}_{t=1}^{T}$ from Eq. (4.22) and then apply a two-symbol generating partition: values $x_i(t) < 1/2$ are mapped to symbol 0, while $x_i(t) \geq 1/2$ are mapped to symbol 1 [177]. This symbolic dynamics preserves the key chaotic features of the original signal [178–180]. The binarized time series for each edge is then used to construct a time-dependent adjacency matrix, yielding the temporal network $\mathcal{S}$. For small values of α, we expect this network to exhibit sensitive dependence on initial conditions.

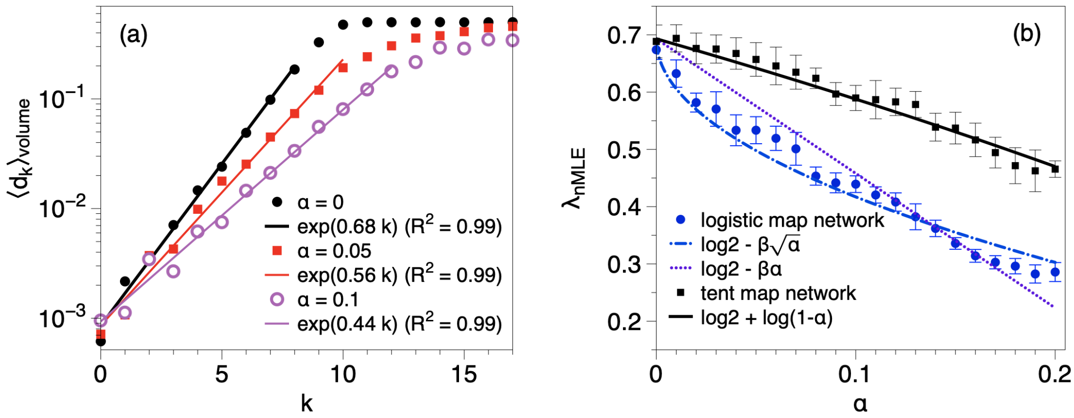

FIGURE 4.12: **Network Lyapunov Exponents from GCM Dynamics.** Panel (a): Semi-log plot of the volume-averaged distance $\langle d_k \rangle_{\text{volume}}$ as a function of the time step k, for a temporal network extracted from the GCM model with coupling constant $\alpha = 0, 0.05, 0.1$. We observe an exponential phase, with different exponents for each value of the coupling constant. The solid lines are the best exponential fits. Panel (b): Estimate of the network maximum Lyapunov exponent λ_{nMLE} vs the coupling constant α for temporal networks generated from a GCM of logistic maps (blue circles) and tent maps (black squares). For each α, a total of 50 initial conditions were considered, and a ball of 100 points for each initial condition was used. Error bars are standard deviations from the average over 50 different temporal network realizations. Blue lines report the theoretical predictions for logistic and tent CMLs [Eq. (4.23)], whereas the black line reports the theoretical prediction for GCM with thermalised mean-field, applicable for tent GCMs only.

In practice, using Wolf/Kantz-type methods to estimate the largest Lyapunov exponent from a single trajectory $\mathcal{S}$ becomes challenging for large n, as very long sequences are needed to observe sufficiently close recurrences. However, in this setting we have access to the underlying GDS defined by Eq. (4.22), allowing us to generate the temporal network from any desired initial condition. To demonstrate that high-dimensional chaotic networks can indeed be constructed and their nMLE estimated, we generate a pair of initial conditions $\{x_i(0)\}$ and $\{x_i'(0)\}$ such that $|x_i'(0) - x_i(0)| < \epsilon$ for small ϵ. For each pair, we generate the corresponding temporal networks and track their network distance over time.

We repeat this process for 100 perturbed replicas per initial condition and for 50 distinct initial conditions to compute volume-averaged distances. Unlike the model from the previous section, the number of edges in each snapshot of this network fluctuates over time, resulting in a larger network phase space. Consequently, the normalization for the distance function is set by the total number of possible edges, $n(n-1)/2$.

Figure 4.12 shows results for a network of $n = 100$ nodes. In panel (a), we plot the average network distance $\langle d_k \rangle_{\text{volume}}$ as a function of time k for three different values of α within the weak coupling regime. In each case, we observe a clear exponential growth, confirming the presence of high-dimensional chaos. For $\alpha = 0$, the observed exponent matches $\ln 2$, as expected. Interestingly, increasing α leads to a decrease in the nMLE and an increase in the saturation time τ.

Panel (b) displays the estimated λ_{nMLE} values (blue circles) as a function of $\alpha \in [0, 0.2]$, showing a systematic decrease. This suggests that as coupling increases, the m degrees of freedom become more interdependent, reducing the overall instability of the system. Theoretical predictions for logistic and tent CMLs based on Eq. (4.23) are also shown (blue lines). For completeness, we perform the same analysis using GCMs constructed from tent maps, where $\lambda_{\max}$ is analytically known. In this case, we use the map $F(x) = 1 - 2|x|$ on the interval $[-1, 1]$, with symbolic partitioning at $x = 0$. The corresponding results are plotted as black squares in panel (b).

In summary, (i) the generated temporal networks exhibit high-dimensional chaos, and the nMLEs reconstructed with our method follow the expected behavior; and (ii) this result validates the robustness of our method in settings where both the structure and the total number of edges evolve over time.

4.4 Conclusions

In the first part of this chapter, we explored how autocorrelation functions can be used to characterize temporal dependencies in labelled temporal networks. The scalar and matrix-valued autocorrelation measures introduced in [28] allow us to quantify the persistence of network structure over time, revealing periodicity, memory, and long-range correlations. These tools were validated on synthetic models—such as noisy periodic networks and temporally correlated DARN processes—as well as on empirical datasets, including human proximity networks and air traffic systems.

While these results highlight how autocorrelation captures patterns of regularity and memory in temporal networks, this perspective does not directly address the question of *stability* of trajectories—that is, whether the network evolution is sensitive to small perturbations in initial conditions. This question lies at the heart of deterministic chaos and is central to understanding the predictability of time-evolving complex systems.

This leads us to the central contribution of the chapter: a novel framework for detecting and quantifying *chaotic dynamics* in temporal networks via a dynamical systems perspective. We propose to interpret a temporal network as the trajectory of an underlying Graph Dynamical System (GDS), even when the governing equations are unknown. Under this interpretation, the natural next step is to ask whether such network trajectories exhibit sensitive dependence on initial conditions—a defining feature of chaos.

To address this, we introduce the notion of the *network Maximum Lyapunov Exponent* (nMLE), and propose a method to estimate it directly from temporal network data. Since the latent GDS is rarely accessible in empirical systems, our algorithm relies on the recurrence properties of the network trajectory in graph space. It generalizes classical Lyapunov exponent estimators (such as the Wolf and Kantz algorithms) from scalar or vector time series to trajectories of graphs.

This approach not only allows us to postulate the existence of chaos in temporal networks, but also provides an operational method to quantify it. We validate our method using several synthetic GDS models where the true Lyapunov exponents are known, and show that our estimator is robust and consistent across different types of network dynamics.

Conceptually, this work is related to studies in statistical physics that examine the stability of evolving structured systems, such as spin lattices. Our perspective bears resemblance to *damage-spreading* and *self-overlap* methods [181, 182], which have been used to study cellular automata [183] and random Boolean networks [184]. At the same time, in nonlinear dynamics, the maximum Lyapunov exponent $\lambda_{\max}$ is a well-established measure of a system's predictability horizon, with its inverse defining the so-called Lyapunov time—the window over which reliable forecasts are possible. Extending this concept to the network domain thus opens the possibility of identifying predictability limits in temporal networks, with potential applications in forecasting [185].

Open Questions and Future Directions

Several conceptual and methodological directions emerge from this contribution:

(i) **Beyond exponential divergence.** While our focus is on exponential separation of trajectories—i.e., sensitive dependence—the method can be generalized to capture other types of dynamical instability, such as algebraic divergence, by suitably adapting the definition of expansion rate.

(ii) **Combining observables for large networks analysis.** Our approach treats the network as a single evolving object in graph space, following the framework introduced in [28]. While this representation is effective for networks of moderate size, in large systems the probability of observing close recurrences in the full graph space becomes very low, making the estimation of divergence rates unreliable. One possible way to address this limitation is to extract a set of informative scalar observables from the network (e.g., link density, clustering coefficient, motif counts), analyze their sensitivity to initial conditions independently, and then combine the results into an indicator of dynamical instability.

(iii) **Full Lyapunov spectrum.** In this chapter we focus on $\lambda_{\max}$, which captures coarse-grained stability. However, computing the full Lyapunov spectrum would provide deeper insight into the multidimensional instability of the graph trajectory. Since the number of degrees of freedom scales with the number of links, this is a challenging but meaningful extension.

(iv) **From labelled to unlabelled networks.** Here, we restricted our analysis to labelled temporal networks—where node identities are preserved across time—because this is typical of most empirical datasets. However, many systems evolve independently of node labels. Extending our framework to unlabelled networks requires the use of permutation-invariant graph distances [186], such as spectral distances or graph kernels [187]. We explore this important extension in the next chapter.

(v) **Applications to empirical systems.** One key strength of our method is that it does not require access to the governing GDS; it can be applied directly to observed temporal network data. This opens up a wide range of applications in physical, biological, economic, and social systems. In particular, it is especially relevant in domains where deterministic dynamics are suspected but not fully known—such as brain activity, biological interactions, or collective behavior [188–191]. In chapter 6, we will explore applications of this method to functional brain networks.

This framework provides a new lens through which to view the dynamics of temporal networks—one that captures their dynamical characteristics at a global scale, going beyond local structural patterns to reveal how the network as a whole evolves.

Chapter 5

Metrics to Characterise the Intrinsic Dynamics of Unlabelled Temporal Networks

"We all need mirrors to remind ourselves who we are."
Leonard Shelby, *Memento* (movie)

5.1 Introduction

In the previous chapter, we introduced a new perspective in which temporal networks are interpreted as trajectories in a graph space, allowing us to study their dynamics at the level of entire network configurations rather than focusing solely on the evolution of individual links. When node identities are known and consistently tracked, each snapshot G_t can be represented by its adjacency matrix A_t, and the full temporal network as a sequence $\mathcal{S}_{\mathcal{G}} \equiv \mathcal{S}_{\mathcal{A}} := (A_1, A_2, \ldots, A_T)$. This allows the use of algebraic tools to characterize network dynamics.

However, in many real-world systems, node labels are difficult to track or even detect reliably. In some cases, such as proximity networks of animal groups or biological swarms [192–194], tracking individual identities is technically challenging. In others, such as epidemic-spreading datasets, privacy constraints may prevent consistent labelling [195–198]. In chemistry, comparing molecular structures involves matching nodes across graphs without a clear one-to-one correspondence—a problem known as molecular similarity [199]. In such cases, each snapshot must be treated as an *unlabelled* graph—a structure defined only up to permutations of node identities. An unlabelled graph of N nodes corresponds to a whole class of $N!$ adjacency matrices under the graph's automorphism group [77]. While it is always possible to assign random labels to construct a surrogate sequence of matrices, doing so disrupts the temporal coherence and compromises the interpretability of any dynamical analysis of the temporal network.

This raises a fundamental question: how can we characterize temporal dynamics in the absence of node identity? The problem is linked to classical graph-theoretical questions, such as isomorphism testing [200–202], and to efforts in defining distances between unlabelled graphs [203–207]. Yet, relatively little attention has been paid to unlabelled *temporal* networks, especially from a dynamical systems perspective.

In this chapter, we address this gap by starting from labelled temporal networks—whether synthetic or empirical—and removing node labels to analyze the resulting unlabelled trajectories. We then ask whether core dynamical features, such as memory, periodicity, or sensitivity to initial conditions, can be recovered using only permutation-invariant information.

We proceed by defining a set of graph-invariant pseudo-distances and use them to construct analogues of autocorrelation functions and Lyapunov exponents for unlabelled network sequences. These tools are validated across a range of temporal processes, from periodic and chaotic dynamics to memory-driven and empirical systems.

The rest of the chapter is organized as follows. In Sec. 5.2, we introduce the graph isomorphism problem and discuss the challenges of working with unlabelled structures, particularly when trying to define meaningful distances between them. In Sec. 5.3, we introduce three permutation-invariant pseudo-distances. Sec. 5.4 then builds on these to define dynamical indicators, which we test on both synthetic and empirical datasets. We conclude in Sec. 5.5 with a discussion of the broader implications of working in the unlabelled regime.

5.2 Comparing Unlabelled Graphs and the Graph Isomorphism Problem

The analysis of unlabelled network trajectories requires a robust way to compare unlabelled graphs. Given two generic unlabelled graphs G and H, the core difficulty is that such graphs do not come with a unique node labelling. As a result, any naive comparison based on adjacency matrices or node-indexed quantities can produce misleading results, since those representations depend on arbitrary labellings.

This issue is deeply connected to the graph isomorphism problem. Two graphs G and H are said to be *isomorphic* if there exists a bijection between their node sets that preserves adjacency, i.e., a relabelling of the nodes of G that maps it exactly onto H. A set of graphs isomorphic to each other is called an *isomorphism class* of graphs. Determining whether two unlabelled graphs are structurally identical is thus equivalent to solving the graph isomorphism problem. Graph isomorphism provides a binary notion of equivalence but does not offer a continuous or graded way to compare graphs that are not strictly isomorphic — a requirement in applications such as dynamical networks, where we often seek to quantify how different two graphs are. Moreover, verifying graph isomorphism is computationally challenging in practice, and the complexity of this task increases rapidly with the number of nodes, making it unsuitable for large-scale or real-time applications.

Several strategies have been proposed to circumvent this challenge:

- **Canonical labelling**, which assigns a unique representative (a canonical form) to each isomorphism class of graphs [203];
- **Graph invariants**, that is, properties that are preserved under node relabelling and can be efficiently computed.

In the next section, we propose a solution based on the second approach: using graph invariants to construct *permutation-invariant pseudo-distances* between unlabelled graphs. While this method is not exact in distinguishing all non-isomorphic graphs, it is computationally efficient and well-suited for comparing graphs in dynamical contexts.

For completeness, we review the canonization approach and discuss its computational and practical limitations in Appendix D.1, and explain why it is not feasible for our purposes.

5.3 A Proposed Solution: Permutation-Invariant Pseudo-Distances

Simple graph invariants which can be obtained in an efficient manner computationally include the following: the degree sequence (and statistical properties derived from it, such as the average degree, the degree distribution, and so on), centrality sequences, the spectrum of the adjacency matrix or the Laplacian matrix, and many others. Below we will construct dissimilarity measures between unlabelled graphs based on some of these properties. More precisely, we will use these invariants to construct *permutation-invariant pseudo-distances* between unlabelled graphs. We stress that distinct graphs can have the same value of simple graph invariants, resulting in a vanishing distance between them. This is the reason why we refer to it as a pseudo-distance between unlabelled graphs. For two given graphs G and H, we use the notation $d(G,H)$, often supplemented by a subscript to indicate the exact type of pseudo-distance.

Of course, the choice of the graph invariant is important, and in a particular application this choice can be informed by the type of graph dynamics at hand. For instance, the total number of edges in a graph is a very simple graph invariant. While this is an important property of the graph, defining a pseudo-distance between two graphs, e.g., as the absolute difference in the number of edges will lack expressivity for graph dynamics in which the total number of edges is preserved.

5.3.1 Three concrete examples of permutation-invariant pseudo-distances

To illustrate the use of permutation-invariant pseudo-distances, we focus on three specific graph invariants: the degree sequence, the spectrum of the adjacency matrix, and the eigenvector centrality vector. In constructing these measures, we assume that the graphs G and H have the same number of nodes—a constraint we leave for future work to relax. Apart from this, the edge sets can be arbitrary, and importantly, these pseudo-distances are applicable to both labelled and unlabelled graphs. Throughout, we consider only simple graphs, i.e., graphs without self-loops or multiple edges between nodes.

Degree sequence pseudo-distance ($\mathbf{d_{deg}}$) – Given two graphs G and H, each with N nodes, we define the degree sequence-based pseudo-distance $d_{\text{deg}}(G,H)$ as a (properly normalized) L_1 distance between the degree sequences:

$$d_{\text{deg}}(G,H) = \frac{1}{N(N-1)} \sum_{i=1}^{N} |k_i^G - k_i^H| \tag{5.1}$$

where k_i^G and k_i^H are the degrees occupying the i-th position of the ordered degree sequence of graphs G and H, respectively. We recall that the degree sequence of a graph is the sequence of the degrees of all nodes, usually sorted in non-increasing order. It encodes information on the graph's connectivity and its heterogeneity. The k_i^H and k_i^G take values between zero and $N-1$, hence the term $1/[N(N-1)]$ in Eq. (5.1) is a normalizing factor ensuring that $d_{\text{deg}}(G,H) \in [0,1]$. The metric $d_{\text{deg}}(G,H)$ is useful for graph dynamics for which the degree sequence changes in time. Since the degree sequence remains invariant under row-column permutations of the adjacency matrix, it is therefore identical for isomorphic graphs. The converse is not necessarily true (two graphs can have the same degree sequence but not be isomorphic to one another), hence while $d_{\text{deg}}(\cdot,\cdot)$ is a distance between degree sequences, it is only a pseudo-distance between graphs.

Spectral pseudo-distance ($\mathbf{d_{spec}}$) – This is the L_1 distance between the spectra of the adjacency matrices of two graphs [208]:

$$d_{\text{spec}}(G,H) = \frac{1}{2N(N-1)} \sum_{i}^{N} |\lambda_i^{A_G} - \lambda_i^{A_H}|, \tag{5.2}$$

where $\lambda_i^{A_G}$ and $\lambda_i^{A_H}$ are the i-th eigenvalues (once they have been ordered) of the adjacency matrices of graphs G and H, respectively. The spectrum of a matrix remains unchanged under row-column permutations and is therefore the same for isomorphic graphs (again, while isomorphic graphs are isospectral, the converse is not necessarily true, hence $d_{\text{spec}}(\cdot,\cdot)$ is pseudo-distance between graphs). The normalization in Eq. (5.2) accounts for the fact that the maximum eigenvalue of the adjacency matrix of a simple graph is always less than or equal to $N-1$ [209]. Additionally, the smallest eigenvalue is greater than or equal to $1-N$. Therefore, the largest possible difference between two eigenvalues is $2(N-1)$.

Eigenvector centrality pseudo-distance ($\mathbf{d_{eig}}$) – We note that d_{deg} uses the degree sequence of the network, also called the degree centrality vector $(k_1,\dots,k_N)$, ordered according to size. The same idea can be extended to centrality properties other than the degrees of the nodes. For illustration, we now consider the so-called eigenvector centrality, i.e. we rank the nodes' importance based on the eigenvector associated with the largest eigenvalue of the adjacency matrix [210, 211]. Let $\mathbf{v}^G = (v_1^G, v_2^G, \dots, v_N^G)$ and $\mathbf{v}^H = (v_1^H, v_2^H, \dots, v_N^H)$ be the eigenvector centrality vectors of graphs G and H. We assume that the components of these vectors are again ordered (i.e. we sort the vectors). We also assume that the vectors are normalised, i.e., $\sqrt{\sum_i (v_i^G)^2} = 1$ and similarly for $\mathbf{v}^H$. Then, we define d_{eig}

as the L_2 distance between the eigenvector centrality vectors:

$$d_{\text{eig}}(G,H) = \frac{1}{2}\sqrt{\sum_{i=1}^{N}\left(v_i^G - v_i^H\right)^2}. \tag{5.3}$$

The prefactor $1/2$ ensures that d_{eig} remains in the interval from zero to one (Because of the normalisation of $\mathbf{v}^G$ and $\mathbf{v}^H$, we have $2 = \|\mathbf{v}^G\| + \|\mathbf{v}^H\| \geq \|\mathbf{v}^G - \mathbf{v}^H\|$ using the triangle inequality, and where $||\cdot||$ is the Euclidean norm). Similarly as before, the sorted eigenvectors remain unchanged under row-column permutations but there exist non-isomorphic graphs with the same eigenvector centrality, hence $d_{\text{eig}}(\cdot,\cdot)$ is a pseudo-distance between graphs.

5.3.2 Exploring the behavior of d_{deg}, d_{spec} and d_{eig}

Before employing graph invariants—and the resulting pseudo-distances—for analyzing the dynamics of unlabelled network trajectories, it is essential to first evaluate their behavior in quantifying dissimilarities between pairs of graphs G and H in a controlled setting. To this end, in Secs. 5.3.2.1 and 5.3.2.2, we introduce two different models that generate labelled graphs G and H, allowing us to tune their level of dissimilarity via a control parameter. This parameter intuitively reflects the magnitude of the perturbation applied to G to obtain H.

As a reference, or ground-truth distance between G and H, we employ a matrix norm: the normalized element-wise L_1-distance between their adjacency matrices,

$$d_{\text{lab}}(G,H) = \frac{1}{\mathcal{Z}}\sum_{i,j=1}^{N}|a_{ij} - b_{ij}|. \tag{5.4}$$

Here, $\mathcal{Z}$ is a normalization factor chosen differently for each model to ensure that d_{lab} lies between zero and one (further details are provided in the following sections). The entries a_{ij} and b_{ij} correspond to the (i,j) elements of the adjacency matrices of G and H, respectively. As before, we assume that G and H have the same number of nodes.

We will first demonstrate that $d_{\text{lab}}(G,H)$ is linearly related to the perturbation parameter used to generate H from G, thereby justifying its role as a ground-truth measure of dissimilarity. We then proceed to remove all label information from G and H, and systematically compare the resulting unlabelled pseudo-distances $d_X(G,H)$—with X representing 'deg', 'spec', or 'eig'—to d_{lab}.

In particular, we aim to assess whether each $d_X(G,H)$ behaves as a non-decreasing function of d_{lab}, whether this relationship is linear or exhibits nonlinear but monotonic behavior, and whether discontinuities arise. These properties will later play a key role when we apply the pseudo-distances to quantify the dynamical behavior of unlabelled network trajectories.

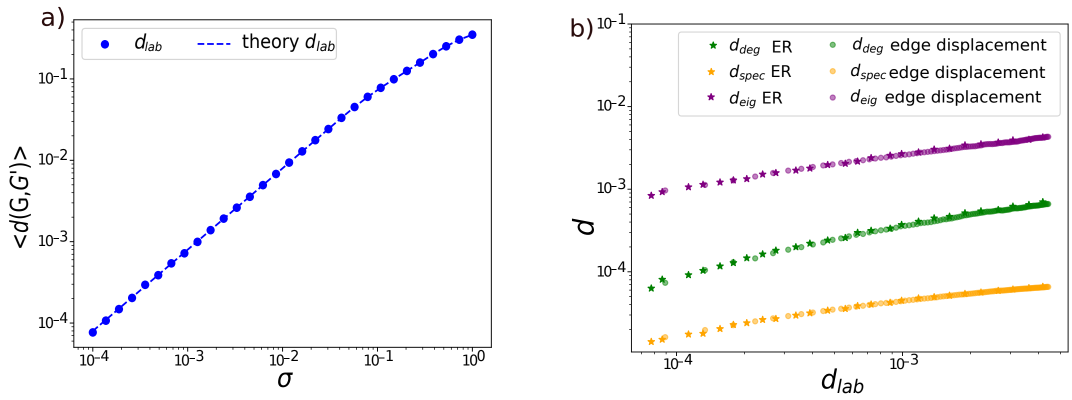

FIGURE 5.1: **Permutation-invariant pseudo-distances and distance between labelled networks** Panel (a) shows numerical results for $\langle d_{\mathrm{lab}}(G, G')\rangle$ as a function of σ (solid circles), see the text for details. The theoretical prediction of Eq. (5.5) is shown in dashed lines. Panel (b) shows log-log plots of scatter plots between the three pseudo-distances d_X and the labelled distance d_{lab}, obtained numerically from the models described in Secs. 5.3.2.1 and 5.3.2.2: data shown as stars is from perturbing Erdős-Rényi graphs (Sec. 5.3.2.1), while solid dots are obtained in the model that displaces edges (Sec. 5.3.2.2). The different colours represent the three different pseudo-distances. The networks consist of 300 nodes and an average of 5000 edges, and each data point is the average over an ensemble of 100 realisations. We observe that the scatter plots obtained via the two different procedures systematically collapse (for each given pseudo-distance). Within the tests performed here, the relation of any given pseudo-distance to the labelled distance appears to be continuous, monotonically increasing, and non-linear.

5.3.2.1 Perturbation using Erdős-Rényi graphs

We start by constructing a reference graph G, which we will then later perturb to construct another graph G' (in the notation of the previous section, $G' \equiv H$). To generate the reference graph, we draw a total of $N(N-1)/2$ standard uniform random variables $x_{ij} \in U(0, 1)$, one for each pair of nodes $i < j$. We then fix a parameter p to a value between zero and one. Nodes i, j are linked in G if and only if $x_{ij} < p$. Thus, G is simply an Erdős-Rényi graph with parameters (N, p). Subsequently, to build the graph G' we 'perturb' the numbers x_{ij}. More specifically, we introduce $x'_{ij} = x_{ij} + \xi_{ij}$, where the ξ_{ij} are *iid* Gaussian random numbers with mean zero and variance σ^2. Nodes i and j are linked in G' if and only if $x'_{ij} < p$. The quantity σ is the model's control parameter, and intuitively it is easy to see that larger values of σ make the perturbation on G to be larger, and thus makes G' 'more different' than G.

We now quantify this using a proper distance between G and G'. To do that, we initially assume that G and G' are labelled. We can then use Eq. (5.4) [in this example we set $\mathcal{Z} = N(N-1)$: this is the maximum value the sum on the right-hand side of that equation can take for graphs of size N]. We can compute the expected distance $d_{\mathrm{lab}}(G, G')$ as a function of σ analytically. This means to average over the ensemble of initial graphs G (with fixed parameter p) and all perturbations ξ_{ij}. The resulting mean distance will depend on the parameter p and on the size of the perturbation, σ. We find (see Appendix D.2) that

$$\begin{aligned} d_{\text{lab}}(\sigma) &= \sigma\sqrt{\frac{2}{\pi}}\left[1-\exp\left(-\frac{p^2}{2\sigma^2}\right)-\exp\left(-\frac{(p-1)^2}{2\sigma^2}\right)\right] \\ &\quad +\frac{1}{2}\left[1-p\cdot\operatorname{erf}\left(\frac{p}{\sqrt{2}\sigma}\right)-(p-1)\cdot\operatorname{erf}\left(\frac{p-1}{\sqrt{2}\sigma}\right)\right]. \end{aligned} \tag{5.5}$$

For small σ this becomes

$$d_{\text{lab}}(\sigma) \approx \sigma\sqrt{\frac{2}{\pi}} + \cdots, \tag{5.6}$$

where the dots represent terms containing factors of the type $\exp\left[-p^2/(2\sigma^2)\right]$, and which hence tend to zero faster than linearly as $\sigma \to 0$. The validity of Eqs. (5.5) and (5.6) is indeed confirmed in simulations, as demonstrated in Fig. 5.1(a). Simulation data are averages over initial graphs G and perturbed graphs G'. We have thus shown that the above construction, combined with varying σ, allows us to parametrically generate pairs of graphs of designated expected distances d_{lab}. Given that d_{lab} increases linearly with σ for small σ, we can now explicitly use d_{lab} as the ground-truth quantifier of the perturbation that distinguishes G from G'.

We are now ready to test the relation between the pseudo-distances $d_X(G, G')$ (which can be measured for the unlabelled versions of G and G') and the ground-truth$d_{\text{lab}}(G, G')$. We have performed extensive numerical simulations and computed $\langle d_{\text{deg}}(G, G')\rangle$, $\langle d_{\text{spec}}(G, G')\rangle$ and $\langle d_{\text{eig}}(G, G')\rangle$ for different choices of σ (the angle bracket denotes an average over realisations of the graphs G and G'). In Fig. 5.1(b) we scatter-plot $\langle d_X(G, G')\rangle$ against $\langle d_{\text{lab}}(\sigma)\rangle$ (the plot is on a log-log scale). We find a monotonically increasing relation, and no discernible discontinuities. However, this relation is not linear.

5.3.2.2 Perturbation through link displacement

A method for constructing a set of graphs $\{H_0, \ldots, H_L\}$ with prescribed pairwise distances (where L is an integer parameter) was proposed in [28, 101] and is described in Sec. 4.3.5. We begin by generating an Erdős-Rényi network H_0 of size N and with parameter p. To construct H_1 we displace one edge in H_0 to a previously vacant location (i.e., we make sure that no double edge is created during the displacement). Subsequently, H_2 is constructed from H_1 by displacing a further edge. This edge is chosen different from the edge that was displaced in the construction of H_1 from H_0, and the place it is displaced to is chosen such that neither H_0 nor H_1 have a edge there. This process continues iteratively until H_L has been constructed. In each step H_n is constructed from H_{n-1} by displacing an edge that is present in all graphs, $H_0, \ldots, H_{n-1}$ to a location that is vacant in all graphs $H_0, \ldots, H_{n-1}$. The set $\{H_0, \ldots, H_L\}$ is referred to as a *dictionary* of graphs in this thesis.

Using Eq. (5.4), it is easy to see that

$$d_{\text{lab}}(H_n, H_m) = \frac{2}{\mathcal{Z}}|n - m| \tag{5.7}$$

In the context of this example, we choose $\mathcal{Z} = 2L$, so that the maximal distance between graphs in the set $H_0, \ldots, H_L$ is $d_{\text{lab}}(H_0, H_L) = 1$.

The above protocol allows us to construct pairs of graphs of different distances d_{lab}. We can then measure the pseudo-distances $d_{\text{deg}}, d_{\text{spec}}$ and d_{eig} between these pairs. Varying $|n - m|$ we can then again produce a parametric plot showing the pseudo-distances between pairs as a function of the distance d_{lab}.

Results are again shown in Fig. 5.1(b). Each marker is an average over realizations of pairs of graphs with the same d_{lab}. The data in the figure shows that the relation between d_X and d_{lab} is the same as that found in Sec. 5.3.2.1. This apparent universality is, however, likely due to the fact that the networks we have used in Sec. 5.3.2.1 and in the current section have the same number of nodes, and a similar average number of edges. Further simulations (not shown here) reveal that the broad shape of the relationship between any one of the pseudo-distances and d_{lab} remains similar when the network size or the edge density is varied, but that the specific details vary (i.e. the parameters of a possible fit). Accordingly, while all functional relationships between d_X and d_{lab} appear to be continuous, non-decreasing and non-linear, we cannot establish a simple and universal quantitative mapping of the form $d_X = f_X(d_{\text{lab}})$.

As a summary, numerical evidence suggests that the pseudo-distances d_X defined above are well-behaved in the sense that they show a monotonic dependence with the ground-truthdistance d_{lab}, and we have not observed discontinuities. As we will describe below, this opens up the prospect of generalizing network autocorrelation functions to unlabelled graphs. The lack of a linear dependence between d_X and d_{lab}, however, suggests that an existing ground-truth sensitivity to initial conditions in a labelled temporal network (marked by exponentially growing labelled distance) cannot necessarily be detected as exponential expansion of d_X from the unlabelled network trajectory. In other words, chaotic unlabelled networks do not necessarily display exponential divergence of nearby conditions, and a more careful analysis is required.

5.4 Quantifying unlabelled network trajectories

In this section, we apply the pseudo-distances defined in Sec. 5.3 to analyze time-varying unlabelled networks. We will explore different types of graph dynamics, e.g., periodic graph orbits subject to noise, autoregressive graphs or chaotic time series of graphs. We also consider real-world temporal networks whose graph dynamics are latent.

Our general procedure consists of the following steps (see Fig. 5.2 for an illustration): First, we generate a labelled temporal network, or we start from an empirical labelled temporal network. Next, we remove the labels for each snapshot to obtain a sequence of unlabelled networks. Finally, we will describe the dynamics of the unlabelled temporal network using metrics based on the set of pseudo-distances defined above. We then compare the resulting behavior with the dynamical description of the labelled temporal networks –which serves as ground truth–, thereby validating the procedure.

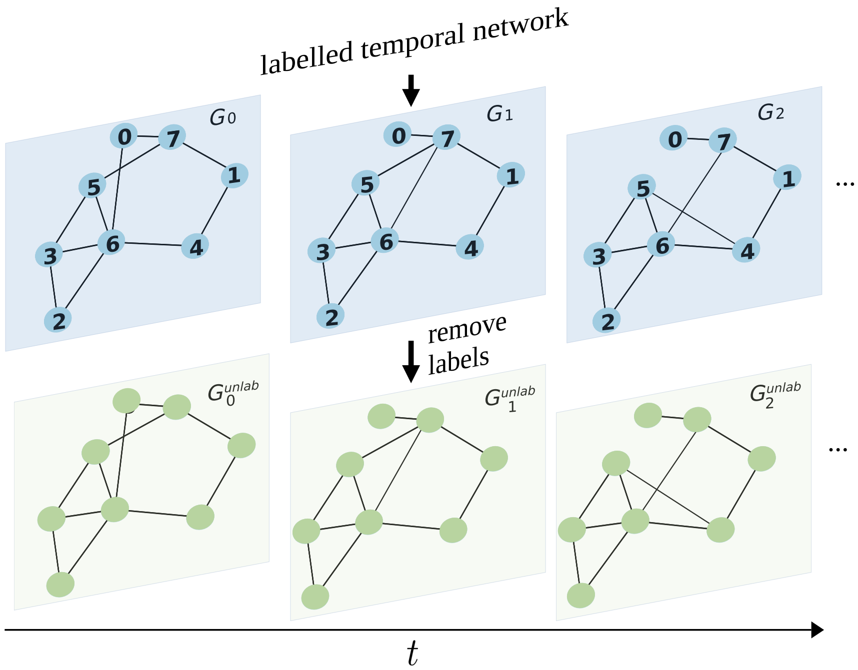

FIGURE 5.2: **Schematic representation of the process for obtaining unlabelled temporal networks.** A labelled temporal network is generated either synthetically, or the starting point is an empirical labelled temporal network. Node labels are removed from each snapshot to simulate the loss of label information, resulting in a sequence of unlabelled networks. These are then analysed with metrics defined in the text.

5.4.1 Sensitivity to initial conditions

We will start by analyzing how the pseudo-distances introduced in Sec. 5.3 behave in cases where the original (labelled) network dynamics is chaotic [101]. Sec. 5.4.1.1 deals with models of low-dimensional chaotic temporal networks, whereas the high-dimensional case is treated in Sec. 5.4.1.2.

5.4.1.1 Low-dimensional chaotic network dynamics

The first analysis focuses on the sensitivity to initial conditions in an artificial temporal network generated from low-dimensional chaotic dynamics. As briefly mentioned in Sec. 5.3.2.2 and previously described in Sec. 4.3, we construct such a network using a chaotic time series and a dictionary-based graph symbolisation, which maps each real value to a labelled graph.

To detect sensitive dependence on initial conditions directly in graph space, we follow the procedure in [101] (and Sec. 4.3.3) and use a network adaptation of

Wolf's algorithm. This method identifies close recurrences in the network trajectory and tracks how distances between initially close states evolve over time. The labelled distance $d_{\mathrm{lab}}(t)$, averaged over recurrent pairs, is shown as a solid blue line in Fig. 5.3(a). We observe an initial exponential growth, $d_{\mathrm{lab}}(t) \sim \exp(\lambda t)$, with $\lambda \approx \ln 2$, consistent with the Lyapunov exponent of the underlying logistic map, followed by a saturation phase.

For comparison, we show $d_{\mathrm{lab}}(t)$ when random node permutations are applied to each snapshot (blue dots in Fig. 5.3(a)), mimicking the scenario in which label information is lost. As expected, this destroys the temporal structure, resulting in a flat curve with $d_{\mathrm{lab}}(t) \approx 0.85$. Shuffling the order of the snapshots produces a similar flat profile (cyan curve), albeit at a slightly lower value, indicating that permuting node labels is a more disruptive intervention than reordering time steps.

In Fig. 5.3(b)–(d), we repeat the analysis using the three pseudo-distances d_X, which operate directly on the unlabelled network sequence and do not rely on node labels. In all cases, we observe an expansion phase, showing that these metrics capture a key signature of chaotic dynamics. However, the growth is not strictly exponential, and the agreement with d_{lab} is only qualitative—consistent with the non-linear relationship between d_X and d_{lab} reported in Sec. 5.3.

That section also suggested that, for specific synthetic models, one might find a monotonic but non-linear function f_X such that $d_X = f_X(d_{\mathrm{lab}})$. If the labelled distance grows exponentially, $d_{\mathrm{lab}}(t) = d(0)\exp(\lambda t)$, then

$$d_X(t) = f_X\left[d(0)\exp(\lambda t)\right], \quad \text{so} \quad \lambda \approx \frac{1}{t}\ln f_X^{-1}\left[d_X(t)\right], \tag{5.8}$$

neglecting additive corrections of order $1/t$. However, since f_X appears to be model-dependent, it must be estimated case-by-case. This prevents the definition of a universal Lyapunov exponent for unlabelled network dynamics that would be valid across different systems.

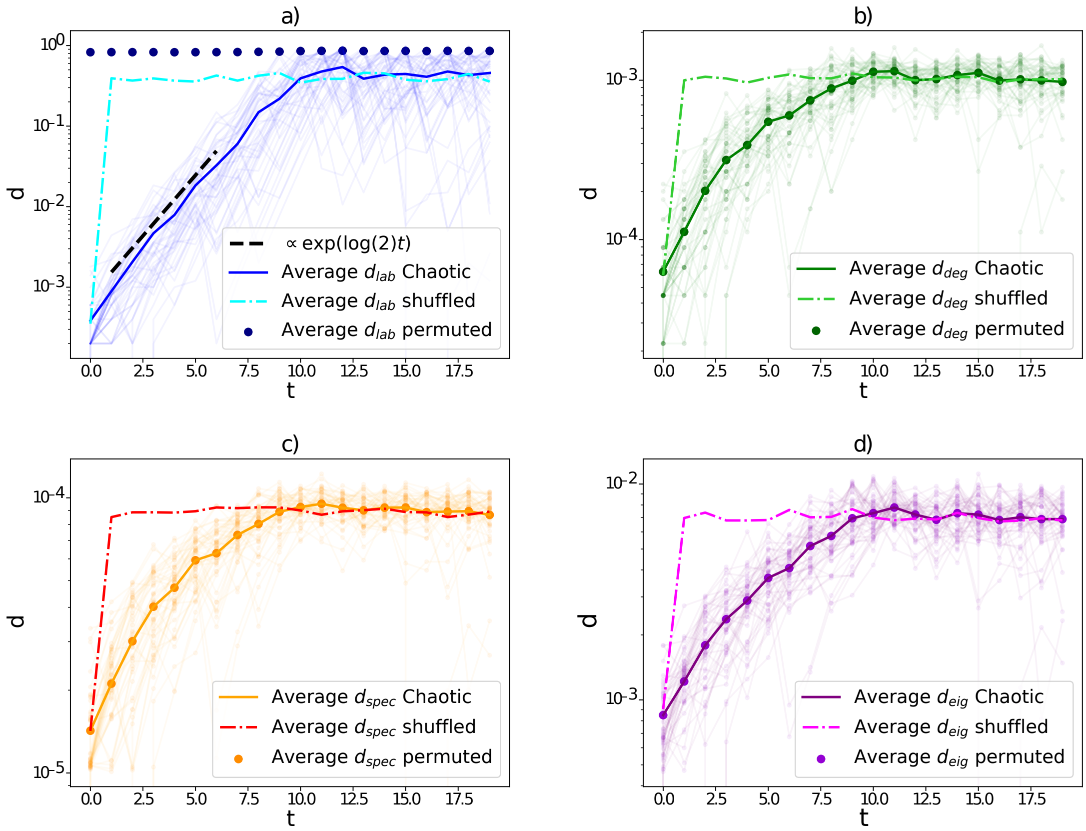

FIGURE 5.3: **Time-evolution of pseudo-distances of initially close (unlabelled) networks evolving via a low-dimensional chaotic dynamics.** Panels [a-d] depict semi-log plots of the different distances of initially close trajectories over time. The network dynamics are generated via a fully chaotic logistic map using the dictionary trick, with a dictionary of $L = 5 \cdot 10^3$ networks, where each network has $N = 300$ nodes (see the text in 5.4.1.1 for details). Each panel highlights in solid line the average distance (average over 50 pairs of close initial conditions), and in solid dots the equivalent result when node labels are shuffled, highlighting that only the pseudo-distances are invariant under random node labelling and are thus genuinely fit to address unlabelled network trajectories. Additionally, we also plot the result after shuffling the snapshot order (dash-dotted line). Panel (a), depicts (for comparison) the labelled distance $d_{\text{lab}}(t)$, that displays a clear exponential phase with an exponent close to $\ln 2$. Panels (b)-(d) depict the expansion as measured by the degree sequence pseudo-distance d_{deg}, the spectral pseudo-distance d_{spec}, and the eigenvector centrality pseudo-distance d_{eig} respectively. All pseudo-distances are capable of capturing the sensitivity to initial conditions although the growth is not strictly exponential, as expected.

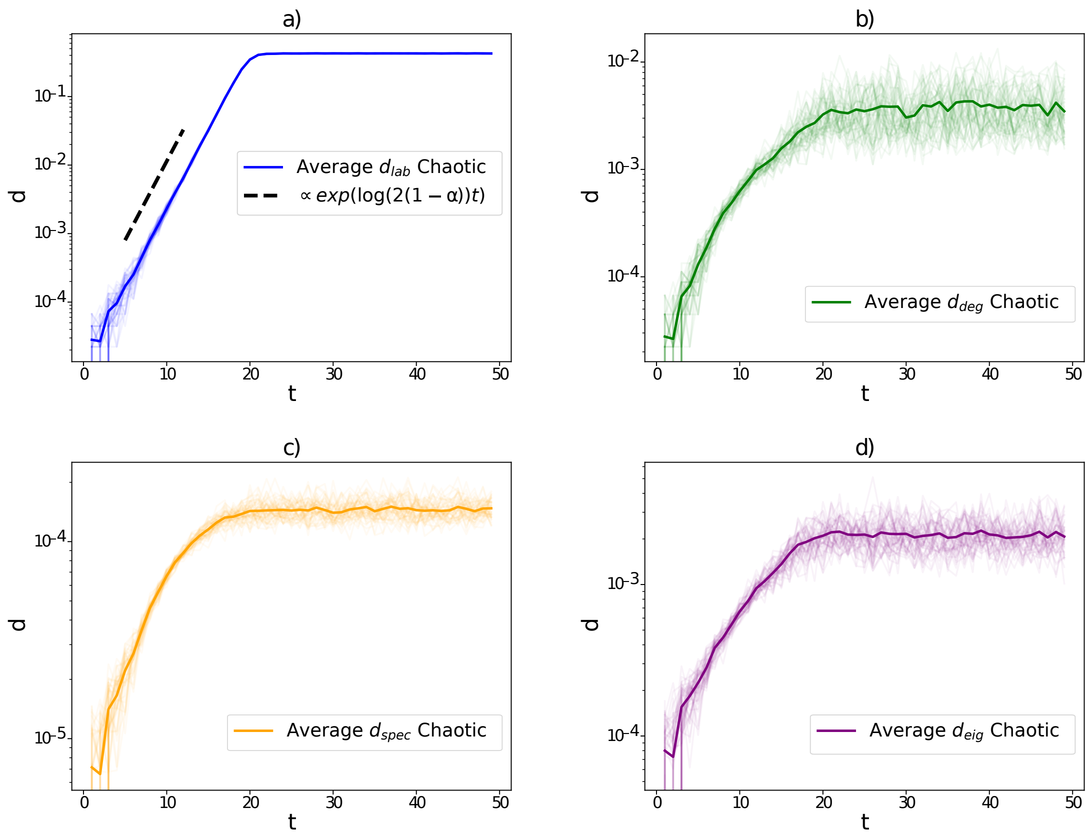

FIGURE 5.4: **Time-evolution of pseudo-distances of initially close (unlabelled) networks evolving via a high-dimensional chaotic dynamics.** This figure is equivalent to Fig. 5.3, except here the dynamics is generated via a system of $m = N(N-1)/2$ globally coupled maps [Eq. (4.22) with $\alpha = 0.15$]. This system describes the (coupled) chaotic evolution of the edges of a network of $N = 300$ nodes. Accordingly, the network trajectories display high-dimensional chaos (see Sec. 5.4.1.2 for details). When measured using the labelled trajectory, $d_{\text{lab}}(t)$ displays a clear exponential phase with an exponent close to the theoretical value $\ln 2(1-\alpha)$. When labels are lost, the pseudo-distances d_X are still capable of capturing the expansion phase, although in these metrics such expansion is not strictly exponential. Panel (a), depicts (for comparison) the labelled distance $d_{\text{lab}}(t)$, that displays a clear exponential phase with an exponent close to $\ln(2(1-\alpha))$. Panels (b)-(d) depict the expansion as measured by the degree sequence pseudo-distance d_{deg}, the spectral pseudo-distance d_{spec}, and the eigenvector centrality pseudo-distance d_{eig} respectively. All pseudo-distances are capable of capturing the sensitivity to initial conditions although the growth is not strictly exponential, as expected.

5.4.1.2 High-dimensional chaotic networks

To validate the previous findings in a higher-dimensional setting, we now repeat the analysis using the globally coupled map (GCM), which was previously introduced in Sec. 4.3.6 for the labelled case. This system involves $m = N(N-1)/2$ coupled variables, each associated with a potential edge x_ℓ of a network of size

N. We recall that the evolution of these edge variables follows

$$x_\ell(t+1) = (1-\alpha)F\big(x_\ell(t)\big) + \frac{\alpha}{m}\sum_{\ell'=1}^{m} F\big(x_{\ell'}(t)\big), \quad \ell = 1,2,\ldots,m, \tag{5.9}$$

where $F(x) = 1 - 2|x|$ is the chaotic tent map, and $\alpha \in [0,1]$ controls the coupling strength. The network trajectory is obtained by binarising the edge states: edge ℓ is present at time t if and only if $x_\ell(t) > \frac{1}{2}$. This construction yields a symbolic network dynamics that inherits the chaotic properties of the underlying GCM, particularly exhibiting turbulent behaviour in the weak coupling regime.

In this setting, since we have full control over the generative dynamics, we can directly initialise pairs of trajectories from nearby initial conditions, bypassing the need to search for recurrences in the temporal network. We then track the evolution of the pseudo-distances $d_X(t)$ between these trajectory pairs. As shown in Fig. 5.4(a), the labelled distance $d_{\text{lab}}(t)$ exhibits exponential growth, consistent with a positive network Lyapunov exponent matching the theoretical prediction for the GCM, $\lambda = \ln[2(1-\alpha)]$ [212–214]. The behavior of the pseudo-distances $d_X(t)$ is reported in Fig. 5.4(b)–(d). As in the low-dimensional case, all three pseudo-distances show a clear expansion phase, indicating sensitivity to initial conditions in the unlabelled domain. However, their growth is not strictly exponential, again reflecting the non-linear and model-dependent relationship between d_X and d_{lab}.

5.4.2 A Network Autocorrelation Function for Unlabelled Network Trajectories

5.4.2.1 Construction of Metrics

As a second approach to characterizing the dynamics of unlabelled temporal network trajectories, we introduce a family of autocorrelation-like functions based on the pseudo-distances defined in Sec. 5.3. These functions are designed to generalize the idea of temporal autocorrelation to the unlabelled setting, where node identities are no longer available.

The classical scalar autocorrelation function $\tilde{c}(\tau)$ for labelled temporal networks, introduced in [28], was reviewed in Sec. 4.2. The key idea behind it is to quantify the similarity between a network snapshot and a time-lagged version of itself. In the unlabelled case, we preserve this idea by replacing the inner product used in $\tilde{c}(\tau)$ (Eq. 4.2) with a similarity measure derived from permutation-invariant pseudo-distances between graphs.

Given a sequence of unlabelled graphs $G_1, G_2, \ldots, G_T$, and a pseudo-distance d_X for $X \in \{\text{deg}, \text{spec}, \text{eig}\}$, we define:

$$c_X(\tau) := 1 - \frac{1}{T-\tau}\sum_{t=1}^{T-\tau} \frac{d_X(G_t, G_{t+\tau})}{\mathcal{J}}, \tag{5.10}$$

where

$$\mathcal{J} = \frac{1}{T(T-1)} \sum_{t,t'=1}^{T} d_X(G_t, G_{t'}) \tag{5.11}$$

is the average pseudo-distance between all pairs of snapshots in the trajectory.

This construction satisfies properties analogous to those of a standard autocorrelation function. In particular, $c_X(0) = 1$ by definition, and $c_X(\tau) \to 0$ for large τ in memoryless systems, where the average lagged distance becomes indistinguishable from the overall average distance $\mathcal{J}$.

In the following, we examine the behavior of these autocorrelation-like functions on synthetic trajectories with periodicity or memory, as well as on real-world temporal networks. We compare this with the scalar autocorrelation function previously introduced for the labelled version of the same temporal network models $\tilde{c}(\tau)$ (Eq. 4.2) [1], highlighting how the unlabelled perspective retains key dynamical signatures.

5.4.2.2 Test on Noisy Periodic Network Trajectories

We begin by testing $c_X(\tau)$ on a noisy periodic temporal network, originally introduced in [28]. While the details of the model were provided earlier in Sec. 4.2.1, here we emphasize that we work with unlabelled graph sequences, obtained by removing the labels from the labelled noisy periodic temporal network. Periodicity is introduced by repeating a sequence of T_{period} independent Erdős-Rényi graphs, which are then perturbed with noise.

In Fig. 5.5, we show the autocorrelation $\tilde{c}(\tau)$ for the labelled version (panel a) and the unlabelled autocorrelation-like functions $c_X(\tau)$ (panels b–d). In all cases, a clear peak appears at $\tau = T_{\text{period}} = 20$, successfully revealing the periodic structure even in the presence of moderate noise. Among the three pseudo-distances, $c_{\text{deg}}(\tau)$ yields the sharpest peak.

[1] we reproduce the graph showing the behavior of $\tilde{c}(\tau)$ for the sake of comparison.

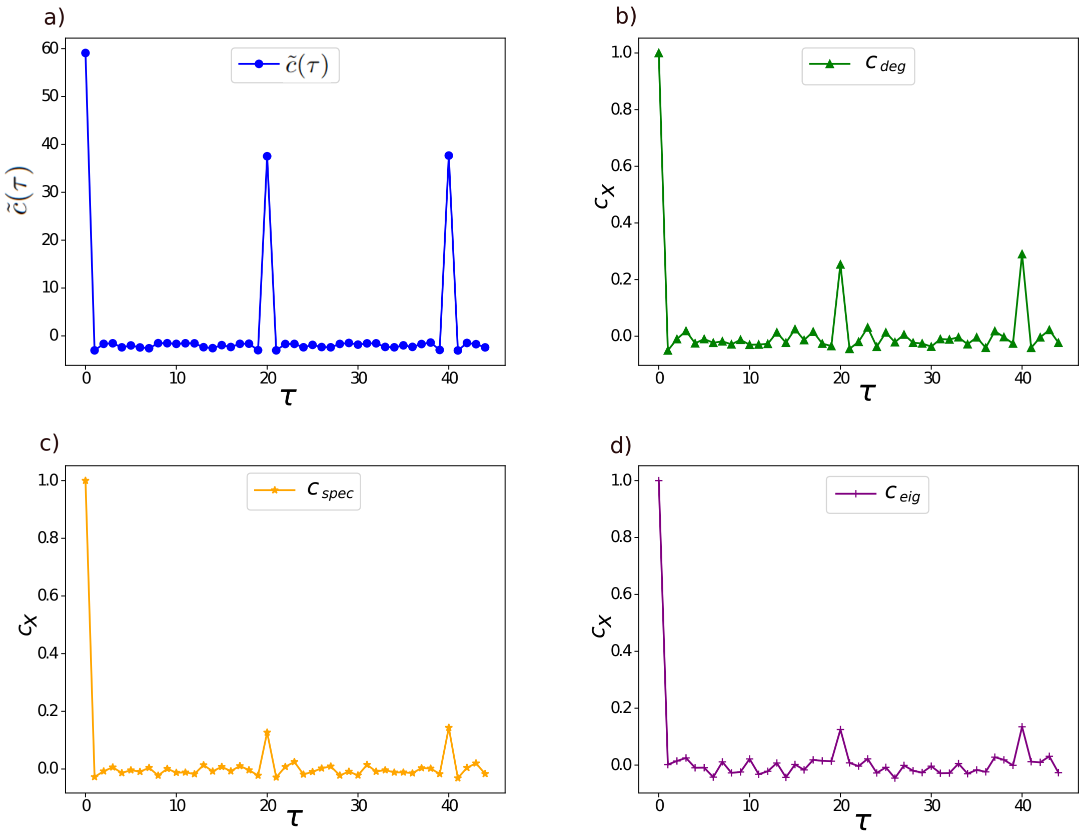

FIGURE 5.5: **Autocorrelation functions for noisy periodic networks.** (a) Scalar autocorrelation function $\tilde{c}(\tau)$ for a labelled network trajectory. (b–d) Autocorrelation-like functions $c_X(\tau)$ computed on the unlabelled version using degree, spectrum, and eigenvector-based distances, respectively. While the labelled case shows stronger peaks, the unlabelled metrics reliably detect the true period.

To quantify detectability, we compute a z-score as in Sec. 4.2.1:

$$z_X = \frac{c_X(T_{\text{period}}) - \mu_X}{\sigma_X}. \tag{5.12}$$

In Fig. 5.6, we plot z_X as a function of noise q. While $\tilde{c}(\tau)$ detects periodicity up to high noise levels ($q \approx 0.9$), the unlabelled functions $c_X(\tau)$ cross the detectability threshold ($z_X > 4$) up to $q \approx 0.4$, still demonstrating robustness in the face of low to moderate noise.

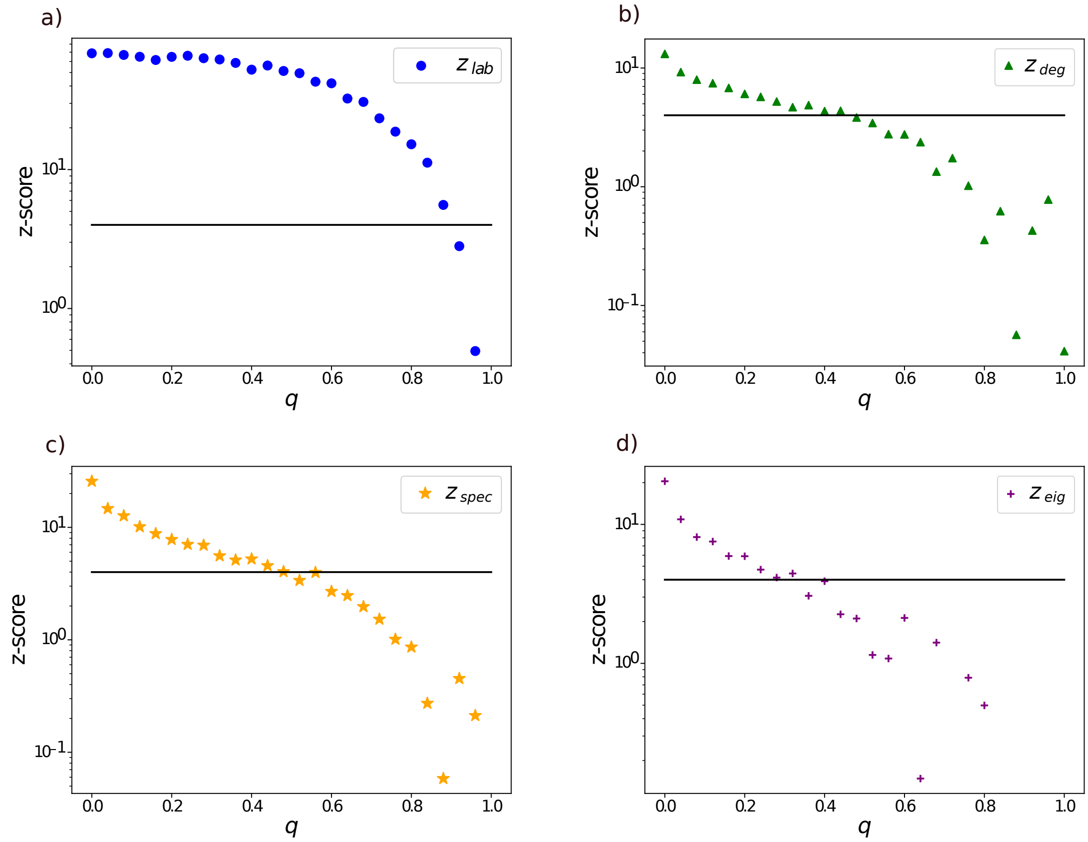

FIGURE 5.6: **Robustness against noise.** z-scores for periodicity detection versus noise level q. Panel (a): labelled autocorrelation. Panels (b–d): unlabelled autocorrelation functions based on different pseudo-distances. The threshold $z = 4$ (horizontal line) marks reliable detection.

5.4.2.3 Test on Discrete Autoregressive Unlabelled Network Trajectories

We next consider a temporal network model with tunable memory: the Discrete Autoregressive Network model of order ρ (DARN(ρ)) [152]. In Sec. 4.2.2, we defined this generative process and analyzed the scalar autocorrelation function for temporal networks generated in the labelled setting. In this section, we remove node labels from these networks and analyze the corresponding autocorrelation-like functions $c_X(\tau)$ defined in Eq. 5.10.

In Fig. 5.7, we compare $\tilde{c}(\tau)$ and $c_X(\tau)$ for several memory lengths ρ. In all cases, both labelled and unlabelled autocorrelation functions exhibit a plateau up to $\tau \approx \rho$, followed by an exponential decay. The pseudo-distances based on degree and spectrum track the labelled behavior closely, while the eigenvector-based measure is again noisier and less reliable.

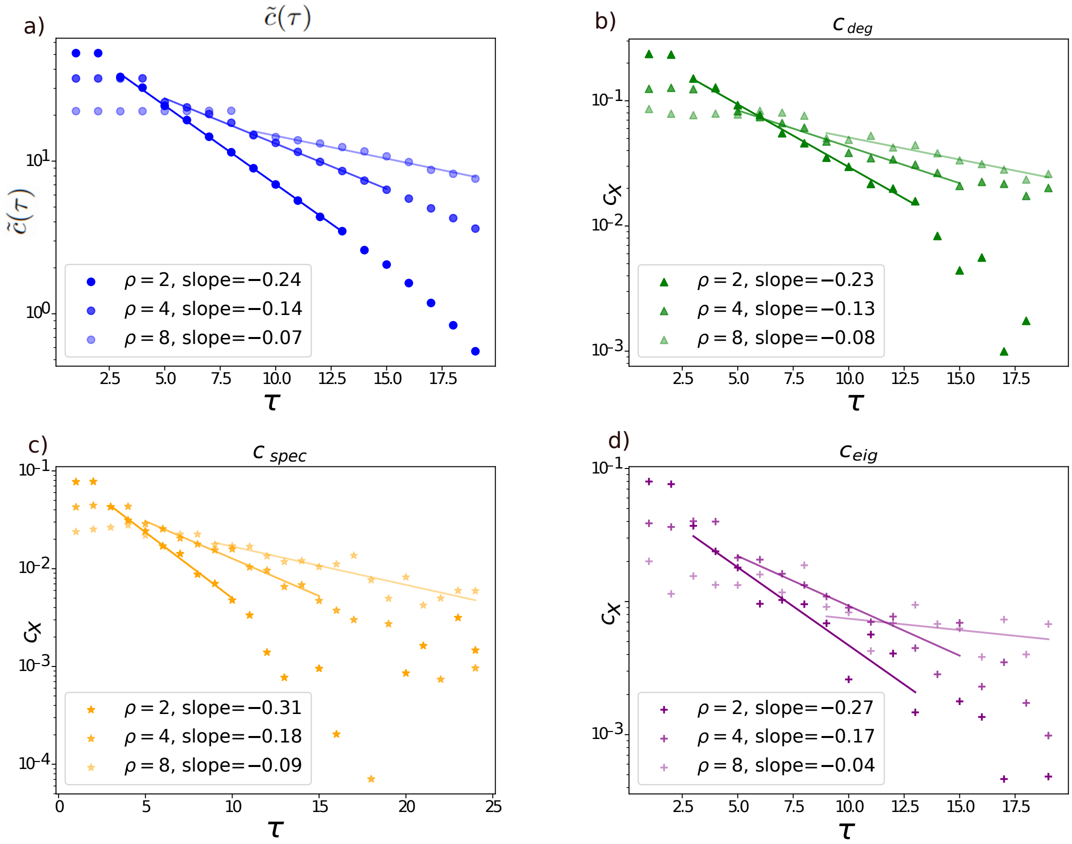

FIGURE 5.7: **Autocorrelation-like functions detect memory in DARN processes.** Semi-log plots of autocorrelation versus lag τ for labelled (a) and unlabelled (b–d) network trajectories, at varying memory length ρ. Unlabelled functions $c_X(\tau)$ successfully recover the flat-then-decay profile characteristic of temporal memory.

5.4.3 Empirical (Unlabelled) Temporal Network Trajectories

Finally, we test our autocorrelation-like functions $c_X(\tau)$ on empirical temporal networks. We consider three datasets already presented in Sec. 4.2.3: (i) US domestic flights in February 2014 [153], (ii) face-to-face contact in a Malawian village [154], and (iii) the SFHH conference in France [155]. While the original data includes node labels, we remove them and evaluate the performance of $c_X(\tau)$ on the resulting unlabelled sequences, comparing them to $\tilde{c}(\tau)$ as a form of validation.

Figure 5.8 summarizes our results. For the flight data, both labelled and unlabelled autocorrelation functions reveal a clear 24-hour periodicity. For the contact datasets, the decay of autocorrelation is slower and indicative of long-range temporal correlations, which are broadly preserved in the unlabelled setting. These results confirm that $c_X(\tau)$ can capture meaningful temporal structure in real-world networks even when node identities are unavailable.

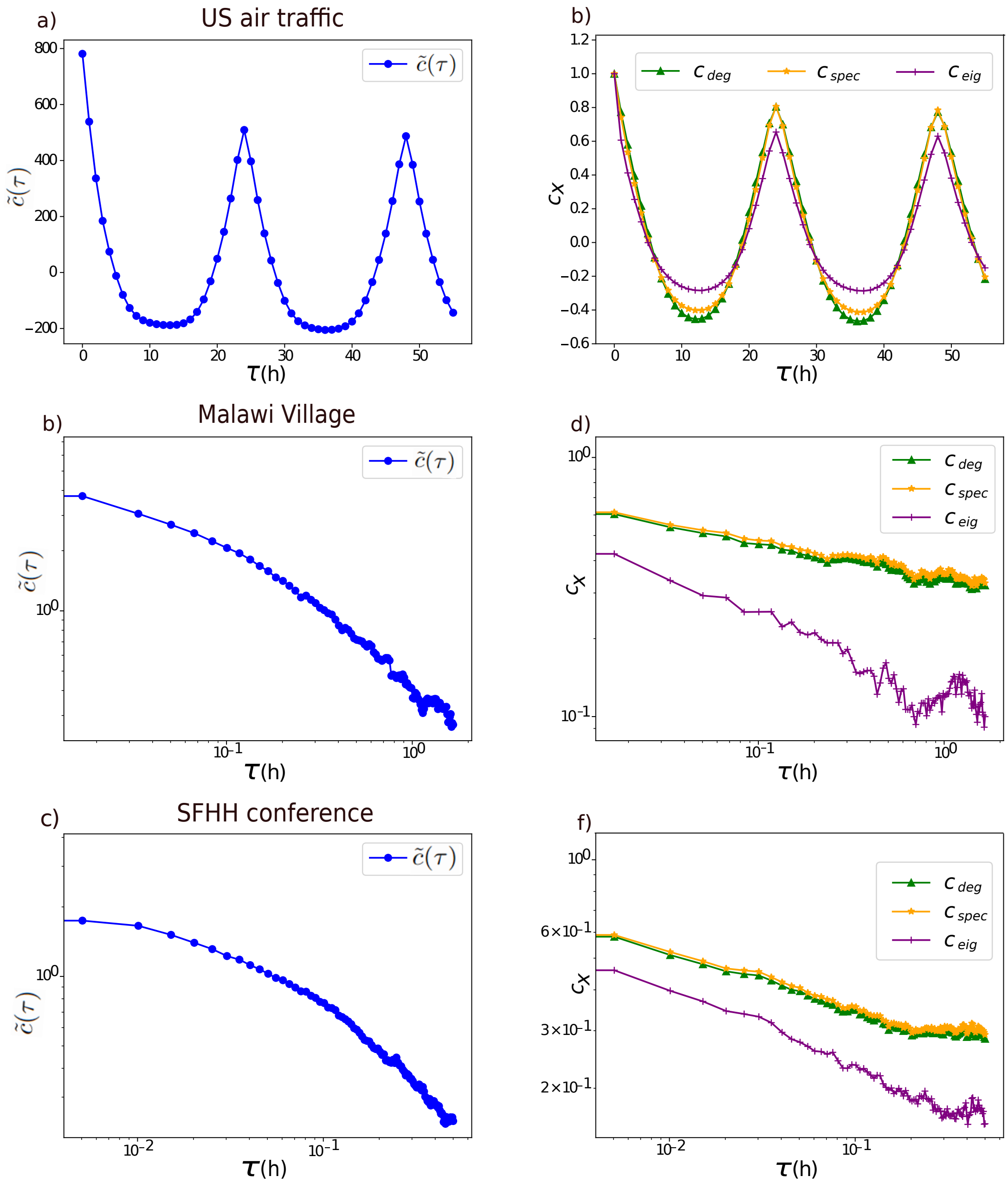

FIGURE 5.8: **Autocorrelation-like functions in empirical networks.** (a,c,e): $\tilde{c}(\tau)$ for labelled trajectories. (b,d,f): $c_X(\tau)$ for corresponding unlabelled versions. (a,b): US flights show strong daily periodicity. (c–f): Human contact networks display slow decay, indicative of long-range memory, recovered also in the unlabelled case.

5.5 Conclusion

In this chapter, we have shown that key dynamical features of labelled temporal networks can be partially recovered even after node labels are removed. By leveraging graph invariants—properties preserved under node relabelling—we extended core dynamical concepts such as autocorrelation functions and sensitivity to initial conditions to the unlabelled setting.

We validated this approach using a variety of generative models for temporal networks, encompassing periodic, correlated, and chaotic dynamics, in both labelled and unlabelled forms. Our results show that certain dynamical signatures, such as chaos (Figs. 5.3 and 5.4), can be qualitatively detected without access to node identities. At the same time, we found that the pseudo-distances available in the unlabelled setting grow sub-linearly relative to true distances between labelled networks [Fig. 5.1(b)], and that this behavior is not universal. As a result, estimating Lyapunov exponents directly from unlabelled trajectories remains challenging. In contrast, the detection of noisy periodicity and linear correlations proved more robust in the unlabelled case.

Taken together, these findings provide a proof of concept for the feasibility of analyzing the temporal structure of network dynamics in the absence of node labels.

More broadly, this work advances our understanding of the relationship between labelled and unlabelled network representations, and opens the door to characterizing dynamical processes directly from unlabelled graph sequences. Future directions include extending our methods to additional graph invariants, or constructing pseudo-distances from combinations of invariants to better capture multiple aspects of network dynamics. This raises the natural question of which combinations are best suited for specific dynamical regimes.

Other avenues for future research include generalizations to higher-order network structures such as hypergraphs [156], as well as applications to empirical unlabelled data. Potential domains include animal migration, flocking, pedestrian motion, and active matter systems in physics and biology. Progress along these lines could help reduce reliance on sophisticated tracking methods in experimental settings.

Chapter 6

Predictability of Temporal Network Dynamics in Normal Ageing and Brain Pathology

"I always thought the joy of reading a book is not knowing what happens next."
Leonard Shelby, *Memento* (movie)

Introduction

Understanding and quantifying spontaneous brain activity—and how it is altered in neurological and psychiatric disorders—remains a major challenge. Traditionally, neuroscience has explored brain dynamics through stimulation-based approaches, using pharmacological agents, electrical stimulation, or cognitive testing. However, it is now well established that unperturbed, spontaneous brain activity is not random [215], but instead carries signatures of underlying structure that can differentiate between healthy and pathological states [216].

Brain activity is typically modeled as a high-dimensional nonlinear dynamical system with both deterministic and stochastic features [217]. Within this framework, brain disorders are interpreted as dynamical regimes—emerging from changes in system control parameters—that produce qualitative shifts in neural activity [218–222]. Accordingly, methods from nonlinear dynamics and time series analysis have long been applied to the study of brain disorders [74, 223–227], yielding insights into both diagnosis and the physiological mechanisms behind altered brain function [228–231].

A central dynamical property in this context is *predictability*—the extent to which future states can be forecast based on past and present ones [232, 233]. Predictability is inherently limited by uncertainty in initial conditions: a system that amplifies initial deviations rapidly cannot be reliably predicted over long times. As we have seen in Sec. 2.1.1.1, this trade-off is well understood in dynamical systems theory [40, 74, 165], where tools like Lyapunov exponents quantify the rate at which uncertainty grows [234]. In systems with a positive largest Lyapunov exponent (LLE), small deviations grow exponentially, making long-term forecasting impossible [74]. In contrast, weakly chaotic or marginally stable systems may show sub-exponential error growth, measurable through generalized Lyapunov exponents [235, 236].

Spontaneous brain activity, like many complex biological systems far from equilibrium, exhibits structure across multiple timescales. This multiscale nature must be accounted for in any dynamical analysis [217]. Statistical, thermodynamical, and dynamical properties vary with scale [237–240]. For example, long-timescale fluctuations show non-trivial ordering [241–246], while short-time behavior tends to relax exponentially, losing memory of past states. Brain activity has also been shown to exhibit scale-dependent time irreversibility [247], reinforcing the need for scale-aware dynamical descriptors.

In addition to temporal complexity, brain dynamics have a rich spatial structure. It is now common to represent this structure as a network [9, 248]. Changes in network topology can reflect transitions between different brain states or regimes [249]. This motivates a description of brain activity as a time-evolving graph—a *network trajectory* [28].

In this chapter, we apply the theoretical framework developed in Chapters 4 and 5 to a biological system, interpreting its temporal network evolution as a trajectory in a latent graph dynamical system. Building on this perspective, we investigate the predictability of spontaneous brain dynamics across ageing and disease, using concepts such as temporal network sensitivity, trajectory divergence, and dynamical metrics, now applied to empirical data [28, 101, 102, 250].

One widely used method in network neuroscience is Dynamic Functional Connectivity (DFC) [251–253], which tracks how functional connectivity between brain regions evolves over time. This method naturally yields a temporal sequence of networks—i.e., a network trajectory. Quantifying the predictability of such trajectories requires tailored tools: dynamical indicators specifically designed to capture instability in temporal network space over multiple scales [101, 102, 250].

Previous studies have computed Lyapunov exponents from EEG and MEG time series in various clinical populations—including depression [254, 255], schizophrenia [254, 256, 257], Parkinson's disease (PD) [258], and Alzheimer's disease (AD) [231, 259, 260]. However, those analyses typically (i) ignored the underlying network structure and (ii) assessed only single time scales or broad frequency bands [261].

In this chapter, we extend those approaches by applying a network-based measure of predictability to brain trajectories reconstructed from real EEG data. This measure builds on the network Lyapunov exponent introduced earlier in the thesis (Sec. 4.3) [101, 250], but generalizes it beyond exponential divergence, allowing for more nuanced forms of trajectory expansion. This makes it suitable for characterizing both strongly and weakly chaotic, as well as stochastic, brain dynamics. We use this framework to analyze spontaneous brain activity in healthy controls, patients with AD and PD, and individuals in preclinical stages of these diseases. By examining trajectory divergence across multiple timescales and frequency bands, we uncover robust, statistically significant differences in predictability patterns between groups. These differences are frequency-dependent and align with other known properties of pathological brain dynamics, such as reduced reversibility and altered network stability.

TABLE 6.1: Demographic data of the participants comprising the data set used in this study.

Subject Group	Size	Men/Women	Avg. Age (SD)	Avg. Years of Education (SD)
Control (young)	18	9 / 9	24.1 (3.68)	15.5 (1.54)
Control (elderly)	19	11 / 8	69.1 (7.25)	10.9 (4.67)
MCI	16	7 / 9	70.4 (5.05)	9.4 (5.88)
AD	19	5 / 14	73.2 (5.68)	8.8 (4.40)
PD-MCI	14	9 / 5	71.1 (6.63)	11.4 (5.18)
PD-D	12	9 / 3	73.0 (6.71)	5.9 (5.21)

6.1 Materials and methods

6.1.1 Control subjects and patients recruiting and selection

This study included ninety-eight subjects, as also analysed in previous works [262, 263]. The subjects are divided into six different groups: healthy (control) young, healthy (control) elderly, patients diagnosed with Mild Cognitive Impairment (MCI), patients diagnosed with Alzheimer's disease with dementia (AD), patients diagnosed with Parkinson's disease with mild cognitive impairment (PD-MCI), and patients diagnosed with Parkinson's disease with dementia (PD-D). Demographic information of the participants is presented in Table 6.1.

Neurologists at Istanbul Medipol University and Dokuz Eylül University performed a clinical diagnosis of the patients, and all patients underwent complete neurological structural magnetic resonance imaging (MRI). An extensive battery of neuropsychological tests was performed for patient groups and elderly subjects. Patients with amnesic MCI were diagnosed according to the NIA-AA criteria [264]; those with PD-MCI, according to the Movement Disorder Society (MDS) Level 2 criteria [265]. Probable PD-D diagnosis was made according to the Movement Disorder Society (MDS) Level 1 criteria [266, 267]. All patients with AD were diagnosed according to the National Institute of Aging-Alzheimer's Association diagnostic guidelines [268]. All participants with PD-MCI and PD-D were using a dopaminergic medication, including levodopa and/or dopamine agonist and/or monoamineoxidase B (MAO-B) inhibitor. All AD patients were on medication with cholinesterase inhibitors, and half of them were also on memantine, a NMDA receptor antagonist. The healthy elderly volunteers included in this study had no neurological abnormality or global cognitive impairment (Mini-mental State Examination (MMSE) score ≥ 27). Inclusion criteria for amnesic MCI included: i) leading an independent life in the community, ii) memory problem as defined by performance ≥ 1.5 standard deviations below that of age and education-matched controls in a set of neuropsychological tests, and iii) no impairment of daily living activities, clinical dementia rating (CDR) score of 0.5. Young healthy controls' inclusion criteria were no presence or history of neurological and psychiatric abnormalities, of drug and alcohol abuse and/or the Mini-mental State Examination (MMSE) score of ≥ 27.

Exclusion criteria for all participants were: i) history of neurological and/or psychiatric abnormalities, including evidence of depression as demonstrated by Yesavage Geriatric Depression Scale scores higher than 13 [269]; ii) presence of non-stabilised medical illnesses; iii) history of severe head injury and alcohol or drug misuse; iv) use of psychoactive drugs or cognitive enhancers, including acetylcholinesterase inhibitors; v) presence of vascular brain lesions, brain tumour as per MRI, and hydrocephalus.

The study conformed to the Declaration of Helsinki's principles. All participants and/or their relatives provided informed consent for the study, which was approved by the local ethical committee (Istanbul Medipol University Ethical Committee, Report No: 10840098-604.01.01-E.8374 and Dokuz Eylül University Ethical Committee, 01.03.2018/3821 GOA and 2018.KB.SAG.084).

6.1.2 Electroencephalographic data recording

EEG activity was recorded with a Brain Products BrainAmp 32 Channel DC system (band limits: 0.001-250 Hz; sampling rate: 500 Hz). The EasyCap with 32 Ag/AgCl electrode montage device with electrodes placed according to the 10/20 system was used. All electrodes' impedances were kept below 10 kΩ. A1 and A2 electrodes placed on the earlobes were used as reference. The electrooculogram was recorded via two Ag/AgCl electrodes placed on the left eye's medial upper and lateral orbital rim. The EEG was recorded in a dimly isolated room in two different centres with identical recording equipment, and the EEG recording procedure was applied precisely in the two centres, Istanbul Medipol University's SABITA Neuroscience Research Center EEG laboratory, and Dokuz Eylül University's Multidisciplinary Brain Dynamics Research Center. For all participants, EEG activity was recorded during 4 minutes in the eyes-open condition and 4 minutes in eyes-closed condition, yielding approximately $240,000$ data points per channel and subject. In the former condition, participants were asked to look at the black screen. Participants were monitored with a video camera throughout the EEG recording session.

The resulting time series were manually inspected for evident errors; an automatic artefact removal procedure was further applied, based on subtracting the two main components of all time series using Independent Component Analysis.

6.1.3 Functional network trajectory reconstruction and distance function

Functional networks representing brain dynamics were reconstructed following standard approaches from the literature [9, 248]. Brain activity was recorded using $N = 30$ brain electrodes, resulting in networks with N nodes, where each node represents a distinct time series corresponding to the signal detected at the electrode location. To enable more robust statistical analyses, each recording was divided into 20 consecutive, non-overlapping time segments. Each segment includes simultaneous signals from all 30 electrodes and is treated as an independent virtual recording. These segments are referred to as "trials." Thus, a trial consists of the full set of 30 time series extracted from one segment of the recording and provides the basis for constructing a corresponding network trajectory. The process for generating a network trajectory from each trial is as follows:

(i) *Correlation calculation:* The signals from the 30 electrodes are analysed over a defined time interval, with a duration determined by the *window length*. For each pair of nodes (electrodes), we compute the Pearson's linear correlation coefficient between the time series within that interval.

(ii) *Network reconstruction:* The resulting pairwise correlations form a weighted, symmetric adjacency matrix, where each entry represents the absolute value of the linear correlation between the corresponding electrode signals. The weights capture the strength of the functional connectivity between nodes.

(iii) *Dynamic evolution:* The analysis window is then shifted by 1 time step, corresponding to 2 milliseconds, and the correlation computation is repeated to generate the next network snapshot. Repeating this process throughout the whole trail yields the full trajectory, composed of $5,000$ functional networks, capturing the temporal evolution of the brain's functional connectivity.

This procedure is illustrated in the cartoon in Fig. 6.1. In total, we analysed 1,960 trials, categorised as follows: 360 from young control participants, 380 from elderly controls, 320 from MCI (potential AD) patients, 380 from AD patients, 280 from PD-MCI patients, and 240 from PD-D patients.

To compare snapshots of a network trajectory, among other distance functions [164, 270], here we define the distance between two arbitrary networks, i.e., two snapshots of the network trajectory, $\mathcal{A}, \mathcal{B}$, as:

$$d(\mathcal{A}, \mathcal{B}) = \frac{2}{N(N-1)} \sum_{i>j} |a_{i,j} - b_{i,j}|, \tag{6.1}$$

This expression is equivalent to the distance introduced in Eq. 4.8 in Section 4.3. Here, $\mathcal{A} = \{a_{ij}\}_{i,j=1}^{N}$ and $\mathcal{B} = \{b_{ij}\}_{i,j=1}^{N}$ are the two weighted adjacency matrices to be compared, N is the number of nodes in each network (here $N = 30$), and i and j take values from 1 to N. Each $a_{i,j}$ and $b_{i,j}$ can take values between zero and one. Consequently, d takes values from 0 (for pairs of identical networks) to 1 (when $|a_{i,j} - b_{i,j}| = 1 \; \forall i, j$, i.e., when each pair $i > j$ has full correlation in one network ($a_{i,j} = 1$) and no correlation in the other ($b_{i,j} = 0$). The value $d = 1$ can only be reached in very artificial situations and is never observed in the experiments examined here. For illustration, sampling networks at random from the trajectories of different subjects and conditions yields an average distance of $\langle d \rangle \approx 0.47$ between networks (standard deviation 0.07).

6.1.4 Quantifying network predictability: a non-parametric expansion rate of close trajectories

In systems with complex dynamics, proximity at an initial time does not guarantee divergence later on. That is, two network trajectories that are initially close in graph space may remain close, cross paths again, or diverge at different rates

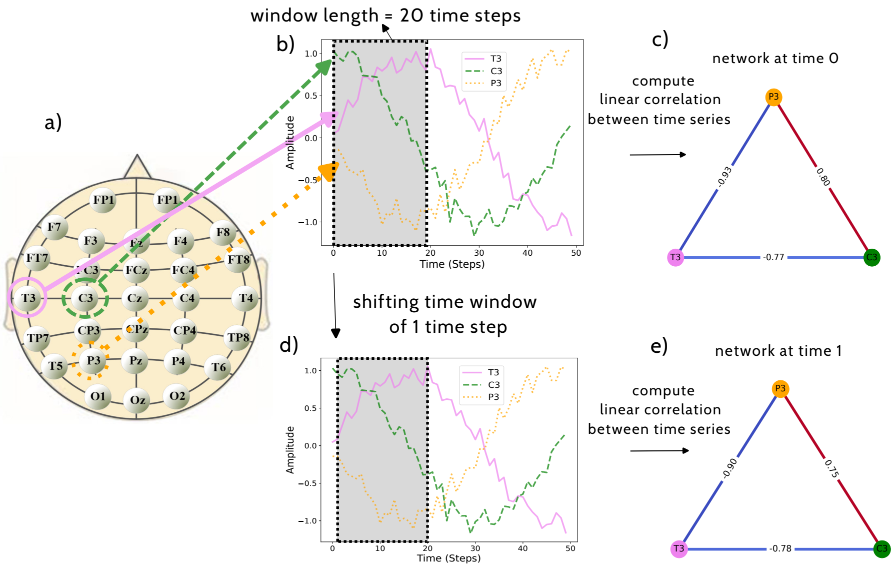

FIGURE 6.1: **Example of the functional brain network reconstruction procedure**. We depict the procedure for the simplified case of three nodes (i.e. three electrodes) and the same number of time series (see panel *a*). In the second step (panel *b*), we consider the time series of the amplitude of the signal and select a time window (grey shaded region) with a length of 20 time steps. Within it, we compute the linear correlation between pairs of the three time series, resulting in correlation values that represent the edge weights in the network at time $t = 0$ (see panel *c*). To obtain the network at the following time step, we shift the time window by 1 time step (2 milliseconds), as shown in panel *d*), and repeat the procedure to construct the network at time $t = 1$ (panel *e*). Iterating this procedure, we obtain the full functional network trajectory.

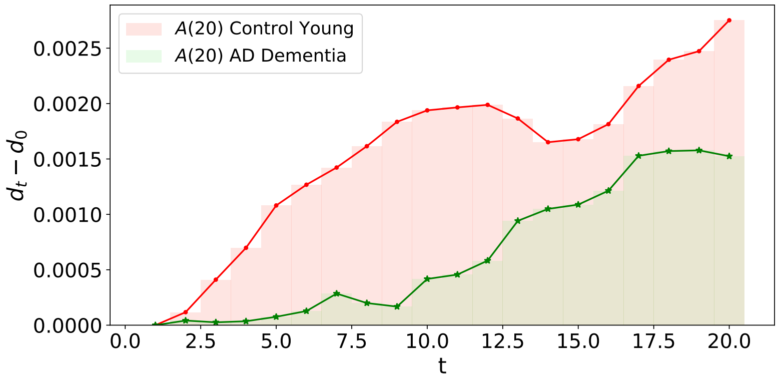

FIGURE 6.2: **Illustration of the non-parametric expansion measure $\mathtt{A}(n)$ for two different pairs of initially close network trajectories**. The plot shows how $\mathtt{A}(n)$ (Eq. 6.2) captures the rate of expansion over time by representing the area below the distance curve between two initially close trajectories, after subtracting the initial distance d_0. In this figure, the area under the curve associated with the young control is larger than that of the AD Dementia patient, indicating that the former exhibits a faster expansion compared to the latter.

depending on the underlying dynamics. For this reason, we cannot assume exponential separation or rely on parametric divergence measures. To capture the actual rate of separation in a data-driven manner, we introduce a non-parametric approach to quantify how two initially close network trajectories evolve over time.

Suppose that two network trajectories are at distance d_0 at time $t = 0$, and denote d_t the distance after t time steps. We define the non-parametric expansion $\mathtt{A}(n)$ as:

$$\mathtt{A}(n) = \frac{1}{n} \sum_{t=1}^{n-1} (d_t - d_0), \tag{6.2}$$

where n is the total number of steps over which the expansion is monitored. Up to a normalization factor (e.g., length of the time step), $\mathtt{A}(n)$ corresponds to the area under the curve of d_t as a function of t, offset by the initial value d_0. A higher value of $\mathtt{A}(n)$ indicates a larger area and, therefore, a faster separation of the initially close trajectories. Figure 6.2 illustrates this measure for two different trajectory pairs, where one pair expands faster than the other.

The algorithm used to compute $\mathtt{A}(n)$ proceeds as follows. First, we select an arbitrary initial condition—a point in the network trajectory, i.e., a network snapshot. We then search the trajectory for a recurrence, i.e., another snapshot that is both sufficiently distant in time (at least 250 time steps, corresponding to 500 milliseconds) and close in graph space to the initial one. This temporal threshold ensures that the two points are not trivially correlated due to proximity in time.

Next, we track the distance between these two network states for the subsequent n steps and compute $\mathtt{A}(n)$ using Eq. (6.2). We repeat this procedure by selecting a new initial condition $n + 1$ steps after the previous one, thereby generating non-overlapping segments. For a trajectory of total length T, this results in approximately T/n independent estimates of $\mathtt{A}(n)$, from which we can construct

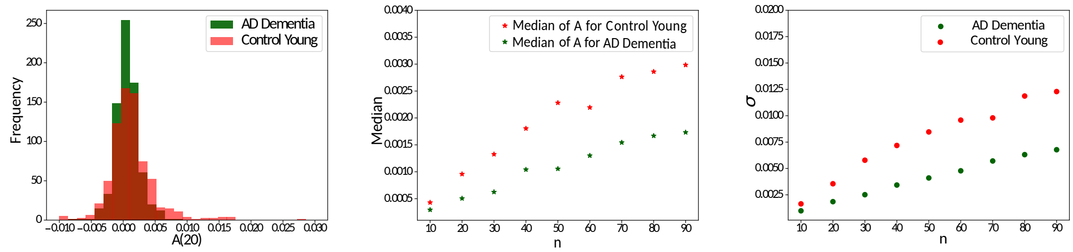

FIGURE 6.3: **Summary Statistics of Non-Parametric Expansion** $\mathtt{A}(n)$ **for Control and AD Dementia**. (Left) Frequency histogram of $\mathtt{A}(n = 20)$, extracted from the network trajectory of a control young individual (red bars) and a patient diagnosed with AD Dementia (green bars). (Middle) Median of the population of $\mathtt{A}$ as a function of the monitoring time n, comparing a control young individual (red points) to a patient diagnosed with AD Dementia (green points). The control individual systematically exhibits a larger median, suggesting that this metric may serve as a distinguishing parameter between cases. (Right) Same as the middle panel, but using the standard deviation $\sigma(\mathtt{A}(n))$ of the population as the metric.

a distribution. Representative distributions for a young control and an AD-D patient are shown in the left panel of Fig. 6.3.

From these distributions, we extract the median and standard deviation. A higher median $\mathtt{A}(n)$ suggests a faster expansion and, by extension, less predictable dynamics. The standard deviation reflects the variability in predictability across time. As seen in Fig. 6.3, these summary statistics allow us to distinguish between participant groups. The right panel shows how the standard deviation $\sigma(\mathtt{A}(n))$ evolves with monitoring time n: it is systematically larger for the control subject than for the patient, and this difference increases with n. While a larger n captures longer-term divergence, it reduces the number of estimates and thus statistical power. To balance resolution and sample size, we set $n = 20$ (equivalent to 40 milliseconds). For clarity, we will refer to $\mathtt{A}(n)$ simply as $\mathtt{A}$ in the remainder.

6.2 Results

6.2.1 Differences in the median of A across groups

We start the analysis by evaluating the differences in the median of $\mathtt{A}$ across groups. Specifically, Fig. 6.4 reports the difference between the median of $\mathtt{A}$ as a function of the window length, i.e. the length of the time interval considered to compute the correlation between pairs of time series, and hence to reconstruct the functional networks. More specifically, for each trial, the distribution of $\mathtt{A}$ is calculated (see left panel of Fig. 6.3), and from it the median is extracted; next, given two groups of subjects, the difference between both sets of medians is encoded through the standardised effect size Cohen's D metric [271]. In Fig. 6.4, the first group corresponds to the one in the column and the second to the one in the row. A positive Cohen's D indicates that the column group has a higher median

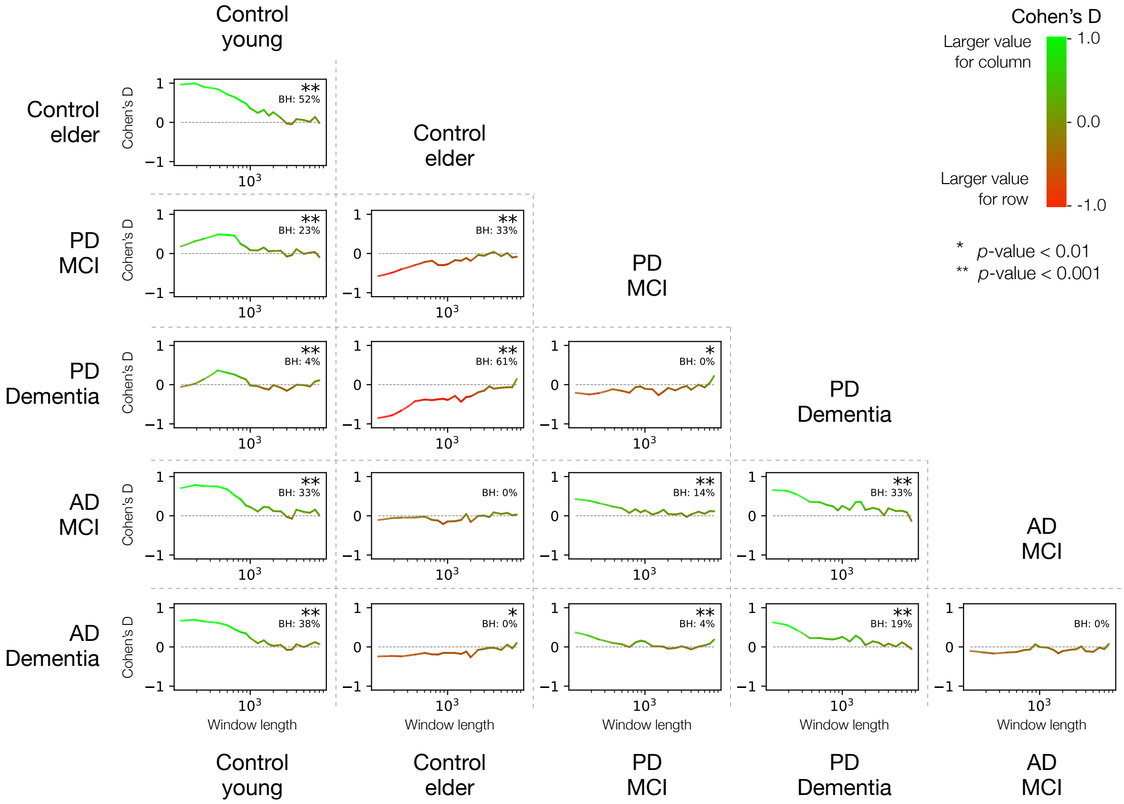

FIGURE 6.4: **Differences of the median of A between groups**. Each panel reports the evolution of the Cohen's D, calculated between the median of the distributions of A of two groups, as a function of the window length. Green points indicate positive Ds, and hence that A is higher for the group marked in the column; red points, negative Ds. The * and ** symbols indicate that the difference is statistically significant at levels 0.01 or 0.001 for at least one value of the window length. Each sub-plot further reports the percentage of window lengths for which a statistically significant difference is observed after a Benjamini-Hochberg (BH) correction.

compared to the row group, while a negative value implies the opposite. For a group to have a higher median than the other, it means that the former exhibits lower predictability.

It can be appreciated that, throughout most pairs, differences appear for short window sizes, usually below 10^3 points (i.e. two seconds). Some relationships are easy to describe; for instance, control young people always show a higher median than the other groups (positive Cohen's D metric corresponding to the control young column in Fig. 6.4) and exhibit the fastest expansions of the dynamics. In other words, networks of control young participants evolve faster, and hence have the lowest predictability. Control elders present a more complicated scenario, in which the functional network dynamics expands slower than for PD MCI and PD Dementia (see the negative Cohen's D), but is nevertheless qualitatively similar to AD. This, in turn, means that PD patients have a more unpredictable dynamic than matched healthy subjects; while AD patients have a similar one.

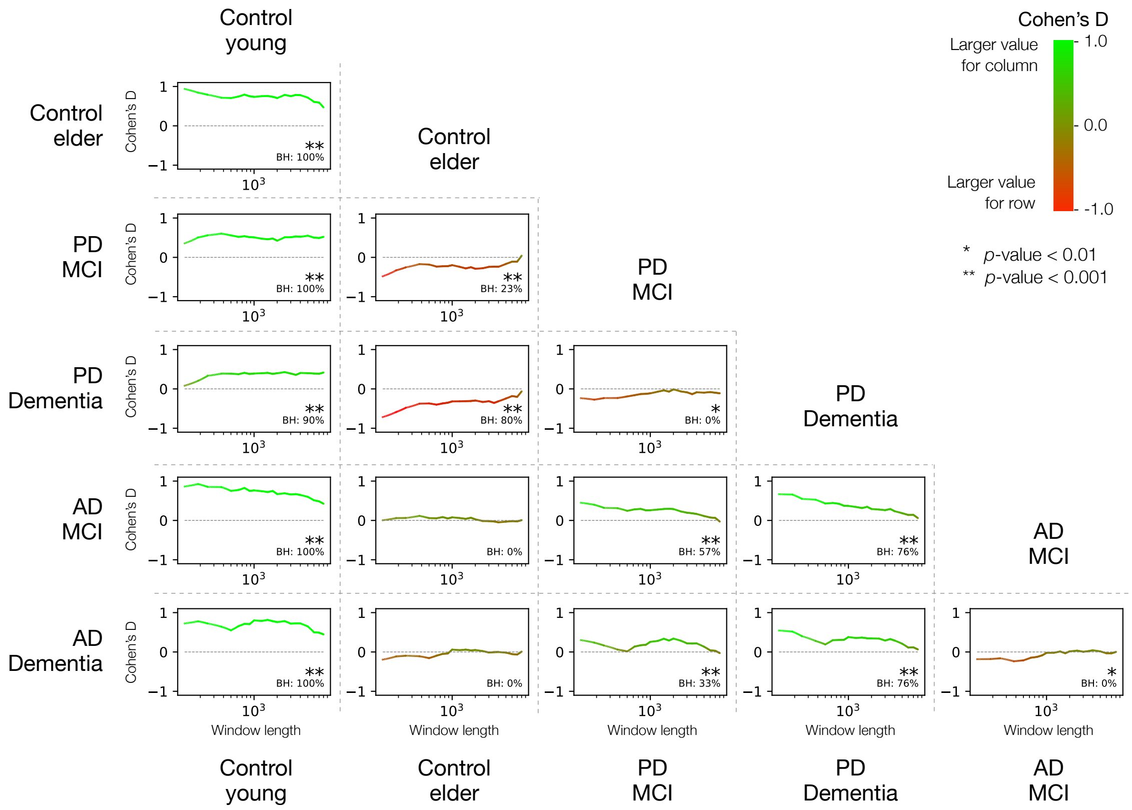

FIGURE 6.5: **Differences of the $\sigma(\mathtt{A})$ between groups.** Each panel reports the Cohen's D (which evaluates here the difference between pairs of groups based on the metric $\sigma(\mathtt{A})$), as a function of the window length over which each functional network snapshot is calculated. Green points ($D > 0$) indicate that the group marked in the column has larger $\sigma(\mathtt{A})$; the opposite ($D < 0$) is true for red points. The * and ** symbols indicate that such difference is statistically significant at levels 0.01 or 0.001 for at least one value of the window length, respectively. Each sub-plot further reports the percentage of window lengths for which a statistically significant difference is observed after a Benjamini-Hochberg (BH) correction.

Note that, in most pairs of subject groups, it is possible to find a statistically significant (with significance level $\alpha < 0.001$) difference between them for at least one value of the window length - as indicated by the ** symbol. This may nevertheless be caused by the large number of window lengths tested in each graph. To complement this information, each sub-plot of Fig. 6.4 reports the percentage of window lengths for which a statistically significant difference is observed, after a Benjamini-Hochberg (BH) procedure for controlling for false discovery rate [272]. Note that values higher than 30% are achieved for many pairs, supporting the statistical significance of the differences here observed.

6.2.2 Differences in the standard deviation of A across groups

We now turn to the analysis based on the second pointwise estimator, $\sigma(\mathtt{A})$, which captures the spread of the distribution. To evaluate group differences in this variability, we compute Cohen's D directly on the standard deviations, treating them as the quantities of interest. These differences are analyzed as a function of the window length—see Fig. 6.5. In this context, a positive value of Cohen's D indicates that the group indicated in the column has a higher average standard deviation, suggesting a greater dispersion in A values and thus more fluctuations in predictability for that group.

The results closely mirror those observed previously in Fig. 6.4: young control subjects show the highest variability in predictability (positive D in the first column), PD patients are less consistent than their matched controls (negative D in the second column), though still more stable than young individuals, and AD patients exhibit a profile qualitatively similar to that of their matched controls. The main difference compared to Fig. 6.4 lies in the fact that significant differences in variability appear across all window lengths—not just at shorter ones—as reflected by the consistently high percentages of BH-corrected significance.

6.2.3 Breakdown into frequency bands

In order to assess whether these patterns hold at different frequencies, we have performed the same computation of the evolution of $\sigma(\mathtt{A})$ by filtering the EEG time series according to five standard frequency bands: θ ($4-8$ Hz), α ($8-13$ Hz), β_1 ($13-20$ Hz), β_2 ($20-30$ Hz), and γ ($30-50$ Hz). Fig. 6.6 then reports the same results as in Fig. 6.5, but with one line for each frequency band (see right legend). Differences between groups in each frequency band are mostly constant across window lengths; or, in other words, the internal ranking of the bands does not substantially change when changing the window length. γ and α bands (respectively yellow and dark blue lines) seem to mostly be at the extreme, i.e., they define the largest and lowest values of D; in many cases, a positive value of D for one of them implies a negative D for the other. Other frequency bands, and most notably θ, have a more intermediate role, with $D \approx 0$ and hence minimal differences across groups.

In order to simplify the interpretation of the previous figure, Fig. 6.7 shows a synthesis of the same results. Specifically, for each pair of groups and frequency band, it shows the median value of the Cohen's D across all considered window lengths.

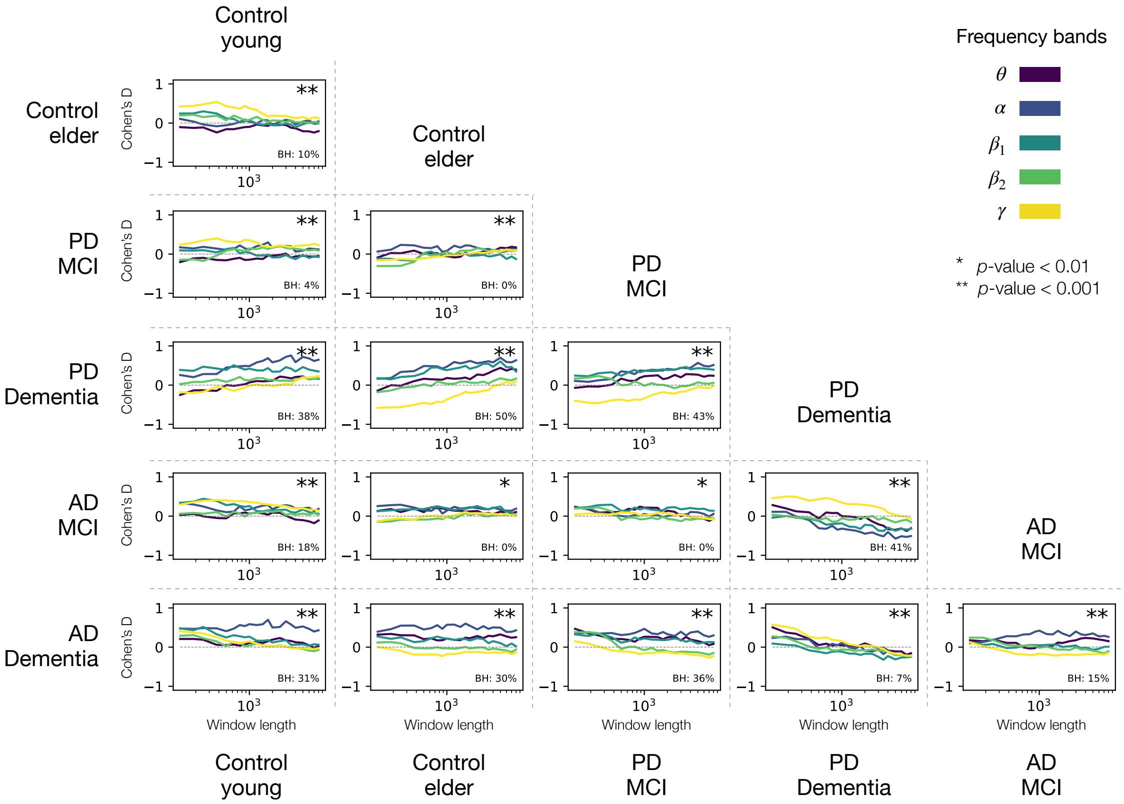

FIGURE 6.6: **Differences in $\sigma(\mathtt{A})$ by frequency bands**. Each panel reports the evolution of the Cohen's D, calculated between the distributions of the $\sigma(\mathtt{A})$ in two groups, as a function of the window length and the frequency band (see right panel). Positive and negative values of D have the same meaning as in Fig. 6.5. The * and ** symbols indicate that such difference is statistically significant at levels 0.01 or 0.001 for at least one value of the window length and one frequency band, respectively. Each sub-plot further reports the percentage of tests for which a statistically significant difference is observed after a Benjamini-Hochberg (BH) correction.

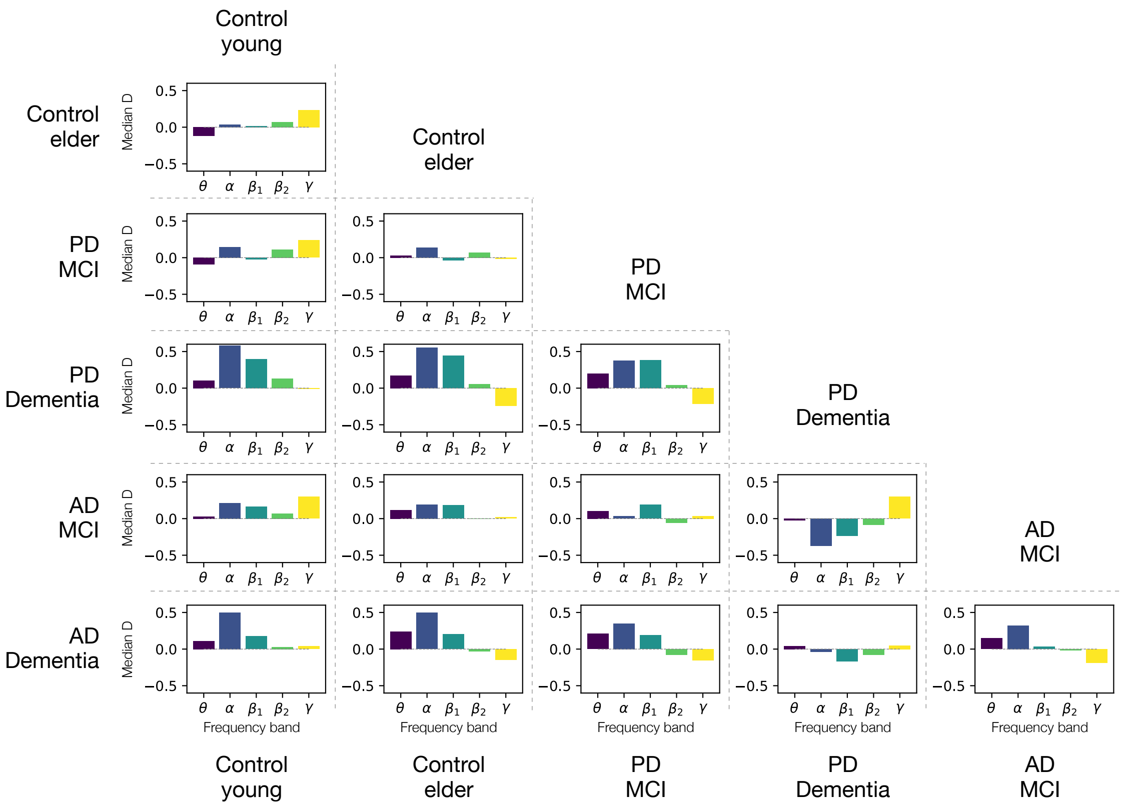

FIGURE 6.7: **Median differences in $\sigma(\mathtt{A})$ by frequency bands.** Each panel reports the median of Cohen's D, which is calculated for $\sigma(\mathtt{A})$, across all values of the window length and for different frequency bands, i.e., the median of each plot in Fig. 6.6. Positive and negative values of D have the same meaning as in Fig. 6.5.

6.3 Discussion

This chapter has explored the dynamics of spontaneous brain activity through the lens of network trajectory predictability. Brain activity presents non-random spatiotemporal structure and is naturally modelled as a high-dimensional networked dynamical system operating at multiple scales. Simultaneously accounting for the spatial and temporal aspects of such a structure is therefore non-trivial, and characterisations of brain activity often concentrate on one aspect, approximating the other. In this chapter, we reconstruct network trajectories [28] of electrical brain activity, we introduce a non-parametric method to quantify the predictability of network trajectories –which generalises network extensions of Lyapunov exponents (Sec. 4.3)– and apply it to characterise spontaneous electrical brain activity in various patient populations as well as in young and elderly healthy control subjects. Our results generally show that this predictability measure significantly differs across these populations and across timescales in a stylised manner. Let us now discuss in some detail these results and their implications, starting from the differences across populations, for then moving to frequencies and time scales.

Overall, fully-fledged dementia appears to be associated with increased median predictability (as compared to young healthy subjects) at short time scales, and with decreased predictability fluctuations at all scales (Figs. 6.4 and 6.5). This interpretation is consistent with the one found in a previous study that followed a different methodology [263], where pathologies were found to be associated with lower stability of links, but with higher persistence of larger-scale features (e.g. node centrality and clustering) and, more generally, with a picture wherein essentially normal function was maintained by compensating the slight decrease in connectivity and increased hub persistence via an overall slower dynamics. These works reflect on the idea that intrinsic variability is a sign of the adaptability of the underlying control networks and therefore of a healthy system [273, 274].

Insofar as our proposed measure quantifies the predictability of network dynamics –and therefore speaks of the connectivity induced by brain activity–, it is interesting to compare these results to previous studies on synchronisation in AD and in its harbinger MCI. Hypersynchrony has been suggested to be associated with MCI [275, 276], particularly for subjects eventually transitioning to dementia [277]. Such an activity would represent a dynamic compensation of the structural damage, preceding the network breakdown characterising fully-fledged AD dementia [275].

However, there are important differences in short term predictability between the paths to AD and PD (Figs. 6.4 and 6.5). In the path to AD dementia, predictability is found to increase progressively (i.e. uncertainty decreases) from young healthy subjects to fully-fledged dementia. Various aspects are worth highlighting. First, while the median predictability tends to differ between young healthy controls and all other conditions from healthy ageing to fully-fledged AD at short but not at long timescales, the standard deviation of such measure differs essentially at all scales. Second, healthy ageing does not appear to be distinguishable from MCI in terms of short-term predictability. However, healthy ageing significantly differs in median predictability from fully-fledged AD, though differences are much less pronounced than those with young healthy subjects, but not in predictability's standard deviation, whereas the opposite pattern characterises differences between MCI and fully-fledged AD, as conditions only differ in terms of predictability's standard deviation.

Predictability also changes in a progressive fashion in the path to PD dementia. Predictability is always lower for the young control group, i.e. dynamical network structure evolves faster than in any other group, while fully-fledged PD dementia appears to be the end point of a continuous increase of predictability from healthy ageing to dementia (see Figs. 6.4 and 6.5). However, there is an important difference with respect to predictability in AD: spatiotemporal brain activity in healthy ageing appears to be significantly more predictable than in both MCI and in fully-fledged PD dementia, particularly for short time-windows, PD patients with dementia turning out to be the least predictable.

Overall, our results suggest that, in addition to what is known to be a progressive tendency to a random but topologically homogeneous spatial structure [278], the dynamical path to dementias is associated with a progressively more regular –i.e., more predictable– temporal structure.

The role of timescales can be appreciated by looking at how our metrics varies (i) as a function of the window length within which networks were reconstructed, and (ii) for different frequency bands (see Fig. 6.6). Both in AD and PD dementia, predictability varies with the size of the time-window used to reconstruct brain networks. On the other hand, frequency-specific analyses (see Figs. 6.6 and 6.7), show that AD Dementia is characterised by higher predictability in the α and β_1 bands, compared to both healthy control groups and AD MCI patients. Interestingly, the predictability of γ band activity increases from young control subjects to elderly control, PD-MCI, and AD-MCI subjects, but does not in PD and AD dementia (see yellow bars in the first column of Fig. 6.7). PD dementia is characterised by similar though more pronounced increases in α and β_1 band predictability and by decreases in γ band predictability.

These results are somehow coherent and complementary with previous results concerning the frequency composition of brain activity in AD and PD. EEG activity in AD is typically characterised by a progressive slowing of brain oscillatory activity [279–282], with increased power in the δ and θ frequency bands, mirrored by decreased power in the α and β (12–30 Hz) bands, as well as complex changes in the γ band [283–286], a pattern already visible prior to fully-fledged AD [280, 287–292]. On the other hand, PD patients' EEG activity generally presents higher θ and δ band power [293], as well as lower power and decreased dynamical connectivity in the β_2 band compared to cognitively unimpaired PD patients, and reduced power in the γ band in central, parietal, and temporal regions. Compared with early-stage PD patients, late-stage PD patients have been associated with reduced power in the β_1 range in the posterior central region, and increased θ and δ power in the left parietal region [294]. Moreover, significant differences in EEG activity was found between patients with PD and AD patients, with more pronounced slowing in PD patients' activity compared to AD patients [295].

Notably, somehow counterintuitively, window length and frequency bands

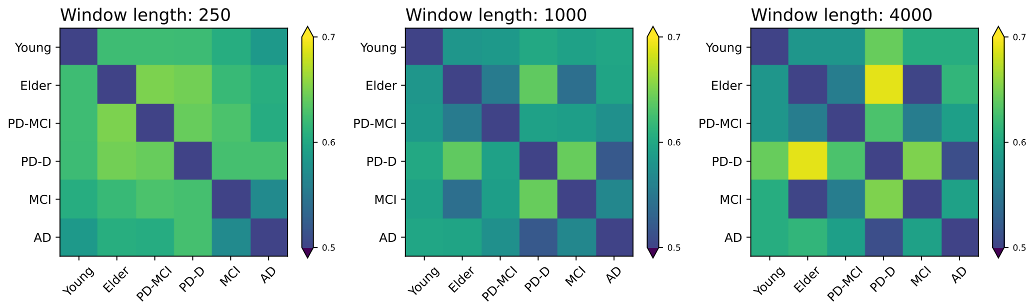

FIGURE 6.8: **Classification score obtained by Random Forest machine learning models while pairwise discriminating the participants' groups**. Each model has been trained with the $\sigma(\mathtt{A})$ obtained from each trial for the five frequency bands considered in this study (see Fig. 6.6); the three panels report the results for different window lengths (see top title). All results correspond to the average accuracy obtained over 100 independent realisations.

are not simply related to each other, and the Cohen's D metric for $\sigma(\mathtt{A})$ across frequency bands remains constant as the window length increases (see Fig. 6.6). Window lengths and frequency bands are therefore capturing two different aspects, though why this is the case and what exactly their respective role is not entirely clear.

To round off the analysis, and since most of the differences between groups are highly statistically significant, it may be tempting to expect that such differences can completely *define* the groups, i.e. that a model on them based could be used to classify or diagnose subjects. We tested this idea by resorting to a random forest machine learning model [296], trained on the $\sigma(\mathtt{A})$ values for the five frequency bands (see Fig. 6.7), and aimed at pairwise discriminating the participants' groups. Results are measured through the accuracy obtained in each classification task, i.e. the fraction of participants whose group has correctly been classified; and their generalisability is further evaluated using a ten-fold cross-validation procedure [297]. In all tasks, a same number of randomly-selected trials is considered for both groups, to ensure balanced data. Results are reported in Fig. 6.8, for three window lengths and for each pair of groups. The obtained classification scores are generally not very high, with a maximum of 0.69 for the task control elder vs. PD Dementia (to be compared to 0.5 that would be expected in a random classification). In other words, in most cases the classification is only moderately better than what would be obtained by chance. The differences presented here do not provide a means to exactly *separate* groups, and instead have a more descriptive value—highlighting alterations in brain dynamics that, by themselves, cannot support predictive tasks. Rather than aiming for classification, the scope of this chapter is to understand and quantify the concept of "predictability of brain trajectories." The fact that the metric succeeds in distinguishing between groups is taken as an indication that it captures meaningful information, but this discriminative power is not the primary objective.

As a final note, it is worth highlighting that the analyses here presented are based on two main methodological choices: (i) linear correlations have been used to reconstruct functional networks, among the many alternatives available in the literature [298], and (ii) a concrete network distance function (Eq. 6.1), among other choices [164]. While these choices were guided by parsimony, we cannot rule out that the use of alternative metrics may elucidate further aspects of the predictability of brain dynamics, as e.g. their connection with the non-linear coupling between brain regions.

This chapter has focused on the brain as a real-world application, and has demonstrated how the methods and concepts introduced in the previous chapters must be adapted in order to properly capture the behaviour of real systems. In doing so, it highlights both the flexibility and the limitations of general-purpose dynamical tools when applied to empirical data, and sets the stage for further refinement of these approaches in the context of complex, biologically grounded systems.

Chapter 7

Conclusion

"We shall go on to the end."
Tommy, *Dunkirk* (movie)

A central aim in the study of complex systems is to understand how rich, collective behaviours arise from the interplay of many interacting components. A key insight of recent decades is that these interactions are not static: they fluctuate, adapt, and reorganise over time. From social groups influenced by shifting external signals, to ecosystems facing seasonal variability, to neural circuits whose functional connectivity changes across cognitive states and disease, the dynamics of the interactions themselves often play as crucial a role as the dynamics of the individual units. This recognition has motivated a broad effort across disciplines to develop theoretical and computational tools that can account for time-varying structures and their impact on collective dynamics.

Within this broad context, this thesis set out to explore how the dynamics of systems with time-varying interactions can be rigorously characterised and understood. We focused on two central questions. First, to examine how temporal variability in external conditions shapes collective behaviour in models of opinion dynamics. Second, to develop and apply dynamical-systems-inspired tools for analysing temporal networks, thereby providing a framework to probe predictability and sensitivity to initial conditions in evolving interacting systems. By addressing these complementary directions, we sought to better characterize the dynamics of networks themselves and to explore how external factors impact collective behaviour in opinion dynamics.

Our first line of research revealed how fluctuating environments shape the outcome of opinion dynamics. By embedding switching processes into the noisy voter model, we showed that the timescale of the environment relative to that of the population crucially determines the emergent collective behaviour. In fast-switching regimes, populations experience an effective averaging of environmental influences, while in slow-switching regimes they retain memory of each environmental state, leading to multimodal stationary distributions and richer transient dynamics. Extending the framework to multiple states and dynamic influencers demonstrated how asymmetry, bias, and structural properties of the environment further shape consensus formation. These results highlight how external variability, often overlooked in static models, can qualitatively alter macroscopic behaviour. Future work in this direction includes the analytical treatment of systems with more than two environmental states in the infinite- or large-population limit, which remains challenging due to the complexity of the underlying piecewise-deterministic dynamics [73, 299]. Another promising avenue is to move beyond the fully connected (mean-field) assumption and incorporate heterogeneous or modular network topologies, to better capture how social structure influences the interplay between intrinsic and environmental noise. Finally, the mathematical framework developed here could be extended to other forms of population dynamics beyond opinion models, such as ecological interactions, evolutionary game dynamics, or cultural transmission, where temporally fluctuating conditions play a central role.

The second line of research focused on the development of tools to quantify the dynamics of temporal networks themselves. Treating a temporal network as a trajectory in graph space, we adapted classical indicators from dynamical systems theory—including autocorrelation functions and Lyapunov exponents—to capture memory and sensitivity to initial conditions. For labelled temporal networks, we demonstrated that Lyapunov exponents computed with adapted Wolf and Kantz algorithms distinguish chaotic from stochastic network evolutions, and we validated the framework through synthetic models. Adapting this methodology to EEG-derived functional brain networks, we showed that predictability varies systematically across neurodegenerative conditions, uncovering subtle yet interpretable signatures of altered brain dynamics. Finally, by generalising the autocorrelation function and Lyapunov exponent to unlabelled temporal networks via permutation-invariant pseudo-distances, we demonstrated that memory effects, periodicity, and sensitivity to initial conditions can be captured even when node identities are unavailable or irrelevant. While chaotic temporal networks exhibit exponential divergence of trajectories in the labelled space, the growth of distances in the unlabelled space is typically sub-exponential, reflecting geometric constraints. Nonetheless, this approach broadens the applicability of dynamical systems concepts to anonymised or intrinsically unlabelled data.

On the methodological side, a systematic comparison between labelled and unlabelled metrics would clarify how dynamical information is compressed or transformed by permutation-invariant representations. Beyond the specific case of autocorrelation and Lyapunov exponents, an important avenue for future research lies in extending other classical indicators from dynamical systems to the network realm. Examples include entropy-based measures of complexity [300–303], correlation dimensions to probe the geometry of trajectories in graph space [304], or recurrence-plot techniques to detect hidden structure and periodicity [305]. Investigating how these metrics translate into temporal networks could provide complementary perspectives on predictability, memory, and chaoticity. At the same time, it is essential to recognize the limitations of such extensions: while some indicators may retain their diagnostic value, others may be fundamentally altered by the discrete and combinatorial nature of graph space. Carefully assessing these boundaries will be key to understanding which dynamical signatures can meaningfully survive the projection from classical state spaces to temporal networks.

Finally, applying the proposed tools to a wider range of empirical systems—from ecological to socio-technical networks—offers a promising avenue to further test their explanatory power. Many real-world systems exhibit temporally structured or seasonally forced patterns of interaction, making them natural candidates for the framework developed here. For instance, epidemic processes unfolding on contact networks shaped by climatic or behavioural cycles [306, 307], mobility and transportation networks driven by daily or weekly rhythms [308], or ecological communities reorganising in response to recurrent environmental forcing [309], they all generate rich temporal structure that can be meaningfully analysed through autocorrelation and Lyapunov-based indicators. Extending the empirical scope in this direction would not only enhance the generalisability of the approach, but also clarify which dynamical signatures remain robust across domains with different sources of variability, timescales, and noise. In this way, the methodology developed in this thesis has the potential to contribute to a broader understanding of predictability and memory in complex systems whose interaction patterns evolve over time.

Importantly, the two perspectives are not isolated: many real-world systems exhibit temporal networks whose very patterns of interaction are themselves shaped by external environments. Examples include social networks adapting to seasonal or political contexts [310, 311], ecological interactions responding to climatic cycles [312, 313], or neural connectivity modulated by external stimuli [314, 315]. Extending the analysis to such scenarios would bridge the gap between externally driven models and intrinsic network dynamics, offering a richer picture of how environmental variability and intrinsic dynamics thus act in concert to shape interaction patterns and, ultimately, collective behaviour.

In light of these findings, we return to the initial question: how can a seemingly simple and linear flow of time give rise to the richness of behaviours observed in complex systems? The work carried out in this thesis suggests that it is not time alone, but the structure and evolution of interactions across time that drive complexity. By developing tools to analyse time-varying interaction systems, this thesis contributes to a growing body of research that seeks to uncover the principles underlying the emergence of order and variability in complex systems.

Acronyms

Acronym	Definition
PDMP	Piecewise Deterministic Markov Process
GDS	Graph Dynamical System
nMLE	Network Maximum Lyapunov Exponent
MLE	Maximum Lyapunov Exponent
TN	Temporal Network
DAR(ρ)	Discrete Autoregressive process of order ρ
DARN(ρ)	Discrete Autoregressive Network model of order ρ
ER	Erdős–Rényi (random graph model)
VM	Voter Model
nVM	Noisy Voter Model
SFHH	Société Française d'Hygiène Hospitalière (2009 conference dataset)
ODE	Ordinary Differential Equation
PDF	Probability Density Function
SF2H	Alternate spelling of SFHH (conference dataset)
AD	Alzheimer's Disease
AD-D	Alzheimer's Disease with Dementia
BH	Benjamini-Hochberg
CDR	Clinical Dementia Rating
DFC	Dynamic Functional Connectivity
EEG	Electroencephalogram
LLE	Largest Lyapunov Exponent
MAO-B	Monoamine Oxidase B
MCI	Mild Cognitive Impairment
MDS	Movement Disorder Society
MMSE	Mini-Mental State Examination
MRI	Magnetic Resonance Imaging
NMDA	N-Methyl-D-Aspartate
PD	Parkinson's Disease
PD-MCI	Parkinson's Disease with Mild Cognitive Impairment
PD-D	Parkinson's Disease with Dementia

Appendix A

Flowcharts of the Wolf and Kantz Algorithms

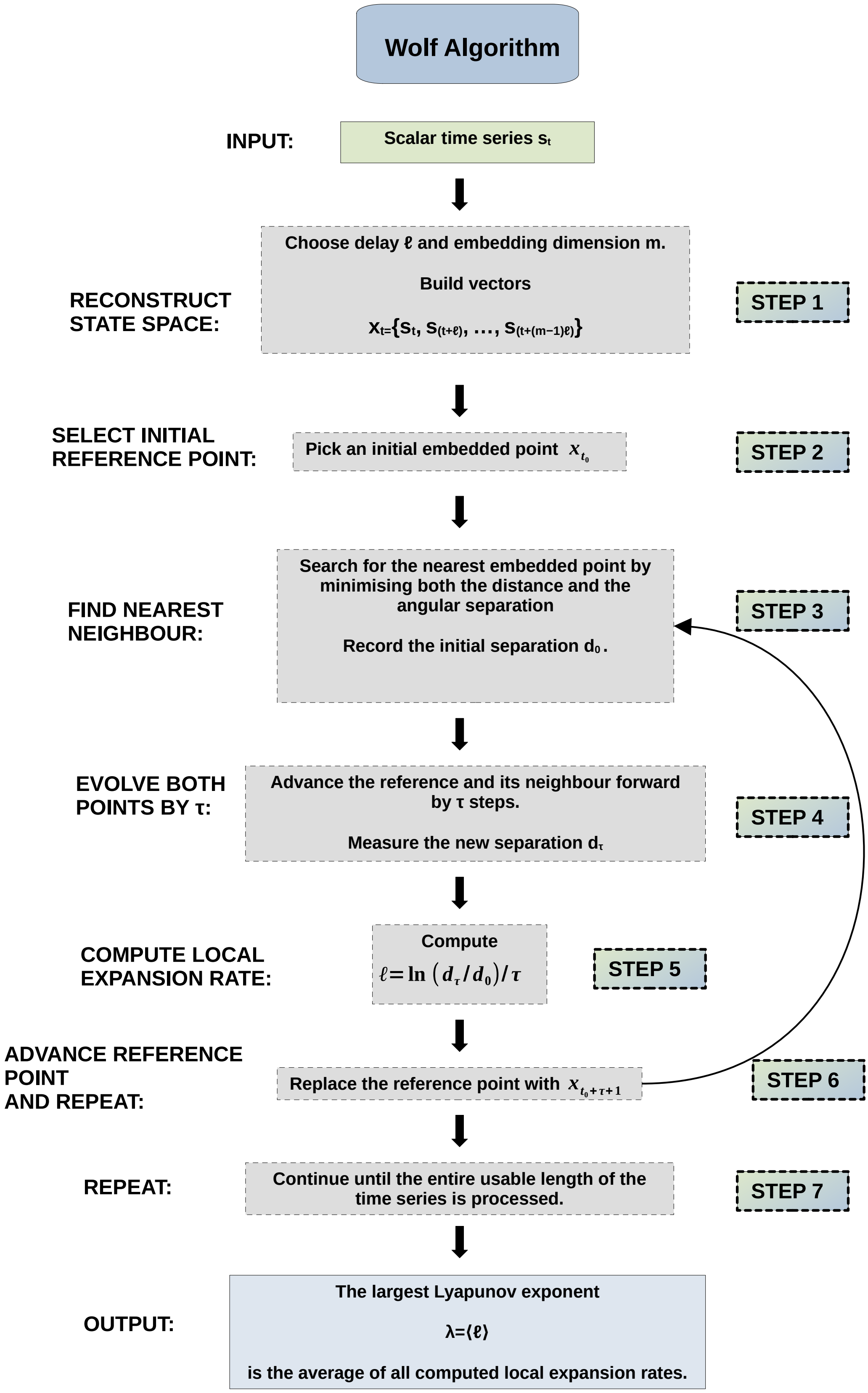

FIGURE A.1: **Flowchart of the Wolf algorithm for estimating Lyapunov exponents.**

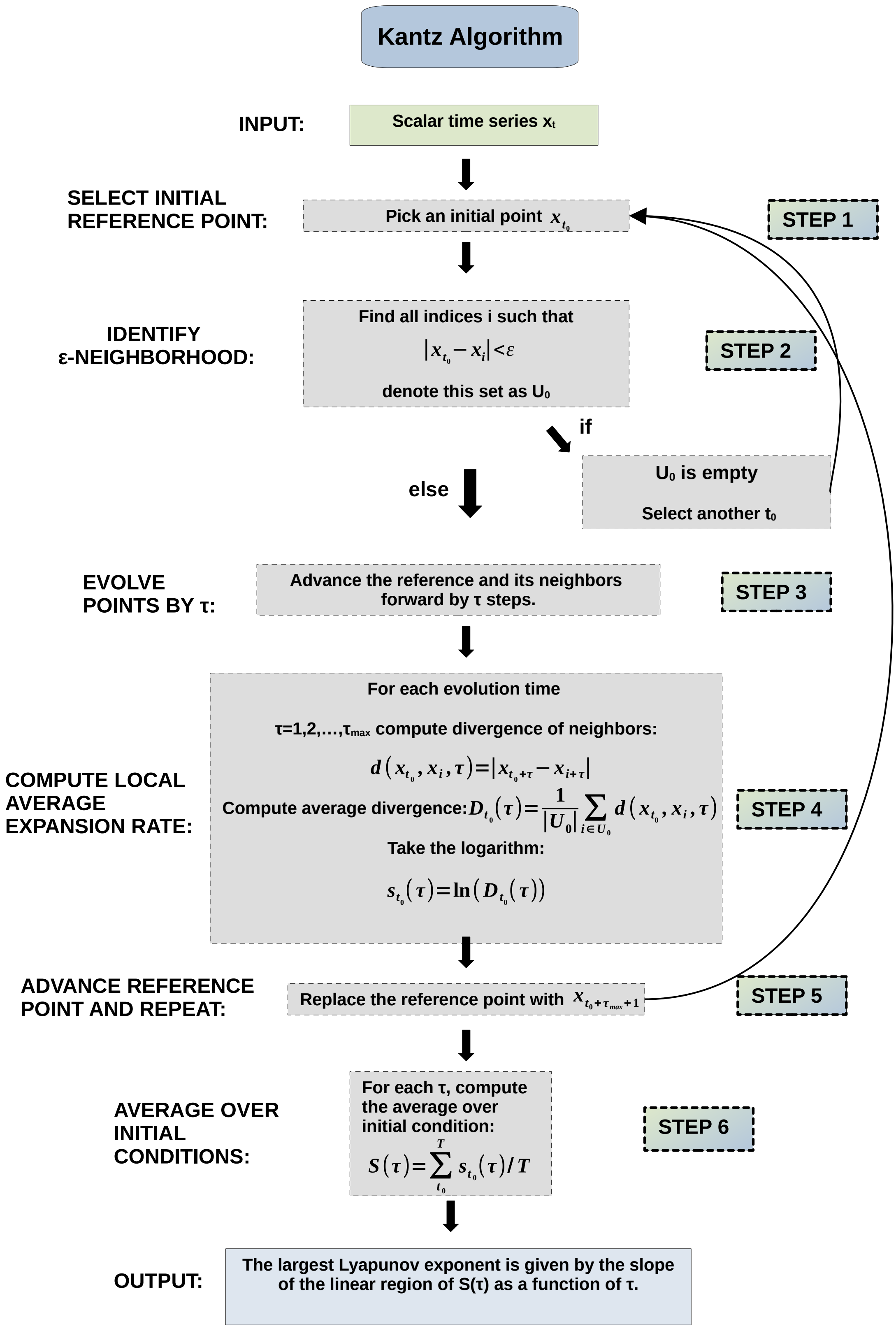

FIGURE A.2: **Flowchart of the Kantz algorithm for estimating the largest Lyapunov exponent.**

Appendix B

Algorithm to determine the stationary state of a PDMP with three environmental states

In this appendix we provide details of the algorithm used to solve Eq.(3.26) when the environment undergoes transitions between three states.

B.1 General theory

Focusing on the PDMP framework with S environmental states we follow [142] and introduce currents

$$J_\sigma(\phi) = \Pi(\phi,\sigma)v_\sigma(\phi) - \int_{\phi_0^*}^{\phi} \lambda \sum_\eta \left(\Pi(\phi',\eta)\mu_{\eta\to\sigma} - \Pi(\phi',\sigma)\mu_{\sigma\to\eta}\right) d\phi'. \tag{B.1}$$

The quantity $J_\sigma(\phi)$ represents the net probability flux into or out of the interval (ϕ_0^*, ϕ) and environmental state σ. This is illustrated in Fig. B.1, the dotted box at the bottom left of the figure highlights the interval (ϕ_0^*, ϕ) at fixed environmental state $\sigma = 0$. The quantity $J_0(\phi)$ is the flux out of this interval, due to either deterministic motion [following $v_0(\phi)$] or to switches of the environment. Further details can be found in [142].

The continuity equation for probability can be expressed as:

$$\partial_t \Pi(\phi,\sigma) = -\partial_\phi J_\sigma(\phi). \tag{B.2}$$

In the stationary state we therefore have

$$\frac{d}{d\phi}\left(\Pi^*(\phi,\sigma)v_\sigma(\phi)\right) - \lambda \sum_\eta \left(\Pi^*(\phi,\eta)\mu_{\eta\to\sigma} - \Pi^*(\phi,\sigma)\mu_{\sigma\to\eta}\right) = 0. \tag{B.3}$$

In the following calculations, we always focus on the stationary state. To keep the notation compact we will omit the asterisk. Stationary implies that the total current vanishes, i.e.,

$$\sum_\sigma J_\sigma(\phi) = 0 \tag{B.4}$$

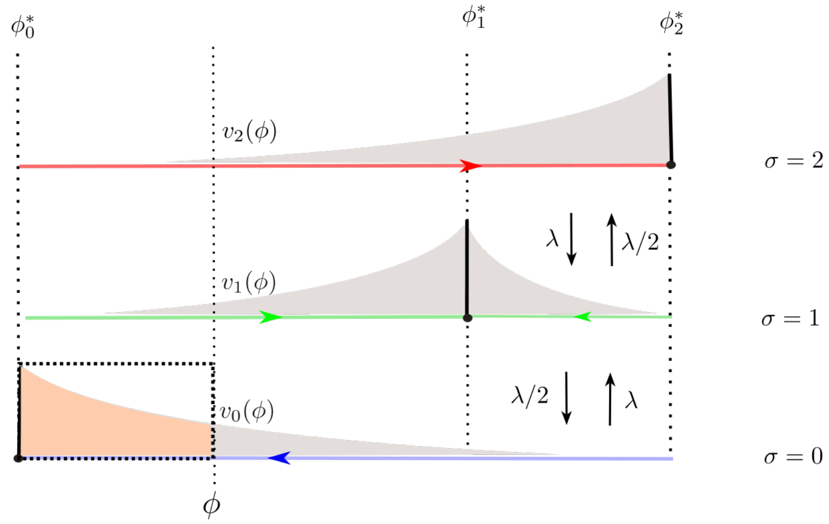

FIGURE B.1: **Physical interpretation of the currents.** Physical interpretation of the currents in Eq. (B.1) for a three states environmental switching as in Sec. 3.3.5.

for all ϕ. Defining

$$\Gamma_\sigma(\phi) = \Pi(\phi, \sigma) v_\sigma(\phi), \tag{B.5}$$

and using Eq. (B.1) this results in

$$\sum_\sigma \Gamma_\sigma(\phi) = 0. \tag{B.6}$$

For any one system, we can therefore pick a particular environmental state τ, and express $\Gamma_\tau(\phi)$ in terms of the $\Gamma_\sigma(\phi)$, $\sigma \neq \tau$,

$$\Gamma_\tau(\phi) = -\sum_{\sigma \neq \tau} \Gamma_\sigma(\phi) \tag{B.7}$$

We can then reduce Eq. (B.3) to the following set of $S - 1$ equations for the Γ_σ with $\sigma \neq \tau$:

$$\frac{d}{d\phi}\Gamma_\sigma(\phi) + \frac{\Gamma_\sigma(\phi)}{v_\sigma(\phi)} \left(\lambda \sum_\eta \mu_{\sigma\to\eta} \right) - \lambda \sum_{\eta \neq \tau} \Gamma_\eta(\phi) \left(\frac{\mu_{\eta\to\sigma}}{v_\eta(\phi)} - \frac{\mu_{\tau\to\sigma}}{v_\tau(\phi)} \right) = 0. \tag{B.8}$$

B.2 Three states

We now consider the case of three environmental states, see Sec. 3.3.5, and in particular Eq. (3.45). After elimination of Γ_1, we can write Eq. (B.8) as

$$\frac{d}{d\phi}\underline{\Gamma}(\phi) = \underline{\underline{\Lambda}}(\phi)\underline{\Gamma}(\phi), \tag{B.9}$$

with $\underline{\Gamma}(\phi) = [\Gamma_0(\phi), \Gamma_2(\phi)]^T$, where the superscript indicates transposition.

The 2×2 matrix $\underline{\underline{\Lambda}}$ is given by

$$\underline{\underline{\Lambda}}(\phi) = -\frac{\lambda}{\lambda_c}\begin{pmatrix} \frac{1}{2(\phi_1^*-\phi)} + \frac{1}{\phi_0^*-\phi} & \frac{1}{2(\phi_1^*-\phi)} \\ \frac{1}{2(\phi_1^*-\phi)} & \frac{1}{2(\phi_1^*-\phi)} + \frac{1}{\phi_2^*-\phi} \end{pmatrix}, \tag{B.10}$$

where ϕ_σ^* is the fixed point of the limiting deterministic dynamics for fixed environment σ [see Eq. (3.24)]. The quantity λ_c is given in Eq. (3.43). The matrix $\underline{\underline{\Lambda}}$ encodes the dynamics of an infinite population combined with a switching environment. We note the singularities at ϕ_0^*, ϕ_1^* and ϕ_2^*.

B.3 Algorithm

We now outline the algorithm we use to solve equation Eq. (B.9) in the domain $\phi \in (\phi_0^*, \phi_2^*)$. A graphical illustration can be found in Fig. B.2. The boundary conditions for the solution will be detailed below.

Due to the singularity of $\underline{\underline{\Lambda}}$ at the internal fixed point $\phi = \phi_1^*$, we divide the domain into two intervals, (ϕ_0^*, ϕ_1^*) and (ϕ_1^*, ϕ_2^*), and first obtain separate solutions on these two subdomains. These are then combined using the boundary conditions.

To numerically integrate Eq. (B.9) we discretise the ϕ-axis into elements of size $\Delta\phi$. Choosing initial conditions $\Gamma_0(\phi_0^* + \Delta\phi) = a_0$ and $\Gamma_2(\phi_0^* + \Delta\phi) = a_2$, we can then forward integrate Eq. (B.9), to obtain $\underline{\Gamma}(\phi_0^* + 2\Delta\phi), \underline{\Gamma}(\phi_0^* + 3\Delta\phi), \ldots, \underline{\Gamma}(\phi_1^* - \Delta\phi)$. This numerical solution will depend on the choice of a_0 and a_2.

Similarly (but independently) we choose final conditions $\Gamma_0(\phi_2^* - \Delta\phi) = b_0$ and $\Gamma_2(\phi_2^* - \Delta\phi) = b_2$ near the right edge of the domain (ϕ_0^*, ϕ_2^*). We then backward integrate Eq. (B.9), to find $\underline{\Gamma}(\phi_2^* - 2\Delta\phi), \underline{\Gamma}(\phi_2^* - 3\Delta\phi), \ldots, \underline{\Gamma}(\phi_1^* + \Delta\phi)$. This numerical solution in turn will depend on the choice of b_0 and b_2.

We now need to determine the right choice for the boundary conditions a_0, a_2, and b_0, b_2. We do this using the following properties of the stationary distribution:

(i) *Overall normalisation.* Noting that Eq. (B.9) is linear in $\underline{\Gamma}$, a multiplication of all of a_0, a_2, b_0, b_2 with a constant factor will simply re-scale the solution. We also recall that $\Gamma_1 = -(\Gamma_0 + \Gamma_2)$ so that Γ_1 undergoes the same re-scaling. The Γ_σ in turn determine the stationary distribution $\Pi(\phi, \sigma)$ [via Eq. (B.5)]. Overall normalisation requires $\sum_\sigma \int d\phi\, \Pi(\phi, \sigma) = 1$. This can be used to fix one of the coefficients a_0, a_2, b_0, b_2.

(ii) *Continuity of* Γ_0 *and* Γ_2 *at the interior fixed point* ϕ_1^*. The velocity fields $v_0(\phi)$

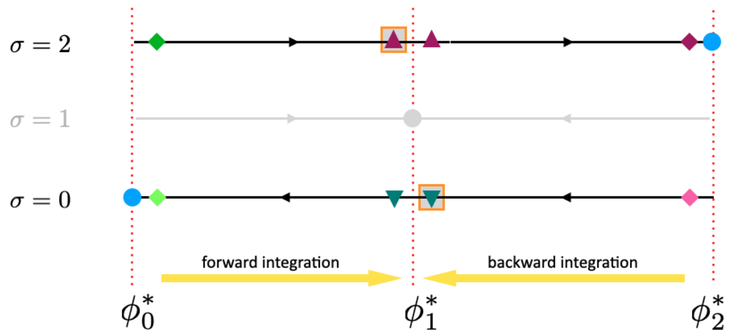

FIGURE B.2: **Illustration of the numerical algorithm used to obtain the stationary distribution of the PDMP for the model with three environmental states (Appendix B.3).** The flow in environment $\sigma = 0$ is directed towards ϕ_0^* (filled circle on the left). In environment $\sigma = 1$ the deterministic flow is towards the internal fixed point ϕ_1^* (filled circle in the centre), and in environment $\sigma = 2$ the system flows towards ϕ_2^*, shown as a filled circle on the right. Using the fact that $\Gamma_0 + \Gamma_1 + \Gamma_2 = 0$, we eliminate Γ_1 (greyed out in the figure). Eqs. (B.9) are then forward-integrated on the interval (ϕ_0^*, ϕ_1^*), starting from an initial condition at $\phi_1^* + \Delta\phi$ (green diamonds) to obtain final values at $\phi_1^* - \Delta\phi$ (triangles). A similar backward-integration is performed starting from $\phi_2^* - \Delta\phi$ (purple and pink diamonds), ending at $\phi_2^* + \Delta\phi$ (triangles). As explained in the text, we impose that the numerical solution approximates the conditions $\Gamma_0(\phi_1^* - \Delta\phi) = \Gamma_0(\phi_1^* + \Delta\phi)$ (green downward triangles), $\Gamma_2(\phi_1^* - \Delta\phi) = \Gamma_2(\phi_1^* + \Delta\phi)$ (purple upward triangles), and $\Gamma_2(\phi_1^* - \Delta\phi) = \Gamma_0(\phi_1^* + \Delta\phi)$ (orange squares).

and $v_2(\phi)$ show no singularity at $\phi = \phi_1^*$. We thus expect Γ_0 and Γ_2 to be continuous at ϕ_1^*. Within the discretisation this translates into

$$\begin{aligned} \Gamma_0(\phi_1^* - \Delta\phi) &= \Gamma_0(\phi_1^* + \Delta\phi), \\ \Gamma_2(\phi_1^* - \Delta\phi) &= \Gamma_2(\phi_1^* + \Delta\phi), \end{aligned} \tag{B.11}$$

up to corrections of order $\Delta\phi$.

(iii) *No-flux condition at ϕ_1^* in environment $\sigma = 1$.* In environment $\sigma = 1$ the flow field is directed towards ϕ_1^*, both from below and from above. This means that

$$\begin{aligned} \Gamma_1(\phi_1^* - \Delta\phi) &\geq 0, \\ \Gamma_1(\phi_1^* + \Delta\phi) &\leq 0. \end{aligned} \tag{B.12}$$

At the same time, the relation $\Gamma_1 = -(\Gamma_0 + \Gamma_2)$ and the conditions in (B.11), imply that $\Gamma_1(\phi_1^* - \Delta\phi) = \Gamma_1(\phi_1^* + \Delta\phi)$. Together with (B.12) this means $\Gamma_1(\phi_1^* \pm \Delta\phi) = 0$, and therefore $\Gamma_0(\phi_1^* \pm \Delta\phi) = -\Gamma_2(\phi_1^* \pm \Delta\phi)$. Using again the conditions in (B.11) this can be written compactly as one single condition $\Gamma_0(\phi_1^* - \Delta\phi) = -\Gamma_2(\phi_1^* + \Delta\phi)$, again to be understood as subject to corrections of order $\Delta\phi$.

In order to impose these conditions we use a gradient-descent algorithm. Specifically, we find the coefficients a_2, b_0 and b_2 such that the function $|\Gamma_2(\phi_1^* -$

$\Delta\phi) - \Gamma_2(\phi_1^* + \Delta\phi)| + |\Gamma_0(\phi_1^* - \Delta\phi) - \Gamma_0(\phi_1^* + \Delta\phi)| + |\Gamma_1(\phi_1^* - \Delta\phi) + \Gamma_2(\phi_1^* + \Delta\phi)|$ is minimised. The last step is then to adjust the remaining coefficient a_0 such that the probability distribution is normalised [item (i) above].

The principles of the algorithm are summarised in Fig. B.2.

B.4 Extension for more than two environments of Lowest-order approximation

In this appendix, we will provide an explicit derivation of Eq. (3.32). This builds on Ref. [142], where a similar calculation is carried out for systems with two environmental states. For the purposes of this appendix, we assume that the environmental switching is independent of the state of population; i.e. the $\mu_{\sigma\to\sigma'}$ do not depend on i.

In the limit of large but finite population size N the master equation (3.13) can be expanded in powers of $1/N$ following for example [69, 142]. Writing $x = i/N$, and retaining leading and sub-leading orders one obtains an equation of the type

$$\frac{d}{dt}\Pi(x,\sigma) = L_\sigma(x)\Pi(x,\sigma) + \lambda\sum_{\sigma'}[\mu_{\sigma'\to\sigma}\Pi(x,\sigma') - \mu_{\sigma\to\sigma'}\Pi(x,\sigma)], \tag{B.13}$$

with Fokker–Planck operators

$$L_\sigma(x) = -\partial_x v_\sigma(x) + \frac{\partial_x^2\omega_\sigma(x)}{2N}, \tag{B.14}$$

where

$$\omega_\sigma(x) = a + \frac{h}{1+\alpha}\left[\alpha\left(z_\sigma + (1-2z_\sigma)\,x\right) + 2x(1-x)\right]. \tag{B.15}$$

Writing $x(t) = \phi(t) + \frac{\xi}{\sqrt{N}}$ one then finds to leading order in the expansion

$$\dot{\phi}(t) = v_\sigma(\phi). \tag{B.16}$$

Additionally making the linear-noise approximation (LNA) [316] sub-leading corrections evolve in time as follows (see [69, 142] for details),

$$\dot{\xi}(t) = v'_\sigma(\phi)\xi + \sqrt{\omega_\sigma(\phi)}\eta(t), \tag{B.17}$$

where $\eta(t)$ is Gaussian white noise of zero mean and unit amplitude. This is a Langevin equation, to be interpreted in the Itō sense. We note that the environment σ retains its time-dependence (via the switching process). Within this expansion and the LNA, the joint distribution for ϕ, ξ and σ, $\Pi(\phi,\xi,\sigma)$, evolves in time as follows,

$$\begin{aligned}\partial_t \Pi(\phi,\xi,\sigma,t) &= -v'_\sigma(\phi)\partial_\xi[\xi\Pi(\phi,\xi,\sigma,t)] - \partial_\phi[v_\sigma(\phi)\Pi(\phi,\xi,\sigma,t)] \\ &+ \frac{\omega_\sigma(\phi)}{2}\partial^2_\xi[\Pi(\phi,\xi,\sigma,t)] + \sum_{\eta\neq\sigma}\lambda\left[\mu_{\eta\to\sigma}\Pi(\phi,\xi,\eta,t) - \mu_{\sigma\to\eta}\Pi(\phi,\xi,\sigma,t)\right].\end{aligned} \tag{B.18}$$

Focusing on the stationary distribution $\Pi^*(\phi,\xi,\sigma)$, and writing $\Pi^*(\phi,\xi,\sigma) = \Pi^*(\xi|\phi,\sigma)\Pi^*(\phi,\sigma)$, we find after summing over environmental states,

$$\begin{aligned}&\sum_\sigma \Big\{\partial_\phi[v_\sigma(\phi)\Pi(\phi,\sigma)\Pi(\xi|\phi,\sigma)] + v'_\sigma(\phi)\Pi(\phi,\sigma)\partial_\xi[\xi\Pi(\xi|\phi,\sigma)] \\ &- \Pi(\phi,\sigma)\frac{\omega_\sigma(\phi)}{2}\partial^2_\xi[\Pi(\xi|\phi,\sigma)]\Big\} = 0.\end{aligned} \tag{B.19}$$

We have omitted the asterisks to keep the notation compact. We stress that Eq. (B.19) and all remaining relations in this section refer to the stationary state.

We follow [142] again, and make the assumption that instantaneous fluctuations about the PDMP trajectory does not depend on the environmental state, i.e., $\Pi(\xi|\phi,\sigma) \simeq \Pi(\xi|\phi)$. We then have

$$\begin{aligned}&\partial_\phi\left[\Pi(\xi|\phi)\left(\sum_\sigma v_\sigma(\phi)\Pi(\phi,\sigma)\right)\right] + \left[\sum_\sigma v'_\sigma(\phi)\Pi(\phi,\sigma)\right]\partial_\xi[\xi\Pi(\xi|\phi)] \\ &-\sum_\sigma\left[\Pi(\phi,\sigma)\frac{\omega_\sigma(\phi)}{2}\right]\partial^2_\xi[\Pi(\xi|\phi)] = 0\end{aligned} \tag{B.20}$$

We further know that $\sum_\sigma v_\sigma(\phi)\Pi(\phi,\sigma) = 0$, and $\Pi(\phi,\sigma) = \Pi(\sigma|\phi)\Pi(\phi)$. Eq. (B.20) can thus be re-written as

$$\begin{aligned}0 &= \sum_\sigma\left[v'_\sigma(\phi)\Pi(\sigma|\phi)\right]\Pi(\phi)\partial_\xi[\xi\Pi(\xi|\phi)] \\ &-\sum_\sigma\left[\Pi(\sigma|\phi)\Pi(\phi)\frac{\omega_\sigma(\phi)}{2}\right]\partial^2_\xi[\Pi(\xi|\phi)],\end{aligned} \tag{B.21}$$

and subsequently as

$$\begin{aligned}&\sum_\sigma\left[v'_\sigma(\phi)\Pi(\sigma|\phi)\right]\partial_\xi[\xi\Pi(\xi|\phi)] \\ &-\sum_\sigma\left[\Pi(\sigma|\phi)\frac{\omega_\sigma(\phi)}{2}\right]\partial^2_\xi[\Pi(\xi|\phi)] = 0.\end{aligned} \tag{B.22}$$

Eq. (B.22) is a stationary Fokker–Planck equation. Its solution is a Gaussian distribution

$$\Pi(\xi|\phi) = \mathcal{A}\exp\left(\frac{\xi^2}{2}\frac{\sum_\sigma v'_\sigma(\phi)\Pi(\sigma|\phi)}{\sum_\sigma \Pi(\sigma|\phi)\frac{\omega_\sigma(\phi)}{2}}\right) \tag{B.23}$$

with $\mathcal{A}$ a normalisation constant. This distribution has mean 0 and variance

$$s^2(\phi) = -\frac{\sum_\sigma \Pi(\sigma|\phi)\omega_\sigma(\phi)}{2\sum_\sigma v'_\sigma(\phi)\Pi(\sigma|\phi)}. \tag{B.24}$$

We note that this object is intrinsically non-negative in our model, given that $v'_\sigma(\phi) < 0$ for all ϕ and σ. Using $\Pi(\sigma|\phi) = \Pi(\phi,\sigma)/\Pi(\phi)$ in the numerator and in the denominator of Eq. (B.24), and cancelling the common factor $\Pi(\phi)$ we find

$$s^2(\phi) = -\frac{\sum_\sigma \Pi(\phi,\sigma)\omega_\sigma(\phi)}{2\sum_\sigma v'_\sigma(\phi)\Pi(\phi,\sigma)}. \tag{B.25}$$

For linear flow as in our model, $v_\sigma(\phi) = \lambda_c(\phi_\sigma - \phi)$, we can further simplify and find the final result

$$s^2(\phi) = \frac{\sum_\sigma \Pi(\phi,\sigma)\omega_\sigma(\phi)}{2\lambda_c \sum_\sigma \Pi(\phi,\sigma)}. \tag{B.26}$$

The distributions $\Pi(\phi,\sigma)$ are known analytically in the case of two-environmental states [see Eq. (3.27)]. For the model with three environmental states we use the numerical method described in Appendix B.3.

Appendix C

Graph distances

Consider two adjacency matrices A, and B, each with binary entries (0 or 1), describing two simple unweighted graphs with n nodes. The so-called *edit distance* [204] is a matrix distance defined as

$$d(A,B) = \sum_{i,j=1}^{n} |a_{ij} - b_{ij}|. \tag{C.1}$$

The object $d(A,B)$ counts the number of entries that are different in A and B.

For simple undirected graphs (symmetric adjacency matrices), we need to account for the fact that the number of edges is only half the number of positive entries of the adjacency matrix, and therefore $d(A,B)/2$ measures the number of edges that exist in one graph but not on the other. We have $d(A,B)/2 = 0$ if and only if $A = B$. It is also easy to see that $d(A,B)/2$ only takes integer values for symmetric adjacency matrices A and B. If $A \neq B$, then $1 \leq d(A,B)/2 \leq n(n-1)/2$. We have $d(A,B)/2 = 1$ when the two graphs are identical except for one edge, which is present in one graph and absent in the other.

One can directly use this unnormalized distance (as we do in Section 4.3.4) or subsequently normalize $d(A,B)$ using different strategies, e.g. one can divide it over $n(n-1)/2$ (as we do in Section 4.3.6), or just divide over the maximum possible distance if further restrictions are imposed between A and B (as we do in Section 4.3.5).

If we further impose that both graphs have the same number of edges, then the lower bound cannot be attained and $2 \leq d(A,B)/2$ when $A \neq B$. This lower bound is reached when we only need a single edge rewiring to get from the first graph to the second. One can thus define the *rewiring distance*

$$d(A,B) = \frac{1}{4} \sum_{i,j=1}^{n} |a_{ij} - b_{ij}|. \tag{C.2}$$

applicable for simple graphs (i.e. no self-links). This quantity measures the total number of rewirings needed to transform A into B when the associated graphs are simple, unweighted, undirected (symmetric adjacency matrices) and have the same number of nodes and edges.

The rewiring distance above is based on the concept of non-overlapping edges, i.e., edges that are present in one graph, but not in the other. Thus, the edit and rewiring distances are based on $|a_{ij} - b_{ij}|$ for the different edges, and hence assign the same importance to the presence or absence of an edge. One can instead construct measures of distance based on the number of links that are present in both networks. If the edge ij is present in both graphs then $a_{ij}b_{ij} = 1$, while this product is zero otherwise. One can prove that the following function is a distance [317]:

$$d(A, B) = 1 - \frac{1}{2|E|} \sum_{i,j=1}^{n} a_{ij}b_{ij}. \tag{C.3}$$

We replicated the analysis in Sec. 4.3.5 for the distance defined above and the results (resulting nMLE) are the same.

Appendix D

Further Analysis on Comparing Unlabelled Graphs

D.1 Graph Canonization

Graph canonization refers to the process of assigning a unique, standardized representation to a graph that is invariant under node relabelling. In essence, it aims to map all isomorphic graphs—i.e., graphs that differ only by the labels of their nodes—into a common form. A perfect canonization algorithm would enable efficient detection of graph isomorphism by reducing the problem to a simple comparison of canonical forms. However, exact canonization is computationally challenging and, for general graphs, remains an open problem in theoretical computer science.

In practice, approximate canonization procedures are often used to reduce the influence of arbitrary node labelling in applications such as graph comparison, classification, or compression. These methods typically rely on structural features of the graph to determine a consistent node ordering, from which a canonical adjacency matrix can be constructed.

In this appendix, we describe a heuristic attempt to approximate graph canonization by using node centrality measures. The goal is to reduce the sensitivity of network representations to random permutations of node labels by enforcing a consistent relabelling scheme.

Our approach is based on assigning a canonical node ordering using *betweenness centrality* [318], a metric that quantifies the importance of a node in terms of its participation in shortest paths. Formally, the betweenness centrality $C_B(v)$ of a node v is defined as

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{D.1}$$

where σ_{st} is the total number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of those paths that pass through node v.

Nodes with higher betweenness are assumed to play a more structurally central role and are ranked earlier in the new labelling.

The method proceeds as follows:

- Generate a graph G;
- Compute the betweenness centrality for all nodes;

- Sort the nodes in decreasing order of centrality and assign ranks accordingly;
- Relabel the nodes based on this ranking to obtain the canonical form $C(G)$;
- Apply the same procedure to a permuted version G' of the graph;
- Compare the adjacency matrices $A(C(G))$ and $A(C(G'))$.

We extend this procedure to a dictionary of graphs $\{G_0, G_1, \ldots\}$ and their randomly permuted versions $\{G'_0, G'_1, \ldots\}$, computing the pairwise distances:

$$d_{\text{lab}}(C(G_0), C(G_i)) \quad \text{and} \quad d_{\text{lab}}(C(G'_0), C(G'_i)). \tag{D.2}$$

As shown in Fig. D.1, the distances between graphs in the dictionary increase rapidly after canonization, quickly reaching levels comparable to those observed between random graphs. This behavior indicates that the structural relationships present in the original sequence are not preserved through the canonization process. The reason lies in the sensitivity of the betweenness-based relabeling: even small rewiring steps in the graph evolution—from G_0 to G_1, for example—can lead to substantial changes in node betweenness rankings. As a result, the canonization reclassifies nodes in a way that effectively shuffles the adjacency matrix, masking the underlying structural continuity and undermining the intended goal of creating a consistent, structure-preserving node ordering.

This result suggests that using betweenness centrality alone is not sufficient to uniquely characterize the structural identity of a graph. Similar issues arise when using alternative centrality measures alone. As such, this heuristic canonization does not provide a reliable foundation for comparing unlabelled graphs.

Future work may explore more robust strategies that combine multiple structural descriptors—such as spectral embeddings, motif counts, or neighborhood trees—or leverage graph neural representations. While the current method provides an intuitive first step, it is not well suited to our purposes, as it fails to adequately capture the continuous changes in graph structure.

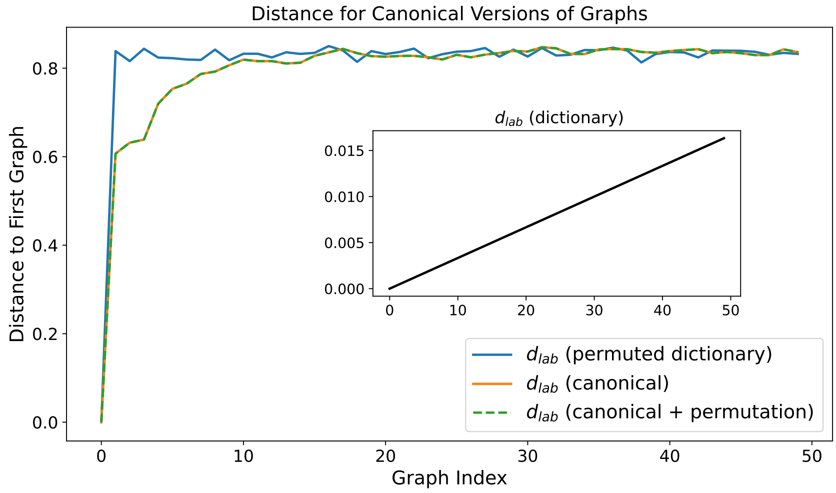

FIGURE D.1: **Comparison of d_{lab} distances to the first graph under different structural modifications.** The main plot shows the evolution of d_{lab} for three cases: permutation of node labels, canonization based on betweenness centrality, and canonization followed by permutation. The inset displays the unmodified dictionary distances in black, serving as a baseline reference. The rapid rise in distance values indicates that the canonization procedure fails to preserve the structure of the graph dictionary.

D.2 Appendix: Analytical calculation of d_{lab} in Eq. (5.5)

The distance d_{lab} in Eq. (5.4) counts the number of egdes that exist either only in G or only in G', but not in the respective other graph. Since all edges in G and G' are constructed independently, we can restrict the discussion to one focal edge. If Z is chosen as the total number of node-pairs, $N(N-1)/2$, the average distance $d_{\text{lab}}(G, G')$ reduces to

$$\langle d_{\text{lab}}(G, G') \rangle = \text{Prob}(x + \xi > p \mid x \leq p) + \text{Prob}(x + \xi \leq p \mid x > p), \tag{D.3}$$

where x is a random number drawn uniformly from the interval between zero and one, and where ξ is a Gaussian random number of mean zero and with variance σ^2. The angle bracket denotes an average over the graphs G and G', in the context of the focal edge this means the average over x and ξ. The quantity $\langle d_{\text{lab}}(G, G') \rangle$ is thus the probability that the focal edge exists in one graph but not in the other.

We write $x' = x + \xi$ and start from the joint probability density $P(x', x)$. Given that x is uniformly distributed in $[0, 1]$ we have

$$P(x', x) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x'-x)^2}{2\sigma^2}\right) & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

To find $P(x' > p \mid x \leq p)$, we integrate the joint pdf:

$$\text{Prob}(x' > p \mid x \leq p) = \int_p^\infty dx' \int_0^p dx\, P(x', x)$$

Similarly, we have

$$\text{Prob}P(x' \leq p \mid x > p) = \int_{-\infty}^p dx' \int_p^1 dx\, P(x', x).$$

These integrals can all be performed using the error function,

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-t^2)\, dt$$

and its integral,

$$\int dx\, \text{erf}(x) = x\, \text{erf}(x) + \frac{e^{-x^2}}{\pi} + \text{const.}$$

We then obtain

$$\begin{aligned} &\text{Prob}(x' > p \mid x \leq p) + \text{Prob}(x' \leq p \mid x > p) \\ = \; &\sigma\sqrt{\frac{2}{\pi}} + \frac{1}{2} - \frac{\sigma}{\sqrt{2\pi}} \left[\exp\left(- \left(\frac{p}{\sqrt{2}\,\sigma} \right)^2 \right) + \exp\left(- \left(\frac{p-1}{\sqrt{2}\sigma} \right)^2 \right) \right] \\ &-\frac{1}{2} \left[p \cdot \text{erf}\left(\frac{p}{\sqrt{2}\,\sigma} \right) + (p-1) \cdot \text{erf}\left(\frac{p-1}{\sqrt{2}\,\sigma} \right) \right]. \end{aligned} \tag{D.4}$$

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